



US012384836B2

(12) **United States Patent**
Cady et al.

(10) **Patent No.: US 12,384,836 B2**

(45) **Date of Patent: *Aug. 12, 2025**

(54) **TREATMENT OF HEADACHE USING ANTI-CGRP ANTIBODIES**

(71) Applicant: **H. LUNDBECK A/S**, Valby (DK)

(72) Inventors: **Roger K. Cady**, Bothell, WA (US);
Jeffrey T. L. Smith, Dublin (IE);
Joseph Hirman, Bothell, WA (US);
Barbara Schaeffler, Bothell, WA (US);
Lahar Mehta, Valby (DK)

(73) Assignee: **H. Lundbeck A/S**, Valby (DK)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/173,124**

(22) Filed: **Feb. 23, 2023**

(65) **Prior Publication Data**

US 2024/0067708 A1 Feb. 29, 2024

Related U.S. Application Data

(60) Division of application No. 16/793,208, filed on Feb. 18, 2020, now Pat. No. 11,639,381, which is a continuation-in-part of application No. PCT/US2020/012781, filed on Jan. 8, 2020.

(60) Provisional application No. 62/789,828, filed on Jan. 8, 2019, provisional application No. 62/872,989, filed on Jul. 11, 2019, provisional application No. 62/842,162, filed on May 2, 2019.

(51) **Int. Cl.**

C07K 16/18 (2006.01)

A61K 9/00 (2006.01)

A61K 31/4172 (2006.01)

A61K 39/00 (2006.01)

A61K 39/395 (2006.01)

A61K 45/06 (2006.01)

A61K 47/26 (2006.01)

A61P 25/06 (2006.01)

A61K 38/00 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 16/18** (2013.01); **A61K 9/0019** (2013.01); **A61K 31/4172** (2013.01); **A61K 39/3955** (2013.01); **A61K 45/06** (2013.01); **A61K 47/26** (2013.01); **A61P 25/06** (2018.01); **A61K 38/00** (2013.01); **A61K 2039/505** (2013.01); **A61K 2039/545** (2013.01); **C07K 2317/24** (2013.01)

(58) **Field of Classification Search**

CPC .. **C07K 16/18**; **C07K 2317/24**; **A61K 9/0019**; **A61K 31/4172**; **A61K 39/3955**; **A61K 45/06**; **A61K 47/26**; **A61K 38/00**; **A61K 2039/505**; **A61K 2039/545**; **A61P 25/06**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,116,964 A	5/1992	Capon et al.
5,266,561 A	11/1993	Cooper et al.
5,364,841 A	11/1994	Cooper et al.
5,585,089 A	12/1996	Queen et al.
5,624,821 A	4/1997	Winter et al.
5,648,260 A	7/1997	Winter et al.
5,942,227 A	8/1999	Cooper et al.
6,180,370 B1	1/2001	Queen et al.
6,313,097 B1	11/2001	Eberlein et al.
6,509,014 B1	1/2003	De Lacharriere et al.
6,521,609 B1	2/2003	Doods et al.
6,737,056 B1	5/2004	Presta
6,956,107 B2	10/2005	Fung et al.
7,279,471 B2	10/2007	Mueller et al.
7,479,488 B2	1/2009	Mueller et al.
7,696,209 B2	4/2010	Mueller et al.
7,700,735 B2	4/2010	Young et al.
7,879,991 B2	2/2011	Vater et al.
7,927,863 B2	4/2011	Cregg et al.
7,935,340 B2	5/2011	Garcia-Martinez et al.
8,007,794 B2	8/2011	Zeller et al.
8,293,239 B2	10/2012	Poulsen et al.
8,298,536 B2	10/2012	Poulsen et al.
8,586,045 B2	11/2013	Zeller et al.
8,597,649 B2	12/2013	Zeller et al.
8,623,366 B2	1/2014	Pios et al.
8,734,802 B1	5/2014	Zeller et al.
9,073,991 B2	7/2015	Allan et al.
9,708,393 B2	7/2017	Russo et al.
9,745,373 B2	8/2017	Kovacevich et al.
9,855,332 B2	1/2018	Russo et al.
10,066,009 B2	9/2018	Kovacevich et al.
10,179,809 B2	1/2019	Kovacevich et al.

(Continued)

FOREIGN PATENT DOCUMENTS

AU	2006313434	5/2007
CA	2611433	12/2006

(Continued)

OTHER PUBLICATIONS

Alstadhaug, Karl B et al. "Preventing and treating medication overuse headache." Pain reports vol. 2,4 e612. Jul. 26, 2017, doi:10.1097/PR9.0000000000000612.

(Continued)

Primary Examiner — Olga N Chernyshev

(74) *Attorney, Agent, or Firm* — Robin L. Teskin; Baker, Donelson, Bearman, Caldwell & Berkowitz PC

(57) **ABSTRACT**

Methods for immediate relief of migraine or headache are provided comprising the administration of an anti-CGRP antagonist antibody to a patient in need thereof.

21 Claims, 64 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited**U.S. PATENT DOCUMENTS**

10,189,895	B2	1/2019	Kovacevich et al.	
10,208,112	B2	2/2019	Kovacevich et al.	
10,214,582	B2	2/2019	Kovacevich et al.	
10,266,587	B2	4/2019	Russo et al.	
10,533,048	B2	1/2020	Kovacevich et al.	
11,639,380	B2 *	5/2023	Cady	A61K 9/0019 424/130.1
11,639,381	B2 *	5/2023	Cady	A61K 47/26 424/133.1
2001/0036647	A1	11/2001	Choudary et al.	
2002/0162125	A1	10/2002	Salmon et al.	
2002/0164707	A1	11/2002	Adamou et al.	
2003/0027213	A1	2/2003	Zhu et al.	
2003/0181462	A1	9/2003	Doods et al.	
2003/0194404	A1	10/2003	Greenfeder et al.	
2004/0110170	A1	6/2004	Pisegna et al.	
2004/0132824	A1	7/2004	Gil et al.	
2005/0234054	A1	10/2005	Mueller et al.	
2006/0183700	A1	8/2006	Vater et al.	
2006/0270045	A1	11/2006	Cregg et al.	
2009/0023644	A1	1/2009	Southard et al.	
2009/0028784	A1	1/2009	Garcia-Martinez et al.	
2009/0220489	A1	9/2009	Zeller et al.	
2010/0152171	A1	6/2010	Rudolf et al.	
2011/0054150	A1	3/2011	Poulsen et al.	
2011/0257371	A1	10/2011	Poulsen et al.	
2011/0305711	A1	12/2011	Allan et al.	
2012/0000192	A1	1/2012	Zeller et al.	
2012/0114741	A1	5/2012	Aung-Din	
2012/0225075	A1	9/2012	Pios et al.	
2012/0294797	A1	11/2012	Kovacevich et al.	
2012/0294802	A1	11/2012	Russo et al.	
2012/0294822	A1	11/2012	Russo et al.	
2013/0216535	A1	8/2013	Zeller et al.	
2013/0295087	A1	11/2013	Poulsen et al.	
2013/0295088	A1	11/2013	Poulsen et al.	
2015/0266948	A1	9/2015	Bigal et al.	
2017/0088612	A1	3/2017	Bigal	
2017/0174754	A1	6/2017	Kovacevich et al.	
2018/0127490	A1	5/2018	Bigal et al.	
2018/0142029	A1	5/2018	Boone et al.	
2018/0161434	A1	6/2018	Russo et al.	
2019/0211085	A1	7/2019	Kovacevich et al.	
2019/0240331	A1	8/2019	Russo et al.	
2019/0367590	A1	12/2019	Russo et al.	
2020/0010537	A1	1/2020	Baker et al.	
2020/0216524	A1	7/2020	Cady et al.	
2020/0216525	A1	7/2020	Cady et al.	

FOREIGN PATENT DOCUMENTS

CA	2626120	12/2012
CN	101309704	11/2008
CN	101979650	2/2011
CN	103421114	12/2013
EA	015526	10/2008
EP	0212432	3/1987
EP	1031350	8/2000
EP	1770091	4/2007
EP	1556020	2/2009
EP	1957106	10/2013
JP	Hei6-87890	3/1994
JP	08-268874	10/1996
JP	2005523418	8/2005
JP	2007517911	7/2007
JP	2009-515942	4/2009
JP	2011046710	3/2011
JP	2011513386	4/2011
JP	2011513387	4/2011
JP	5123197	1/2013
JP	2014-517699	7/2014
JP	2017-515579	6/2017
KR	10-1250049	4/2013
RU	2329062	7/2008

WO	WO 1996/0004928	2/1996
WO	WO 97/09046	3/1997
WO	WO 98/09630	3/1998
WO	WO 98/11128	3/1998
WO	WO 98/56779	12/1998
WO	WO 00/18764	4/2000
WO	WO 2001/022972	4/2001
WO	WO 2003/045424	6/2003
WO	WO 2003/093472	11/2003
WO	WO 03/104236	12/2003
WO	WO 2004/003019	1/2004
WO	WO 2004/014351	2/2004
WO	WO 2004/050683	6/2004
WO	WO 2004/058184	7/2004
WO	WO 2004/082602	9/2004
WO	WO 2004/082605	9/2004
WO	WO 2004/082678	9/2004
WO	WO 2004/083187	9/2004
WO	WO 2004/087649	10/2004
WO	WO 2004/091514	10/2004
WO	WO 2004/092166	10/2004
WO	WO 2004/092168	10/2004
WO	WO 2004/096122	11/2004
WO	WO 2004/097421	11/2004
WO	WO 2005/009962	2/2005
WO	WO 2005/040395	5/2005
WO	WO 2005/041757	5/2005
WO	WO 2005/070444	8/2005
WO	WO 2005/100360	10/2005
WO	WO 2006/077212	7/2006
WO	WO 2007/025212	3/2007
WO	WO 2007/048026	4/2007
WO	WO 2007/054800	5/2007
WO	WO 2007/054809	5/2007
WO	WO 2007/061676	5/2007
WO	WO 2007/076336	7/2007
WO	WO 2007/141285	12/2007
WO	WO 2008/011190	1/2008
WO	2008144757	11/2008
WO	WO 2009/109908	9/2009
WO	WO 2009/109911	9/2009
WO	WO 2010/075238	7/2010
WO	WO 2011/024113	3/2011
WO	WO 2011/156324	12/2011
WO	2012162243	11/2012
WO	2015143409	9/2015
WO	2015173539	11/2015
WO	2016171742	10/2016
WO	2016205037	12/2016
WO	2017186928	11/2017
WO	2018/055574	3/2018
WO	2020146527	7/2020

OTHER PUBLICATIONS

[No Author Attributed] Clinical Trial No. LY2951742, started Mar. 2015, "A Study of LY2951742 in Participants With Episodic Cluster Headache," from ClinicalTrials.gov [database online], Retrieved from the Internet: <<https://clinicaltrials.gov/ct2/show/study/NCT02397473?term=LY2951742&rank=9>>, retrieved Sep. 3, 2016. 6 pages.

[No Author Attributed] [machine translated from website] "Dysfunction of the temporomandibular joint," as published on the Colgate-Palmolive Company website [online], Retrieved from the Internet: <<http://www.colgate.ru/ru/oc/oral-health/conditions/temporomandibular-disorder>> 2017; 7 pages.

Androulakis, X Michelle et al. "Central Executive and Default Mode Network Intranet work Functional Connectivity Patterns in Chronic Migraine." Journal of neurological disorders vol. 6,5 (2018): 393. doi:10.4172/2329-6895.1000393.

Carlsen, Louise Ninett, et al. "Complete detoxification is the most effective treatment of medication overuse headache: a randomized controlled open-label trial." Cephalalgia 38.2 (2018): 225-236.

Cevoli, Sabina, et al. "Family history for chronic headache and drug overuse as a risk factor for headache chronification." Headache: The Journal of Head and Face Pain 49.3 (2009): 412-418.

(56)

References Cited**OTHER PUBLICATIONS**

- Chen, Zhiye, et al. "Altered functional connectivity architecture of the brain in medication overuse headache using resting state fMRI." *The Journal of Headache and Pain* 18.1 (2017): 1-9.
- Iranian Office Action dated Apr. 15, 2022, for Pat. Appl. No. 140050140003002468, filed Jun. 15, 2021 entitled "Treatment of Medication Overuse Headache Using AntiCGRP or Anti-CGRP-R Antibodies".
- Iranian Office Action dated Feb. 7, 2022, for Pat. Appl. No. 140050140003002305, filed Jun. 9, 2021 entitled "Acute Treatment and Rapid Treatment of Headache Using Anti-CGRP Antibodies".
- Ferrari, Anna, et al. "Need for analgesics/drugs of abuse: a comparison between headache patients and addicts by the Leeds Dependence Questionnaire (LDQ)." *Cephalalgia* 26.2 (2006): 187-193.
- Ferraro, Stefania et al. "In medication overuse headache, fMRI shows long-lasting dysfunction in midbrain areas." *Headache* vol. 52,10 (2012): 1520-34. doi:10.1111/j.1526-4610.2012.02276.x.
- Find, Ninette Louise et al. "Medication overuse headache in Europe and Latin America: general demographic and clinical characteristics, referral pathways and national distribution of painkillers in a descriptive, multinational, multicenter study." *The journal of headache and pain* 17.1 (2016): 1-12.
- Fuh, Jong-Ling et al. "Does medication overuse headache represent a behavior of dependence?." *Pain* vol. 119,1-3 (2005): 49-55. doi:10.1016/j.pain.2005.09.034.
- Fumal, Arnaud, et al. "Orbitofrontal cortex involvement in chronic analgesic-overuse headache evolving from episodic migraine." *Brain* 129.2 (2006): 543-550.
- Grande, Ragnhild Berling, et al. "The Severity of Dependence Scale detects people with medication overuse: the Akershus study of chronic headache." *Journal of Neurology, Neurosurgery & Psychiatry* 80.7 (2009): 784-789.
- Kopruszinski et al. "Prevention of stress-or nitric oxide donor-induced medication overuse headache by a calcitonin gene-related peptide antibody in rodents." *Cephalalgia*. May 2017;37(6):560-70.
- Lai, Tzu-Hsien, et al. "Gray matter changes related to medication overuse in patients with chronic migraine." *Cephalalgia* 36.14 (2016): 1324-1333.
- Lundqvist, C., et al. "An adapted Severity of Dependence Scale is valid for the detection of medication overuse: the Akershus study of chronic headache." *European Journal of Neurology* 18.3 (2011): 512-518.
- Lundqvist, Christofer, et al. "The severity of dependence score correlates with medication overuse in persons with secondary chronic headaches. The Akershus study of chronic headache." *Pain®* 148.3 (2010): 487-491.
- Newman-Norlund, Roger D., et al. "Cortical and subcortical changes following sphenopalatine ganglion blocks in chronic migraine with medication overuse headache: a preliminary longitudinal study." *Women's midlife health* 6.1 (2020): 1-8.
- Riederer, Franz, et al. "Decrease of gray matter volume in the midbrain is associated with treatment response in medication-overuse headache: possible influence of orbitofrontal cortex." *Journal of Neuroscience* 33.39 (2013): 15343-15349.
- Riederer, Franz, et al. "Grey matter changes associated with medication-overuse headache: correlations with disease related disability and anxiety." *The world journal of biological psychiatry* 13.7 (2012): 517-525.
- Torta, D. M., et al. "Nucleus accumbens functional connectivity discriminates medication-overuse headache." *NeuroImage: Clinical* 11 (2016): 686-693.
- "Headache Classification Committee of the International Headache Society (IHS) The International Classification of Headache Disorders, 3rd edition." *Cephalalgia : an international journal of headache* vol. 38,1 (2018): 1-211. doi:10.1177/0333102417738202.
- Scuteri et al. "New trends in migraine pharmacology: targeting calcitonin gene-related peptide (CGRP) with monoclonal antibodies." *Frontiers in pharmacology*. Apr. 9, 2019;10:363.
- Winner et al. "Effects of Intravenous Eptinezumab vs Placebo on Headache Pain and Most Bothersome Symptom When Initiated During a Migraine Attack: A Randomized Clinical Trial." *JAMA*. Jun. 15, 2021;325(23):2348-56.
- Goadsby et al. *Pathophysiology of Migraine: A Disorder of Sensory Processing*. Physiological reviews. Apr. 1997 (2):553-622, 1997.
- Messlinger et al. "The Big CGRP Flood-sources, Sinks and Signalling Sites in the Trigemino-vascular System," *The Journal of Headache and Pain*. Dec. 2018;19(1):1-7.
- Kumar et al. "Protective role of α -calcitonin gene-related peptide in cardiovascular diseases." *Frontiers in physiology*. Jul. 2, 2019;10:821.
- Van Dongen et al. "Migraine biomarkers in cerebrospinal fluid: A systematic review and meta-analysis." *Cephalalgia*. Jan. 2017;37(1):48-63.
- Christensen et al. "Migraine induction with calcitonin gene-related peptide in patients from erenumab trials." *The Journal of Headache and Pain*. Dec. 2018;19(1):1-9.
- Covasala et al. "Calcitonin gene-related peptide receptors in rat trigeminal ganglion do not control spinal trigeminal activity." *Journal of neurophysiology*. Jul. 15, 2012;108(2):431-40.
- Storer et al. "Calcitonin gene-related peptide (CGRP) modulates nociceptive trigemino-vascular transmission in the cat." *British journal of pharmacology*. Aug. 2004;142(7):1171-81.
- Burstein et al. "The neurobiology of photophobia." *Journal of neuro-ophthalmology: the official journal of the North American Neuro-Ophthalmology Society*. Mar. 2019;39(1):94.
- Wang et al. "Monoclonal antibody exposure in rat and cynomolgus monkey cerebrospinal fluid following systemic administration." *Fluids and Barriers of the CNS*. Dec. 2018;15(1):1-0.
- Kelman L. "Pain characteristics of the acute migraine attack." *Headache: The Journal of Head and Face Pain*. Jun. 2006;46(6):942-53.
- "Cluster Headache," Wolff's Headache 1974, p. 348.
- "Highlights of Prescribing Information" BLA STN 103000/5215—FDA Approved Labeling Text, Botox Package Insert, Oct. 2010, 25 pages.
- "Teva to Acquire Labrys Biologics, Inc.: Novel Migraine Prophylaxis Treatment Adds Significant New Dimension to Teva's Growing Pain Care Franchise" "Business Wire Jun. 3, 2014," 4 pages.
- "TMJ Disorders," National Institute of Dental and Craniofacial Research, NIH Publication No. 15-3487, Apr. 2015. 20 pages.
- Abdiche YN, et al. "Probing the binding mechanism and affinity of tanezumab, a recombinant humanized anti-NGF monoclonal antibody, using a repertoire of biosensors," *Protein Sci*. Aug. 2008;17(8):1326-35.
- Adwanikar H, et al. Spinal CGRP1 receptors contribute to supraspinally organized pain behavior and pain-related sensitization of amygdala neurons. *Pain*. Nov. 2007;132(1-2):53-66. Epub Mar. 1, 2007.
- Akerman S, et al. "Nitric oxide synthase inhibitors can antagonize neurogenic and calcitonin gene-related peptide induced dilation of dural meningeal vessels," *Br J Pharmacol*. Sep. 2002;137(1):62-8.
- Akerman, S., et al. "Pearls and pitfalls in experimental in vivo models of migraine: dural trigemino-vascular nociception," *Cephalalgia*. Jun. 2013;33(8):577-92.
- Alder Biopharmaceuticals Inc., "Alder Presents Positive ALD403 Clinical Data at European Headache and Migraine Trust International Congress," Press Release, Sep. 15, 2016.
- Alder Biopharmaceuticals Inc., "Alder Presents Positive Clinical Data for ALD403 at the 17th Congress of the International Headache Society" Press Release, May 15, 2015. (3 pages).
- Alder Biopharmaceuticals Inc., "Alder Reports Phase 2b Trial of ALD403 Meets Primary and Secondary Endpoints Demonstrating Migraine Prevention in Patients with Chronic Migraine," Press Release, Mar. 28, 2016. (4 pages).
- Alder Biopharmaceuticals Inc., "Alder Reports Positive Top-Line 24-Week Data Demonstrating Persistent Migraine Prevention in Phase 2b Study of ALD403 in Patients with Chronic Migraine" Press Release, Jul. 25, 2016. (3 pages).
- Alder Biopharmaceuticals Inc., "Data From Proof-of-Concept Clinical Trial of ALD403, a Monoclonal Antibody Against CGRP for the Prevention of Migraine, to be Presented at 56th Annual Scientific Meeting of the American Headache Society," Press Release, Jun. 26, 2014. 2 pages.

(56)

References Cited

OTHER PUBLICATIONS

- Almagro JC et al. "Chapter 13 Antibody Engineering: Humanization, Affinity Maturation, and Selection Techniques." *Therapeutic Monoclonal Antibodies: From Bench to Clinic* (Zhiqiang An (Editor)) Oct. 2009: 311-34.
- Amara SG, et al. "Expression in brain of a messenger RNA encoding a novel neuropeptide homologous to calcitonin gene-related peptide." *Science*. Sep. 13, 1985;229(4718):1094-7.
- Ambalavanar R., et al. "Deep tissue inflammation upregulates neuropeptides and evokes nociceptive behaviors which are modulated by a neuropeptide antagonist." *Pain*. Jan. 2006;120(1-2):53-68. Epub Dec. 13, 2005.
- Amrutar DV. "Calcitonin gene-related peptide (CGRP) uptake and release in rat dura mater, trigeminal ganglion and trigeminal nucleus caudalis." PHD thesis, Faculty of Health and Medical Sciences University of Copenhagen, Academic advisor: Inger Jansen-Olesen and Jes Olesen, Submitted: Feb. 20, 2013.
- An Z. "Therapeutic Monoclonal Antibodies: From Bench to Clinic." Wiley & Sons, Inc., 2009 Chapter 31, 711-62.
- Andersen DC, et al. "Production technologies for monoclonal antibodies and their fragments." *Curr Opin Biotechnol*. Oct. 2004;15(5):456-62.
- Andrew DP, et al. "Monoclonal antibodies distinguishing alpha and beta forms of calcitonin gene-related peptide." *J Immunol Methods*. Nov. 6, 1990;134(1):87-94.
- Antibody Structure and Function, Chapter 4 of Elgert's Immunology: Understanding the Immune System, pp. 58-78. Wiley 1998.
- Aoki KR. "Review of a proposed mechanism for the antinociceptive action of botulinum toxin type A." *Neurotoxicology*, Oct. 2005;26(5):785-93.
- Aoki-Nagase T, et al. "Attenuation of antigen-induced airway hyperresponsiveness in CGRP-deficient mice." *Am J Physiol Lung Cell Mol Physiol*. Nov. 2002;283(5):L963-70.
- Armour KL, et al. "Recombinant human IgG molecules lacking Fc gamma receptor I binding and monocyte triggering activities." *Eur J Immunol*. Aug. 1999;29(8):2613-24.
- Arulmani U, et al. "Calcitonin gene-related peptide and its role in migraine pathophysiology." *Eur J Pharmacol*. Oct. 1, 2004;500(1-3):315-30.
- Arulmani U, et al. "Experimental migraine models and their relevance in migraine therapy." *Cephalalgia*. Jun. 2006;26(6):642-59.
- Arulmozhi DK, et al. "Migraine: current concepts and emerging therapies." *Vascul Pharmacol*. Sep. 2005;43(3):176-87.
- Asgar, MS, et al. "Evidence for a vascular factor in migraine." *Ann Neurol*. Apr. 2011;69(4):635-45.
- Ashina M. "Vascular changes have a primary role in migraine." *Cephalalgia*. Apr. 2012;32(5):428-30.
- Ashina M, et al. "Evidence for increased plasma levels of calcitonin gene-related peptide in migraine outside of attacks." *Pain*. May 2000;86(1-2):133-8.
- Ashina M, et al. "Peaks and pitfalls in human pharmacological models of migraine: 30 years' experience." *Cephalalgia*. Jun. 2013;33(8):540-53.
- Ashina M, et al. "Plasma levels of calcitonin gene-related peptide in chronic tension-type headache." *Neurology*. Nov. 14, 2000;55(9):1335-40.
- Ashina M. "Calcitonin gene-related peptide in tension-type headache." *ScientificWorldJournal*. Jun. 7, 2002;2:1527-31.
- Aziz Q., "Visceral hypersensitivity: fact or fiction." *Gastroenterology*. Aug. 2006; 131(2):661-4.
- Bagdy, G, et al. "Headache-type adverse effects of NO donors: vasodilation and beyond." *Br J Pharmacol*. May 2010;160(1):20-35.
- Balint RF, et al. "Antibody engineering by parsimonious mutagenesis." *Gene*. Dec. 27, 1993;137(1):109-18.
- Barker JN, et al. "Progress in psoriasis. Psoriasis: from gene to clinic. London, UK, Dec. 5-7, 1996." *Mol Med Today*. May 1997;3(5):193-4.
- Batra SK, et al. "Pharmacokinetics and biodistribution of genetically engineered antibodies." *Curr Opin Biotechnol*. Dec. 2002; 13(6):603-8.
- Baxter LT, et al. "Biodistribution of monoclonal antibodies: scale-up from mouse to human using a physiologically based pharmacokinetic model." *Cancer Res*. Oct. 15, 1995;55(20):4611-22.
- Bell RD, et al. "Breaching the blood-brain barrier for drug delivery." *Neuron*. Jan. 8, 2014;81(1):1-3.
- Benarroch EE. "CGRP: sensory neuropeptide with multiple neurologic implications." *Neurology*. Jul. 19, 2011;77(3):281-7.
- Benemei S, et al. "CGRP receptors in the control of pain and inflammation." *Curr Opin Pharmacol*. Feb. 2009;9(1):9-14.
- Benemei S, et al. "Migraine." *Handb Exp Pharmacol*. 2009;(194):75-89.
- Benemei S, et al. "Pain pharmacology in migraine: focus on CGRP and CGRP receptors." *Neurol Sci*. May 2007;28 Suppl 2:S89-93.
- Benincosa LJ, et al. "Pharmacokinetics and Pharmacodynamics of a Humanized Monoclonal Antibody to Factor IX in Cynomolgus Monkeys." *J Pharmacol Exp Ther*. Feb. 2000;292(2):810-6.
- Bennett AD, et al. "Alleviation of mechanical and thermal allodynia by CGRP(8-37) in a rodent model of chronic central pain." *Pain*. May 2000;86(1-2):163-75.
- Benschop U.S. Appl. No. 60/753,044, filed Dec. 22, 2005, File History, 48 pages.
- Biacore 3000 Instrument Handbook, Mar. 1999. 201 pages.
- Bigal and Krymchanski, "Emerging drugs for migraine prophylaxis and treatment." *Med. Gen. Med*. 2006;8(2):31.
- Bigal M. "Clinical Trials Update—2012; Year in Review—A Comment" *Headache*. Jun. 2013;53(6):1003-4.
- Bigal ME, et al. "Emerging drugs for migraine prophylaxis and treatment." *MedGenMed*. May 4, 2006;8(2):31.
- Bigal ME, et al. "Ergotamine and dihydroergotamine: a review." *Curr Pain Headache Rep*. Feb. 2003;7(1):55-62.
- Bigal ME, et al. "Headache prevention outcome and body mass index." *Cephalalgia*. Apr. 2006;26(4):445-50.
- Bigal ME, et al. "Migraine in the Triptan Era: Lessons From Epidemiology, Pathophysiology, and Clinical Science." *Headache*. Feb. 2009;49 Suppl 1:S21-33.
- Bigal ME, et al. "Migraine in the triptan era: progresses achieved, lessons learned and future developments." *Arq Neuropsiquiatr*. Jun. 2009;67(2B):559-69.
- Bigal ME, et al. "Modifiable risk factors for migraine progression." *Headache*. Oct. 2006;46(9):1334-43.
- Bigal ME, et al. "Monoclonal Antibodies for Migraine: Preventing Calcitonin Gene-Related Peptide Activity." *CNS Drugs*. May 2014;28(5):389-99.
- Bigal ME, et al. "New developments in migraine prophylaxis." *Expert Opin Pharmacother*. Apr. 2003;4(4):433-43.
- Bigal ME, et al. "New migraine preventive options: an update with pathophysiological considerations." *Rev Hosp Clin Fac Med Sao Paulo*. Nov.-Dec. 2002;57(6):293-8.
- Bigal ME, et al. "Obesity and migraine: a population study." *Neurology*. Feb. 28, 2006;66(4):545-50.
- Bigal ME, et al. "Obesity is a risk factor for transformed migraine but not chronic tension-type headache." *Neurology*. Jul. 25, 2006;67(2):252-7.
- Bigal ME, et al. "Prophylactic migraine therapy: emerging treatment options." *Curr Pain Headache Rep*. Jun. 2004;8(3):178-84.
- Bigal ME, et al. "Safety and tolerability of LBR-101, a humanized monoclonal antibody that blocks the binding of CGRP to its receptor: Results of the Phase 1 program." *Cephalalgia*. Dec. 23, 2013;34(7):483-492.
- Bigal ME, et al. "Safety, tolerability, and efficacy of TEV-48125 for preventive treatment of high-frequency episodic migraine: a multicentre, randomised, double-blind, placebo-controlled, phase 2b study." *Lancet Neurol*. Nov. 2015;14(11):1081-90.
- Bigal ME, et al. "The preventive treatment of migraine." *Neurologist*. Jul. 2006;12(4):204-13.
- Bigal ME, et al. "The triptans." *Expert Rev Neurother*. May 2009;9(5):649-59.
- Bigal, ME "Glutamate Receptor Antagonists." *Headache Currents*, 1:20-21. Jul. 2004.
- Birder L, et al. "Neural control of the lower urinary tract: peripheral and spinal mechanisms." *Neurol Urodyn*. 2010;29(1):128-39.

(56)

References Cited

OTHER PUBLICATIONS

- Boeckh M, et al. "Phase 1 Evaluation of the Respiratory Syncytial Virus-Specific Monoclonal Antibody Palivizumab in Recipients of Hematopoietic Stem Cell Transplants," *J Infect Dis.* Aug. 1, 2001;184(3):350-4.
- Bolay H, et al. "Intrinsic brain activity triggers trigeminal meningeal afferents in a migraine model," *Nat Med.* Feb. 2002;8(2):136-42.
- Brain SD, et al. "CGRP receptors: a headache to study, but will antagonists prove therapeutic in migraine?" *Trends Pharmacol Sci.* Feb. 2002;23(2):51-3.
- Brain SD, et al. "Vascular actions of calcitonin gene-related peptide and adrenomedullin," *Physiol Rev.* Jul. 2004;84(3):903-34.
- Brekke OH, et al. "Therapeutic Antibodies For Human Diseases At The Dawn Of The Twenty-First Century," *Nat Rev Drug Discov.* Jan. 2003;2(1):52-62.
- Brorson K, et al. "Mutational analysis of avidity and fine specificity of anti-levan antibodies," *J Immunol.* Dec. 15, 1999;163(12):6694-701.
- Brüggemann M, et al. "The Immunogenicity Of Chimeric Antibodies," *J Exp Med.* Dec. 1, 1989;170(6):2153-7.
- Brummell DA, et al. "Probing the combining site of an anti-carbohydrate antibody by saturation-mutagenesis: role of the heavy-chain CDR3 residues," *Biochemistry.* Feb. 2, 1993;32(4):1180-7.
- Buckley TL, et al. "The partial inhibition of inflammatory responses induced by capsaicin using the Fab fragment of a selective calcitonin gene-related peptide antiserum in rabbit skin," *Neuroscience.* Jun. 1992;48(4):963-8.
- Burks EA, "In vitro scanning saturation mutagenesis of an antibody binding pocket," *Proc Natl Acad Sci U S A.* Jan. 21, 1997;94(2):412-7.
- Buzzi MG, et al. "The antimigraine drug, sumatriptan (GR43175), selectively blocks neurogenic plasma extravasation from blood vessels in dura mater," *Br J Pharmacol.* Jan. 1990;99(1):202-6.
- Carter PJ. "Potent antibody therapeutics by design," *Nat Rev Immunol.* May 2006;6(5):343-57.
- Casset F, et al. "A peptide mimetic of an anti-CD4 monoclonal antibody by rational design," *Biochem Biophys Res Commun.* Jul. 18, 2003;307(1):198-205.
- Castañó A, et al. "Headache in symptomatic intracranial hypertension secondary to leptospirosis: a case report," *Cephalalgia.* Apr. 2005;25(4):309-11.
- Cernuda-Morollón E, et al. "CGRP and VIP levels as predictors of efficacy of Onabotulinumtoxin type A In chronic migraine," *Headache.* Jun. 2014;54(6):987-95.
- Chancellor MB, et al. "Neurophysiology of stress urinary incontinence," *Rev Urol.* 2004;6 Suppl 3:819-28.
- Charbit, A et al. "Dopamine: what's new in migraine?" *Curr Opin Neurol.* Jun. 2010;23(3):275-81.
- Charles A, "Migraine is not primarily a vascular disorder," *Cephalalgia.* Apr. 2012;32(5):431-2.
- Chauhan M, et al. "Studies on the effects of the N-terminal domain antibodies of calcitonin receptor-like receptor and receptor activity-modifying protein 1 on calcitonin gene-related peptide-induced vasorelaxation in rat uterine artery," *Biol Reprod.* Jun. 2004;70(6):1658-63.
- Chen JT, et al. "Menopausal flushes and calcitonin-gene-related peptide," *Lancet.* Jul. 3, 1993;342(8862):49.
- Chen Y, et al. "Selection and analysis of an optimized anti-VEGF antibody: crystal structure of an affinity-matured Fab in complex with antigen," *J Mol Biol.* Nov. 5, 1999;293(4):865-81.
- Cheung B et al. "Adrenomedullin: Its Role in the Cardiovascular System," *Semin Vasc Med.* May 2004;4(2):129-34.
- Chowdhury PS, et al. "Tailor-made antibody therapeutics," *Methods.* May 2005;36(1):11-24.
- Chuang YC, et al. "Intraprostatic botulinum toxin a injection inhibits cyclooxygenase-2 expression and suppresses prostatic pain on capsaicin induced prostatitis model in rat," *J Urol.* Aug. 2008;180(2):742-8.
- Chuang YC, et al. "Urodynamic and immunohistochemical evaluation of intravesical botulinum toxin A delivery using liposomes," *J Urol.* Aug. 2009;182(2):786-92.
- Cianchetti C. "The role of the neurovascular scalp structures in migraine," *Cephalalgia.* Jul. 2012; 32(10):778-84.
- Colcher D, et al. "Pharmacokinetics and biodistribution of genetically-engineered antibodies," *Q J Nucl Med.* Dec. 1998;42(4):225-41.
- Colman PM. "Effects of amino acid sequence changes on antibody-antigen interactions," *Res Immunol.* Jan. 1994;145(1):33-6.
- Conner AC, et al. "Interaction of calcitonin-gene-related peptide with its receptors," *Biochem Soc Trans.* Aug. 2002;30(4):451-5.
- Conner AC, et al. "Ligand binding and activation of the CGRP receptor," *Biochem Soc Trans.* Aug. 2007;35(Pt 4):729-32.
- Connor K M et al: "Randomized, controlled trial of telcagepant for the acute treatment of migraine," *Neurology* Sep. 22, 2009, vol. 73, No. 12, Sep. 22, 2009 (Sep. 22, 2009), pp. 970-977, XP002732737, ISSN: 1526-632X.
- Correia IR. "Stability of IgG isotypes in serum," *MAbs.* May-Jun. 2010;2(3):221-32.
- Cottrell GS, et al. "Localization of calcitonin receptor-like receptor (CLR) and receptor activity-modifying protein 1 (RAMPI) in human gastrointestinal tract," *Peptides.* Jun. 2012;35(2):202-11.
- Covell DG, et al. "Pharmacokinetics of monoclonal immunoglobulin G1, F(ab)2, and Fab' in mice," *Cancer Res.* Aug. 1986;46(8):3969-78.
- Cutrer F. "Pathophysiology of Migraine," *Semin Neurol.* Apr. 2006;26(2):171-80.
- Cutrer F. "Pathophysiology of Migraine," *Semin Neurol.* Apr. 2010;30(2):120-30.
- Dakhama A, et al. "Calcitonin gene-related peptide: role in airway homeostasis," *Curr Opin Pharmacol.* Jun. 2004;4(3):215-20.
- Davies J, et al. "Affinity improvement of single antibody VH domains: residues in all three hypervariable regions affect antigen binding," *Immunotechnology.* Sep. 1996;2(3):169-79.
- Davis CD et al. "The Tortuous Road to an Ideal CGRP Function Blocker for the Treatment of Migraine," *Curr Top Med Chem.* 2008;8(16):1468-79.
- Davietov B, et al. "Beyond Botox: advantages and limitations of individual botulinum neurotoxins," *Trends Neurosci.* Aug. 2005;28(8):446-52.
- De Pascalis R, et al. "Grafting of "abbreviated" complementarity-determining regions containing specificity-determining residues essential for ligand contact to engineer a less immunogenic humanized monoclonal antibody," *J Immunol.* Sep. 15, 2002;169(6):3076-84.
- Delafay L, et al. "Interactive involvement of brain derived neurotrophic factor, nerve growth factor, and calcitonin gene related peptide in colonic hypersensitivity in the rat," *Gut.* Jul. 2006;55(7):940-5. *Epub Jan. 9, 2006.*
- Denekas T, et al. "Inhibition of stimulated meningeal blood flow by a calcitonin gene-related peptide binding mirror-image RNA oligonucleotide," *Br J Pharmacol.* Jun. 2006;148(4):536-43.
- Deng R et al. "Projecting human pharmacokinetics of therapeutic antibodies from nonclinical data," *MAbs.* Jan.-Feb. 2011;3(1):61-6.
- Derosa G, et al. "Optimizing combination treatment in the management of type 2 diabetes," *Vasc Health Risk Manag.* 2007;3(5):665-71.
- Diamond S, et al. "Patterns of diagnosis and acute and preventive treatment for migraine in the United States: results from the American Migraine Prevalence and Prevention study," *Headache.* Mar. 2007;47(3):355-63.
- Diener HC, et al. "Utility of topiramate for the treatment of patients with chronic migraine in the presence or absence of acute medication overuse," *Cephalalgia.* Oct. 2009;29(10):1021-7.
- Dockray et al., "Immunoneutralization studies with calcitonin gene-related peptide," *Ann. NY Acad Sci.* 1992;657:258-67.
- Dodick D, et al. "Cluster Headache: Diagnosis, Management and Treatment," *Wolff's Headache* 2001, p. 283.
- Dodick DW, et al. "Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial," *Lancet Neurol.* Nov. 2014;13(11):1100-7.

(56)

References Cited**OTHER PUBLICATIONS**

- Doggrell S. "Migraine and beyond: cardiovascular therapeutic potential for CGRP modulators," *Expert Opin Investig Drugs*. Jun. 2001;10(6):1131-8.
- Dolgin E. "Antibody drugs set to revive flagging migraine target," *Nat Rev Drug Discov*. Apr. 2013;12(4):249-50.
- Doods H, et al. "Pharmacological profile of BIBN4096BS, the first selective small molecule CGRP antagonist," *Br J Pharmacol*. Feb. 2000;129(3):420-3.
- Doods, H et al. "CGRP antagonists: unravelling the role of CGRP in migraine," *Trends Pharmacol Sci*. Nov. 2007;28(11):580-7.
- Dooley JS, et al. "Antibiotics in the treatment of biliary infection," *Gut*. Sep. 1984;25(9):988-98.
- Drake AW, et al. "Characterizing high-affinity antigen/antibody complexes by kinetic- and equilibrium-based methods," *Anal Biochem*. May 1, 2004;328(1):35-43.
- Dressler and Saberi. "Botulinum toxin: mechanisms of action," *Eur. Neurol*. 2005;53:3-9.
- Dressler D, et al. "Botulinum toxin: mechanisms of action," *Arq Neuropsiquiatr*. Mar. 2005;63(1):180-5.
- Dufner P, et al. "Harnessing phage and ribosome display for antibody optimisation," *Trends Biotechnol*. Nov. 2006;24(11):523-9. Epub Sep. 26, 2006.
- Durham P. "CGRP-receptor antagonists—a fresh approach to migraine therapy?" *N Engl J Med*. Mar. 11, 2004;350(11):1073-5.
- Durham Paul L et al: "Calcitonin Gene-Related Peptide (CGRP) Receptor Antagonists in the Treatment of Migraine", *CNS Drugs*, vol. 24, No. 7, 2010, pp. 539-548.
- Durham PL et al. "New insights into the molecular actions of serotonergic antimigraine drugs," *Pharmacol Ther*. Apr.-May 2002;94(1-2):77-92.
- Durham PL, et al. "Regulation of calcitonin gene-related peptide secretion from trigeminal nerve cells by botulinum toxin type A: implications for migraine therapy," *Headache*. Jan. 2004;44(1):35-42; discussion 42-3.
- Durham PL. "Calcitonin Gene-Related Peptide (CGRP) and Migraine," *Headache*. Jun. 2006;46 Suppl 1:S3-8.
- Durham PL. "Inhibition of calcitonin gene-related peptide function: a promising strategy for treating migraine," *Headache*. Sep. 2008;48(8):1269-75.
- Edvinsson L et al. "Blockade of CGRP receptors in the intracranial vasculature: a new target in the treatment of headache," *Cephalalgia*. Aug. 2004;24(8):611-22.
- Edvinsson L et al. "CGRP Receptor Antagonism and Migraine," *Neurotherapeutics*. Apr. 2010;7(2):164-75.
- Edvinsson L et al. "Extracerebral manifestations in migraine. A peptidergic involvement?" *J Intern Med*. Oct. 1990;228(4):299-304.
- Edvinsson L et al. "Neurobiology in primary headaches," *Brain Res Brain Res Rev*. Jun. 2005;48(3):438-56.
- Edvinsson L et al. "Perivascular neuropeptides (NPY, VIP, CGRP and SP) in human brain vessels after subarachnoid hemorrhage," *Acta Neurol Scand*. Nov. 1994;90(5):324-30.
- Edvinsson L et al. "The blood-brain barrier in migraine treatment," *Cephalalgia*. Dec. 2008;28(12):1245-58.
- Edvinsson L et al: "New drugs in migraine treatment and prophylaxis: telcagepant and topiramate", *The Lancet*, the Lancet Publishing Group, GB, vol. 376, No. 9741, Aug. 21, 2010 (Aug. 21, 2010), pp. 645-655.
- Edvinsson L, et al. "Calcitonin gene-related peptide and cerebral blood vessels: distribution and vasomotor effects," *J Cereb Blood Flow Metab*. Dec. 1987;7(6):720-8.
- Edvinsson L, et al. "Inhibitory effect of BIBN4096BS, CGRP(8-37), a CGRP antibody and an RNA-Spiegelmer on CGRP induced vasodilatation in the perfused and non-perfused rat middle cerebral artery," *Br J Pharmacol*. Mar. 2007;150(5):633-40. Epub Jan. 22, 2007.
- Edvinsson L, et al. "Innervation of the human middle meningeal artery; immunohistochemistry, ultrastructure, and role of endothelium for vasomotility," *Peptides*. 1998;19(7):1213-25.
- Edvinsson L, et al. "Neuropeptides in migraine and cluster headache," *Cephalalgia*. Oct. 1994;14(5):320-7.
- Edvinsson L. "Aspects on the Pathophysiology of Migraine and Cluster Headache," *Pharmacol Toxicol*. Aug. 2001;89(2):65-73.
- Edvinsson L. "Calcitonin Gene-Related Peptide (CGRP) and the Pathophysiology of Headache Therapeutic Implications," *CNS Drugs*. 2001;15(10):745-53.
- Edvinsson L. "CGRP blockers in migraine therapy: where do they act?" *Br J Pharmacol*. Dec. 2008;155(7):967-9.
- Edvinsson L. "CGRP-receptor antagonism in migraine treatment," *Lancet*. Dec. 20, 2008;372(9656):2089-90.
- Edvinsson L. "Clinical Data on the CGRP Antagonist BIBN4096BS for Treatment of Migraine Attacks," *CNS Drug Rev*. 2005 Spring;11(1):69-76.
- Edvinsson L. "Innervation and effects of dilatory neuropeptides on cerebral vessels. New aspects," *Blood Vessels*. 1991;28(1-3):35-45.
- Edvinsson L. "Neuronal Signal Substances as Biomarkers of Migraine," *Headache*. Jul.-Aug. 2006;46(7):1088-94.
- Edvinsson L. "New therapeutic target in primary headaches—blocking the CGRP receptor," *Expert Opin Ther Targets*. Jun. 2003;7(3):377-83.
- Edvinsson L. "Novel migraine therapy with calcitonin gene-regulated peptide receptor antagonists," *Expert Opin Ther Targets*. Sep. 2007;11(9):1179-88.
- Edvinsson L: "CGRP blockers in migraine therapy: where do they act?", *British Journal of Pharmacology*, vol. 155, No. 7, Dec. 2008 (Dec. 2008). pp. 967-969.
- Edvinsson Lars: "CGRP-receptor antagonism in migraine treatment," *Lancet* Dec. 20, 2008, vol. 372, No. 9656, Dec. 20, 2008 (Dec. 20, 2008), pp. 2089-2090.
- Eftekhari S et al. "Differentiation of Nerve Fibers Storing CGRP and CGRP Receptors in the Peripheral Trigeminovascular System," *J Pain*. Nov. 2013;14(11):1289-303.
- Elshourbagy NA, et al. "Molecular cloning and characterization of the porcine calcitonin gene-related peptide receptor," *Endocrinology*. Apr. 1998;139(4):1678-83.
- Emerick GT. "Migraines in the Presence of Glaucoma, Recent advances in diagnosis and management," *Glaucoma Today*, Sep./Oct. 2008, 21-23.
- Escott et al., "Effect of a calcitonin gene-related peptide antagonist (CGRP8-37) on skin vasodilatation and oedema induced by stimulation of the rat saphenous nerve," *Br. J. Pharmacol*. 1993;110:772-6.
- Escott KJ, et al. "Trigeminal ganglion stimulation increases facial skin blood flow in the rat: a major role for calcitonin gene-related peptide." *Brain Res*. Jan. 9, 1995;669(1):93-9.
- Esfandiyari T. "The Role Of Calcitonin Gene-Related Peptide (CGRP) In Colonic Inflammation, And Secretion In The Rat Distal Colon," Thesis, University of Calagary, Department of Neuroscience and Gastrointestinal Sciences. 1999. 145 pages.
- Evans BN, et al. "CGRP-RCP, a novel protein required for signal transduction at calcitonin gene-related peptide and adrenomedullin receptors," *J Biol Chem*. Oct. 6, 2000;275(40):31438-43.
- Evans RW, et al. "Target doses and titration schedules for migraine preventive medications," *Headache*. Jan. 2006;46(1):160-4.
- Evans RW. "Exploding head syndrome followed by sleep paralysis: a rare migraine aura," *Headache*. Apr. 2006;46(4):682-3.
- Everitt DE et al. "The Pharmacokinetics, Antigenicity, and Fusion-Inhibition Activity of RSHZ19, a Humanized Monoclonal Antibody to Respiratory Syncytial Virus, in Healthy Volunteers," *J Infect Dis*. Sep. 1996;174(3):463-9.
- Faraci FM, et al. "Vascular responses of dura mater," *Am J Physiol*. Jul. 1989;257(1 Pt 2):H157-61.
- Farinelli, I et al. "Future drugs for migraine," *Intern Emerg Med*. Oct. 2009;4(5):367-73.
- Feuerstein G et al. "Clinical perspectives of calcitonin gene related peptide pharmacology," *Can J Physiol Pharmacol*. Jul. 1995;73(7):1070-4.
- File History U.S. Appl. No. 60/736,623, filed Nov. 14, 2005, Zeller, et al. Antagonist Antibodies Directed Against Calcitonin Gene-Related Peptide and Methods Using Same. 110 pages.
- Fischer MJ et al. "The Nonpeptide Calcitonin Gene-Related Peptide Receptor Antagonist BIBN4096BS Lowers the Activity of Neurons

(56)

References Cited**OTHER PUBLICATIONS**

- with Meningeal Input in the Rat Spinal Trigeminal Nucleus," *J Neurosci.* Jun. 22, 2005;25(25):5877-83.
- Fischer MJ. "Calcitonin gene-related peptide receptor antagonists for migraine," *Expert Opin Investig Drugs.* Jul. 2010;19(7):815-23.
- Forssman B, et al. "Atenolol for migraine prophylaxis," *Headache.* Jul. 1983;23(4):188-90.
- Forster ER, et al. "The role of calcitonin gene-related peptide in gastric mucosal protection in the rat," *Exp Physiol.* Jul. 1991;76(4):623-6.
- Friend PJ, et al. "Phase I study of an engineered aglycosylated humanized CD3 antibody in renal transplant rejection," *Transplantation.* Dec. 15, 1999;68(11):1632-7.
- Frøbert Y, et al. "A sensitive sandwich enzyme immunoassay for calcitonin gene-related peptide (CGRP): characterization and application," *Peptides.* 1999;20(2):275-84.
- Galitsky BA, et al. "Predicting amino acid sequences of the antibody human VH chains from its first several residues," *Proc Natl Acad Sci U S A.* Apr. 28, 1998;95(9):5193-8.
- Gallai V, et al. "Vasoactive peptide levels in the plasma of young migraine patients with and without aura assessed both interictally and ictally," *Cephalalgia.* Oct. 1995;15(5):384-90.
- Gangula PR, et al. "Increased blood pressure in alpha-calcitonin gene-related peptide/calcitonin gene knockout mice," *Hypertension.* Jan. 2000;35(1 Pt 2):470-5.
- Gearing D, et al. "A fully caninised anti-NGF monoclonal antibody for pain relief in dogs," *BMC Vet Res.* Nov. 9, 2013;9:226.
- Geppetti P et al. "Antidromic vasodilatation and the migraine mechanism," *J Headache Pain.* Mar. 2012;13(2):103-11.
- Geppetti P et al. "CGRP and migraine: neurogenic inflammation revisited," *J Headache Pain.* Apr. 2005;6(2):61-70.
- Geppetti P et al. "Novel therapeutic targets," *Neurol Sci.* May 2006;27 Suppl 2:S111-4.
- Giamberardino MA, et al. "Emerging drugs for migraine treatment," *Expert Opin Emerg Drugs.* Mar. 2015;20(1):137-47.
- Gillies S et al. "Improving the efficacy of antibody-interleukin 2 fusion proteins by reducing their interaction with Fc receptors," *Cancer Res.* May 1, 1999;59(9):2159-66.
- Giniatullin R et al. "Molecular Mechanisms of Sensitization of Pain-transducing P2X3 Receptors by the Migraine Mediators CGRP and NGF," *Mol Neurobiol.* Feb. 2008;37(1):83-90.
- Glennie MJ, et al. "Clinical trials of antibody therapy," *Immunol Today.* Aug. 2000;21(8):403-10.
- Glover V, et al. "Can the vascular and neurogenic theories of migraine finally be reconciled?" *Trends Pharmacol Sci.* Jan. 1989;10(1):1-3.
- Goadsby PJ et al. "Release of vasoactive peptides in the extracerebral circulation of humans and the cat during activation of the trigeminovascular system," *Ann Neurol.* Feb. 1988;23(2):193-6.
- Goadsby PJ, et al. "Migraine—current understanding and treatment," *N Engl J Med.* Jan. 24, 2002;346(4):257-70.
- Goadsby PJ, et al. "Vasoactive peptide release in the extracerebral circulation of humans during migraine headache," *Ann Neurol.* Aug. 1990;28(2):183-7.
- Goadsby PJ. "Advances in the understanding of headache," *Br Med Bull.* Oct. 5, 2005;73-74:83-92. Print 2005.
- Goadsby PJ. "Calcitonin gene-related peptide antagonists as treatments of migraine and other primary headaches," *Drugs.* 2005;65(18):2557-67.
- Goadsby PJ. "Can we develop neurally acting drugs for the treatment of migraine?" *Nat Rev Drug Discov.* Sep. 2005;4(9):741-50.
- Goadsby PJ. "Headache: a good year for research," *Lancet Neurol.* Jan. 2006;5(1):5-6.
- Goadsby PJ. "Migraine Pathophysiology," *Headache.* Apr. 2005;45 Suppl 1:S14-24.
- Goadsby PJ. "New targets in the acute treatment of headache," *Curr Opin Neurol.* Jun. 2005;18(3):283-8.
- Goadsby PJ. "The vascular theory of migraine—a great story wrecked by the facts," *Brain.* Jan. 2009;132(Pt 1):6-7.
- Goadsby PJ, et al. "Randomized, double-blind, placebo-controlled trial of ALD403, an anti-CGRP antibody in the prevention of frequent episodic migraine." 56th Annual Scientific Meeting of the American Headache Society, Jun. 2014. 4 pages.
- Gómez-Foix AM, et al., "Anti-insulin effects of amylin and calcitonin-gene-related peptide on hepatic glycogen metabolism," *Biochem J.* Jun. 15, 1991;276 (Pt 3):607-10.
- Green LL, et al. "Antigen-specific human monoclonal antibodies from mice engineered with human Ig heavy and light chain YACs," *Nat Genet.* May 1994;7(1):13-21.
- Grunenberger F. "[Calcitonin gene-related peptide (CGRP): a vasodilator neuropeptide with many potential applications]" *Pathol Biol (Paris).* Dec. 1993;41(10):936-42.
- Gupta S et al. "Evidence for CGRP re-uptake in rat dura mater encephali," *Br J Pharmacol.* Dec. 2010;161(8):1885-98.
- Gupta S et al. "Intravital microscopy on a closed cranial window in mice: a model to study trigeminovascular mechanisms involved in migraine," *Cephalalgia.* Nov. 2006;26(11):1294-303.
- Gupta S et al. "Potential role of female sex hormones in the pathophysiology of migraine," *Pharmacol Ther.* Feb. 2007;113(2):321-40.
- Gupta S et al. "The relevance of preclinical research models for the development of antimigraine drugs: focus on 5-HT(1B/1D) and CGRP receptors," *Pharmacol Ther.* Oct. 2010;128(1):170-90.
- Hakala JM, et al. "Modelling constrained calcitonin gene-related peptide analogues," *Protein Eng.* Feb. 1996;9(2):143-8.
- Halimi S, et al. "Combination treatment in the management of type 2 diabetes: focus on vildagliptin and metformin as a single tablet," *Vasc Health Risk Manag.* 2008;4(3):481-92.
- Hanes J et al. "Picomolar affinity antibodies from a fully synthetic naive library selected and evolved by ribosome display," *Nat Biotechnol.* Dec. 2000;18(12):1287-92.
- Hansen JM, et al. "Calcitonin gene-related peptide triggers migraine-like attacks in patients with migraine with aura," *Cephalalgia.* Oct. 2010;30(10):1179-86.
- Hargreaves R. "New Migraine and Pain Research," *Headache.* Apr. 2007;47 Suppl 1:S26-43.
- Hatcher JP, et al. "Biologics: the next-generation therapeutics for analgesia?" *Expert Rev Neurother.* Nov. 2011;11(11):1653-8.
- Hay D et al. "A comparison of the actions of BIBN4096BS and CGRP(8-37) on CGRP and adrenomedullin receptors expressed on SK-N-MC, L6, col. 29 and Rat 2 cells," *Br J Pharmacol.* Sep. 2002;137(1):80-6.
- Hay D et al. "International Union of Pharmacology. LXIX. Status of the Calcitonin Gene-Related Peptide Subtype 2 Receptor," *Pharmacol Rev.* Jun. 2008;60(2):143-5.
- Hay D et al. "The pharmacology of CGRP-responsive receptors in cultured and transfected cells," *Peptides.* Nov. 2004;25(11):2019-26.
- Hay D et al. "The Preclinical Pharmacology of BIBN4096BS, a CGRP Antagonist," *Cardiovasc Drug Rev.* 2005 Spring;23(1):31-42.
- Hay D. "What Makes a CGRP2 Receptor?" *Clin Exp Pharmacol Physiol.* Oct. 2007;34(10):963-71.
- Hay DL, et al. "CL/RAMP2 and CL/RAMP3 produce pharmacologically distinct adrenomedullin receptors: a comparison of effects of adrenomedullin22-52, CGRP8-37 and BIBN4096BS," *Br J Pharmacol.* Oct. 2003;140(3):477-86. Epub Aug. 26, 2003.
- Hershey JC, et al. "Investigation of the species selectivity of a nonpeptide CGRP receptor antagonist using a novel pharmacodynamic assay," *Regul Pept.* Apr. 15, 2005;127(1-3):71-7.
- Hill RG et al. "Neuropeptide and Kinin Antagonists," *Handb Exp Pharmacol.* 2007;(177):181-216.
- Hillmen P, et al. "Effect of eculizumab on hemolysis and transfusion requirements in patients with paroxysmal nocturnal hemoglobinuria," *N Engl J Med.* Feb. 5, 2004;350(6):552-9.
- Hinton PR, et al. "Engineered human IgG antibodies with longer serum half-lives in primates," *J Biol Chem.* Feb. 20, 2004;279(8):6213-6.
- Hirsch S et al. "The CGRP receptor antagonist BIBN4096BS peripherally alleviates inflammatory pain in rats," *Pain.* May 2013;154(5):700-7.

(56)

References Cited

OTHER PUBLICATIONS

- Ho TW et al. "CGRP and its receptors provide new insights into migraine pathophysiology." *Nat Rev Neurol*. Oct. 2010;6(10):573-82.
- Ho TW, et al. "Impact of recent prior opioid use on rizatriptan efficacy. A post hoc pooled analysis," *Headache*. Mar. 2009;49(3):395-403.
- Ho TW, et al. "Randomized controlled trial of the CGRP receptor antagonist telcagepant for migraine prevention," *Neurology*. Sep. 9, 2014;83(11):958-66.
- Ho TW, et al. "Efficacy and tolerability of MK-0974 (telcagepant), a new oral antagonist of calcitonin gene-related peptide receptor, compared with zolmitriptan for acute migraine: a randomised, placebo-controlled, parallel-treatment trial," *Lancet*. Dec. 20, 2008;372(9656):2115-23.
- Hoff AO et al. "Increased bone mass is an unexpected phenotype associated with deletion of the calcitonin gene," *J Clin Invest*. Dec. 2002;110(12):1849-57.
- Hoffmann J, et al. "New Agents for Acute Treatment of Migraine: CGRP Receptor Antagonists, iNOS Inhibitors," *Curr Treat Options Neurol*. Feb. 2012;14(1):50-9.
- Holland J et al. "Calcitonin Gene-Related Peptide Reduces Brain Injury in a Rat Model of Focal Cerebral Ischemia," *Stroke*. Oct. 1994;25(10):2055-8; discussion 2058-9.
- Holliger P, et al. "Engineered antibody fragments and the rise of single domains," *Nat Biotechnol*. Sep. 2005;23(9):1126-36.
- Holm P, et al. "Functional mapping and single chain construction of the anti-cytokeratin 8 monoclonal antibody TS1," *Mol Immunol*. Feb. 2007;44(6):1075-84. Epub Sep. 20, 2006.
- Holman JJ, et al. "Human alpha- and beta-CGRP and rat alpha-CGRP are coronary vasodilators in the rat." *Peptides*. Mar.-Apr. 1986;7(2):231-5.
- Holt LJ, et al. "Domain antibodies: proteins for therapy," *Trends Biotechnol*. Nov. 2003;21(11):484-90.
- Holzer P et al. "Afferent Nerve-Mediated Protection Against Deep Mucosal Damage in the Rat Stomach," *Gastroenterology*. Apr. 1990;98(4):838-48.
- Holzer P et al. "Sensory neurons mediate protective vasodilatation in rat gastric mucosa," *Am J Physiol*. Mar. 1991;260(3 Pt 1):G363-70.
- Holzer P et al. "Stimulation Of Afferent Nerve Endings By Intragastric Capsaicin Protects Against Ethanol-Induced Damage Of Gastric Mucosa," *Neuroscience*. Dec. 1988;27(3):981-7.
- Holzer P. "Implications of tachykinins and calcitonin gene-related peptide in inflammatory bowel disease," *Digestion*. Jul.-Aug. 1998;59(4):269-83.
- Holzer P. "Capsaicin: Cellular Targets, Mechanisms of Action, and Selectivity for Thin Sensory Neurons," *Pharmacol Rev*. Jun. 1991;43(2):143-201.
- Hong KW, et al. "Effect of omega-conotoxin GVIA and omega-agatoxin IVA on the capsaicin-sensitive calcitonin gene-related peptide release and autoregulatory vasodilation in rat pial arteries," *J Cereb Blood Flow Metab*. Jan. 1999;19(1):53-60.
- Hong KW, et al. "Pharmacological coupling and functional role for CGRP receptors in the vasodilation of rat pial arterioles," *Am J Physiol*. Jan. 1996;270(1 Pt 2):H317-23.
- Hong KW, et al. "Pharmacological evidence that calcitonin gene-related peptide is implicated in cerebral autoregulation," *Am J Physiol*. Jan. 1994;266(1 Pt 2):H11-6.
- Hoogenboom HR, et al. "Multi-subunit proteins on the surface of filamentous phage: methodologies for displaying antibody (Fab) heavy and light chains," *Nucleic Acids Res*. Aug. 11, 1991;19(15):4133-7.
- Hoogenboom HR. "Selecting and screening recombinant antibody libraries," *Nat Biotechnol*. Sep. 2005;23(9):1105-16.
- Hopkins, CR. "ACS Chemical Neuroscience Molecule Spotlight on Telcagepant (MK-0974)," *ACS Chem Neurosci*. Jul. 20, 2011;2(7):334-5.
- Hu H, et al. "Acute migraine treatment with rizatriptan in real world settings—focusing on treatment strategy, effectiveness, and behavior." *Headache*. Feb. 2009;49 Suppl 1:S34-42.
- Hubbard JA, et al. "Identification of the epitopes of calcitonin gene-related peptide (CGRP) for two anti CGRP monoclonal antibodies by 2D NMR," *Protein Sci*. Sep. 1997;6(9):1945-52.
- Hudson PJ, et al. "Engineered antibodies," *Nat Med*. Jan. 2003;9(1):129-34.
- Hughes SR et al. "A calcitonin gene-related peptide (CGRP) antagonist (CGRP8-37) inhibits microvascular responses induced by CGRP and capsaicin in skin," *Br J Pharmacol*. Nov. 1991;104(3):738-42.
- Hurley D. "CGRP Drug Improves Wellness on Headache-Free Days, Study Finds," *Neurology Today*, p. 31, Jul. 2016.
- Hwang WY, et al. "Immunogenicity of engineered antibodies," *Methods*. May 2005;36(1):3-10.
- Ibrahimi K, et al. "Development of an experimental model to study trigeminal nerve-mediated vasodilation on the human forehead," *Cephalalgia*. Jan. 3, 2014;34(7):514-522.
- Idusogie EE, et al. "Mapping of the C1q Binding Site on Rituxan, a Chimeric Antibody with a Human IgG1 Fc," *J Immunol*. Apr. 15, 2000;164(8):4178-84.
- Iovino M, et al. "Safety, tolerability and pharmacokinetics of BIBN 4096 BS, the first selective small molecule calcitonin gene-related peptide receptor antagonist, following single intravenous administration in healthy volunteers," *Cephalalgia*. Aug. 2004;24(8):645-56.
- Janeway CA et al. "Immuno Biology: The Immune System in Health and Disease." Current Biology Ltd./Garland Publishing Inc. 1994 Glossary p. G:2.
- Jang YJ, et al. "The structural basis for DNA binding by an anti-DNA autoantibody," *Mol Immunol*. Dec. 1998;35(18):1207-17.
- Jansen-Olesen I, et al. "In-depth characterization of CGRP receptors in human intracranial arteries," *Eur J Pharmacol*. Nov. 28, 2003;481(2-3):207-16.
- Jones PT, et al. "Replacing the complementarity-determining regions in a human antibody with those from a mouse," *Nature*. May 29-Jun. 4, 1986;321(6069):522-5.
- Juaneda C, et al. "The molecular pharmacology of CGRP and related peptide receptor subtypes," *Trends Pharmacol Sci*. Nov. 2000;21(11):432-8.
- Juhasz G, et al. "NO-induced migraine attack: strong increase in plasma calcitonin gene-related peptide (CGRP) concentration and negative correlation with platelet serotonin release," *Pain*. Dec. 2003;106(3):461-70.
- Juhl L, et al. "Effect of two novel CGRP-binding compounds in a closed cranial window rat model," *Eur J Pharmacol*. Jul. 12, 2007;567(1-2):117-24.
- Julia V, et al. "Tachykinergic mediation of viscerosensitive responses to acute inflammation in rats: role of CGRP," *Am J Physiol*. Jan. 1997;272(1 Pt 1):G141-6.
- Jung ST, et al. "Bypassing glycosylation: engineering aglycosylated full-length IgG antibodies for human therapy," *Curr Opin Biotechnol*. Dec. 2011;22(6):858-67.
- Kaiser EA, et al. "CGRP and migraine: could PACAP play a role too?" *Neuropeptides*. Dec. 2013;47(6):451-61.
- Kapoor K, et al. "Effects of BIBN4096BS on cardiac output distribution and on CGRP induced carotid haemodynamic responses in the pig," *Eur J Pharmacol*. Aug. 15, 2003;475(1-3):69-77.
- Kapoor K, et al. "Effects of the CGRP receptor antagonist BIBN4096BS on capsaicin-induced carotid haemodynamic changes in anaesthetised pigs," *Br J Pharmacol*. Sep. 2003;140(2):329-38.
- Kapoor, K. "Novel Potential Antimigraine Compounds: Carotid and Systemic Haemodynamic Effects in a Porcine Model of Migraine." Thesis, Erasmus University, Rotterdam, With summary in Dutch. 2003. 157 pages.
- Karasek C., et al. "Characterization of the intrinsic binding features of three anti-CGRP therapeutic antibodies effective in preventing migraine: a comparative pre-clinical case study of ALD403, LY-2951742, TEV-48125." 5th European Headache and Migraine Trust International Congress, Sep. 2016. 4 pages.
- Kato K, et al. "CGRP antagonists enhance gastric acid secretion in 2-h pylorus-ligated rats," *Peptides*. 1995;16(7):1257-62.

(56)

References Cited

OTHER PUBLICATIONS

- Kawamura M, et al. "Antinociceptive effect of intrathecally administered antiserum against calcitonin gene-related peptide on thermal and mechanical noxious stimuli in experimental hyperalgesic rats." *Brain Res. Sep. 11, 1989*;497(1):199-203.
- Kaymakçalan Z, et al. "Comparisons of affinities, avidities, and complement activation of adalimumab, infliximab, and etanercept in binding to soluble and membrane tumor necrosis factor," *Clin Immunol. May 2009*;131(2):308-16.
- Keates AC, et al. "CGRP upregulation in dorsal root ganglia and ileal mucosa during *Clostridium difficile* toxin A-induced enteritis," *Am J Physiol. Jan. 1998*;274(1 Pt 1):G196-202.
- Kennel SJ, et al. "Direct Binding of Radioiodinated Monoclonal Antibody to Tumor Cells: Significance of Antibody Purity and Affinity for Drug Targeting or Tumor Imaging," *Hybridoma. 1983*;2(3):297-310.
- Kim SJ, et al. "Antibody Engineering for the Development of Therapeutic Antibodies," *Mol Cells. Aug. 31, 2005*;20(1):17-29.
- Kipriyanov S, et al. "Generation and Production of Engineered Antibodies," *Mol Biotechnol. Jan. 2004*;26(1):39-60.
- Kipriyanov S. "Generation of Antibody Molecules Through Antibody Engineering" from *Methods in Molecular Biology*, vol. 207: Recombinant Antibodies for Cancer Therapy Methods and Protocols, 2003 pp. 3-25.
- Knotkova H, et al. "Imaging intracranial plasma extravasation in a migraine patient: a case report," *Pain Med. May-Jun. 2007*;8(4):383-7.
- Kobayashi D, et al. "Calcitonin Gene-Related Peptide Mediated Neurogenic Vasorelaxation in the Isolated Canine Lingual Artery," *Jpn J Pharmacol. Apr. 1995*;67(4):329-39.
- Kobayashi H, et al. "Tryptophan H33 plays an important role in pyrimidine (6-4) pyrimidone photoproduct binding by a high-affinity antibody." *Protein Eng. Oct. 1999*;12(10):879-84.
- Krymchantowski AV, et al. "New and emerging prophylactic agents for migraine," *CNS Drugs. 2002*;16(9):611-34.
- Krymchantowski AV, et al. "Rizatriptan in migraine," *Expert Rev Neurother. Sep. 2005*;5(5):597-603.
- Krymchantowski AV, et al. "Rizatriptan vs. rizatriptan plus trimebutine for the acute treatment of migraine: a double-blind, randomized, cross-over, placebo-controlled study," *Cephalalgia. Jul. 2006*;26(7):871-4.
- Krymchantowski AV, et al. "Topiramate plus nortriptyline in the preventive treatment of migraine: a controlled study for nonresponders," *J Headache Pain. Jan. 2012*;13(1):53-9.
- Kumar S, et al. "Molecular cloning and expression of the Fabs of human autoantibodies in *Escherichia coli*. Determination of the heavy or light chain contribution to the anti-DNA/cardiophilin activity of the Fab," *J Biol Chem. Nov. 10, 2000*;275(45):35129-36.
- Kunkel RS, et al. "Surgical treatment of chronic migrainous neuralgia," *Cleve Clin Q. 1974 Winter*;41(4):189-92.
- Kuraishi Y, et al. "Antinociception induced in rats by intrathecal administration of antiserum against calcitonin gene-related peptide." *Neurosci Lett. Oct. 17, 1988*;92(3):325-9.
- Kurosawa M, et al. "Increase of meningeal blood flow after electrical stimulation of rat dura mater encephali: mediation by calcitonin gene-related peptide," *Br J Pharmacol. Apr. 1995*;114(7):1397-402.
- Kuus-Reichel K, et al. "Will Immunogenicity Limit the Use, Efficacy, and Future Development of Therapeutic Monoclonal Antibodies?" *Clin Diagn Lab Immunol. Jul. 1994*;1(4):365-72.
- Lambrecht N, et al. "Role of calcitonin gene-related peptide and nitric oxide in the gastroprotective effect of capsaicin in the rat," *Gastroenterology. May 1993*;104(5):1371-80.
- Lance J. "Migraine Pain Originates from Blood Vessels," *Headache Pathogenesis; Monoamines, Neuropeptides, Purines, and Nitric Oxide*, edited by J. Olesen and L. Edvinsson, Lippincott-Raven Publishers, Philadelphia, 1997. Chapter 1, pp. 3-9.
- Lassen LH, et al. "CGRP may play a causative role in migraine." *Cephalalgia. Feb. 2002*;22(1):54-61.
- Lassen LH, et al. "Involvement of calcitonin gene-related peptide in migraine: regional cerebral blood flow and blood flow velocity in migraine patients," *J Headache Pain. Jun. 2008*;9(3):151-7.
- Lazzeria M, et al. "The Challenge of the Overactive Bladder: From Laboratory to New Drugs," *European Association of Urology*, vol. 5, Issue 6, Dec. 2007, pp. 250-258.
- Lee CV, et al. "High-affinity human antibodies from phage-displayed synthetic Fab libraries with a single framework scaffold," *J Mol Biol. Jul. 23, 2004*;340(5):1073-93.
- Leighton B, et al. "Pancreatic amylin and calcitonin gene-related peptide cause resistance to insulin in skeletal muscle in vitro," *Nature. Oct. 13, 1988*;335(6191):632-5.
- Levêque D, et al. "Pharmacokinetics of therapeutic monoclonal antibodies used in oncology," *Anticancer Res. May-Jun. 2005*;25(3c):2327-43.
- Levy D, et al. "A critical view on the role of migraine triggers in the genesis of migraine pain," *Headache. Jun. 2009*;49(6):953-7.
- Levy D, et al. "Calcitonin gene-related peptide does not excite or sensitize meningeal nociceptors: implications for the pathophysiology of migraine," *Ann Neurol. Nov. 2005*;58(5):698-705.
- Levy D, et al. "Migraine pain and nociceptor activation—where do we stand?" *Headache. May 2010*;50(5):909-16.
- Levy D, et al. "The vascular theory of migraine: leave it or love it?" *Ann Neurol. Apr. 2011*;69(4):600-1.
- Li DS, et al. "Role of calcitonin gene-related peptide in gastric hyperemic response to intragastric capsaicin," *Am J Physiol. Oct. 1991*;261(4 Pt 1):G657-61.
- Lin HC, et al. "Immunoneutralization of Calcitonin Gene-Related Peptide (CGRP) During Inhibition of Intestinal Transit by Fat," *Gastroenterology* vol. 114, No. 4, 1998. 1 page. Abstract No. G3253.
- Lin YS, et al. "Preclinical pharmacokinetics, interspecies scaling, and tissue distribution of a humanized monoclonal antibody against vascular endothelial growth factor," *J Pharmacol Exp Ther. Jan. 1999*;288(1):371-8.
- Link AS, et al. "Treatment of migraine attacks based on the interaction with the trigemino-cerebrovascular system," *J Headache Pain. Feb. 2008*;9(1):5-12.
- Lipton RB, et al. "CGRP antagonists in the acute treatment of migraine," *Lancet Neurol. Jun. 2004*;3(6):332.
- Lipton RB, et al. "Headache: triumphs in translational research," *Lancet Neurol. Jan. 2005*;4(1):11-2.
- Lipton RB, et al. "Moving forward—essential questions for the next 10 years," *Headache. Feb. 2009*;49 Suppl 1:S43-6.
- Little M, et al. "Of mice and men: hybridoma and recombinant antibodies," *Immunol Today. Aug. 2000*;21(8):364-70.
- Lonberg N, et al. "Antigen-specific human antibodies from mice comprising four distinct genetic modifications," *Nature. Apr. 28, 1994*;368(6474):856-9.
- Lonberg N, et al. "Human antibodies from transgenic animals," *Nat Biotechnol. Sep. 2005*;23(9):1117-25.
- Longoni M, et al. "Inflammation and excitotoxicity: role in migraine pathogenesis," *Neurol Sci. May 2006*;27 Suppl 2:S107-10.
- Louis SM, et al. "Antibodies to calcitonin-gene related peptide reduce inflammation induced by topical mustard oil but not that due to carrageenin in the rat," *Neurosci Lett. Jul. 31, 1989*;102(2-3):257-60.
- Louis SM, et al. "Immunization with calcitonin gene-related peptide reduces the inflammatory response to adjuvant arthritis in the rat," *Neuroscience. 1990*;39(3):727-31.
- Louis SM, et al. "The role of substance P and calcitonin gene-related peptide in neurogenic plasma extravasation and vasodilatation in the rat," *Neuroscience. 1989*;32(3):581-6.
- MacCallum RM, et al. "Antibody-antigen interactions: contact analysis and binding site topography," *J Mol Biol. Oct. 11, 1996*;262(5):732-45.
- MacGregor EA. "Migraine in pregnancy and lactation; a clinical review," *J Fam Plann Reprod Health Care. Apr. 2007*;33(2):83-93.
- Majima M, et al. "Roles of calcitonin gene-related peptide in enhancement of angiogenesis," *Inflammation and Regeneration* vol. 31 No. 2 Mar. 2011, 146-150.

(56)

References Cited

OTHER PUBLICATIONS

- Mallee JJ, et al. "Receptor activity-modifying protein 1 determines the species selectivity of non-peptide CGRP receptor antagonists." *J Biol Chem.* Apr. 19, 2002;277(16):14294-8.
- Marcelo E. Bigal et al: "Calcitonin Gene-Related Peptide (CGRP) and Migraine Current Understanding and State of Development", *Headache*, vol. 53, No. 8, Sep. 12, 2013 (Sep. 12, 2013), pp. 1230-1244.
- Mareska M, et al. "Lambert-Eaton myasthenic syndrome," *Semin Neurol.* Jun. 2004;24(2):149-53.
- Marquez de Prado B and Russo AF, "CGRP receptor antagonists: A new frontier of anti-migraine medications," *Drug Discov Today Ther Strateg.* 2006 Winter;3(4):593-597.
- Marshall I, et al. "Human and rat alpha-CGRP but not calcitonin cause mesenteric vasodilatation in rats." *Eur J Pharmacol.* Apr. 16, 1986;123(2):217-22.
- Martínez-Sáenz A, et al. "Role of calcitonin gene-related peptide in inhibitory neurotransmission to the pig bladder neck," *J Urol.* Aug. 2011;186(2):728-35.
- Maynard JA, et al. "Protection against anthrax toxin by recombinant antibody fragments correlates with antigen affinity," *Nat Biotechnol.* Jun. 2002;20(6):597-601.
- McCafferty J, et al. "Phage antibodies: filamentous phage displaying antibody variable domains," *Nature.* Dec. 6, 1990;348(6301):552-4.
- McCulloch J, et al. "Calcitonin gene-related peptide: functional role in cerebrovascular regulation," *Proc Natl Acad Sci U S A.* Aug. 1986;83(15):5731-5.
- McLatchie LM, et al. "RAMPs regulate the transport and ligand specificity of the calcitonin-receptor-like receptor," *Nature.* May 28, 1998;393(6683):333-9.
- Mehrotra S, et al. "Current and prospective pharmacological targets in relation to antimigraine action," *Naunyn Schmiedeberg's Arch Pharmacol.* Oct. 2008;378(4):371-94.
- Mense S. "Pathophysiology of low back pain and the transition to the chronic state—experimental data and new concepts." *Schmerz.* Dec. 2001;15(6):413-7.
- Messlinger K, et al. "Neuropeptide effects in the trigeminal system: pathophysiology and clinical relevance in migraine," *Keio J Med.* 2011;60(3):82-9.
- Messlinger K. "Migraine: where and how does the pain originate?" *Exp Brain Res.* Jun. 2009;196(1):179-93.
- Messlinger, et al. "Inhibition of neurogenic blood flow increases in the rat cranial dura matter by a CGRP-binding Spiegelmer," *Cephalalgia*, No. F022 2004.
- Middlemiss DN. "Direct evidence for an interaction of beta-adrenergic blockers with the 5-HT receptor," *Nature.* May 19, 1977;267(5608):289-90.
- Middlemiss DN. "Stereoselective blockade at [3H]5-HT binding sites and at the 5-HT autoreceptor by propranolol," *Eur J Pharmacol.* Jun. 1, 1984;101(3-4):289-93.
- Mirick GR, et al. "A review of human anti-globulin antibody (HAGA, HAMA, HACA, HAHA) responses to monoclonal antibodies. Not four letter words," *Q J Nucl Med Mol Imaging.* Dec. 2004;48(4):251-7.
- Molina JM, et al. "Induction of insulin resistance in vivo by amylin and calcitonin gene-related peptide," *Diabetes.* Feb. 1990;39(2):260-5.
- Moore CK, et al. "Urological Applications of Botulinum Toxin." *Female Urology: A Practical Clinical Guide.* 2007 Chapter 14:213-217.
- Moore EL, et al. "Targeting a family B GPCR/RAMP receptor complex: CGRP receptor antagonists and migraine," *Br J Pharmacol.* May 2012;166(1):66-78.
- Morara S, et al. "Monoclonal antibodies reveal expression of the CGRP receptor in Purkinje cells, interneurons and astrocytes of rat cerebellar cortex," *Neuroreport.* Nov. 16, 1998;9(16):3755-9.
- Morell A, et al. "Metabolic properties of IgG subclasses in man." *J Clin Invest.* Apr. 1970;49(4):673-80.
- Morrison SL, et al. "Chimeric human antibody molecules: mouse antigen-binding domains with human constant region domains," *Proc Natl Acad Sci U S A.* Nov. 1984;81(21):6851-5.
- Moskowitz MA, "Neurogenic inflammation in the pathophysiology and treatment of migraine," *Neurology.* Jun. 1993;43(6 Suppl 3):S16-20.
- Moskowitz MA, et al. "CGRP: blood flow and more?" *Cephalalgia.* Aug. 1996;16(5):287.
- Moskowitz MA. "Pathophysiology of headache—past and present," *Headache.* Apr. 2007;47 Suppl 1:S58-63.
- Mould DR, et al. "A population pharmacokinetic-pharmacodynamic analysis of single doses of clenoliximab in patients with rheumatoid arthritis," *Clin Pharmacol Ther.* Sep. 1999;66(3):246-57.
- Mountain A, et al. "Engineering antibodies for therapy," *Biotechnol Genet Eng Rev.* 1992;10:1-142.
- Muff R, et al. "Calcitonin, calcitonin gene-related peptide, adrenomedullin and amylin: homologous peptides, separate receptors and overlapping biological actions," *Eur J Endocrinol.* Jul. 1995;133(1):17-20.
- Mulder PK, et al. "Differential expression of alpha-CGRP and beta-CGRP by primary sensory neurons and enteric autonomic neurons of the rat." *Neuroscience.* Apr. 1988;25(1):195-205.
- Mullins MW, et al. "Characterization of a calcitonin gene-related peptide (CGRP) receptor on mouse bone marrow cells." *Regul Pept.* Nov. 19, 1993;49(1):65-72.
- Nakamura-Craig M, et al. "Effect of neurokinin A, substance P and calcitonin gene related peptide in peripheral hyperalgesia in the rat paw." *Neurosci Lett.* Mar. 11, 1991;124(1):49-51.
- Naot D, et al. "The role of peptides and receptors of the calcitonin family in the regulation of bone metabolism," *Bone.* Nov. 2008;43(5):813-8.
- Negro A, et al. "CGRP receptor antagonists: an expanding drug class for acute migraine?" *Expert Opin Investig Drugs.* Jun. 2012;21(6):807-18.
- Newman R, et al. "Modification of the Fc region of a primatized IgG antibody to human CD4 retains its ability to modulate CD4 receptors but does not deplete CD4(+) T cells in chimpanzees," *Clin Immunol.* Feb. 2001;98(2):164-74.
- Ng-Mak DS, et al. "Migraine treatment with rizatriptan and almotriptan: a crossover study." *Headache.* May 2009;49(5):655-62.
- Nippon Rinsho, "Recent Development of Calcitonin Gene-related Peptide (CGRP) receptor antagonist," 2005, vol. 63, Suppl.10, pp. 263-266 [Original With English Translation].
- Nishimoto N, et al. "Anti-interleukin-6 receptor antibody therapy in rheumatic diseases," *Endocr Metab Immune Disord Drug Targets.* Dec. 2006;6(4):373-81.
- Oates PJ, et al. "Studies on the mechanism of ethanol-induced gastric damage in rats." *Gastroenterology.* Jan. 1988;94(1):10-21.
- Ober RJ, et al. "Visualizing the site and dynamics of IgG salvage by the MHC class I-related receptor, FcRn," *J Immunol.* Feb. 15, 2004;172(4):2021-9.
- O'Connell JP, et al. "On the role of the C-terminus of alpha-calcitonin-gene-related peptide (alpha CGRP). The structure of des-phenylalaninamide37-alpha CGRP and its interaction with the CGRP receptor," *Biochem J.* Apr. 1, 1993;291 (Pt 1):205-10.
- Oh-hashii Y, et al. "Elevated sympathetic nervous activity in mice deficient in alphaCGRP." *Circ Res.* Nov. 23, 2001;89(11):983-90.
- Olesen J, et al. "Calcitonin gene-related peptide receptor antagonist BIBN 4096 BS for the acute treatment of migraine." *N Engl J Med.* Mar. 11, 2004;350(11):1104-10.
- Olesen J, et al. "Chapter 16: Calcitonin Gene-Related Peptide and Other Peptides." *The Headaches Third Edition.* Lippincott Williams & Wilkins 2006 159-164.
- Olesen J, et al. "Chapter 31: CGRP Involvement in Migraines." *The Headaches Third Edition.* Lippincott Williams & Wilkins 2006 289-99.
- Olesen J, et al. "Emerging migraine treatments and drug targets," *Trends Pharmacol Sci.* Jun. 2011;32(6):352-9.
- Olesen J, et al. "Finding new drug targets for the treatment of migraine attacks," *Cephalalgia.* Sep. 2009;29(9):909-20.
- Olesen J, et al. "Migraine: a research field matured for the basic neurosciences," *Trends Neurosci.* Jan. 1991;14(1):3-5.

(56)

References Cited**OTHER PUBLICATIONS**

- Olesen J, et al. "Origin of pain in migraine; evidence for peripheral sensitisation," *Lancet Neurol.* Jul. 2009;8(7):679-90.
- Olesen J. "Migraine: A neural pathway for photophobia in migraine," *Nat Rev Neurol.* May 2010;6(5):241-2.
- Ondo WG, et al. "Botulinum toxin A for chronic daily headache: a randomized, placebo-controlled, parallel design study," *Cephalalgia.* Jan. 2004;24(1):60-5.
- O'Sullivan J, et al. "Migraine development, treatments, research advances, and anesthesia implications," *AANA J.* Feb. 2006;74(1):61-9.
- Ottosson A, et al. "Release of histamine from dural mast cells by substance P and calcitonin gene-related peptide," *Cephalalgia.* May 1997;17(3):166-74.
- Pabst MA, et al. "Ablation of capsaicin sensitive afferent nerves impairs defence but not rapid repair of rat gastric mucosa," *Gut.* Jul. 1993;34(7):897-903.
- Panconesi A, et al. "Migraine pain: reflections against vasodilatation," *J Headache Pain.* Oct. 2009;10(5):317-25.
- Panka DJ, et al. "Defining the structural correlates responsible for loss of arsonate affinity in an IDCR antibody isolated from an autoimmune mouse," *Mol Immunol.* Aug. 1993;30(11):1013-20.
- Paone DV, et al. "Calcitonin gene-related peptide receptor antagonists for the treatment of migraine: a patent review," *Expert Opin Ther Pat.* Dec. 2009;19(12):1675-713.
- Papadopoulos N, et al. "Binding and neutralization of vascular endothelial growth factor (VEGF) and related ligands by VEGF Trap, ranibizumab and bevacizumab," *Angiogenesis.* Jun. 2012;15(2):171-85.
- Papp K, et al. "The treatment of moderate to severe psoriasis with a new anti-CD11a monoclonal antibody," *J Am Acad Dermatol.* Nov. 2001;45(5):665-74.
- Pavlou AK, et al. "Recombinant protein therapeutics—success rates, market trends and values to 2010," *Nat Biotechnol.* Dec. 2004;22(12):1513-9.
- Peroutka SJ, et al. "Neurogenic inflammation and migraine: implications for the therapeutics," *Mol Interv.* Oct. 2005;5(5):304-11.
- Peskar BM, et al. "A monoclonal antibody to calcitonin gene-related peptide abolishes capsaicin-induced gastroprotection," *Eur J Pharmacol.* Nov. 30, 1993;250(1):201-3.
- Petersen KA, et al. "BIBN4096BS antagonizes human alpha-calcitonin gene related peptide-induced headache and extracerebral artery dilatation," *Clin Pharmacol Ther.* Mar. 2005;77(3):202-13.
- Petersen KA, et al. "Effect of hypotension and carbon dioxide changes in an improved genuine closed cranial window rat model," *Cephalalgia.* Jan. 2005;25(1):23-9.
- Petersen KA, et al. "Inhibitory effect of BIBN4096BS on cephalic vasodilatation induced by CGRP or transcranial electrical stimulation in the rat," *Br J Pharmacol.* Nov. 2004;143(6):697-704.
- Petersen KA, et al. "Presence and function of the calcitonin gene-related peptide receptor on rat pial arteries investigated in vitro and in vivo," *Cephalalgia.* Jun. 2005;25(6):424-32.
- Petersen KA, et al. "The effect of nonpeptide CGRP-antagonist, BIBN4096BS on human-alphaCGRP induced headache and hemodynamics in healthy volunteers," *Cephalalgia*, vol. 23, extract from Abstracts of the XI Congress of the International Headache Society, p. 725. 2003.
- Petkova SB, et al. "Enhanced half-life of genetically engineered human IgG1 antibodies in a humanized FcRn mouse model: potential application in humorally mediated autoimmune disease," *Int Immunol.* Dec. 2006; 18(12):1759-69.
- Pietrobon D, et al. "Pathophysiology of migraine," *Annu Rev Physiol.* 2013;75:365-91.
- Plessas IN, et al. "Migraine-like episodic pain behavior in a dog: can dogs suffer from migraines?" *J Vet Intern Med.* Sep.-Oct. 2013;27(5):1034-40.
- Plourde V, et al. "CGRP antagonists and capsaicin on celiac ganglia partly prevent postoperative gastric ileus," *Peptides.* Nov.-Dec. 1993;14(6):1225-9.
- Poyner DR, et al. "International Union of Pharmacology. XXXII. The mammalian calcitonin gene-related peptides, adrenomedullin, amylin, and calcitonin receptors," *Pharmacol Rev.* Jun. 2002;54(2):233-46.
- Presta L. "Antibody engineering for therapeutics," *Curr Opin Struct Biol.* Aug. 2003;13(4):519-25.
- Presta LG, et al. "Engineering therapeutic antibodies for improved function," *Biochem Soc Trans.* Aug. 2002;30(4):487-90.
- Prewett M, et al. "The biologic effects of C225, a chimeric monoclonal antibody to the EGFR, on human prostate carcinoma," *J Immunother Emphasis Tumor Immunol.* Nov. 1996;19(6):419-27.
- Qing-Hui Niu, et al. "Expression of mast cell and calcitonin gene related peptides in the mucosa of irritable bowel syndrome," *World Chinese Journal of Digestology*, Jan. 18, 2009 p. 213-217; ISSN 1099-3079.
- Raddant AC, et al. "Calcitonin gene-related peptide in migraine: intersection of peripheral inflammation and central modulation," *Expert Rev Mol Med.* Nov. 29, 2011;13:e36.
- Ramadan NM, et al. "New and future migraine therapy," *Pharmacol Ther.* Oct. 2006;112(1):199-212.
- Ramadan NM. "Acute treatments: future developments," *Curr Med Res Opin.* 2001;17 Suppl 1:s81-6.
- Ramos ML, et al. "AMG 334 CGRP antibody for migraine: time to celebrate?" *Lancet Neurol.* Apr. 2016;15(4):347-9.
- Rapoport AM, Bigal ME, et al. "Naratriptan in the preventive treatment of refractory chronic migraine," In Olsen J, Silberstein SD, Tfelt-Hansen P, eds. *Preventive Pharmacotherapy of Headache Disorders.* Copenhagen: Oxford University Press, 2004, Chapter 31.
- Rapoport AM, et al. "Intranasal medications for the treatment of migraine and cluster headache," *CNS Drugs.* 2004;18(10):671-85.
- Rapoport AM, et al. "Levetiracetam in the preventive treatment of transformed migraine: A prospective, open-label, pilot study," *Curr Ther Res Clin Exp.* May 2005;66(3):212-21.
- Rapoport AM, et al. "Migraine preventive therapy: current and emerging treatment options," *Neurol Sci.* May 2005;26 Suppl 2:111-20.
- Rapoport AM, et al. "Preventive migraine therapy: what is new," *Neurol Sci.* Oct. 2004;25 Suppl 3:8177-85.
- Raybould HE, et al. "Selective ablation of spinal afferent neurons containing CGRP attenuates gastric hyperemic response to acid," *Peptides.* Mar.-Apr. 1992;13(2):249-54.
- Reasbeck PG, et al. "Calcitonin gene-related peptide: enteric and cardiovascular effects in the dog," *Gastroenterology.* Oct. 1988;95(4):966-71.
- Recober A, et al. "Calcitonin gene-related peptide: A molecular link between obesity and migraine?" *Drug News Perspect.* Mar. 2010;23(2):112-7.
- Recober A, et al. "Calcitonin gene-related peptide: an update on the biology," *Curr Opin Neurol.* Jun. 2009;22(3):241-6.
- Recober A, et al. "Olcegepant, a non-peptide CGRP1 antagonist for migraine treatment," *IDrugs.* Aug. 2007;10(8):566-74.
- Recober A, et al. "Role of calcitonin gene-related peptide in light-averse behavior: implications for migraine," *J Neurosci.* Jul. 8, 2009;29(27):8798-804.
- Reddy MP, et al. "Elimination of Fc receptor-dependent effector functions of a modified IgG4 monoclonal antibody to human CD4," *J Immunol.* Feb. 15, 2000;164(4):1925-33.
- Reff ME, et al. "A review of modifications to recombinant antibodies: attempt to increase efficacy in oncology applications," *Crit Rev Oncol Hematol.* Oct. 2001;40(1):25-35.
- Reff ME, et al. "Depletion of B cells in vivo by a chimeric mouse human monoclonal antibody to CD20," *Blood.* Jan. 15, 1994;83(2):435-45.
- Reichert JM, et al. "Monoclonal antibody successes in the clinic," *Nat Biotechnol.* Sep. 2005;23(9):1073-8.
- Reinshagen M, et al. "Calcitonin gene-related peptide mediates the protective effect of sensory nerves in a model of colonic injury," *J Pharmacol Exp Ther.* Aug. 1998;286(2):657-61.
- Reuter U, et al. "Experimental models of migraine," *Funct Neurol.* 2000;15 Suppl 3:9-18.
- Reuter U. "Anti-CGRP antibodies: a new approach to migraine prevention," *Lancet Neurol.* Sep. 2014;13(9):857-9.

(56)

References Cited

OTHER PUBLICATIONS

- Rolston RK, et al., "Intravenous calcitonin gene-related peptide stimulates net water secretion in rat colon in vivo," *Dig Dis Sci*. Apr. 1989;34(4):612-6.
- Roon KI, et al. "No acute antimigraine efficacy of CP-122,288, a highly potent inhibitor of neurogenic inflammation: results of two randomized, double-blind, placebo-controlled clinical trials," *Ann Neurol*. Feb. 2000;47(2):238-41.
- Roopenian DC, et al. "FoRn: the neonatal Fc receptor comes of age," *Nat Rev Immunol*. Sep. 2007;7(9):715-25.
- Roque AC, et al. "Antibodies and genetically engineered related molecules: production and purification," *Biotechnol Prog*. May-Jun. 2004;20(3):639-54.
- Roskos LK, et al. "The Clinical Pharmacology of Therapeutic Monoclonal Antibodies," *Drug Development Research* 2004 61:108-120.
- Rother RP, et al. "Discovery and development of the complement inhibitor eculizumab for the treatment of paroxysmal nocturnal hemoglobinuria," *Nat Biotechnol*. Nov. 2007;25(11):1256-64.
- Rovero P, et al., "CGRP antagonist activity of short C-terminal fragments of human alpha CGRP, CGRP(23-37) and CGRP(19-37)," *Peptides*. Sep.-Oct. 1992;13(5):1025-7.
- Rudikoff S, et al. "Single amino acid substitution altering antigen-binding specificity," *Proc Natl Acad Sci U S A*. Mar. 1982;79(6):1979-83.
- Ruiz-Gayo M, et al. "Vasodilatory effects of cholecystokinin: new role for an old peptide?" *Regul Pept*. Dec. 10, 2006;137(3):179-84.
- Russo AF, et al., "A Potential Preclinical Migraine Model: CGRP-Sensitized Mice," *Mol Cell Pharmacol*. 2009;1(5):264-270.
- Russo AF. "Calcitonin gene-related peptide (CGRP): a new target for migraine," *Annu Rev Pharmacol Toxicol*. 2015;55:533-52.
- Russo. "CGRP Meeting Abstract Book," The 4th International Meeting on CGRP. Copenhagen, Sep. 2001, 71 pages.
- Russo. "CGRP Meeting Abstract Book," Joint International Symposium on Calcitonin Gene-Related Peptide, Amylin and Calcitonin; 4th Symposium on Adrenomedullin and Proadrenomedullin N-20 Peptide, Zurich, Switzerland, Mar. 2004. 38 pages.
- Ryan AM, et al. "Preclinical safety evaluation of rhuMAbVEGF, an antiangiogenic humanized monoclonal antibody," *Toxicol Pathol*. Jan.-Feb. 1999;27(1):78-86.
- Ryan S. "Medicines for migraine," *Arch Dis Child Educ Pract Ed*. Apr. 2007;92(2):ep50-5.
- Saleh MN, et al. "Phase I trial of the chimeric anti-GD2 monoclonal antibody ch14.18 in patients with malignant melanoma," *Hum Antibodies Hybridomas*. Jan. 1992;3(1):19-24.
- Salonen R. et al. "Triptans: do they differ?" *Curr Pain Headache Rep*. Apr. 2002;6(2):133-9.
- Salvatore CA, et al. "Pharmacological characterization of MK-0974 [N-[(3R,6S)-6-(2,3-difluorophenyl)-2-oxo-1-(2,2,2-trifluoroethyl)azepan-3-yl]-4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)piperidine-1-carboxamide], a potent and orally active calcitonin gene-related peptide receptor antagonist for the treatment of migraine," *J Pharmacol Exp Ther*. Feb. 2008;324(2):416-21.
- Sams-Nielsen A, et al. "Pharmacological evidence for CGRP uptake into perivascular capsaicin sensitive nerve terminals," *Br J Pharmacol*. Mar. 2001;132(5):1145-53.
- Saphire EO, et al. "Crystal structure of a neutralizing human IGG against HIV-1: a template for vaccine design," *Science*. Aug. 10, 2001;293(5532):1155-9.
- Schaible HG, et al. "Mechanisms of pain in arthritis," *Ann NY Acad Sci*. Jun. 2002;966:343-54.
- Schelstraete C, et al. "CGRP antagonists: hope for a new era in acute migraine treatment," *Acta Neurol Belg*. Dec. 2009;109(4):252-61.
- Schier R, et al. "Isolation of high-affinity monomeric human anti-c-erbB-2 single chain Fv using affinity-driven selection," *J Mol Biol*. Jan. 12, 1996;255(1):28-43.
- Schier R, et al. "Isolation of picomolar affinity anti-c-erbB-2 single-chain Fv by molecular evolution of the complementarity determining regions in the center of the antibody binding site," *J Mol Biol*. Nov. 8, 1996;263(4):551-67.
- Schifter S. "Circulating concentrations of calcitonin gene-related peptide (CGRP) in normal man determined with a new, highly sensitive radioimmunoassay," *Peptides*. Mar.-Apr. 1991;12(2):365-9.
- Schindler M, et al. "Binding properties of the novel, non-peptide CGRP receptor antagonist radioligand, [(3)H]BIBN4096BS," *Eur J Pharmacol*. May 10, 2002;442(3):187-93.
- Schoenen J, et al. "Almotriptan and its combination with aceclofenac for migraine attacks: a study of efficacy and the influence of auto-evaluated brush allodynia," *Cephalalgia*. Oct. 2008;28(10):1095-105.
- Schreiber CP. "The pathophysiology of migraine," *Dis Mon*. Oct. 2006;52(10):385-401.
- Schwenger N, et al. "Interaction of calcitonin gene-related peptide, nitric oxide and histamine release in neurogenic blood flow and afferent activation in the rat cranial dura mater," *Cephalalgia*. Jun. 2007;27(6):481-91.
- Schytz HW, et al. "What have we learnt from triggering migraine?" *Curr Opin Neurol*. Jun. 2010;23(3):259-65.
- Seike M, et al. "Increased synthesis of calcitonin gene-related peptide stimulates keratinocyte proliferation in murine UVB-irradiated skin," *J Dermatol Sci*. Feb. 2002;28(2):135-43.
- Selenko N, et al. "CD20 antibody (C2B8)-induced apoptosis of lymphoma cells promotes phagocytosis by dendritic cells and cross-priming of CD8+ cytotoxic T cells," *Leukemia*. Oct. 2001;15(10):1619-26.
- Seong J, et al. "Radiation-induced alteration of pain-related signals in an animal model with bone invasion from cancer," *Ann NY Acad Sci*. Dec. 2004;1030:179-86.
- Seybold VS. "The role of peptides in central sensitization," *Handb Exp Pharmacol*. 2009;(194):451-91.
- Shaw NE, et al. "The effect of monoclonal antibodies to calcitonin gene-related peptide (CGRP) on CGRP-induced vasodilatation in pig coronary artery rings," *Br J Pharmacol*. May 1992;106(1):196-8.
- Sheets MD, et al. "Efficient construction of a large nonimmune phage antibody library: the production of high-affinity human single-chain antibodies to protein antigens," *Proc Natl Acad Sci U S A*. May 26, 1998;95(11):6157-62.
- Sheffell FD, et al. "Naratriptan in the preventive treatment of refractory transformed migraine: a prospective pilot study," *Headache*. Nov.-Dec. 2005;45(10):1400-6.
- Shen YT, et al. "Functional role of alpha-calcitonin gene-related peptide in the regulation of the cardiovascular system," *J Pharmacol Exp Ther*. Aug. 2001;298(2):551-8.
- Shevel E. "The extracranial vascular theory of migraine—a great story confirmed by the facts," *Headache*. Mar. 2011;51(3):409-17.
- Shields RL, et al. "High resolution mapping of the binding site on human IgG1 for Fc gamma RI, Fc gamma RII, Fc gamma RIII, and FcRn and design of IgG1 variants with improved binding to the Fc gamma R," *J Biol Chem*. Mar. 2, 2001;276(9):6591-604.
- Shulkes A, et al. "Production of calcitonin gene related peptide, calcitonin and PTH-related protein by a prostatic adenocarcinoma," *Clin Endocrinol (Oxf)*. May 1991;34(5):387-93.
- Silberstein S, et al. "Botulinum toxin type A as a migraine preventive treatment. For the Botox Migraine Clinical Research Group," *Headache*. Jun. 2000;40(6):445-50.
- Silberstein SD, "Practice parameter: evidence-based guidelines for migraine headache (an evidence-based review): report of the Quality Standards Subcommittee of the American Academy of Neurology," *Neurology*. Sep. 26, 2000;55(6):754-62.
- Silberstein SD. "Emerging target-based paradigms to prevent and treat migraine," *Clin Pharmacol Ther*. Jan. 2013;93(1):78-85.
- Silverman AJ, et al. "Mast cells migrate from blood to brain," *J Neurosci*. Jan. 1, 2000;20(1):401-8.
- Simmons LC, et al. "Expression of full-length immunoglobulins in *Escherichia coli*: rapid and efficient production of aglycosylated antibodies," *J Immunol Methods*. May 1, 2002;263(1-2):133-47.
- Sixt ML, et al. "Calcitonin gene-related peptide receptor antagonist olcegepant acts in the spinal trigeminal nucleus," *Brain*. Nov. 2009;132(Pt 11):3134-41.

(56)

References Cited

OTHER PUBLICATIONS

- Skofitsch G, et al. "Comparative immunohistochemical distribution of amylin-like and calcitonin gene related peptide like immunoreactivity in the rat central nervous system," *Can J Physiol Pharmacol*. Jul. 1995;73(7):945-56.
- Smillie SJ, et al. "Calcitonin gene-related peptide (CGRP) and its role in hypertension," *Neuropeptides*. Apr. 2011;45(2):93-104.
- Smith KA, et al. "Demystified . . . recombinant antibodies," *J Clin Pathol*. Sep. 2004;57(9):912-7.
- Smith TW, et al. "Reversal of advanced digoxin intoxication with Fab fragments of digoxin-specific antibodies," *N Engl J Med*. Apr. 8, 1976;294(15):797-800.
- Smith-Gill SJ, et al. "Contributions of immunoglobulin heavy and light chains to antibody specificity for lysozyme and two haptens," *J Immunol*. Dec. 15, 1987;139(12):4135-44.
- Solomon S. "Major therapeutic advances in the past 25 years," *Headache*. Apr. 2007;47 Suppl 1:S20-2.
- Song MK, et al. "Light chain of natural antibody plays a dominant role in protein antigen binding," *Biochem Biophys Res Commun*. Feb. 16, 2000;268(2):390-4.
- Spetz AC, et al. "Momentary increase in plasma calcitonin gene-related peptide is involved in hot flashes in men treated with castration for carcinoma of the prostate," *J Urol*. Nov. 2001;166(5):1720-3.
- Sprenger T, et al. "Migraine pathogenesis and state of pharmacological treatment options," *BMC Med*. Nov. 16, 2009;7:71.
- Stensrud P, et al. "Comparative trial of Tenormin (atenolol) and Inderal (propranolol) in migraine," *Headache*. Jul. 1980;20(4):204-7.
- Storer RJ, et al. "Calcitonin gene-related peptide (CGRP) modulates nociceptive trigeminovascular transmission in the cat," *Br J Pharmacol*. Aug. 2004;142(7):1171-81.
- Stovner LJ, et al. "New drugs for migraine," *J Headache Pain*. Dec. 2009;10(6):395-406.
- Strassman AM, et al. "On the origin of headaches," *Endeavour*. 1997;21(3):97-100.
- Strassman AM, et al. "Response properties of dural nociceptors in relation to headache," *J Neurophysiol*. Mar. 2006;95(3):1298-306.
- Subramanian KN, et al. "Safety, tolerance and pharmacokinetics of a humanized monoclonal antibody to respiratory syncytial virus in premature infants and infants with bronchopulmonary dysplasia," *MEDI-493 Study Group, Pediatr Infect Dis J*. Feb. 1998;17(2):110-5.
- Tam SH, et al. "Abciximab (ReoPro, chimeric 7E3 Fab) demonstrates equivalent affinity and functional blockade of glycoprotein IIb/IIIa and alpha(v)beta3 integrins," *Circulation*. Sep. 15, 1998;98(11):1085-91.
- Tamura M, et al. "Structural correlates of an anticarcinoma antibody: identification of specificity-determining residues (SDRs) and development of a minimally immunogenic antibody variant by retention of SDRs only," *J Immunol*. Feb. 1, 2000;164(3):1432-41.
- Tan et al. "Demonstration of the neurotransmitter role of calcitonin gene-related peptides (CGRP) by immunoblockade with anti-CGRP monoclonal antibodies," *Br J Pharmacol*. Mar. 1994;111(3):703-10.
- Tan KK, et al. "Calcitonin gene-related peptide as an endogenous vasodilator: immunoblockade studies in vivo with an anti-calcitonin gene-related peptide monoclonal antibody and its Fab' fragment," *Clin Sci (Lond)*. Dec. 1995;89(6):565-73.
- Tanaka H, et al. "Inhibition of calcitonin gene-related peptide (CGRP) has the potential to extend first-phase insulin secretion," *Exp Clin Endocrinol Diabetes*. May 2013;121(5):280-5.
- Taylor AW, et al. "Suppression of nitric oxide generated by inflammatory macrophages by calcitonin gene-related peptide in aqueous humor," *Invest Ophthalmol Vis Sci*. Jul. 1998;39(8):1372-8.
- Tedstone, et al. "The effect of islet amyloid polypeptide (amylin) and calcitonin gene-related peptide on glucose removal in the anaesthetized rat and on insulin secretion from rat pancreatic islets in vitro," *Biosci Rep*. Aug. 1990;10(4):339-45.
- Tepper SJ, Bigal ME, et al. "Botulinum toxin type A in the treatment of refractory headache," In Olsen J, Silberstein SD, Tfelt-Hansen P, eds. *Preventive Pharmacotherapy of Headache Disorders*. Copenhagen: Oxford University Press, 2004, Chapter 20.
- Tepper SJ, et al. "Botulinum neurotoxin type A in the preventive treatment of refractory headache: a review of 100 consecutive cases," *Headache*. Sep. 2004;44(8):794-800.
- Tepper SJ, et al. "Clinical and preclinical rationale for CGRP-receptor antagonists in the treatment of migraine," *Headache*. Sep. 2008;48(8):1259-68.
- Tepper SJ, et al. "Mechanisms of action of the 5-HT1B/1D receptor agonists," *Arch Neurol*. Jul. 2002;59(7):1084-8.
- Teva Pharmaceutical Industries Ltd., Press Release, "Teva to Acquire Labrys Biologics, Inc.," Jun. 3, 2014. 4 pages.
- Tfelt-Hansen P, et al. "Effervescent metoclopramide and aspirin (Migravess) versus effervescent aspirin or placebo for migraine attacks: a double-blind study," *Cephalalgia*. Jun. 1984;4(2):107-11.
- Tfelt-Hansen P, et al. "Possible site of action of CGRP antagonists in migraine," *Cephalalgia*. Apr. 2011;31(6):748-50.
- Tfelt-Hansen PC. "Verisimilitude (or "truthlikeness") as an alternative to pro and cons: migraine and cluster headache mechanisms," *J Headache Pain*. Oct. 2010;11(5):379-89.
- Theoharides TC, et al. "The role of mast cells in migraine pathophysiology," *Brain Res Brain Res Rev*. Jul. 2005;49(1):65-76.
- Thomas TC, et al. "Inhibition of complement activity by humanized anti-C5 antibody and single-chain Fv," *Mol Immunol*. Dec. 1996;33(17-18):1389-401.
- Tjen-A-Looi S, et al. "CGRP and somatostatin modulate chronic hypoxic pulmonary hypertension," *Am J Physiol*. Sep. 1992;263(3 Pt 2):H681-90.
- Toda M, et al. "Neuronal system-dependent facilitation of tumor angiogenesis and tumor growth by calcitonin gene-related peptide," *Proc Natl Acad Sci U S A*. Sep. 9, 2008;105(36):13550-5.
- Todd J. Schwedt et al: "14th International Headache Congress: Basic Science Highlights", *Headache*, vol. 50, No. 3, Mar. 1, 2010 (Mar. 1, 2010), pp. 520-526.
- Tokuda Y, et al. "Dose escalation and pharmacokinetic study of a humanized anti-HER2 monoclonal antibody in patients with HER2/neu-overexpressing metastatic breast cancer," *Br J Cancer*. Dec. 1999;81(8):1419-25.
- Tsujikawa K, et al. "Hypertension and dysregulated proinflammatory cytokine production in receptor activity-modifying protein 1-deficient mice," *Proc Natl Acad Sci U S A*. Oct. 16, 2007;104(42):16702-7.
- Tumer LC, et al. "A neural shift theory of migraine," *Neuroepidemiology*. 1993; 12(4):249-50.
- Tvedskov JF, et al. "No increase of calcitonin gene-related peptide in jugular blood during migraine," *Ann Neurol*. Oct. 2005;58(4):561-8.
- Tzabazis AZ, et al. "Antihyperalgesic effect of a recombinant herpes virus encoding antisense for calcitonin gene-related peptide," *Anesthesiology*. Jun. 2007; 106(6):1196-203.
- Uhr M, et al. "Penetration of endogenous steroid hormones corticosterone, cortisol, aldosterone and progesterone into the brain is enhanced in mice deficient for both mdra and mdr1b P-glycoproteins," *J Neuroendocrinol*. Sep. 2002;14(9):763-9.
- Unger J. "Migraine headaches: a historical prospective, a glimpse into the future, and migraine epidemiology," *Dis Mon*. Oct. 2006;52(10):367-84.
- Vajdos FF, et al. "Comprehensive functional maps of the antigen-binding site of an anti-ErbB2 antibody obtained with shotgun scanning mutagenesis," *J Mol Biol*. Jul. 5, 2002;320(2):415-28.
- Van der Schueren BJ, et al. "Calcitonin gene-related peptide8-37 antagonizes capsaicin-induced vasodilation in the skin: evaluation of a human in vivo pharmacodynamic model," *J Pharmacol Exp Ther*. Apr. 2008;325(1):248-55.
- Van Rossum D, et al. "Neuroanatomical localization, pharmacological characterization and functions of CGRP, related peptides and their receptors," *Neurosci Biobehav Rev*. Sep. 1997;21(5):649-78.
- Vater A, et al. "Short bioactive Spiegelmers to migraine-associated calcitonin gene-related peptide rapidly identified by a novel approach: tailored-SELEX," *Nucleic Acids Res*. Nov. 1, 2003;31(21):e130.
- Vaughan TJ, et al. "Human antibodies with sub-nanomolar affinities isolated from a large non-immunized phage display library," *Nat Biotechnol*. Mar. 1996;14(3):309-14.

(56)

References Cited

OTHER PUBLICATIONS

- Villalón CM, et al. "The role of CGRP in the pathophysiology of migraine and efficacy of CGRP receptor antagonists as acute antimigraine drugs," *Pharmacol Ther.* Dec. 2009;124(3):309-23.
- Vincent A, et al. "Molecular targets for autoimmune and genetic disorders of neuromuscular transmission," *Eur J Biochem.* Dec. 2000;267(23):6717-28.
- Vogler B, et al. "Role of melatonin in the pathophysiology of migraine: implications for treatment," *CNS Drugs.* 2006;20(5):343-50.
- Volcy M, et al. "Botulinum toxin A for the treatment of greater occipital neuralgia and trigeminal neuralgia: a case report with pathophysiological considerations," *Cephalalgia.* Mar. 2006;26(3):336-40.
- Von Mehren M, et al. "Monoclonal antibody therapy for cancer," *Annu Rev Med.* 2003;54:343-69.
- Wachter C, et al. "Visceral vasodilatation and somatic vasoconstriction evoked by acid challenge of the rat gastric mucosa: diversity of mechanisms," *J Physiol.* Jul. 15, 1995;486 (Pt 2):505-16.
- Wacnik PW, et al. "Tumor-induced mechanical hyperalgesia involves CGRP receptors and altered innervation and vascularization of DsRed2 fluorescent hindpaw tumors." *Pain.* May 2005;115(1-2):95-106.
- Waeber C, et al. "Migraine as an inflammatory disorder." *Neurology.* May 24, 2005;64(10 Suppl 2):S9-15.
- Walker CS, et al. "Mice lacking the neuropeptide alpha-calcitonin gene-related peptide are protected against diet-induced obesity," *Endocrinology.* Sep. 2010;151(9):4257-69.
- Walker CS, et al. "Regulation of signal transduction by calcitonin gene-related peptide receptors," *Trends Pharmacol Sci.* Oct. 2010;31(10):476-83.
- Ward ES, et al. "Binding activities of a repertoire of single immunoglobulin variable domains secreted from *Escherichia coli*." *Nature.* Oct. 12, 1989;341(6242):544-6.
- Weir AN, et al. "Formulating antibody fragments to mediate specific therapeutic functions," *Biochem Soc Trans.* Aug. 2002;30(4):512-6.
- Welch KM, et al. "Mismatch in how oestrogen modulates molecular and neuronal function may explain menstrual migraine," *Neurol Sci.* May 2006;27 Suppl 2:S190-2.
- Werther WA, et al. "Humanization of an anti-lymphocyte function-associated antigen (LFA)-1 monoclonal antibody and reengineering of the humanized antibody for binding to rhesus LFA-1," *J Immunol.* Dec. 1, 1996;157(11):4986-95.
- Wick EC, et al. "Transient receptor potential vanilloid 1, calcitonin gene-related peptide, and substance P mediate nociception in acute pancreatitis." *Am J Physiol Gastrointest Liver Physiol.* May 2006;290(5):G959-69. Epub Jan. 6, 2006.
- Willats WG. "Phage display: practicalities and prospects," *Plant Mol Biol.* Dec. 2002;50(6):837-54.
- Williamson DJ, et al. "Intravital microscope studies on the effects of neurokinin agonists and calcitonin gene-related peptide on dural vessel diameter in the anaesthetized rat," *Cephalalgia.* Jun. 1997;17(4):518-24.
- Williamson DJ, et al. "Neurogenic inflammation in the context of migraine," *Microsc Res Tech.* May 1, 2001;53(3):167-78.
- Williamson DJ, et al. "Sumatriptan inhibits neurogenic vasodilation of dural blood vessels in the anaesthetized rat—intravital microscope studies," *Cephalalgia.* Jun. 1997;17(4):525-31.
- Williamson DJ, et al. "The anti-migraine 5-HT_{1B/1D} agonist rizatriptan inhibits neurogenic dural vasodilation in anaesthetized guinea-pigs," *Br J Pharmacol.* Aug. 2001;133(7):1029-34.
- Williamson DJ, et al. "The novel anti-migraine agent rizatriptan inhibits neurogenic dural vasodilation and extravasation," *Eur J Pharmacol.* Jun. 5, 1997;328(1):61-4.
- Wimalawansa SJ, et al. "Comparative study of distribution and biochemical characterization of brain calcitonin gene-related peptide receptors in five different species," *Neuroscience.* May 1993;54(2):513-9.
- Wimalawansa SJ, et al. "Validation, role in perioperative assessment, and clinical applications of an immunoradiometric assay for human calcitonin," *Peptides.* 1995;16(2):307-12.
- VVimalawansa SJ. "Amylin, calcitonin gene-related peptide, calcitonin, and adrenomedullin: a peptide superfamily," *Crit Rev Neurobiol.* 1997;11(2-3):167-239.
- Wimalawansa SJ. "Calcitonin gene-related peptide and its receptors: molecular genetics, physiology, pathophysiology, and therapeutic potentials," *Endocr Rev.* Oct. 1996;17(5):533-85.
- Wimalawansa SJ. "Effects of in vivo stimulation on molecular forms of circulatory calcitonin and calcitonin gene-related peptide in man," *Mol Cell Endocrinol.* May 28, 1990;71(1):13-9.
- Winkler K, et al. "Changing the antigen binding specificity by single point mutations of an anti-p24 (HIV-1) antibody," *J Immunol.* Oct. 15, 2000;165(8):4505-14.
- Winter G, et al. "Making antibodies by phage display technology," *Annu Rev Immunol.* 1994;12:433-55.
- Wong G, et al., "Safety and tolerability of LBR-101, a humanized monoclonal antibody that blocks the binding of CGRP to its receptor," *Labrys Biologics Poster*, 1 p. 2013 International Headache Congress.
- Wong HC, et al. "Monoclonal antibody to rat alpha-CGRP: production, characterization, and in vivo immunoneutralization activity," *Hybridoma.* Feb. 1993;12(1):93-106.
- Wong HC, et al. "Preparation of a monoclonal antibody to rat alpha-CGRP for in vivo immunoneutralization of peptides." *Ann N Y Acad Sci.* Jun. 30, 1992;657:525-7.
- Wu D, et al. "Development and potential of non-peptide antagonists for calcitonin-gene-related peptide (CGRP) receptors: evidence for CGRP receptor heterogeneity," *Biochem Soc Trans.* Aug. 2002;30(4):468-73.
- Wu H, et al. "Humanization of a murine monoclonal antibody by simultaneous optimization of framework and CDR residues," *J Mol Biol.* Nov. 19, 1999;294(1):151-62.
- Wu H, et al. "Humanized antibodies and their applications," *Methods.* May 2005;36(1):1-2.
- Wyon Y, et al. "Postmenopausal women with vasomotor symptoms have increased urinary excretion of calcitonin gene-related peptide," *Maturitas.* Nov. 16, 1998;30(3):289-94.
- Xu, F.T. Study on the Mechanism of SP and CGRP in the Chronic Pain and Knee Joint. Master Thesis. Guangxi Medical University. May 2005. (In Chinese with English abstract).
- Yallampalli C, et al. "Calcitonin gene-related peptide in pregnancy and its emerging receptor heterogeneity," *Trends Endocrinol Metab.* Aug. 2002;13(6):263-9.
- Yoshikawa R, et al. "Suppression of ovalbumin-induced allergic diarrhea by diminished intestinal peristalsis in RAMP1-deficient mice." *Biochem Biophys Res Commun.* Jul. 8, 2011;410(3):389-93.
- Yu LC, et al. "Roles of calcitonin gene-related peptide and its receptors in pain-related behavioral responses in the central nervous system," *Neurosci Biobehav Rev.* Sep. 2009;33(8):1185-91.
- Zeller J, et al. "CGRP function-blocking antibodies inhibit neurogenic vasodilatation without affecting heart rate or arterial blood pressure in the rat." *Br J Pharmacol.* Dec. 2008;155(7):1093-103. doi: 10.1038/bjp.2008.334. Epub Sep. 8, 2008.
- Zhang L. et al. "Arthritic calcitonin/alpha calcitonin gene-related peptide knockout mice have reduced nociceptive hypersensitivity." *Pain.* Jan. 2001;89(2-3):265-73.
- Zhang M, et al. "Rheumatoid factor specificity of a VH3-encoded antibody is dependent on the heavy chain CDR3 region and is independent of protein A binding." *J Immunol.* Sep. 1, 1998;161(5):2284-9.
- Zhuang X, et al. "Brain mast cell degranulation regulates blood-brain barrier," *J Neurobiol.* Dec. 1996;31(4):393-403.
- Zittel et al., "Role of spinal afferents and calcitonin gene-related peptide in the postoperative gastric ileus in anesthetized rats," *Ann Surg.* Jan. 1994;219(1):79-87.
- Zittel TT, et al. "Calcitonin gene-related peptide and spinal afferents partly mediate postoperative colonic ileus in the rat," *Surgery.* May 1998;123(5):518-27.

(56)

References Cited**OTHER PUBLICATIONS**

Zuckier LS, et al. "Chimeric human-mouse IgG antibodies with shuffled constant region exons demonstrate that multiple domains contribute to in vivo half-life," *Cancer Res.* Sep. 1, 1998;58(17):3905-8.

Misura, et al. "The Eptinezumab: CGRP Complex Structure and Characterization of the Ligand Binding Interface," poster Presented at the American Headache Society (AHS) 61st Annual Scientific Meeting Jul. 11-14, 2019.

Rita Costa A, Elisa Rodrigues M, Henriques M, Azeredo J, Oliveira R. Guidelines to cell engineering for monoclonal antibody production. *Eur J Pharm Biopharm.* 2010;74(2):127-138. doi:10.1016/j.ejpb.2009.10.002.

Potgieter TI, Cukan M, Drummond JE, et al. Production of monoclonal antibodies by glycoengineered *Pichia pastoris*. *J Biotechnol.* 2009;139(4):318-325. doi:10.1016/j.jbiotec.2008.12.015.

Trill JJ, Shatzman AR, Ganguly S. Production of monoclonal antibodies in COS and CHO cells. *Curr Opin Biotechnol.* 1995;6(5):553-560. doi:10.1016/0958-1669(95)80092-1.

Alder Biopharmaceuticals. "Alder BioPharmaceuticals announces positive eptinezumab Phase 3 results for prevention of frequent episodic migraine." (2017).

Dodick, David W., et al. "Eptinezumab demonstrated efficacy in sustained prevention of episodic and chronic migraine beginning on day 1 after dosing." *Headache: The Journal of Head and Face Pain* 60.10 (2020): 2220-2231.

Dodick, David W., et al. "Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial." *The lancet neurology* 13.11 (2014): 1100-1107.

Edvinsson, L. "The Trigemino-vascular pathway: role of CGRP and CGRP receptors in migraine." *Headache.* 57 (Suppl 2): 47-55." (2017).

George, Judy. "Eptinezumab Effective in Chronic Migraine: Intravenous CGRP blocker shows rapid treatment effect," *MedPageToday*, Apr. 27, 2018.

Lee, Mi Ji, et al. "Feasibility of serum CGRP measurement as a biomarker of chronic migraine: a critical reappraisal." *The journal of headache and pain* 19.1 (2018): 1-8.

Maasumi, Kasra, Rebecca L. Michael, and Alan M. Rapoport. "CGRP and migraine: the role of blocking calcitonin gene-related peptide ligand and receptor in the management of migraine." *Drugs* 78 (2018): 913-928.

Marmura MJ, et al. Preventive migraine treatment with eptinezumab reduced acute headache medication and headache frequency to below diagnostic thresholds in patients with chronic migraine and medication-overuse headache. *Headache: The Journal of Head and Face Pain.* Oct. 2021;61(9):1421-31.

Peters, Golden L. "Migraine overview and summary of current and emerging treatment options." *Am J Manag Care* 25.2 Suppl (2019): S23-S34.

Raffaelli, Bianca, and Uwe Reuter. "The biology of monoclonal antibodies: focus on calcitonin gene-related peptide for prophylactic migraine therapy." *Neurotherapeutics* 15.2 (2018): 324-335.

Silberstein, S. D., et al. "Eptinezumab results for the prevention of episodic migraine over one year in the PROMISE-1 (PREvention of migraine via intravenous Eptinezumab safety and efficacy-1) trial." *Headache.* vol. 58. No. 8. 111 River St, Hoboken 07030-5774, NJ USA: Wiley, 2018. p. 1298.

Singh SR, Zhang J, O'Dell C, Hsieh MC, Goldstein J, Liu J, Srivastava A. Effect of polysorbate 80 quality on photostability of a monoclonal antibody. *Aaps PharmSciTech.* Jun. 2012;13:422-30.

Tepper, Stewart J. "CGRP and headache: a brief review." *Neurological Sciences* 40 (2019): 99-105.

The Department of Health and Human Services U.S. Food and Drug Administration, The Pediatric Exclusivity Provision, Jan. 2001 Status Report to Congress (Year 2001).

The International Classification of Headache Disorders, second edition, *Cephalalgia*, 24(Suppl 1) 2004 (Year: 2004).

Warne, Nicholas W. "Development of high concentration protein biopharmaceuticals: the use of platform approaches in formulation development." *European journal of pharmaceuticals and biopharmaceuticals* 78.2 (2011): 208-212.

Dodick, David, et al. "A single intravenous administration of ALD403 (eptinezumab) reduces use of triptans among patients with chronic migraine." *Cephalalgia.* vol. 37. 1 Olivers Yard, 55 City Road, London EC1Y 1SP, England: Sage Publications Ltd, 2017.

Lipton, Richard B et al. "Patient-identified most bothersome symptom in preventive migraine treatment with eptinezumab: A novel patient-centered outcome." *Headache* vol. 61.5 (2021): 766-776. doi:10.1111/head.14120.

Lipton, Richard B et al. "Evaluating the clinical utility of the patient-identified most bothersome symptom measure from PROMISE-2 for research in migraine prevention." *Headache* vol. 62.6 (2022): 690-699. doi:10.1111/head.14295.

Brandes, Jan Lewis, et al. "Effects of fremanezumab on the use of acute headache medication and associated symptoms of migraine in patients with episodic migraine." *Cephalalgia* 40.5 (2020): 470-477.

Munjal, Sagar, et al. "Most Bothersome Associated Migraine Symptom: Results from 2017 Migraine in America Symptoms and Treatment (MAST) Study (p. 3 10-017)." *Neurology* 92.15 supplement (2019): p. 3-10.

Silberstein, Stephen D., et al. "Fremanezumab for the preventive treatment of chronic migraine." *New England Journal of Medicine* 377.22 (2017): 2113-2122.

Database Embase [online] Jan. 1, 2018 (Jan. 1, 2018), Silberstein S: "The impact of fremanezumab on medication overuse in patients with chronic migraine2018", Database accession No. EMB-624431011*.

* cited by examiner

Figure 1A - Heavy Chain Protein Sequence

Sequence Name	FR1	CDR1	FR2	CDR2
Ab1	QSLSESGGRLVTPGTPLTITCTVSGLDLS	SYMQ	WVRQAPGKGLEWIG	VIGINDNTYYASWAKG
Ab2	EVQLVESGGGLVQPGGSLRLSCAVSGLDLS	SYMQ	WVRQAPGKGLEWVG	VIGINDNTYYASWAKG
Ab3	EVQLVESGGGLVQPGGSLRLSCAVSGLDLS	SYMQ	WVRQAPGKGLEWVG	VIGINDNTYYASWAKG
Ab4	QSLSESGGRLVTPGTPLTITCTVSGIDLS	GYMN	WVRQAPGKGLEWIG	VIGINGATYYASWAKG
Ab5	EVQLVESGGGLVQPGGSLRLSCAVSGIDLS	GYMN	WVRQAPGKGLEWVG	VIGINGATYYASWAKG
Ab6	EVQLVESGGGLVQPGGSLRLSCAVSGIDLS	GYMN	WVRQAPGKGLEWVG	VIGINGATYYASWAKG
Ab7	QEQLKESGGRLVTPGTSLTITCTVSGIDLS	NHYMQ	WVRQAPGKGLEWIG	VVGINGRTYYASWAKG
Ab8	EVQLVESGGGLVQPGGSLRLSCAVSGIDLS	NHYMQ	WVRQAPGKGLEWVG	VVGINGRTYYASWAKG
Ab9	QSLSESGGRLVTPGTPLTITCTVSGIGLS	SYMQ	WVRQSPGRGLEWIG	VIGSDGKTYATWAKG
Ab10	EVQLVESGGGLVQPGGSLRLSCAVSGIGLS	SYMQ	WVRQAPGKGLEWVG	VIGSDGKTYATWAKG
Ab11	QSLSESGGRLVTPGGSITITCTVSGIDVT	NYMQ	WVRQAPGKGLEWIG	VIGVNGKRYIASWAKG
Ab12	EVQLVESGGGLVQPGGSLRLSCAVSGIDVT	NYMQ	WVRQAPGKGLEWVG	VIGVNGKRYIASWAKG
Ab13	QSVESGGGLVQPEGSLITCTASGFDFS	SNAMW	WVRQAPGKGLEWIG	CIYNGDGSYYASWVNG
Ab14	EVQLVESGGGLVQPGGSLRLSCAVSGIGLS	SYMQ	WVRQAPGKGLEWVG	VIGSDGKTYATWAKG

Figure 1B - Heavy Chain Protein Sequence

Sequence Name	FR3	CDR3	FR4
Ab1	RFTISRASSTTVDLKMTSLTTEDTATYFCAR	GDI	WGPGLTVTVSS
Ab2	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab3	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab4	RFTISKTSSTTVDLKMTSLTTEDTATYFCAR	GDI	WGPGLTVTVSS
Ab5	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab6	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab7	RFTISKTSSTTVDLKMTSLTTEDTATYFCAR	GDI	WGPGLTVTVSS
Ab8	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab9	RFTISKTSSTTVDLRMASLTEDTATYFCTR	GDI	WGPGLTVTVSS
Ab10	RFTISRDNKTTVYLQMNSLRAEDTAVYFCTR	GDI	WGQGLTVTVSS
Ab11	RFTISKTSSTTVDLKMTSLTTEDTATYFCAR	GDI	WGPGLTVTVSS
Ab12	RFTISRDNKTTVYLQMNSLRAEDTAVYFCAR	GDI	WGQGLTVTVSS
Ab13	RFSISKTSSTTVTLQNSLTIVADTATYFCAR	DLDL	WGPGLTVTVSS
Ab14	RFTISRDNKTTVYLQMNSLRAEDTAVYFCTR	GDI	WGQGLTVTVSS

Figure 1C - Heavy Chain Protein Sequence

Sequence Name	Constant Region
Ab1	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab2	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab3	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab4	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab5	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab6	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab7	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab8	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab9	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab10	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab11	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab12	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab13	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV
Ab14	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSNVSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV

Figure 1D - Heavy Chain Protein Sequence

Sequence Name	Constant Region
Ab1	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab2	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab3	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab4	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab5	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab6	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab7	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab8	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab9	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab10	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab11	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab12	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab13	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
Ab14	NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA

Figure 1E - Heavy Chain Protein Sequence

Sequence Name	Constant Region
Ab1	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab2	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab3	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab4	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab5	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab6	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab7	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab8	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab9	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab10	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab11	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab12	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab13	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF
Ab14	KTKPREEQYASTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLTCLVKGF

Figure 1F - Heavy Chain Protein Sequence

Sequence Name	Constant Region	(SEQ ID NO: 1)
Ab1	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 1)
Ab2	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 41)
Ab3	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 81)
Ab4	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 121)
Ab5	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 161)
Ab6	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 201)
Ab7	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 241)
Ab8	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 281)
Ab9	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 321)
Ab10	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 361)
Ab11	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 401)
Ab12	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 441)
Ab13	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 481)
Ab14	YPSDIAVWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCVMHEALHNHYTQKSLSLSPGK	(SEQ ID NO: 521)

Figure 2A - Light Chain Protein Sequence

Sequence Name	FR1	CDR1	FR2	CDR2
Ab1	QVLTQTASPVSAAVGSTVTINC	QASQSVYDNNYLA	WYQQKPGQPPKQLIY	STSTLAS
Ab2	QVLTQSPSSLSASVGDRVTTINC	QASQSVYDNNYLA	WYQQKPGKVPKQLIY	STSTLAS
Ab3	QVLTQSPSSLSASVGDRVTTINC	QASQSVYDNNYLA	WYQQKPGKVPKQLIY	STSTLAS
Ab4	QVLTQTTPSPVSAAVGSTVTINC	QASQSVYHNTYLA	WYQQKPGQPPKQLIY	DASTLAS
Ab5	QVLTQSPSSLSASVGDRVTTINC	QASQSVYHNTYLA	WYQQKPGKVPKQLIY	DASTLAS
Ab6	QVLTQSPSSLSASVGDRVTTINC	QASQSVYHNTYLA	WYQQKPGKVPKQLIY	DASTLAS
Ab7	QVLTQTASPVSAAVGSTVTINC	QASQSVYNNYLA	WYQQKPGQPPKQLIY	STSTLAS
Ab8	QVLTQSPSSLSASVGDRVTTINC	QASQSVYNNYLA	WYQQKPGKVPKQLIY	STSTLAS
Ab9	QVLTQTTPSPVSAAVGSTVTINC	QASQSVYNNYLA	WYQQKPGQPPKQLIY	STSTLAS
Ab10	QVLTQSPSSLSASVGDRVTTINC	QASQSVYNNYLA	WYQQKPGKVPKQLIY	STSTLAS
Ab11	QVLTQTASPVSFPAVGSTVTINC	PASQSVYNNYLA	WYQQKPGQPPKQLIY	STSTLAS
Ab12	QVLTQSPSSLSASVGDRVTTINC	PASQSVYNNYLA	WYQQKPGKVPKQLIY	STSTLAS
Ab13	AIVMTQTTPSSKSPVPGDVTINC	QASESLYNNALA	WFQQKPGQPPKRLIY	DASKLAS
Ab14	QVLTQSPSSLSASVGDRVTTINC	QASQSVYNNYLA	WYQQKPGKVPKQLIY	STSTLAS

Figure 2B - Light Chain Protein Sequence

Sequence Name	FR3	CDR3	FR4
Ab1	GVSSRFKSGSGTQFTLTISDLECAATAIYC	LGSYDCSSGDCVF	FGGTEVVVKR
Ab2	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSSGDCVF	FGGTTKVEIKR
Ab3	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSSGDCVF	FGGTTKVEIKR
Ab4	GVPSRFSGSGTQFTLTISGVQCNDAAAIYC	LGSYDCTNGDCVF	FGGTEVVVKR
Ab5	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCTNGDCVF	FGGTTKVEIKR
Ab6	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCTNGDCVF	FGGTTKVEIKR
Ab7	GVSSRFKSGSGTQFTLTISDVQCDDAAAIYC	LGSYDCSTGDCVF	FGGTEVVVKR
Ab8	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSTGDCVF	FGGTTKVEIKR
Ab9	GVSSRFSGSGTQFTLTISDVQCDDAAAIYC	LGSYDCSRGDCVF	FGGTEVVVKR
Ab10	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSRGDCVF	FGGTTKVEIKR
Ab11	GVSSRFKSGSGTQFTLTISDVQCDDAAAIYC	LGSYDCSNGDCVF	FGGTEVVVKR
Ab12	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSNGDCVF	FGGTTKVEIKR
Ab13	GVPSRFSGSGTQFTLTISGVQCDDAAAIYC	GGYRSDSDGVA	FAGGTEVVVKR
Ab14	GVPSRFSGSGTDFTLTISLQPEDVATYYC	LGSYDCSRGDCVF	FGGTTKVEIKR

Figure 2C - Light Chain Protein Sequence

Sequence Name	Constant Region
Ab1	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab2	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab3	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab4	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab5	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab6	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab7	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab8	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab9	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab10	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab11	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab12	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab13	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA
Ab14	TVAAPSVFIFFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKKHKVYA

Figure 2D - Light Chain Protein Sequence

Sequence Name	Constant Region
Ab1	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 21)
Ab2	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 61)
Ab3	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 101)
Ab4	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 141)
Ab5	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 181)
Ab6	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 221)
Ab7	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 261)
Ab8	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 301)
Ab9	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 341)
Ab10	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 381)
Ab11	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 421)
Ab12	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 461)
Ab13	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 501)
Ab14	CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 541)

Figure 3A - Heavy Chain DNA Sequence

Sequence Name	FR1
Ab1	CAGTCCGCTGGAGGAGTCCGGGGTGCCTGGTCAAGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGACTCGACCTCAGT
Ab2	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGACTCGACCTCAGT
Ab3	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGACTCGACCTCAGT
Ab4	CAGTCCGCTGGAGGAGTCCGGGGTGCCTGGTCAAGCCTGGGACACCCCTGACACTCACCTGTTCCCTCTCTGGCATCGACCTCAGT
Ab5	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGACCTCAGT
Ab6	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGACCTCAGT
Ab7	CAGGACAGCTGAAGGAGTCCGGGGTGCCTGGTCAAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGACCTCAGT
Ab8	GAGGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGACCTCAGT
Ab9	CAGTCCGCTGGAGGAGTCCGGGGTGCCTGGTCAAGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGAAATCGGCTCAGT
Ab10	GAGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGGCTCAGT
Ab11	CAGTCCGCTGGAGGAGTCCGGGGTGCCTGGTCAAGCCTGGAGGATCCCTGACACTCACCTGCACAGTCTCTGGAAATCGACCTCAGT
Ab12	GAGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGACCTCAGT
Ab13	CAGTCCGCTGGAGGAGTCCGGGGGAGGCTGGTCCAGCCTGAGGGATCCCTGACACTCACCTGCACAGCCTCTGGAAATCGACTTTCAGT
Ab14	GAGTGCAGCTTGTGGAGTCTGGGGGAGGCTGGTCCAGCCTGGGGGTCCCTGAGACTCTCCTGTGCAGTCTCTGGAAATCGGCTCAGT

Figure 3B - Heavy Chain DNA Sequence

Sequence Name	CDR1	FR2
Ab1	AGCTACTACATGCAA	TGGTCCGCCAGGCTCCAGGGAAGGGGTGGAATGGATCGGA
Ab2	AGCTACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab3	AGCTACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab4	GGCTACTACATGAAC	TGGTCCGCCAGGCTCCAGGGAAGGGGTGGAATGGATCGGA
Ab5	GGCTACTACATGAAC	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab6	GGCTACTACATGAAC	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab7	AACCACTACATGCAA	TGGTCCGCCAGGCTCCAGGGAAGGGGTGGAGTGGATCGGA
Ab8	AACCACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab9	AGCTACTACATGCAG	TGGTCCGCCAGTCTCCAGGGAAGGGGTGGAATGGATCGGA
Ab10	AGCTACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab11	AACCTACTATATGCAA	TGGTCCGCCAGGCTCCAGGGAAGGGGTGGATGGATCGGA
Ab12	AACCTACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA
Ab13	AGCAATGCAATGTGG	TGGTCCGCCAGGCTCCAGGGAAGGGGTGGAGTGGATCGGA
Ab14	AGCTACTACATGCAA	TGGTCCGTCCAGGCTCCAGGGAAGGGGTGGAGTGGTTCGGA

Figure 3C - Heavy Chain DNA Sequence

Sequence Name	CDR2
Ab1	GTCATTGGTATTAAATGATAAACACATACTACGCGAGCTGGCGAAAGGC
Ab2	GTCATTGGTATCAATGATAAACACATACTACGCGAGCTGGCGAAAGGC
Ab3	GTCATTGGTATCAATGATAAACACATACTACGCGAGCTGGCGAAAGGC
Ab4	GTCATTGGTATTAATGGTCCACATACTACGCGAGCTGGCGAAAGGC
Ab5	GTCATTGGTATTAATGGTCCACATACTACGCGAGCTGGCGAAAGGC
Ab6	GTCATTGGTATTAATGGTCCACATACTACGCGAGCTGGCGAAAGGC
Ab7	GTCGTTGGTATTAATGGTCCACATACTACGCGAGCTGGCGAAAGGC
Ab8	GTCGTTGGTATCAATGGTCCACATACTACGCGAGCTGGCGAAAGGC
Ab9	GTCATTGGTAGTGATGGTAAGACATACTACGCGACCTGGCGAAAGGC
Ab10	GTCATTGGTAGTGATGGTAAGACATACTACGCGACCTGGCGAAAGGC
Ab11	GTCATTGGTGTGAATGGTAAGAGATACTACGCGAGCTGGCGAAAGGC
Ab12	GTCATTGGTGTGAATGGTAAGAGATACTACGCGAGCTGGCGAAAGGC
Ab13	TGCATTACAATGGTATGGCAGACATACTACGCGAGCTGGTGAATGGC
Ab14	GTCATTGGTAGTGATGGTAAGACATACTACGCGACCTGGCGAAAGGC

Figure 3D - Heavy Chain DNA Sequence

Sequence Name	FR3
Ab1	CGATTCACCATCTCCAGAGCCTCGTGGACCAACGGTGGATCTGAAAATGACCAGTCTGACAACCGAGGACACGGCCACCTATTTCTGTGCCAGA
Ab2	CGATTCACCATCTCCAGAGACAAATCCAAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab3	CGATTCACCATCTCCAGAGACAAATCCAAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab4	CGATTCACCATCTCCAAAACCTCGTCGACCACGGTGGATCTGAAAATGACCAGTCTGACAACCGAGGACACGGCCACCTATTTCTGTGCCAGA
Ab5	CGATTCACCATCTCCAGAGACAAATCCAAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab6	CGATTCACCATCTCCAGAGACAAATCCAAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab7	CGATTCACCATCTCCAGAACCTCGTCGACCACGGTGGATCTGAAAATGACCAGGCTGACAACCGAGGACACGGCCACCTATTTCTGTGCCAGA
Ab8	CGATTCACCATCTCCAGAGACAAATTCGAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab9	CGATTCACCATCTCCAGACCTCGTCGACCACGGTGGATCTGGAATGGCCAGTCTGACAACCGAGGACACGGCCACCTATTTCTGTACCAGA
Ab10	CGATTCACCATCTCCAGAGACAAATTCGAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCTAGA
Ab11	CGATTCACCATCTCCAAAACCTCGTCGACCACGGTGGATCTGAAAATGACCAGTCTGACAACCGAGGACACGGCCACCTATTTCTGTGCCAGA
Ab12	CGATTCACCATCTCCAGAGACAAATTCGAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTGCCAGA
Ab13	CGATTCTCCATCTCCAAAACCTCGTCGACCACGGTGAATCTGGAACCTGAATAGTCTGACAGTCCGGGACACGGCCACCTATTTATTTGTGCCAGA
Ab14	CGATTCACCATCTCCAGAGACAAATTCGAAGACCACGGTGTATCTTCAAATGAACAGCCTGAGAGCTGAGGACACTGCTGTGTAATTTCTGTACCAGA

Figure 3E - Heavy Chain DNA Sequence

Sequence Name	CDR3	FR4	Constant Region
Ab1	GGGACATC	TGGGGCCAGGCACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab2	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab3	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab4	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab5	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab6	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab7	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab8	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab9	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab10	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab11	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab12	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab13	GATCTTGACTTG	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
Ab14	GGGACATC	TGGGGCCAGGGACCTCGTCAACGGTCTCGAGC	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC

Figure 3F - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab2	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab3	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab4	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab5	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab6	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab7	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab8	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab9	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab10	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab11	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab12	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab13	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC
Ab14	AAGAGCACCTCTGGGGCCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAACCGGTGACGGTGTGTTGGAACCTCAGGCGGC

Figure 3G - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab2	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab3	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab4	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab5	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab6	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab7	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab8	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab9	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab10	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab11	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab12	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab13	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC
Ab14	CTGACCAGGGCGTGACACACCTTCCGGGTGTCTCTACAGTCTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGC

Figure 3H - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab2	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab3	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab4	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab5	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab6	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab7	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab8	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab9	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab10	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab11	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab12	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab13	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA
Ab14	TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAAGCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAATCTTGTGACAAA

Sequence	Name	Constant	Region
	Ab1	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab2	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab3	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab4	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab5	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab6	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab7	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab8	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab9	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab10	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab11	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab12	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab13	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG
	Ab14	ACTCACATGCCACCGTGCCAGCACCTGAAC	TCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAAGGACACCCCTCATG

Sequence	Name	Constant	Region
Ab1	Ab1	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab2	Ab2	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab3	Ab3	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab4	Ab4	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab5	Ab5	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab6	Ab6	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab7	Ab7	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab8	Ab8	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab9	Ab9	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab10	Ab10	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab11	Ab11	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab12	Ab12	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab13	Ab13	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	
Ab14	Ab14	ATCTCCGGACCCCTGAGGTCACATGCGTGTGGTGGACGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGGTG	

Figure 3K - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab2	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab3	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab4	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab5	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab6	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab7	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab8	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab9	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab10	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab11	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab12	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab13	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC
Ab14	GAGGTGCATAATGCCAAGACAAAAGCCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACACAGGAC

Figure 3L - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab2	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab3	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab4	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab5	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab6	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab7	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab8	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab9	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab10	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab11	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab12	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab13	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG
Ab14	TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAG

Figure 3M - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab2	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab3	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab4	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab5	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab6	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab7	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab8	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab9	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab10	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab11	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab12	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab13	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC
Ab14	CCCCGAGAACCCACAGGTGTACACCCCTGCCCCCATCCCGGAGGAGATGACCAAGAACACAGGTGAGCTGACCTGGCTGGTCAAAGGCTTC

Figure 3N - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab2	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab3	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab4	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab5	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab6	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab7	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab8	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab9	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab10	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab11	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab12	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab13	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC
Ab14	TATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGAGCCGAGAACAACTACAAGACCACGGCTCCCGTGGCTGGACTCCGACGGC

Figure 30 - Heavy Chain DNA Sequence

Sequence Name	Constant Region
Ab1	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab2	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab3	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab4	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab5	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab6	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab7	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab8	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab9	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab10	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab11	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab12	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab13	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG
Ab14	TCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTCATGCAATGAGGCTCTG

Figure 3P - Heavy Chain DNA Sequence

Sequence Name	Constant Region	(SEQ ID NO: 11)
Ab1	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 11)
Ab2	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 51)
Ab3	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 91)
Ab4	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 131)
Ab5	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 171)
Ab6	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 211)
Ab7	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 251)
Ab8	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 291)
Ab9	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 331)
Ab10	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 371)
Ab11	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 411)
Ab12	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 451)
Ab13	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 491)
Ab14	CACAACCACTACACGCAGAGAAGAGCCTCTCCCTGTCTCCGGTAAATGA	(SEQ ID NO: 531)

Figure 4A - Light Chain DNA Sequence

Sequence Name	FR1
Ab1	CAAGTGTGACCCAGACTGCATCCCGTGTCTGCAGCTGTGGGAAGCACAGTCAACATCAATTGC
Ab2	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab3	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab4	CAAGTGTGACCCAGACTCCATCCCGTGTCTGCAGCTGTGGGAAGCACAGTCAACATCAATTGC
Ab5	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab6	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab7	CAAGTGTGACCCAGACTGCATCCCGTGTCTGCAGCTGTGGGAAGCACAGTCAACATCAATTGC
Ab8	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab9	CAAGTGTGACCCAGACTCCATCCCGTGTCTGCAGCTGTGGGAAGCACAGTCAACATCAATTGC
Ab10	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab11	CAGGTGCTGACCCAGACTGCATCCCGTGTCTGCAGCTGTGGGAAGCACAGTCAACATCAATTGC
Ab12	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC
Ab13	GCCATCGTGATGACCCAGACTCCATCTTCCAAAGTCTGTCCCTGTGGGAGACACAGTCAACATCAATTGC
Ab14	CAAGTGTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCAATTGC

Figure 4B - Light Chain DNA Sequence

Sequence Name	CDR1	FR2
Ab1	CAGGCCAGTCAGAGTGTATTATGATAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCAACTGATCTAT
Ab2	CAGGCCAGTCAGAGTGTATTATGATAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab3	CAGGCCAGTCAGAGTGTATTATGATAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab4	CAGGCCAGTCAGAGTGTATTATCATAACACCTACCTGGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCAACTGATCTAT
Ab5	CAGGCCAGTCAGAGTGTATTATCATAACACCTACCTGGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab6	CAGGCCAGTCAGAGTGTATTATCATAACACCTACCTGGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab7	CAGGCCAGTCAGAGTGTATTATTAATTACAACCTACCTTGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCAACTGATCTAT
Ab8	CAGGCCAGTCAGAGTGTATTACAAATTACAACCTTGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab9	CAGGCCAGTCAGATGTATTATAATAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCAACTGATCTAT
Ab10	CAGGCCAGTCAGATGTATTACAAATAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab11	CGGCCAGTCAGAGTGTATTATAAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCAACTGATCTAT
Ab12	CGGCCAGTCAGAGTGTATTATAAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT
Ab13	CAGGCCAGTCAGAGTCTTTATAATAACAACCGCCTTGCC	TGGTATCAGCAGAAAACCCAGGGCAGCCTCCCAGCGCCTGATCTAT
Ab14	CAGGCCAGTCAGATGTATTATAAACAACCTACCTAGCC	TGGTATCAGCAGAAAACCCAGGGAAAGTTCCCTAAGCAACTGATCTAT

Figure 4C - Light Chain DNA Sequence

Sequence Name	CDR2	FR3
Ab1	TCTACATCCACTCTGGCATCT	GGGCTCTCATCGGGTTCAAAGGCAGTGGATCTGGGACACAGTTCACCTCTCACCA
Ab2	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab3	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab4	GATGATCCACTCTGGCGTCT	GGGCTCCCATCTCGGGTTTCAGGGCAGTGGATCTGGGACACAGTTCACCTCTCACCA
Ab5	GATGATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab6	GATGATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab7	TCTACATCCACTCTGGCATCT	GGGCTCTCATCGCGATTCAAAGGCAGTGGATCTGGGACACAGTTCACCTCTCACCA
Ab8	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab9	TCTACGTCACACTCTGGCATCT	GGGCTCTCATCGCGATTCAAAGGCAGTGGATCTGGGACACAGTTCACCTCTCACCA
Ab10	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab11	TCTACATCCACTCTGGCATCT	GGGCTCTCATCGGGTTCAAAGGCAGTGGATCTGGGACACAGTTCACCTCTCACCA
Ab12	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA
Ab13	GATGATCCAAACTGGCATCT	GGGCTCCCATCTCGGGTTTCAGTGGCGGTGGTCTGGGACACAGTTCACCTCTCACCA
Ab14	TCTACATCCACTCTGGCATCT	GGGCTCCCATCTCGTTTTCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCA

Figure 4D - Light Chain DNA Sequence

Sequence Name	FR3	CDR3
Ab1	TCAGGACCTGGAGTGTGCCGATGCTGCCACTTACTACTGT	CTAGGCAGTTATGATTCTAGTAGTGGTGATTTGTTTGT
Ab2	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTAGGCAGTTATGATTCTAGTAGTGGTGATTTGTTTGT
Ab3	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTAGGCAGTTATGATTCTAGTAGTGGTGATTTGTTTGT
Ab4	TCAGGGCGTGCAGTGTAAAGATGCTGCCGCTTACTACTGT	CTGGGCAGTTATGATTCTAGTAAAGTGATTTGTTTGT
Ab5	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTAAAGTGATTTGTTTGT
Ab6	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTAAAGTGATTTGTTTGT
Ab7	TCAGGACGTGCAGTGTGACGATGCTGCCACTTACTACTGT	CTAGGCAGTTATGACCTAGTACTCTGGTGATTTGTTTGT
Ab8	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTACTCTGGTGATTTGTTTGT
Ab9	TCAGGACGTGCAGTGTGACGATGCTGCCACTTACTACTGT	CTAGGCAGTTATGATTCTAGTCTCGTGGTGATTTGTTTGT
Ab10	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTCTCGTGGTGATTTGTTTGT
Ab11	TCAGGACGTGCAGTGTGACGATGCTGCCACTTACTACTGT	CTAGGCAGTTATGATTCTAGTAAAGTGATTTGTTTGT
Ab12	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTAAAGTGATTTGTTTGT
Ab13	TCAGTGGCGTGCAGTGTGACGATGCTGCCACTTACTACTGT	GGAGGCTACAGAAAGTAGTGTGTGATGGTGTGCT
Ab14	TCAGCAGCCTGCAGCCTGAAGATGTTGCAACTTATTACTGT	CTGGGCAGTTATGATTCTAGTCTCGTGGTGATTTGTTTGT

Figure 4E - Light Chain DNA Sequence

Sequence Name	FR4	Constant Region
Ab1	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab2	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab3	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab4	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab5	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab6	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab7	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab8	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab9	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab10	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab11	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab12	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab13	TTCCGGCGGAGGACCGAGGTGGTGGTCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG
Ab14	TTCCGGCGGAGGAACCAAGGTGGAAATCAAACGT	ACGGTGGCTGCACCATCTGTCTTCATCTTCCGGCCATCTGATGAGCAGTTG

Figure 4F - Light Chain DNA Sequence

Sequence Name	Constant Region
Ab1	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab2	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab3	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab4	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab5	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab6	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab7	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab8	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab9	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab10	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab11	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab12	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab13	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC
Ab14	AAATCTGGAACCTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCAAAAGTACAGTGAAGTGGATAACGCC

Figure 4G - Light Chain DNA Sequence

Sequence Name	Constant Region
Ab1	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab2	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab3	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab4	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab5	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab6	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab7	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab8	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab9	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab10	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab11	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab12	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab13	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG
Ab14	TCCAATCGGGTAACTCCACGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACTACAGCCTCAGCAGCACCCCTGACGCTGAG

Figure 4H - Light Chain DNA Sequence

Sequence Name	Constant Region
Ab1	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab2	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab3	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab4	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab5	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab6	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab7	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab8	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab9	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab10	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab11	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab12	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab13	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC
Ab14	CAAAGCAGACTACGAGAAACACAAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTGCCCGTCACAAAGAGCTTCAAC

Figure 4I - Light Chain DNA Sequence

Sequence Name	
Ab1	ACGGGAGAGTGTTAG (SEQ ID NO: 31)
Ab2	ACGGGAGAGTGTTAG (SEQ ID NO: 71)
Ab3	ACGGGAGAGTGTTAG (SEQ ID NO: 111)
Ab4	ACGGGAGAGTGTTAG (SEQ ID NO: 151)
Ab5	ACGGGAGAGTGTTAG (SEQ ID NO: 191)
Ab6	ACGGGAGAGTGTTAG (SEQ ID NO: 231)
Ab7	ACGGGAGAGTGTTAG (SEQ ID NO: 271)
Ab8	ACGGGAGAGTGTTAG (SEQ ID NO: 311)
Ab9	ACGGGAGAGTGTTAG (SEQ ID NO: 351)
Ab10	ACGGGAGAGTGTTAG (SEQ ID NO: 391)
Ab11	ACGGGAGAGTGTTAG (SEQ ID NO: 431)
Ab12	ACGGGAGAGTGTTAG (SEQ ID NO: 471)
Ab13	ACGGGAGAGTGTTAG (SEQ ID NO: 511)
Ab14	ACGGGAGAGTGTTAG (SEQ ID NO: 551)

Figure 5
Heavy Chain Protein Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab1	1-109	2	30-34	4	49-64	6	96-98	8
Ab2	1-111	42	31-35	44	50-65	46	98-100	48
Ab3	1-111	82	31-35	84	50-65	86	98-100	88
Ab4	1-109	122	30-34	124	49-64	126	96-98	128
Ab5	1-111	162	31-35	164	50-65	166	98-100	168
Ab6	1-111	202	31-35	204	50-65	206	98-100	208
Ab7	1-110	242	31-35	244	50-65	246	97-99	248
Ab8	1-111	282	31-35	284	50-65	286	98-100	288
Ab9	1-109	322	30-34	324	49-64	326	96-98	328
Ab10	1-111	362	31-35	364	50-65	366	98-100	368
Ab11	1-109	402	30-34	404	49-64	406	96-98	408
Ab12	1-111	442	31-35	444	50-65	446	98-100	448
Ab13	1-111	482	30-34	484	49-65	486	97-100	488
Ab14	1-111	522	31-35	524	50-65	526	98-100	528

Figure 6
Heavy Chain Protein Sequence Features

Antibody	FR1 Coordinates	SEQ ID NO:	FR2 Coordinates	SEQ ID NO:	FR3 Coordinates	SEQ ID NO:	FR4 Coordinates	SEQ ID NO:	Constant Region Coordinates	SEQ ID NO:
Ab1	1-29	3	35-48	5	65-95	7	99-109	9	110-439	10
Ab2	1-30	43	36-49	45	66-97	47	101-111	49	112-441	50
Ab3	1-30	83	36-49	85	66-97	87	101-111	89	112-441	90
Ab4	1-29	123	35-48	125	65-95	127	99-109	129	110-439	130
Ab5	1-30	163	36-49	165	66-97	167	101-111	169	112-441	170
Ab6	1-30	203	36-49	205	66-97	207	101-111	209	112-441	210
Ab7	1-30	243	36-49	245	66-96	247	100-110	249	111-440	250
Ab8	1-30	283	36-49	285	66-97	287	101-111	289	112-441	290
Ab9	1-29	323	35-48	325	65-95	327	99-109	329	110-439	330
Ab10	1-30	363	36-49	365	66-97	367	101-111	369	112-441	370
Ab11	1-29	403	35-48	405	65-95	407	99-109	409	110-439	410
Ab12	1-30	443	36-49	445	66-97	447	101-111	449	112-441	450
Ab13	1-29	483	35-48	485	66-96	487	101-111	489	112-441	490
Ab14	1-30	523	36-49	525	66-97	527	101-111	529	112-441	530

Figure 7
Light Chain Protein Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab1	1-113	22	23-35	24	51-57	26	90-102	28
Ab2	1-113	62	23-35	64	51-57	66	90-102	68
Ab3	1-113	102	23-35	104	51-57	106	90-102	108
Ab4	1-113	142	23-35	144	51-57	146	90-102	148
Ab5	1-113	182	23-35	184	51-57	186	90-102	188
Ab6	1-113	222	23-35	224	51-57	226	90-102	228
Ab7	1-113	262	23-35	264	51-57	266	90-102	268
Ab8	1-113	302	23-35	304	51-57	306	90-102	308
Ab9	1-113	342	23-35	344	51-57	346	90-102	348
Ab10	1-113	382	23-35	384	51-57	386	90-102	388
Ab11	1-113	422	23-35	424	51-57	426	90-102	428
Ab12	1-113	462	23-35	464	51-57	466	90-102	468
Ab13	1-113	502	24-36	504	52-58	506	91-102	508
Ab14	1-113	542	23-35	544	51-57	546	90-102	548

Figure 8

Light Chain Protein Sequence Features

Antibody	FR1 Coordinates	SEQ ID NO:	FR2 Coordinates	SEQ ID NO:	FR3 Coordinates	SEQ ID NO:	FR4 Coordinates	SEQ ID NO:	Constant Region Coordinates	SEQ ID NO:
Ab1	1-22	23	36-50	25	58-89	27	103-113	29	114-219	30
Ab2	1-22	63	36-50	65	58-89	67	103-113	69	114-219	70
Ab3	1-22	103	36-50	105	58-89	107	103-113	109	114-219	110
Ab4	1-22	143	36-50	145	58-89	147	103-113	149	114-219	150
Ab5	1-22	183	36-50	185	58-89	187	103-113	189	114-219	190
Ab6	1-22	223	36-50	225	58-89	227	103-113	229	114-219	230
Ab7	1-22	263	36-50	265	58-89	267	103-113	269	114-219	270
Ab8	1-22	303	36-50	305	58-89	307	103-113	309	114-219	310
Ab9	1-22	343	36-50	345	58-89	347	103-113	349	114-219	350
Ab10	1-22	383	36-50	385	58-89	387	103-113	389	114-219	390
Ab11	1-22	423	36-50	425	58-89	427	103-113	429	114-219	430
Ab12	1-22	463	36-50	465	58-89	467	103-113	469	114-219	470
Ab13	1-23	503	37-51	505	59-90	507	103-113	509	114-219	510
Ab14	1-22	543	36-50	545	58-89	547	103-113	549	114-219	550

Figure 9

Heavy Chain DNA Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab1	1-327	12	88-102	14	145-192	16	286-294	18
Ab2	1-333	52	91-105	54	148-195	56	292-300	58
Ab3	1-333	92	91-105	94	148-195	96	292-300	98
Ab4	1-327	132	88-102	134	145-192	136	286-294	138
Ab5	1-333	172	91-105	174	148-195	176	292-300	178
Ab6	1-333	212	91-105	214	148-195	216	292-300	218
Ab7	1-330	252	91-105	254	148-195	256	289-297	258
Ab8	1-333	292	91-105	294	148-195	296	292-300	298
Ab9	1-327	332	88-102	334	145-192	336	286-294	338
Ab10	1-333	372	91-105	374	148-195	376	292-300	378
Ab11	1-327	412	88-102	414	145-192	416	286-294	418
Ab12	1-333	452	91-105	454	148-195	456	292-300	458
Ab13	1-333	492	88-102	494	145-195	496	289-300	498
Ab14	1-333	532	91-105	534	148-195	536	292-300	538

Figure 10
Heavy Chain DNA Sequence Features

	FR1 Coordinates	SEQ ID NO:	FR2 Coordinates	SEQ ID NO:	FR3 Coordinates	SEQ ID NO:	FR4 Coordinates	SEQ ID NO:	Constant Region Coordinates	SEQ ID NO:
Antibody										
Ab1	1-87	13 103-144		15 193-285		17 295-327		19 328-1320		20
Ab2	1-90	53 106-147		55 196-291		57 301-333		59 334-1326		60
Ab3	1-90	93 106-147		95 196-291		97 301-333		99 334-1326		100
Ab4	1-87	133 103-144		135 193-285		137 295-327		139 328-1320		140
Ab5	1-90	173 106-147		175 196-291		177 301-333		179 334-1326		180
Ab6	1-90	213 106-147		215 196-291		217 301-333		219 334-1326		220
Ab7	1-90	253 106-147		255 196-288		257 298-330		259 331-1323		260
Ab8	1-90	293 106-147		295 196-291		297 301-333		299 334-1326		300
Ab9	1-87	333 103-144		335 193-285		337 295-327		339 328-1320		340
Ab10	1-90	373 106-147		375 196-291		377 301-333		379 334-1326		380
Ab11	1-87	413 103-144		415 193-285		417 295-327		419 328-1320		420
Ab12	1-90	453 106-147		455 196-291		457 301-333		459 334-1326		460
Ab13	1-87	493 103-144		495 196-288		497 301-333		499 334-1326		500
Ab14	1-90	533 106-147		535 196-291		537 301-333		539 334-1326		540

Figure 11
Light Chain DNA Sequence Features

Antibody	Variable Region Coordinates	SEQ ID NO:	CDR1 Coordinates	SEQ ID NO:	CDR2 Coordinates	SEQ ID NO:	CDR3 Coordinates	SEQ ID NO:
Ab1	1-339	32	67-105	34	151-171	36	268-306	38
Ab2	1-339	72	67-105	74	151-171	76	268-306	78
Ab3	1-339	112	67-105	114	151-171	116	268-306	118
Ab4	1-339	152	67-105	154	151-171	156	268-306	158
Ab5	1-339	192	67-105	194	151-171	196	268-306	198
Ab6	1-339	232	67-105	234	151-171	236	268-306	238
Ab7	1-339	272	67-105	274	151-171	276	268-306	278
Ab8	1-339	312	67-105	314	151-171	316	268-306	318
Ab9	1-339	352	67-105	354	151-171	356	268-306	358
Ab10	1-339	392	67-105	394	151-171	396	268-306	398
Ab11	1-339	432	67-105	434	151-171	436	268-306	438
Ab12	1-339	472	67-105	474	151-171	476	268-306	478
Ab13	1-339	512	70-108	514	154-174	516	271-306	518
Ab14	1-339	552	67-105	554	151-171	556	268-306	558

Figure 12
Light Chain DNA Sequence Features

Antibody	FR1 Coordinates	SEQ ID NO: FR2 Coordinates	SEQ ID NO: FR3 Coordinates	SEQ ID NO: FR4 Coordinates	SEQ ID NO: Constant Region Coordinates	SEQ ID NO:
Ab1	1-66	33 106-150	35 172-267	37 307-339	39 340-660	40
Ab2	1-66	73 106-150	75 172-267	77 307-339	79 340-660	80
Ab3	1-66	113 106-150	115 172-267	117 307-339	119 340-660	120
Ab4	1-66	153 106-150	155 172-267	157 307-339	159 340-660	160
Ab5	1-66	193 106-150	195 172-267	197 307-339	199 340-660	200
Ab6	1-66	233 106-150	235 172-267	237 307-339	239 340-660	240
Ab7	1-66	273 106-150	275 172-267	277 307-339	279 340-660	280
Ab8	1-66	313 106-150	315 172-267	317 307-339	319 340-660	320
Ab9	1-66	353 106-150	355 172-267	357 307-339	359 340-660	360
Ab10	1-66	393 106-150	395 172-267	397 307-339	399 340-660	400
Ab11	1-66	433 106-150	435 172-267	437 307-339	439 340-660	440
Ab12	1-66	473 106-150	475 172-267	477 307-339	479 340-660	480
Ab13	1-69	513 109-153	515 175-270	517 307-339	519 340-660	520
Ab14	1-66	553 106-150	555 172-267	557 307-339	559 340-660	560

FIG. 13

Responders at all three time-points

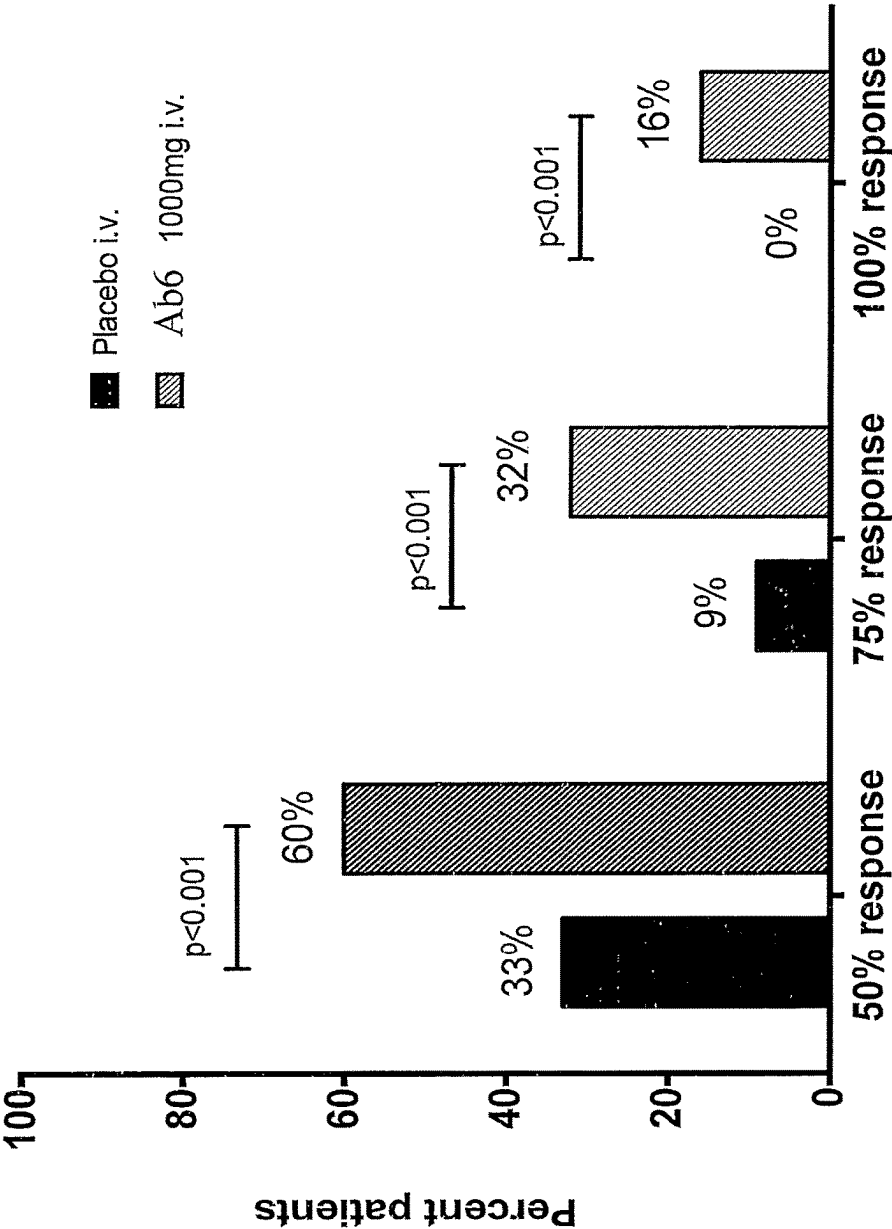


FIG. 14

Median % change from baseline: migraine days per month

Median ($\pm$ IQR) percentage change from baseline in migraine days per month: AB6 versus Placebo

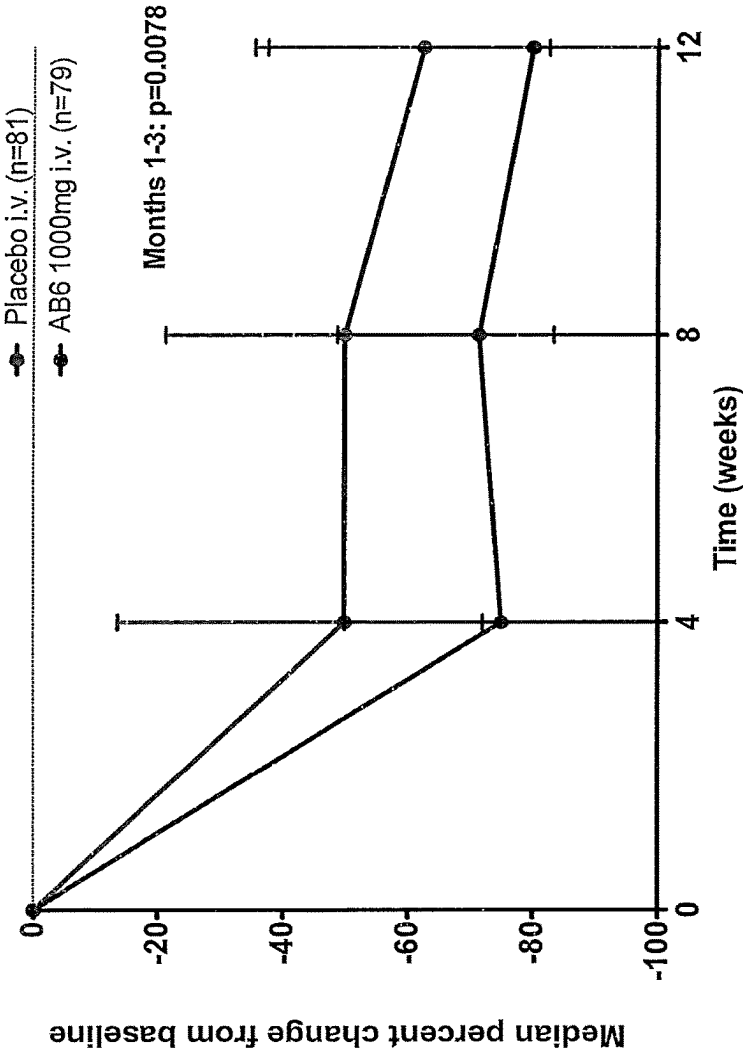


FIG. 15

Median % change from baseline: migraine episodes per month

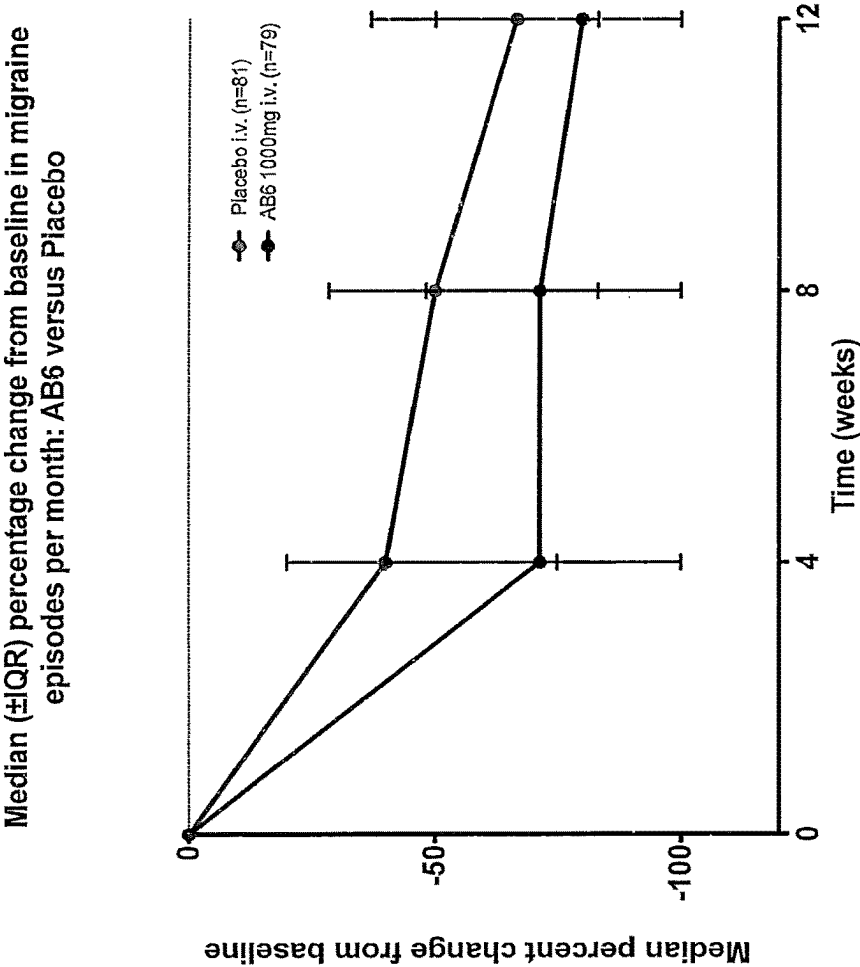


FIG. 16

Median % change from baseline: migraine hours per month

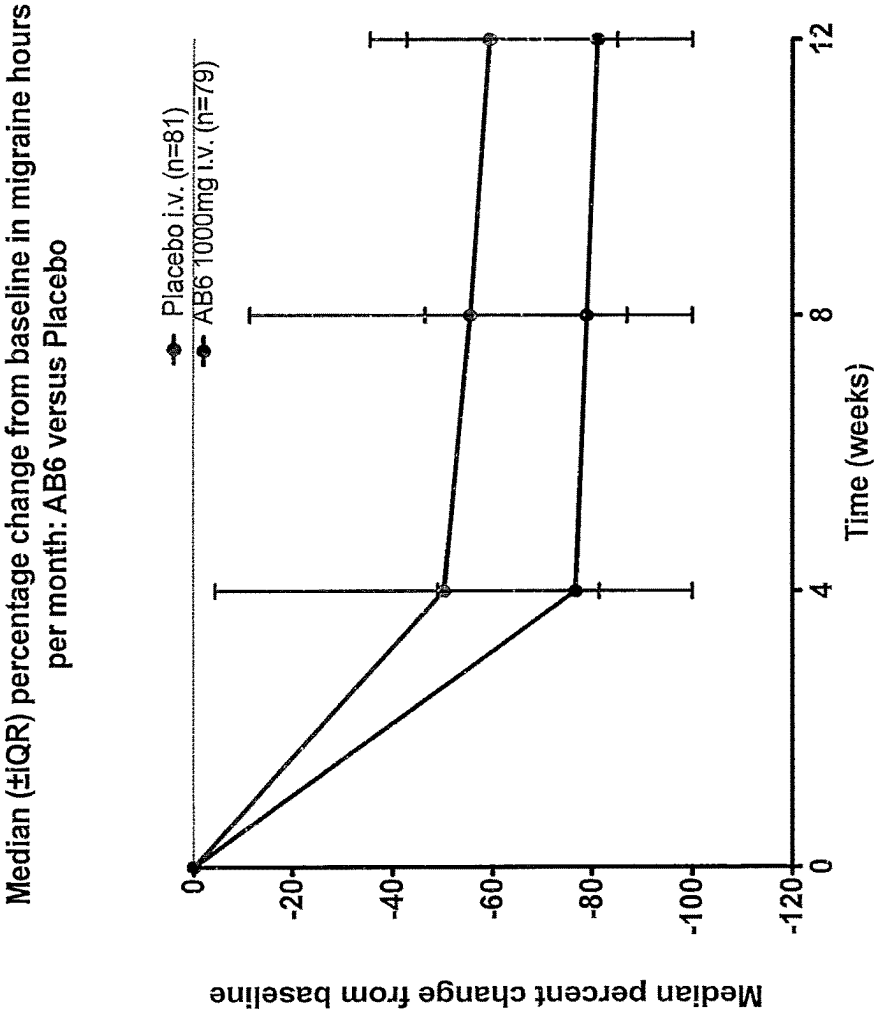


FIG. 17

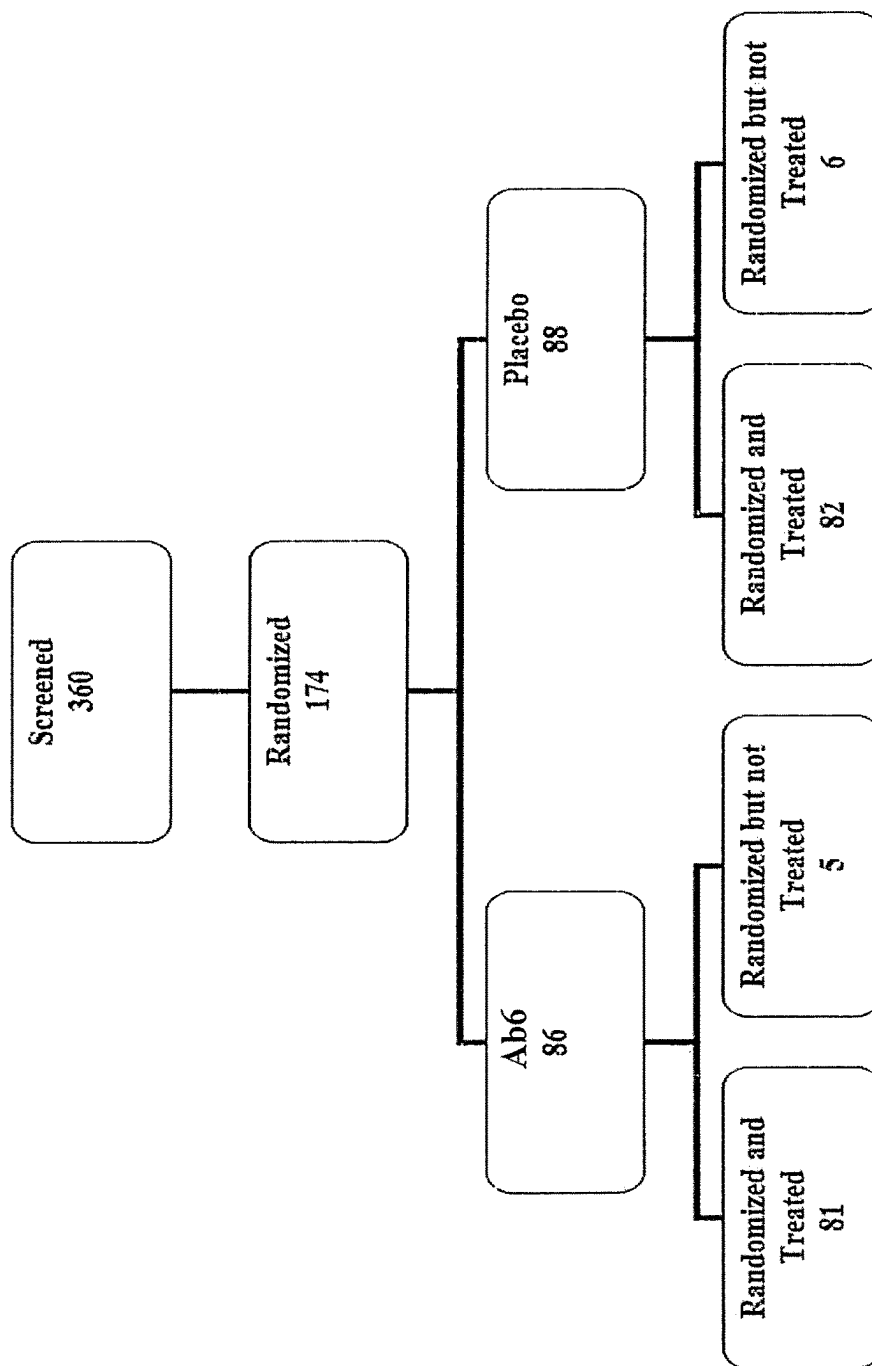


FIG. 18

Mean Change Baseline HIT-6 score

Mean ($\pm$ SD) absolute change from baseline in Headache Impact Test (HIT-6) Score: AB6 vs Placebo

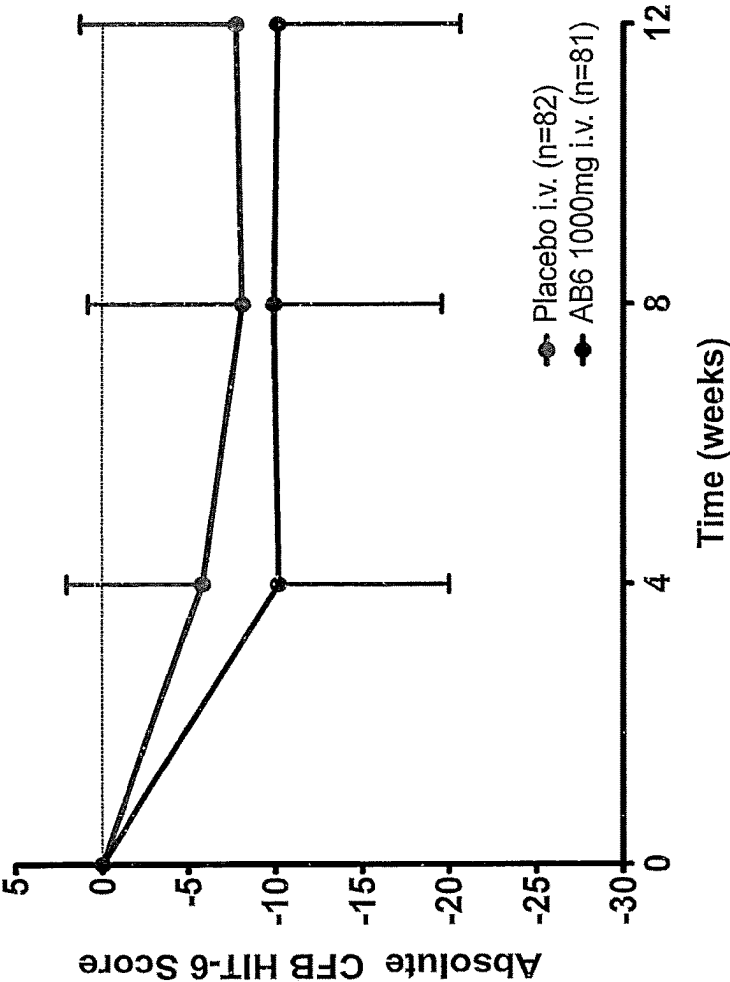


FIG. 19

HIT-6 Responder Analysis

Percent patients who are some or little/none life
impact for headache impact score (HIT-6) versus time:
AB6 vs Placebo

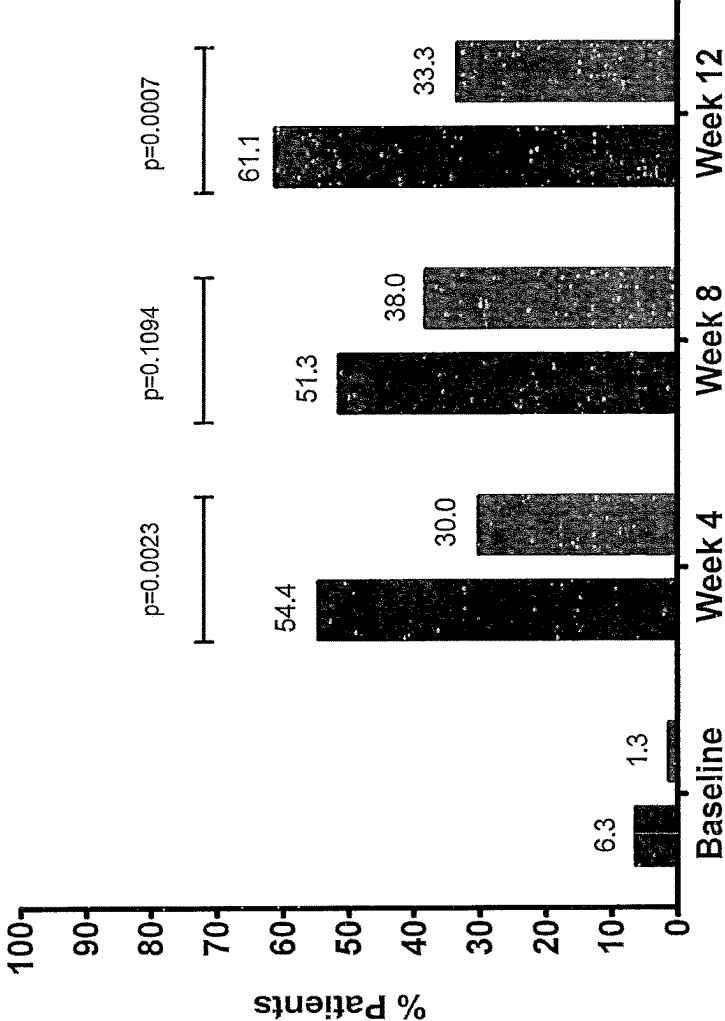


FIG. 20. PK Profile

Ab6 1000 mg I.V.
Mean +/- SD

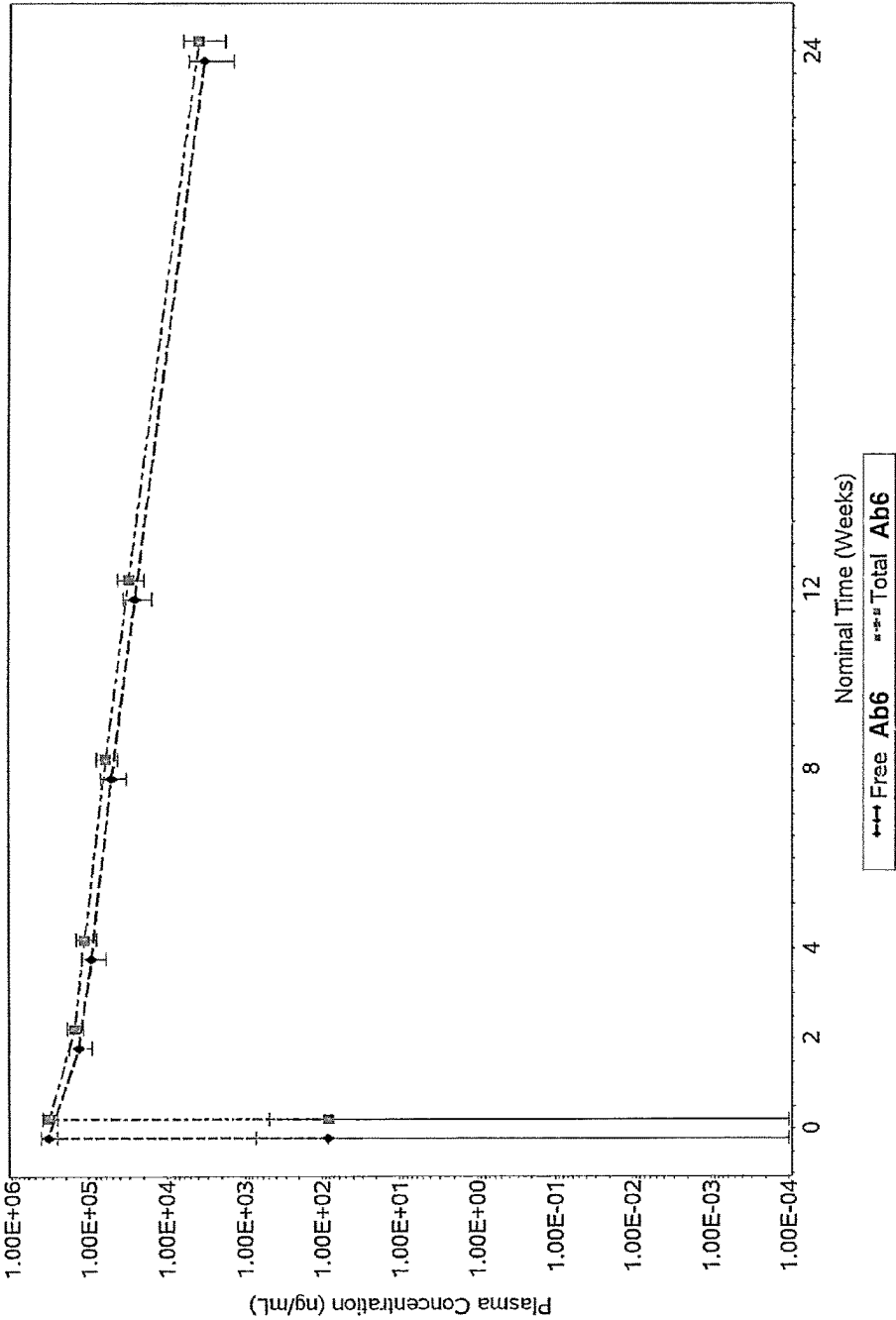


FIG. 21. PK Parameters

Plasma Free Ab6*

	C _{max} (μg/mL)	AUC _{0-∞} (mg*hr/mL)	Half-Life (Days)	V _z (L)	CL (mL/hr)
N	81	78	78	78	78
Mean	336	219	31	5.2	5.0
SD	80	64	8	2.1	1.5

* - Following 1000 mg Ab6 IV single-dose

FIG. 22

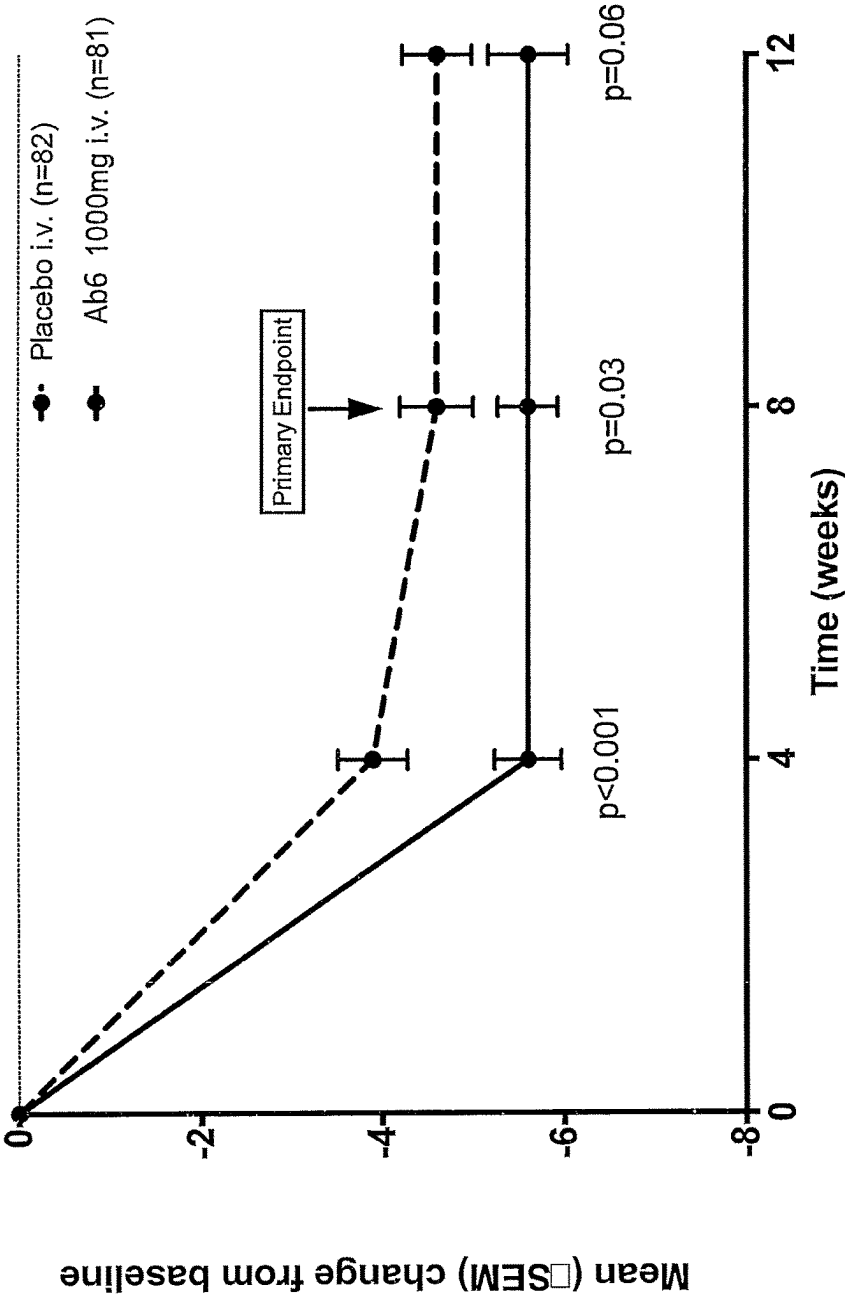


FIG. 23

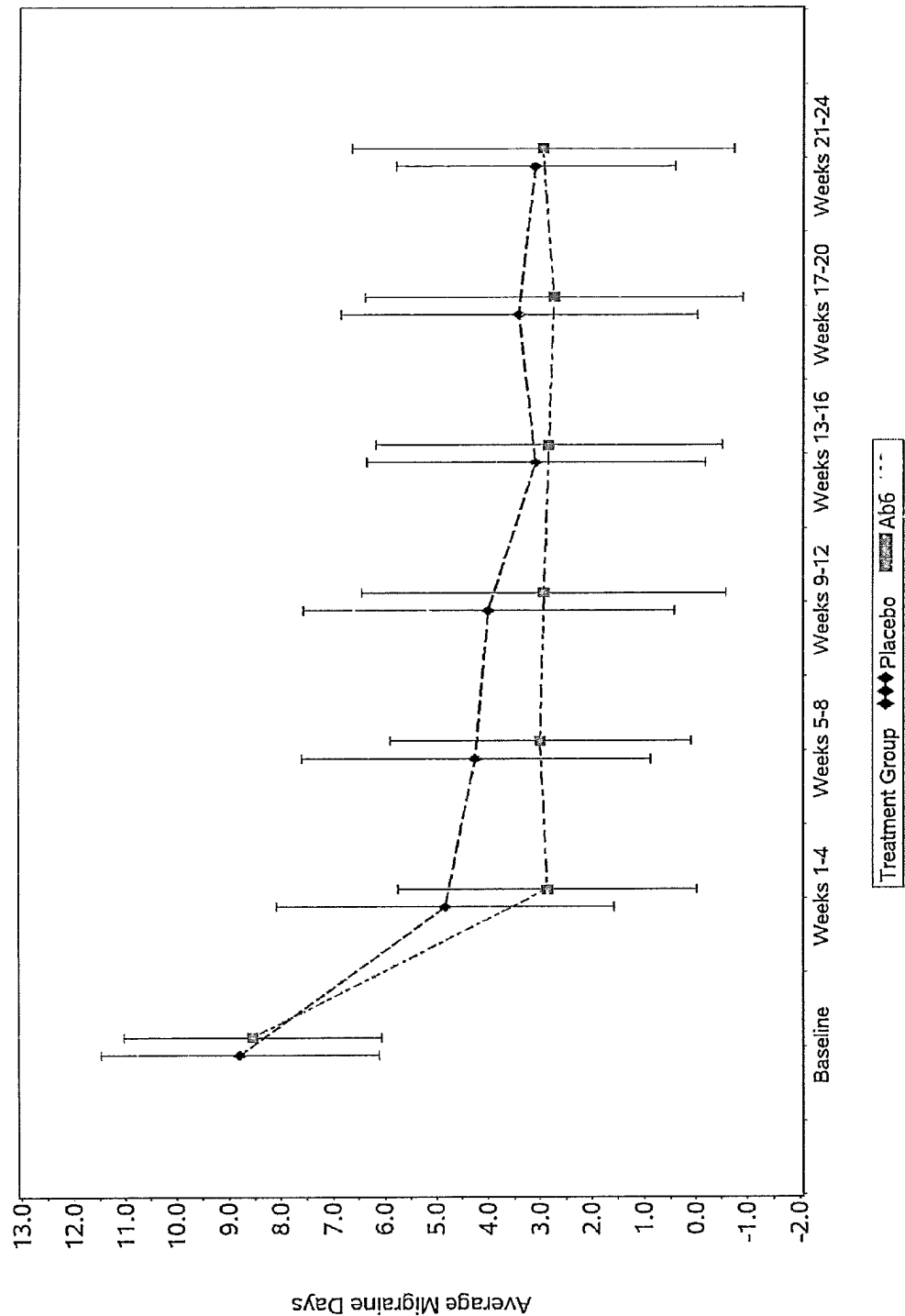


FIG. 24

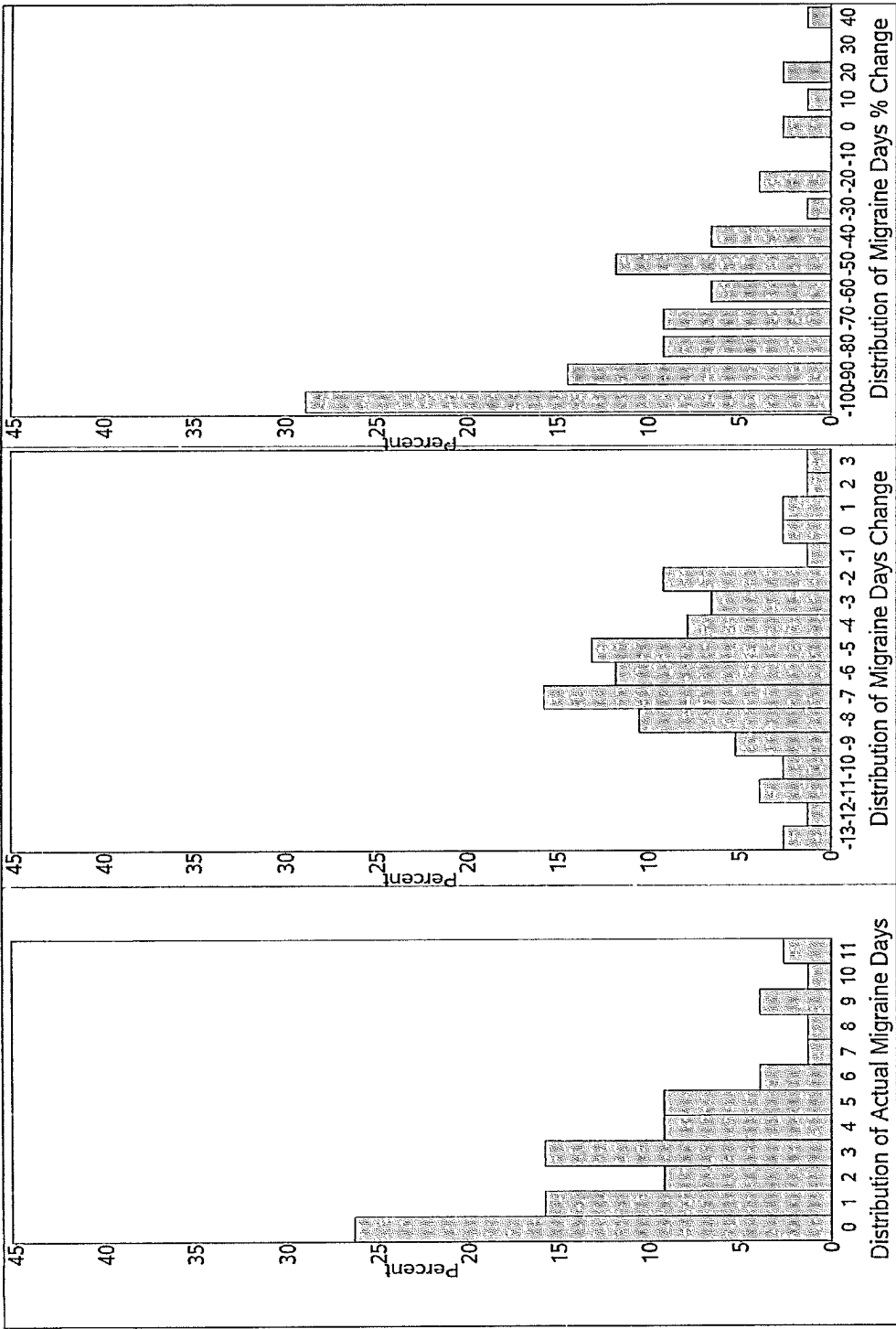


FIG. 25

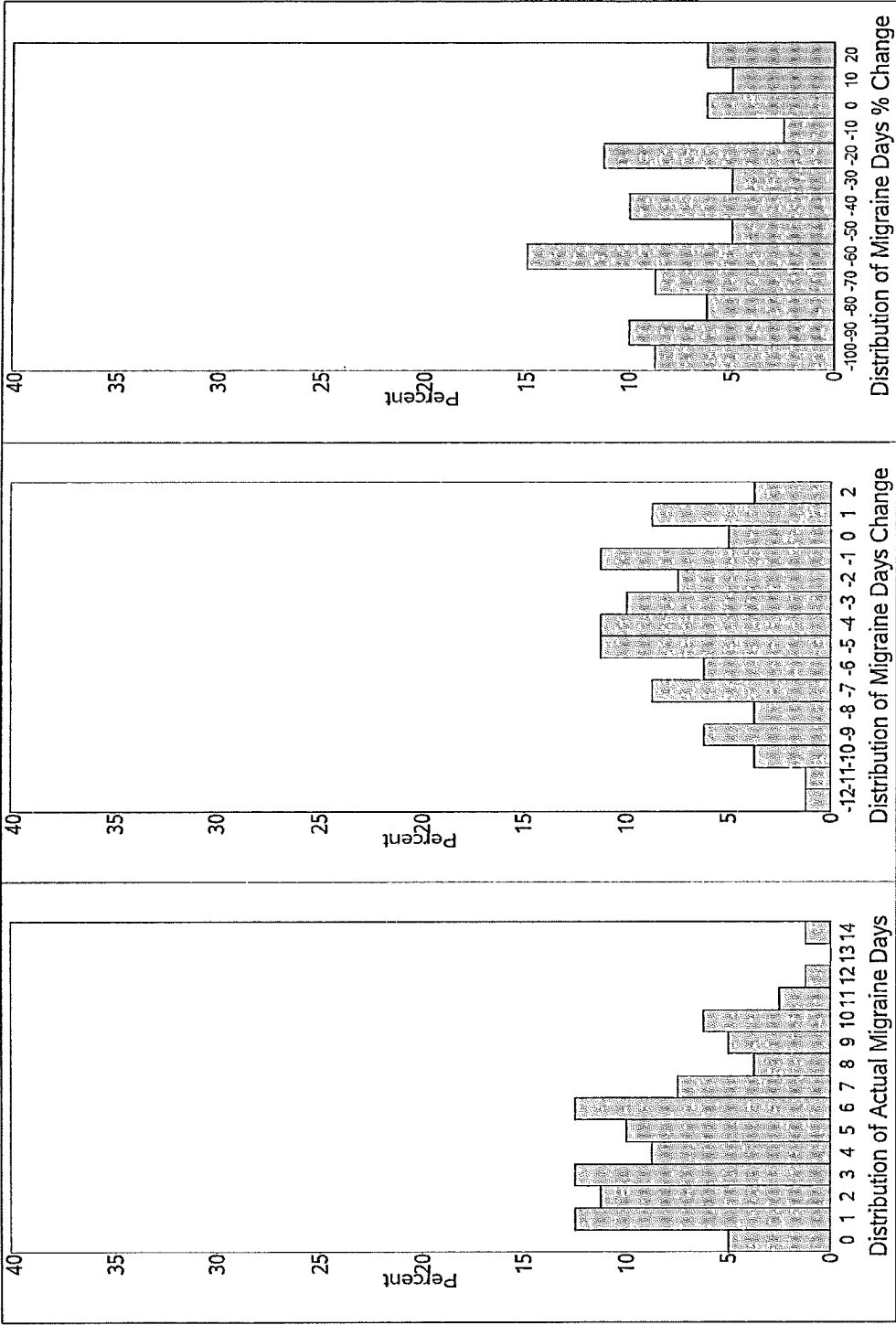


FIG. 26

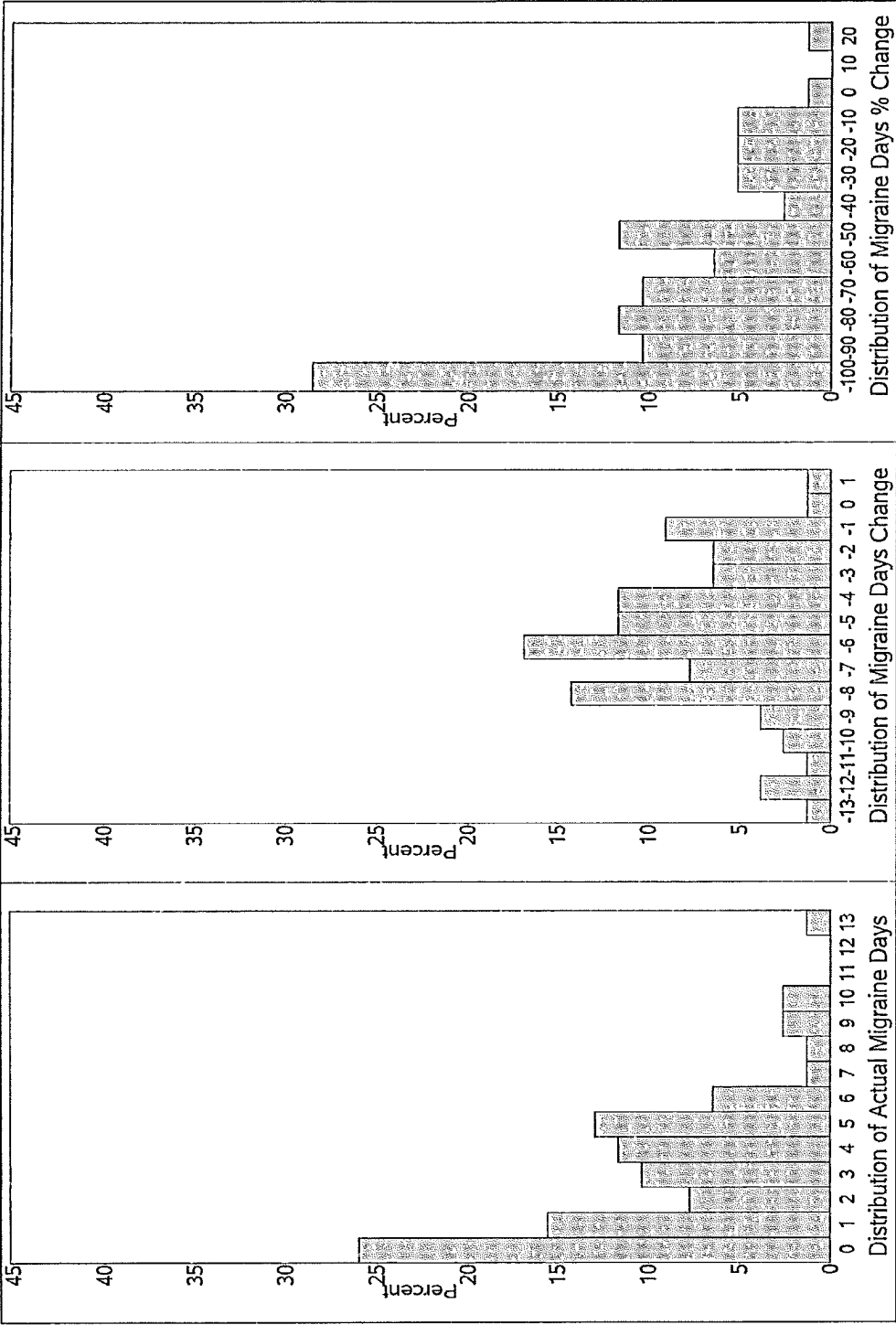


FIG. 27

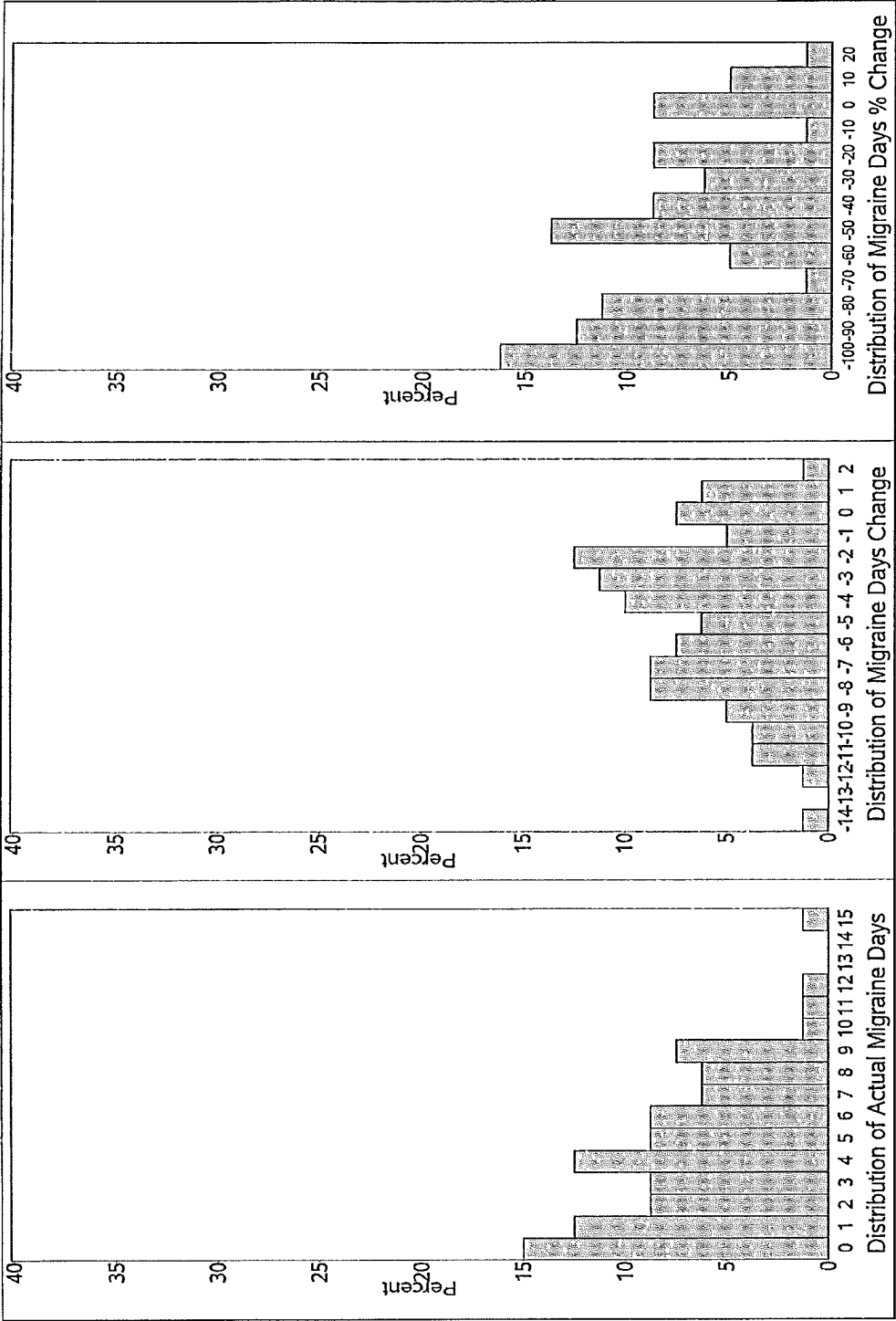


FIG. 28

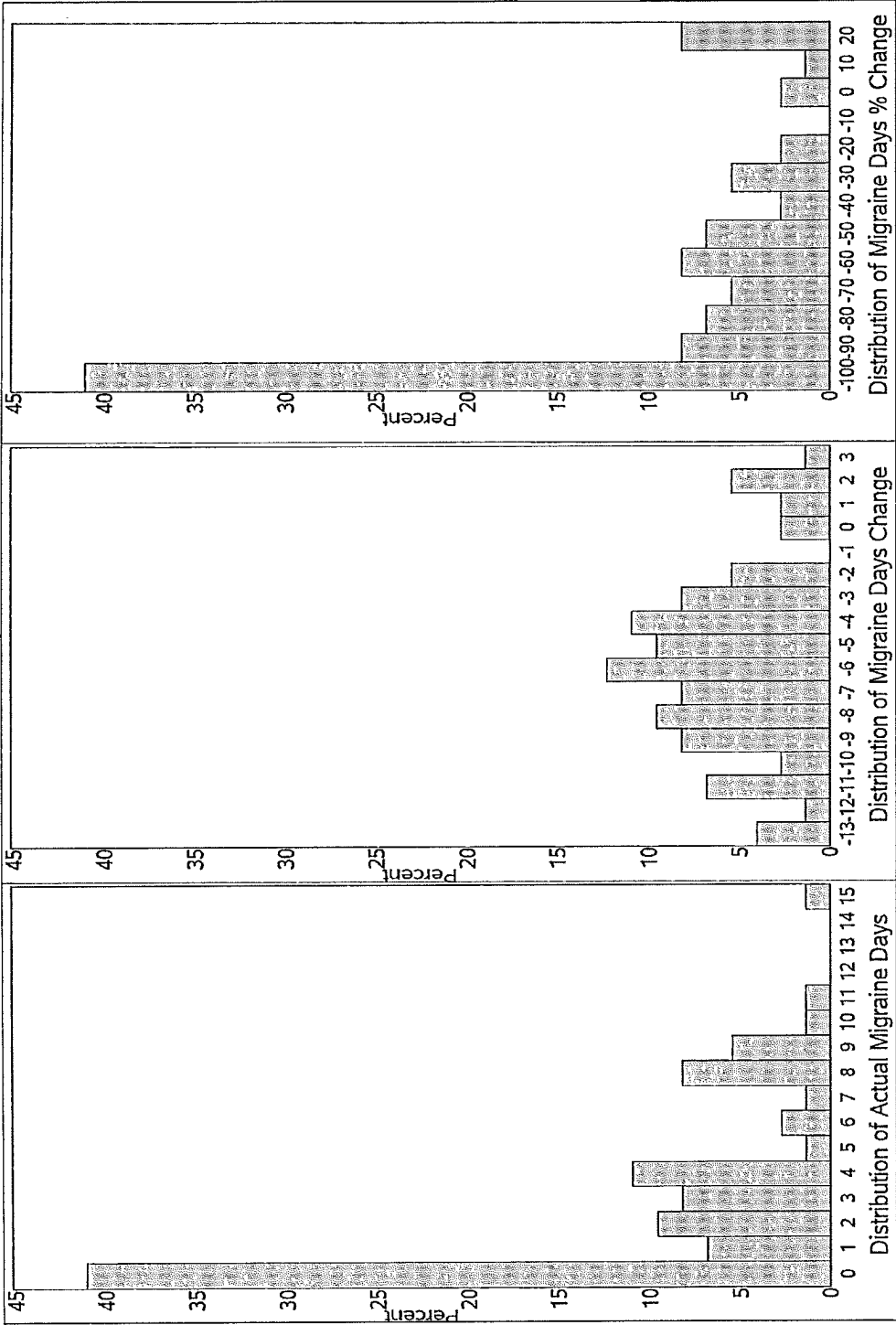


FIG. 29

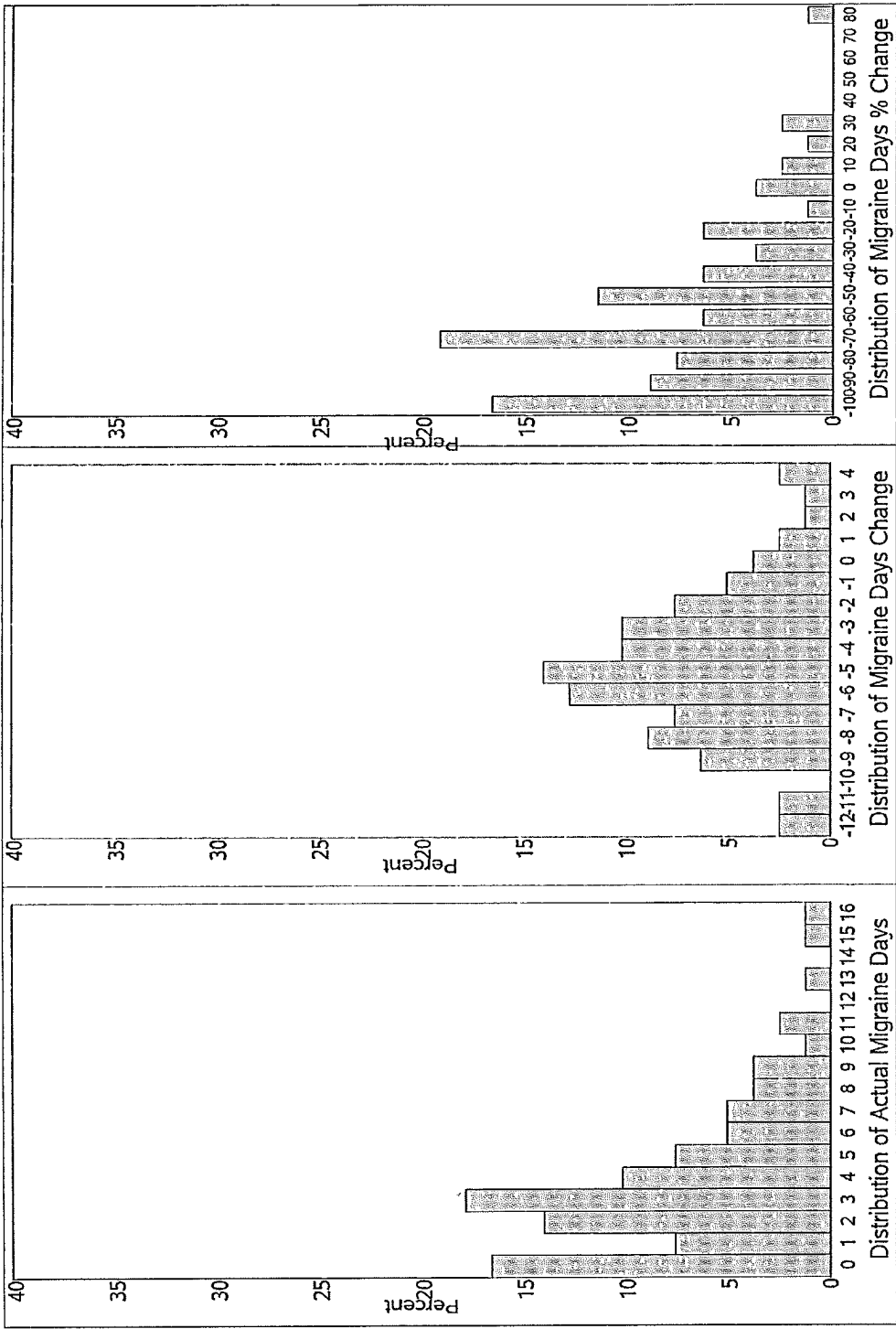


FIG. 30

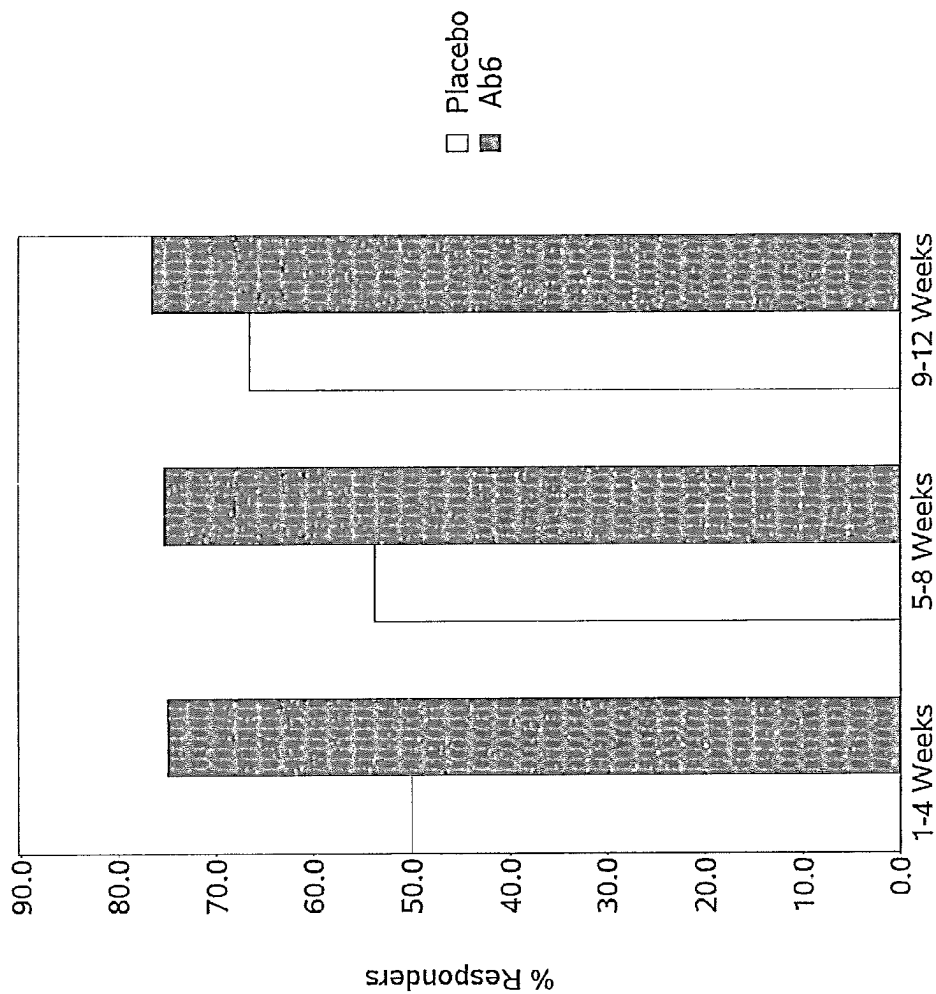


FIG. 31

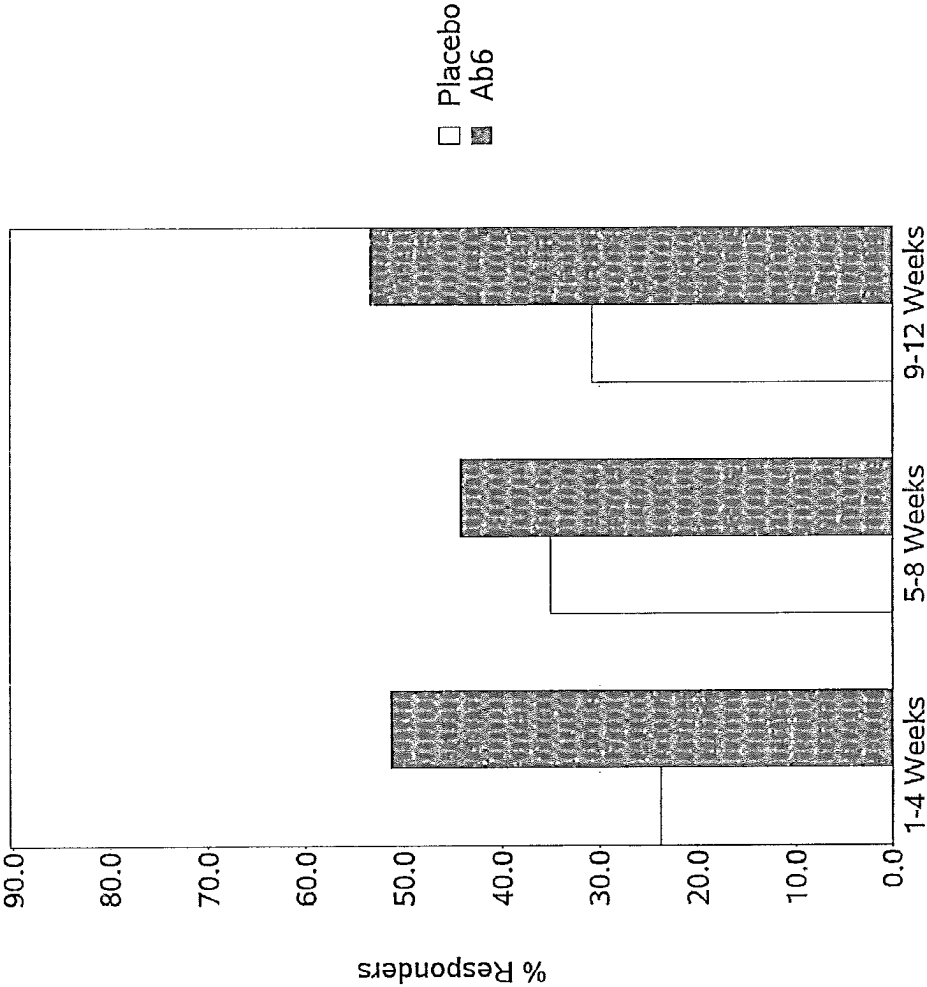


FIG. 32

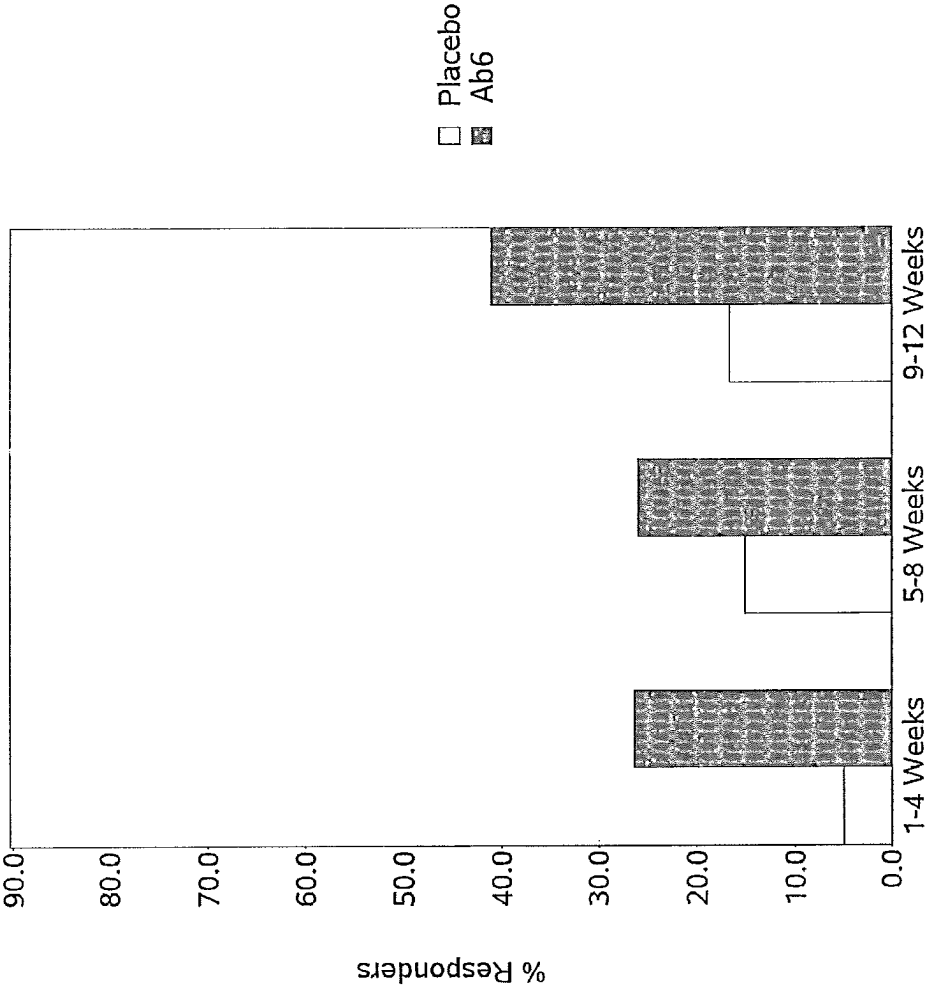


FIG. 33

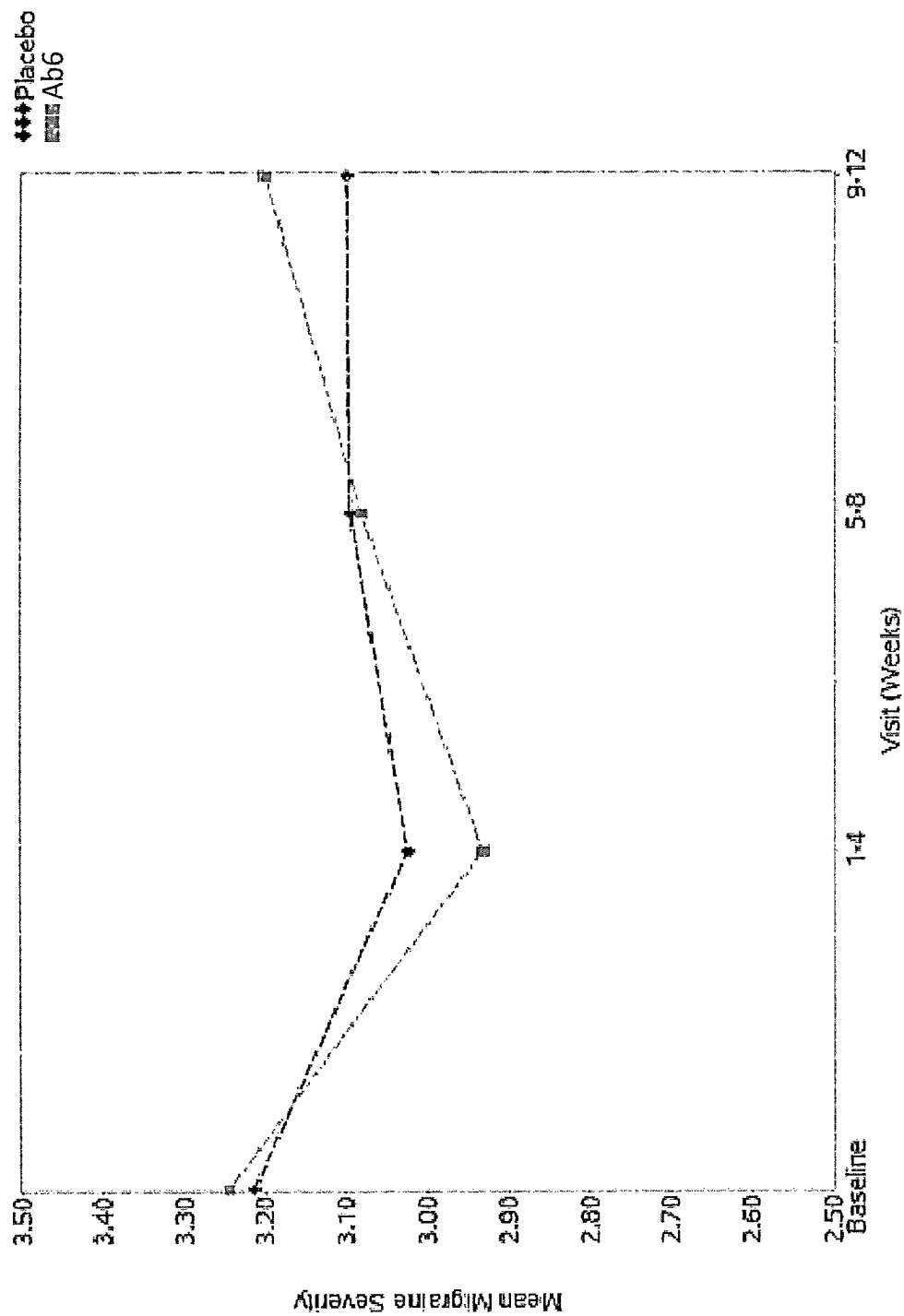


FIG. 34

Mean ($\pm$ SD) Change from Baseline In Study Endpoints						
Endpoint	Weeks 1-4		Weeks 5-8		Weeks 9-12	
	Placebo i.v. (n=82)	Ab6 1000mg i.v. (n=81)	Placebo i.v. (n=82)	Ab6 1000mg i.v. (n=81)	Placebo i.v. (n=82)	Ab6 1000mg i.v. (n=81)
Migraine Days	-3.9 (3.5)	-5.6 (3.3) ¹	-4.6 (3.6)	-5.6 (3.0) ²	-4.6 (3.5)	-5.6 (4.0) ³
Migraine Episodes	-3.0 (2.7)	-3.7 (2.4)	-3.7 (2.9)	-3.8 (2.2)	-3.7 (2.8)	-3.9 (2.6)
Migraine Hours	-33.7 (41.8)	-58.0 (49.1)	-36.1 (45.9)	-54.4 (48.3)	-37.1 (40.0)	-54.6 (60.5)
Average Migraine Severity ⁴	-0.16 (0.58)	-0.31 (0.58)	-0.10 (0.54)	-0.16 (0.50)	-0.08 (0.54)	-0.11 (0.43)
Headache Frequency	-4.0 (3.8)	-5.6 (3.4)	-5.0 (3.7)	-5.3 (3.5)	-5.1 (3.7)	-5.9 (3.8)
HIT-6 score	-5.8 (7.8)	-10.2 (9.8)	-8.1 (8.9)	-9.9 (9.7)	-7.7 (9.0)	-10.1 (10.6)
MSQ RFP	19.9 (23.8)	29.3 (24.3)	25.2 (24.8)	28.8 (24.7)	22.2 (23.1)	28.5 (24.5)
MSQ RFR	16.3 (23.2)	21.1 (23.9)	20.2 (22.1)	20.9 (23.3)	18.0 (20.5)	21.4 (23.1)
MSQ EF	19.4 (27.6)	25.1 (28.3)	21.2 (25.1)	23.8 (25.8)	21.1 (25.1)	23.1 (26.8)

¹p<0.001; ²p=0.03; ³p=0.06; ⁴Severity measured on a 4 point scale with 1= mild and 4 = severe

FIG. 35

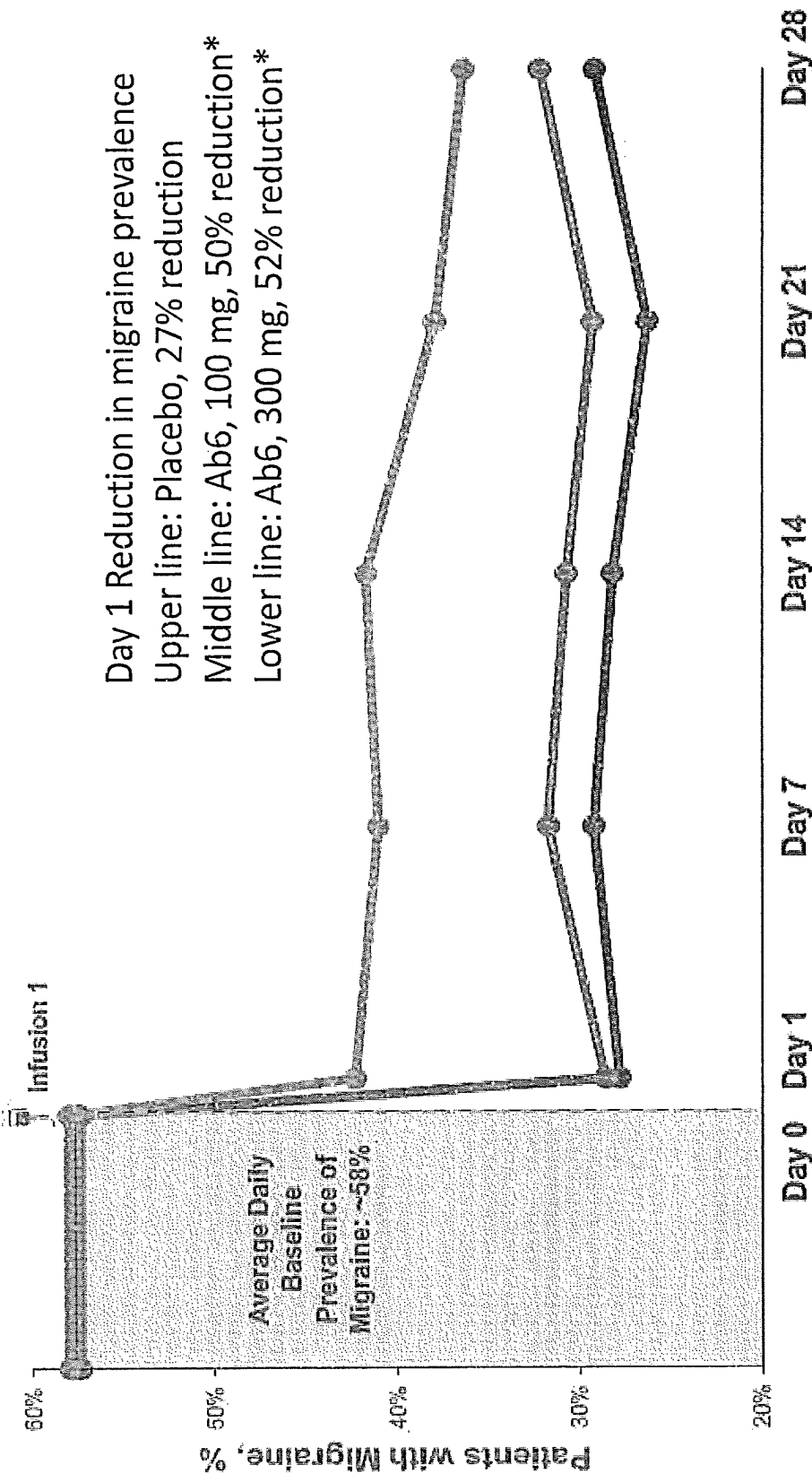


FIG. 36. Chronic migraine $\geq 50\%$ responder rates

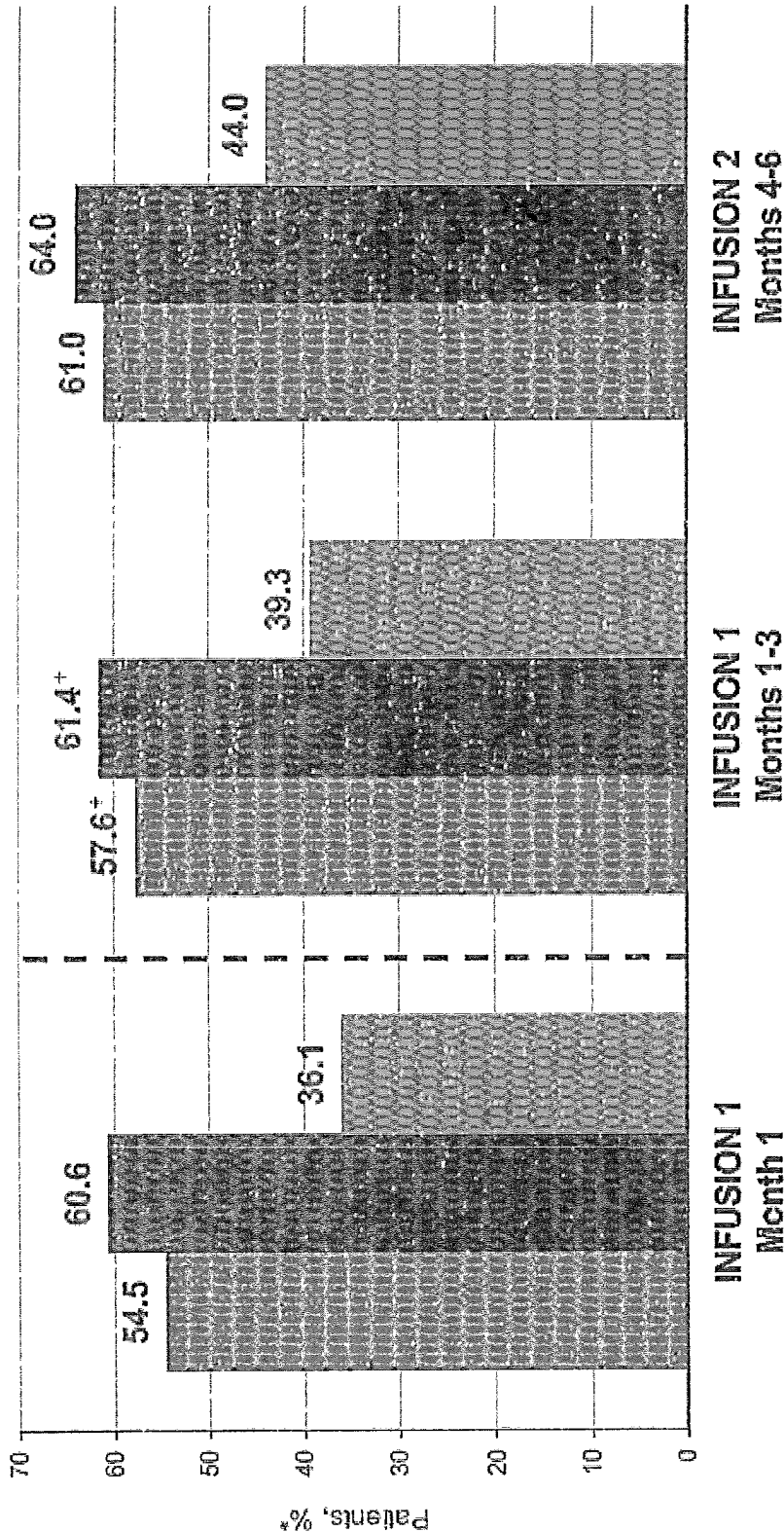


FIG. 37. Chronic migraine $\geq 75\%$ responder rates

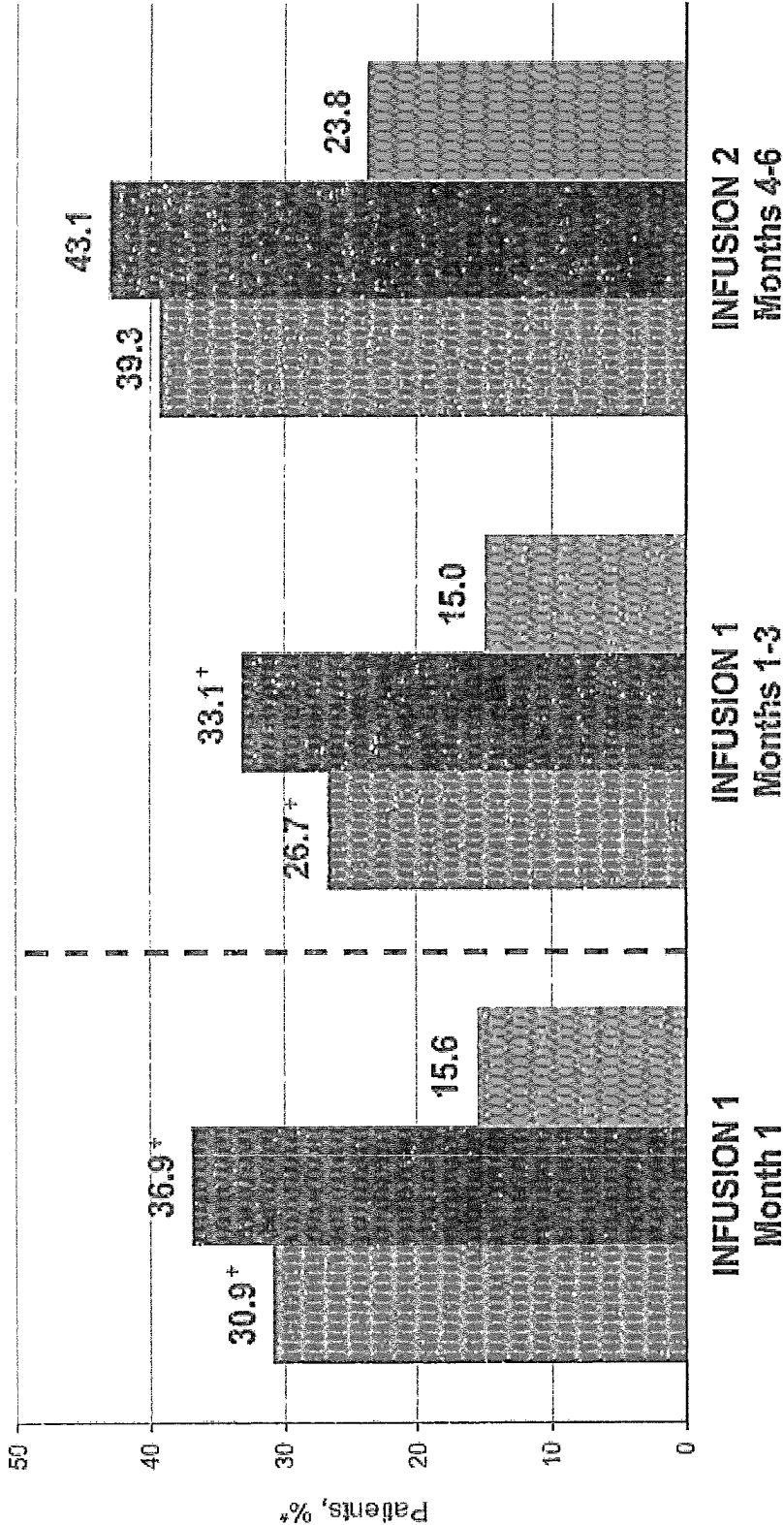


FIG. 38. Chronic migraine 100% responder rates

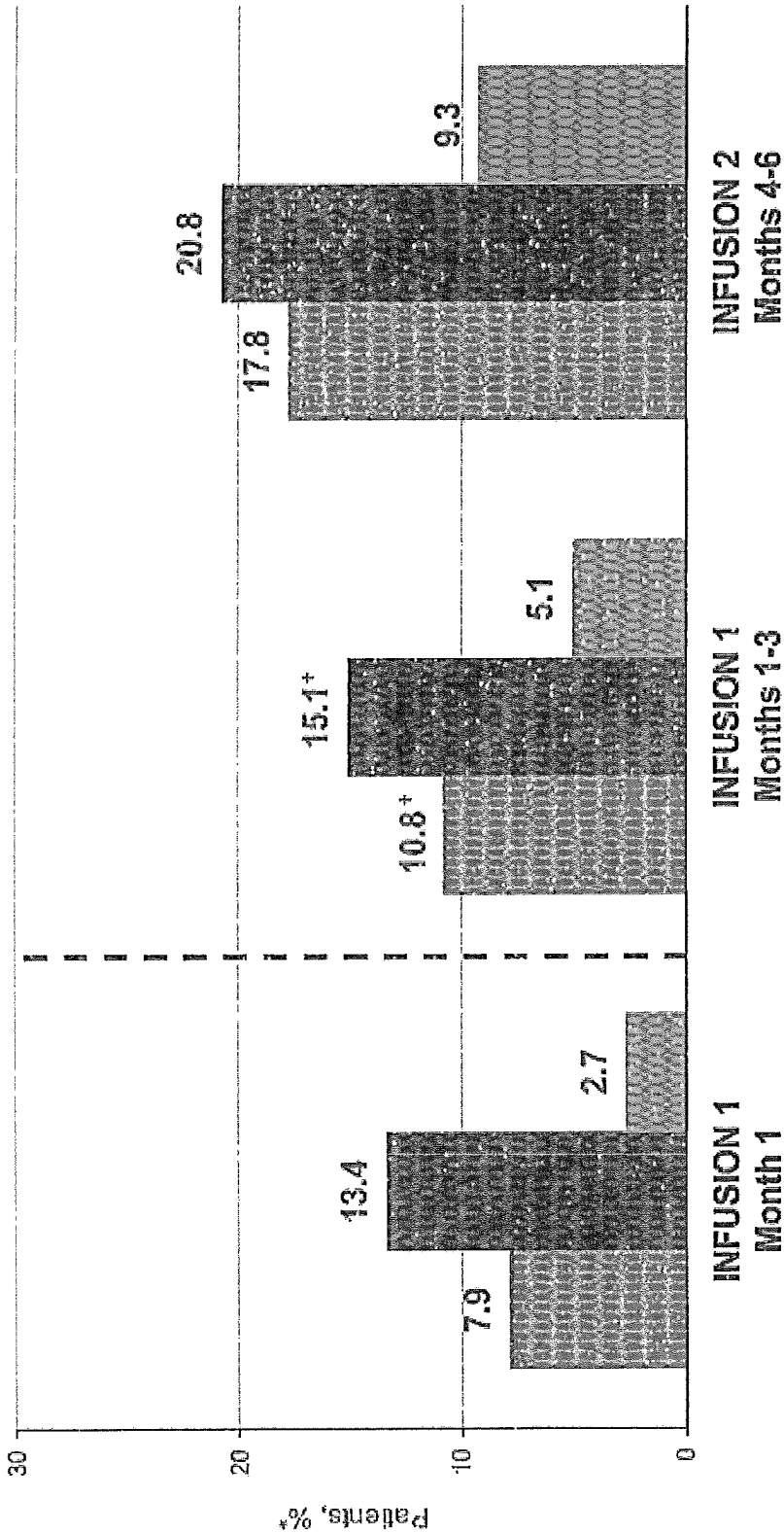


FIG. 39

	Placebo	100 mg	300 mg
Subjects, n	366	356	350
Mean age, years (SD)	39.6 (11.3)	41.0 (11.7)	41.0 (10.4)
Mean BMI, kg/m ² (SD)	27.0 (5.6)	26.4 (5.0)	26.3 (5.0)
Female, %	89	86	90
Mean years from migraine diagnosis	17.0	18.3	19.0
Mean duration of chronic migraine, years (SD)	11.6 (10.9)	11.6 (11.7)	12.4 (11.2)
≥1 prophylactic medication, n (%) [*]	163 (44.5)	161 (45.2)	155 (44.3)
Mean migraine days/month (SD)	16.2 (4.6)	16.1 (4.6)	16.1 (4.8)
Mean headache days/month (SD)	20.6 (3.0)	20.4 (3.1)	20.4 (3.2)

FIG. 40. Difference from placebo in change from baseline in mean migraine days (MMD) over months 1-3 by baseline subgroup

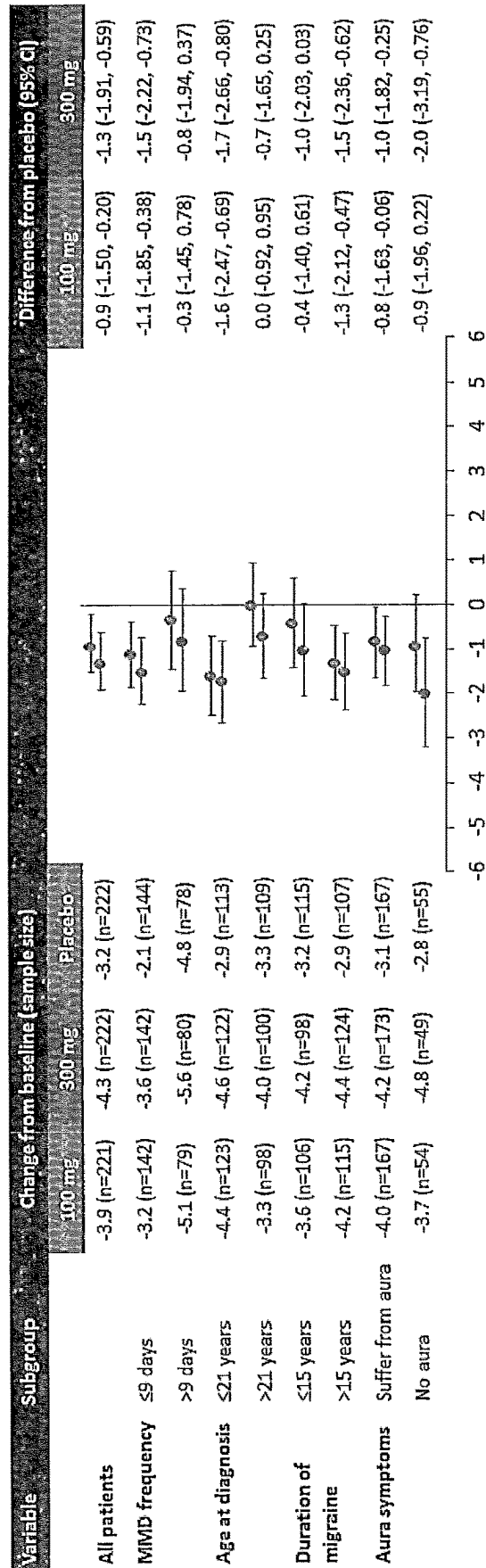


FIG. 41. Difference from placebo in change from baseline in mean migraine days (MMD) over months 1-3 by baseline subgroup

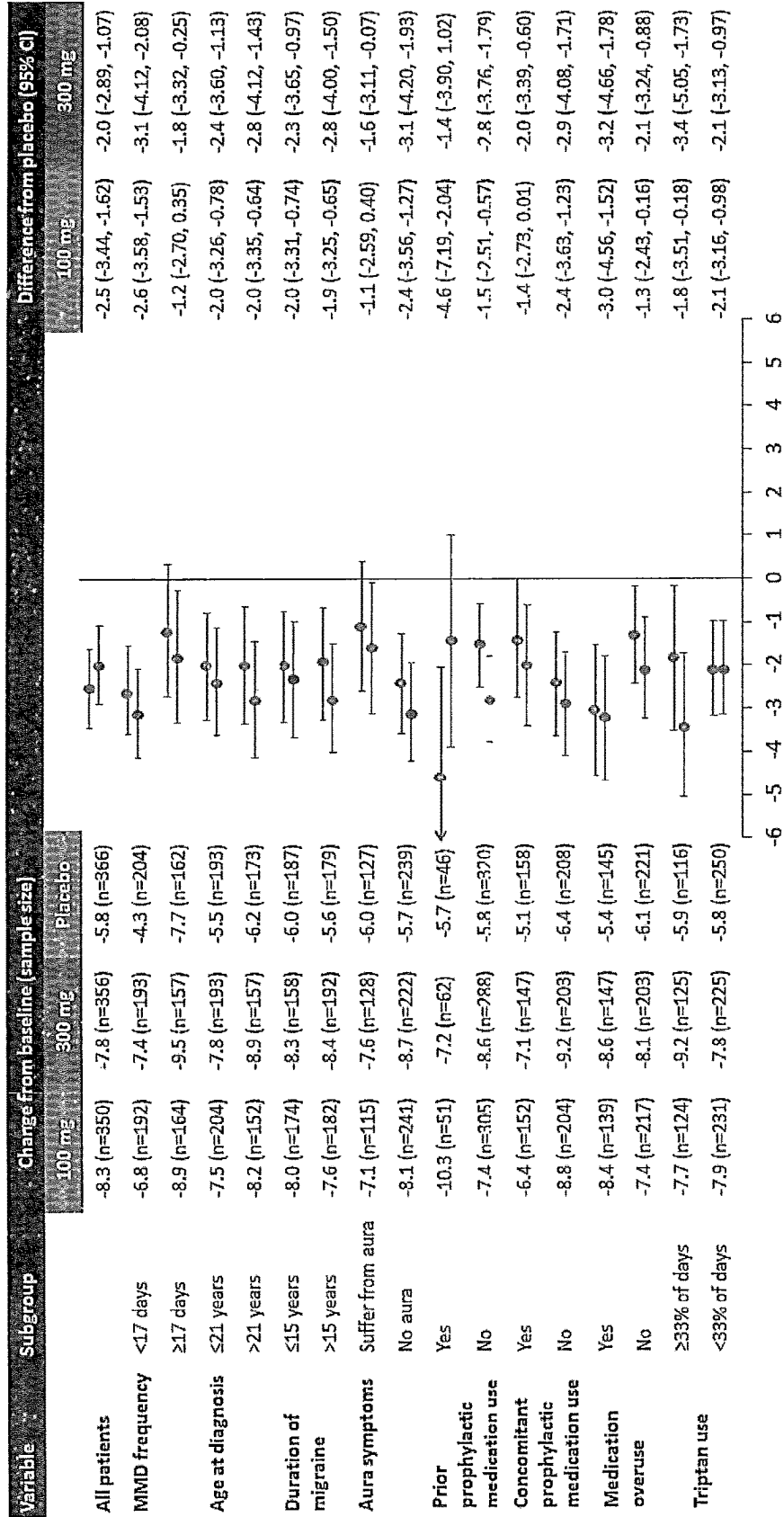


FIG. 42.

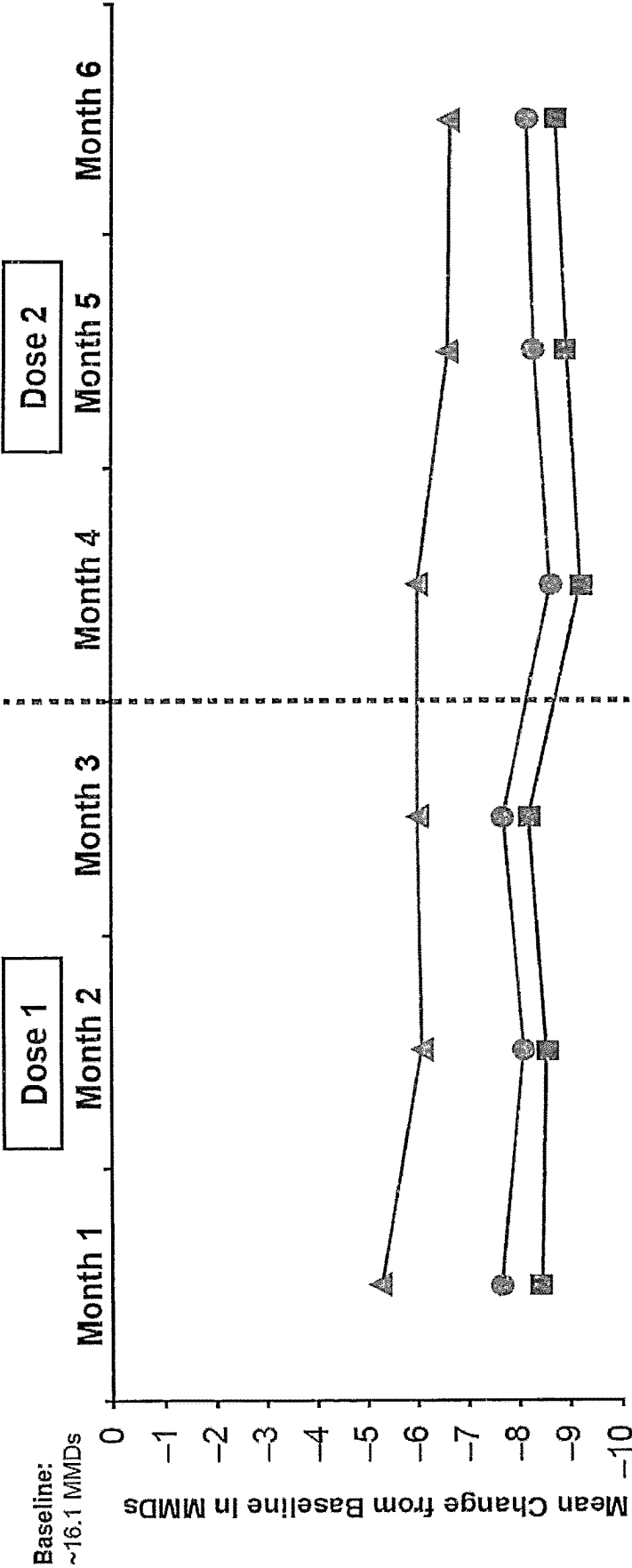


FIG. 43.

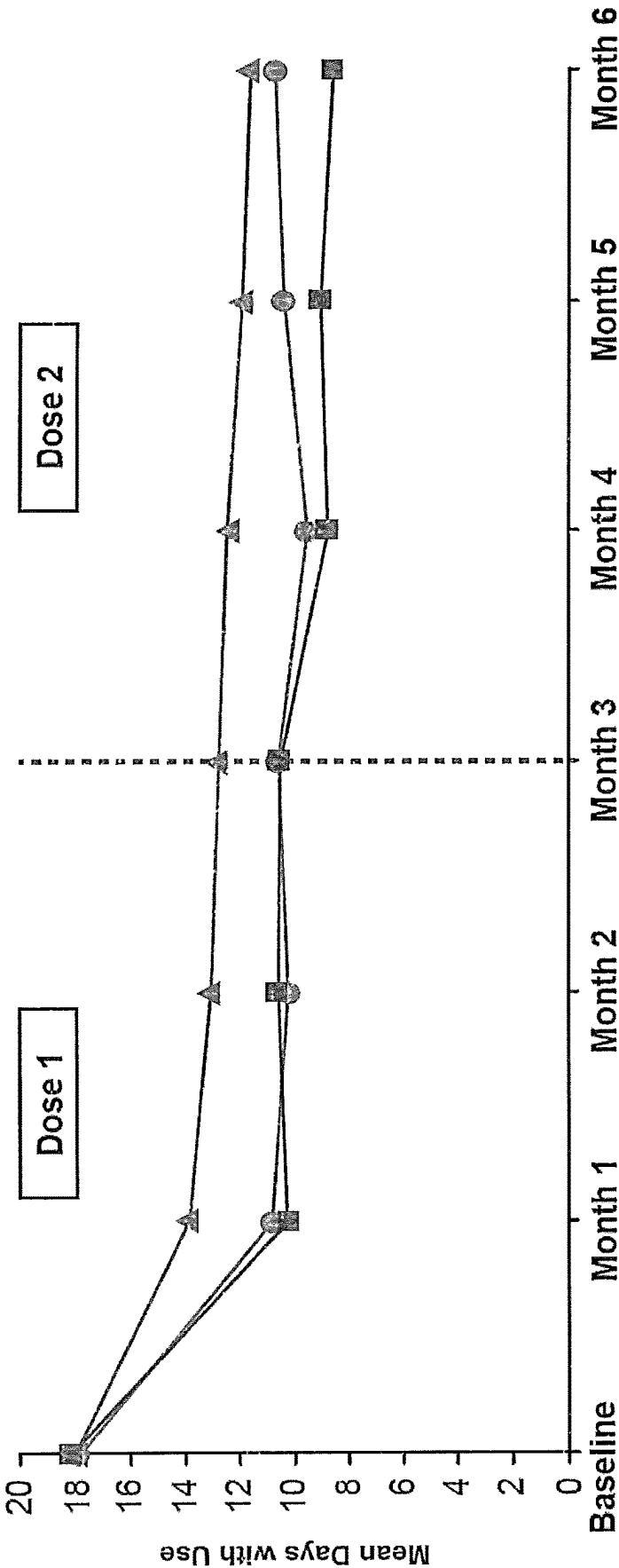


FIG. 44.

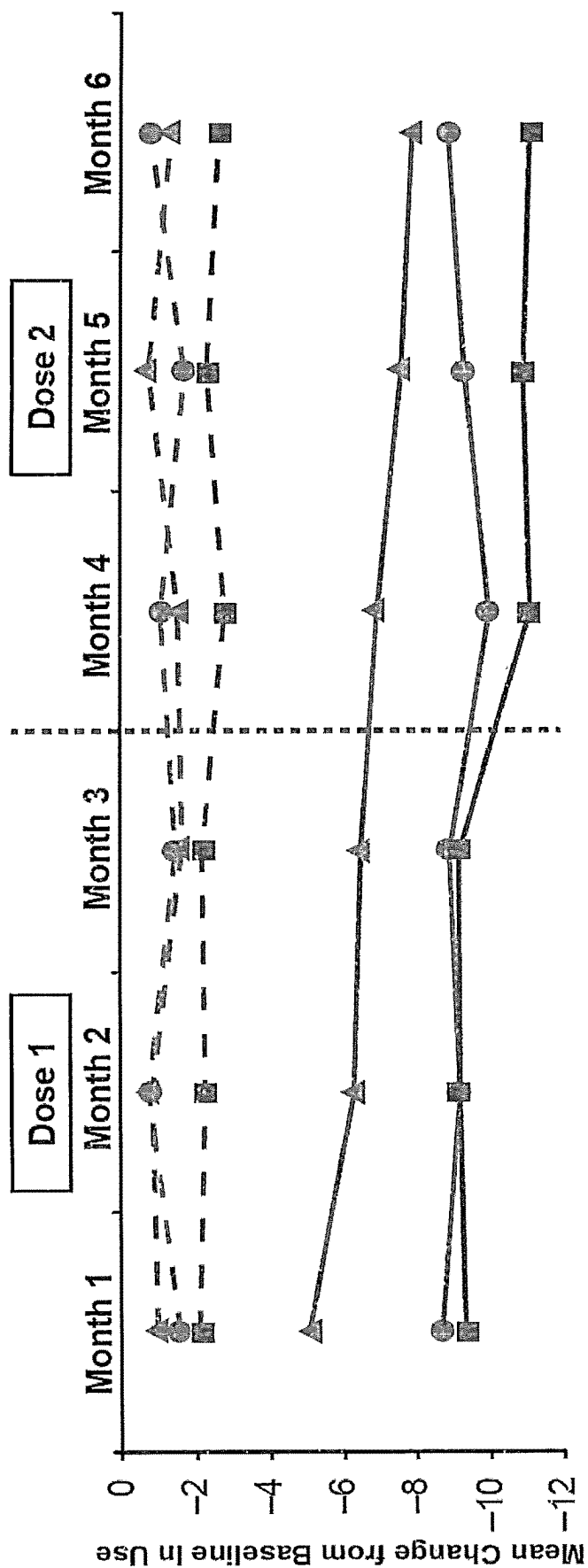


FIG. 45.

	Month 1			Month 6		
	Ab6 100 mg	Ab6 400 mg	Placebo	Ab6 100 mg	Ab6 400 mg	Placebo
Baseline use						
1-9 days/month, n	37	49	49	37	49	49
≥10 days/month, n	264	265	260	264	265	260
≥1 day/month, mean (SD)	18.3 (9.05)	18.4 (9.61)	17.9 (8.60)	18.3 (9.05)	18.4 (9.61)	17.9 (8.60)
Post-baseline use, mean (SD)						
≥1 day/month	10.7 (9.39)	10.2 (9.87)	13.8 (9.52)	10.8 (11.18)	8.6 (9.97)	11.5 (10.16)
Change from baseline, mean (SD)						
≥1 day/month	-7.8 (8.08)	-8.3 (7.64)	-4.5 (7.46)	-8.1 (9.90)	-9.6 (9.92)	-7.0 (9.39)
1-9 days/month	-1.5 (4.44)	-2.3 (4.34)	-1.0 (5.29)	-0.8 (6.63)	-2.6 (4.57)	-1.3 (4.83)
≥10 days/month	-8.7 (8.08)	-9.4 (7.62)	-5.1 (7.63)	-8.9 (9.88)	-11.1 (10.10)	-7.9 (9.64)
Percent change from baseline, mean (SD)						
≥1 day/month	-42.6 (39.98)	-47.0 (40.90)	-22.4 (52.02)	-40.7 (60.66)	-52.9 (48.97)	-34.7 (58.48)
1-9 days/month	-31.8 (67.95)	-47.3 (65.38)	-9.5 (100.52)	1.4 (132.84)	-45.0 (73.05)	-11.2 (108.44)
≥10 days/month	-44.1 (34.24)	-47.0 (34.73)	-24.8 (36.17)	-45.3 (44.91)	-54.5 (42.52)	-38.5 (44.63)

FIG. 46.

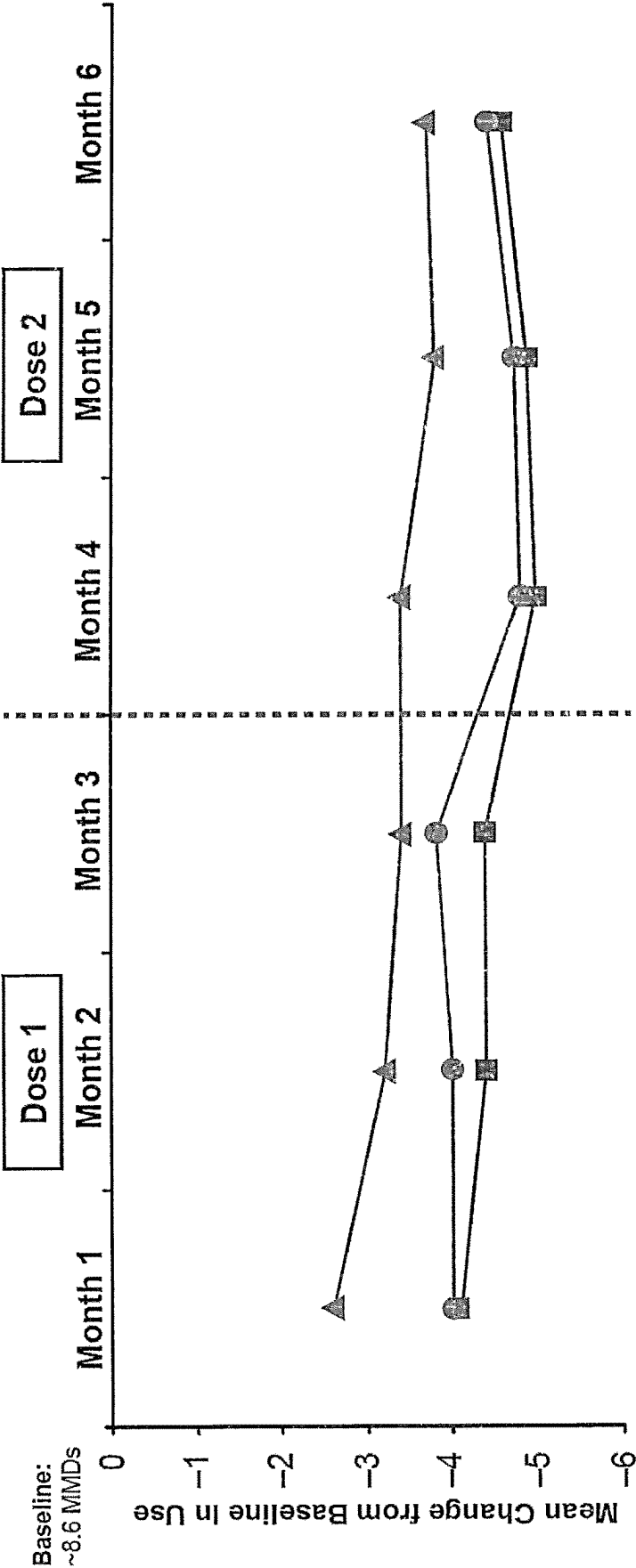


FIG. 47.

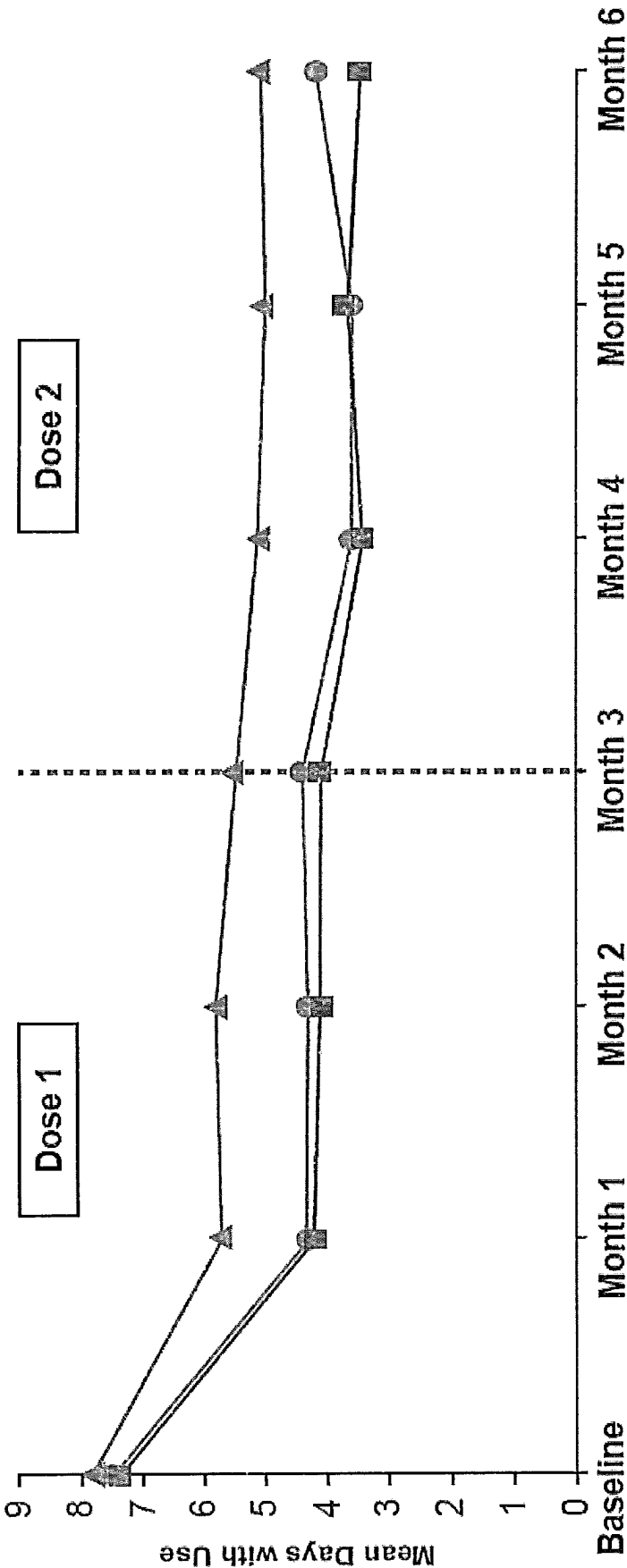


FIG. 48.

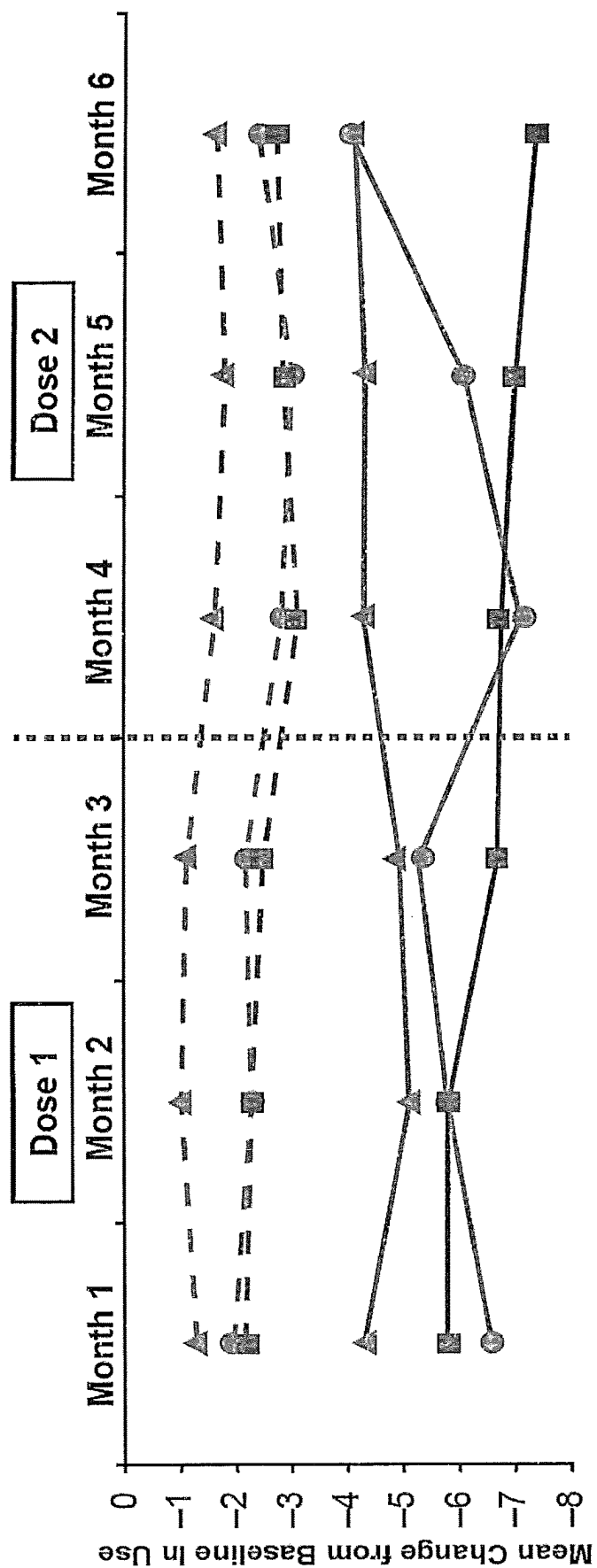
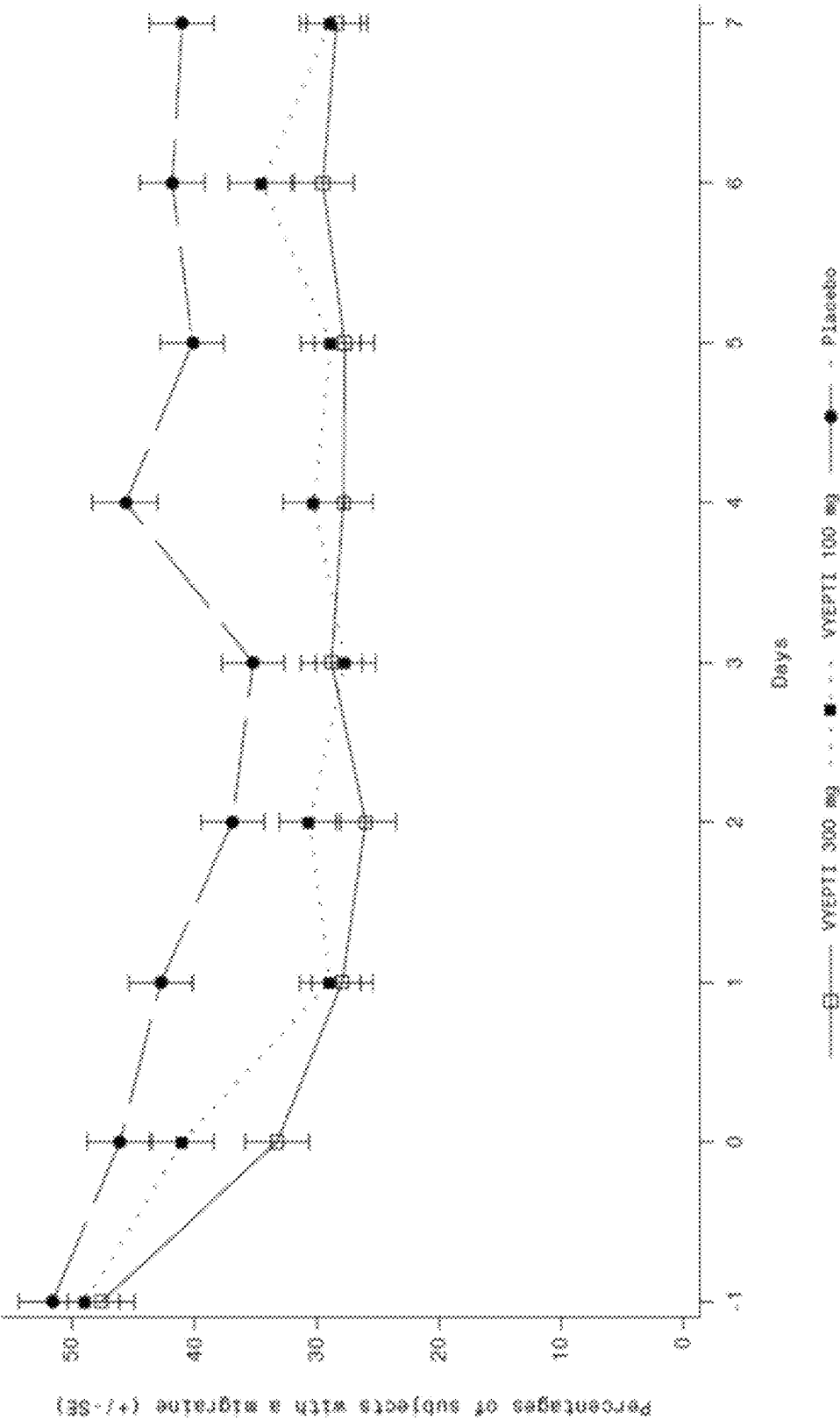


FIG. 49. Summary of Acute Medication Days by Subgroups of Episodic Migraine Patients with Baseline Acute Medication Use

	Month 1			Month 6		
	Ab6 100 mg	Ab6 400 mg	Placebo	Ab6 100 mg	Ab6 400 mg	Placebo
Baseline use						
1-9 days/month, n	117	111	108	117	111	108
≥10 days/month, n	42	41	44	42	41	44
≥1 day/month, mean (SD)	7.5 (4.97)	7.5 (4.58)	7.8 (4.98)	7.5 (4.97)	7.5 (4.58)	7.8 (4.98)
Post-baseline use, mean (SD)						
≥1 day/month	4.3 (3.99)	4.2 (4.45)	5.7 (5.04)	4.2 (5.87)	3.5 (3.92)	5.1 (5.19)
Change from baseline, mean (SD)						
≥1 day/month	-3.3 (4.14)	-3.2 (4.20)	-2.2 (4.68)	-2.8 (4.92)	-4.1 (4.60)	-2.3 (4.69)
1-9 days/month	-2.0 (2.91)	-2.2 (3.57)	-1.3 (3.10)	-2.4 (3.11)	-2.7 (3.83)	-1.6 (3.52)
≥10 days/month	-6.6 (5.11)	-5.8 (4.66)	-4.3 (6.82)	-4.0 (8.60)	-7.4 (4.60)	-4.1 (6.60)
Percent change from baseline, mean (SD)						
≥1 day/month	-36.9 (63.96)	-39.4 (77.71)	-22.4 (60.27)	-45.4 (62.28)	-50.9 (59.88)	-22.5 (95.61)
1-9 days/month	-33.9 (72.22)	-37.0 (88.45)	-19.7 (64.62)	-50.1 (59.65)	-48.2 (68.26)	-18.2 (107.55)
≥10 days/month	-45.1 (30.26)	-45.9 (34.95)	-29.1 (47.94)	-29.2 (69.14)	-57.2 (32.59)	-33.9 (52.53)

FIG 50



1

TREATMENT OF HEADACHE USING ANTI-CGRP ANTIBODIES

RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 16/793,208, filed Feb. 18, 2020, which is a Continuation-in-part of Int'l Nat'l Appl. No. PCT/US2020/012781, filed Jan. 8, 2020, which claims priority to Provisional Appl. No. 62/872,989, filed Jul. 11, 2019, Provisional Appl. No. 62/842,162, filed May 2, 2019, and Provisional Appl. No. 62/789,828, filed Jan. 8, 2019, the disclosures of each of which are hereby incorporated by reference in their entireties,

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

The contents of the electronic sequence listing (1143257.009203.xml; Size: 771,623 bytes; and Date of Creation: Feb. 22, 2023) is herein incorporated by reference in its entirety.

BACKGROUND

Field

This invention pertains to methods of treatment of headache disorders, such as migraine, using antibodies and fragments thereof (including Fab fragments) that specifically bind to human Calcitonin Gene Related Peptide (hereinafter "CGRP"). The invention also pertains to immediate treatment of headache, e.g., chronic migraine, using antibodies and fragments thereof (including Fab fragments) that specifically bind to human Calcitonin Gene Related Peptide (hereinafter "CGRP").

Description of Related Art

Calcitonin Gene Related Peptide (CGRP) is produced as a multifunctional neuropeptide of 37 amino acids in length. Two forms of CGRP, the CGRP-alpha and CGRP-beta forms, exist in humans and have similar activities. CGRP-alpha and CGRP-beta differ by three amino acids in humans, and are derived from different genes. CGRP is released from numerous tissues such as trigeminal nerves, which when activated release neuropeptides within the meninges, mediating neurogenic inflammation that is characterized by vasodilation, vessel leakage, and mast-cell degradation. Durham, P. L., *New Eng. J. Med.*, 350 (11):1073-75 (2004). Biological effects of CGRP are mediated via the CGRP receptor (CGRP-R), which consists of a seven-transmembrane component, in conjunction with receptor-associated membrane protein (RAMP). CGRP-R further requires the activity of the receptor component protein (RCP), which is essential for an efficient coupling to adenylate cyclase through G proteins and the production of cAMP. Doods, H., *Curr. Op. Invest. Drugs*, 2(9):1261-68 (2001).

Migraines are neurovascular disorder affecting approximately 10% of the adult population in the U.S., and are typically accompanied by intense headaches. CGRP is believed to play a prominent role in the development of migraines. In fact several companies, i.e., Amgen, Eli Lilly, Teva and Alder Biopharmaceuticals (recently acquired by Lundbeck A/S) have developed anti-CGRP and anti-CGRP-R antibodies for use in treating or preventing migraine headaches. The present assignee has previously

2

filed patent applications related to anti-CGRP antibodies and uses thereof including published PCT Application WO/2012/162243 filed May 21, 2012 entitled "ANTI-CGRP COMPOSITIONS AND USE THEREOF", published PCT Application WO/2012/162257 filed May 21, 2012, entitled "USE OF ANTI-CGRP ANTIBODIES AND ANTIBODY FRAGMENTS TO PREVENT OR INHIBIT PHOTOPHOBIA OR LIGHT AVERSION IN SUBJECTS IN NEED THEREOF, ESPECIALLY MIGRAINE SUFFERERS" published PCT Application WO/2012/162253, filed May 21, 2012, entitled "USE OF ANTI-CGRP OR ANTI-CGRP-R ANTIBODIES OR ANTIBODY FRAGMENTS TO TREAT OR PREVENT CHRONIC AND ACUTE FORMS OF DIARRHEA" and published PCT Application WO/2015/003122, filed Jul. 3, 2014, entitled "REGULATION OF GLUCOSE METABOLISM USING ANTI-CGRP ANTIBODIES" all of which applications are incorporated by reference in their entirety.

BRIEF SUMMARY

In one aspect, the present disclosure provides a method for treatment of migraine or headache in a patient in the need of immediate relief of migraine or headache symptoms or for prevention of migraine or headache in a patient in need of immediate preventative treatment of migraine or headache, comprising intravenous administering to a patient in need 100 or 300 mg of an anti-CGRP antibody comprising the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

In some aspects, said patient may exhibit at least one headache and/or migraine symptom at the time of administration.

In some aspects, said at least one headache and/or migraine symptom may comprise one or more of pain, nausea, photophobia, or phonophobia.

In some aspects, said at least one headache and/or migraine symptom may comprise head pain.

In some aspects, the most bothersome symptom may be alleviated after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

In some aspects, said patient may no longer have a migraine after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

In some aspects, said anti-CGRP antibody may comprise the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222.

In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232.

In some aspects, said anti-CGRP antibody may comprise the variable heavy chain polypeptide of SEQ ID NO: 202.

In some aspects, said anti-CGRP antibody may comprise the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221.

In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231.

In some aspects, said anti-CGRP antibody may comprise the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

In some aspects, said anti-CGRP antibody may comprise the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some aspects, said intravenous administration may be infused over a period of approximately 30 min to 60 minutes.

In some aspects, the headache or migraine symptoms may decline or may be abolished immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

In some aspects, said patient may be headache free 2 hours post-completion of infusion.

In some aspects, said method may further comprise intravenously administering 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

In some aspects, said method may further comprise intravenously administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

In some aspects, said anti-CGRP antibody may be comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

In some aspects, said formulation may comprise or may consist of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

In some aspects, said formulation may comprise or may consist of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Poly-

sorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

In some aspects, said formulation may comprise or may consist of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

In some aspects, said formulation may comprise or may consist of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

In some aspects, said formulation may comprise or may consist of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

In some aspects, said L-Histidine in said formulation comprises a mixture of L-Histidine and L-Histidine monohydrate. Said 3.1 mg of histidine in said formulation may comprise a mixture of L-Histidine (1 mg) and L-Histidine monohydrate (2.8 mg), which in the final formulation sums up to 3.1 mg L-histidine free base.

In some aspects, said formulation may be comprised in a 100 mg/mL single-dose vial wherein each mL contains 100 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

In some aspects, said formulation may be comprised in a 300 mg/mL single-dose vial wherein each mL contains 300 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

In some aspects, said migraine or headache may be selected from the group comprising acute migraine or headache, migraines with or without aura, chronic migraine, episodic migraine, chronic/episodic migraine, hemiplegic migraines, cluster headaches, migrainous neuralgia, chronic headaches, tension headaches, general headaches, headaches due to an underlying structural problem in the head or neck, sinus headaches (such as for example associated with sinusitis), and allergy-induced headaches or migraines.

In some aspects, said patient may exhibit a pain level of at least 2 on the VRS-4 at the time of administration of said antibody.

In some aspects, said patient may exhibit a pain level of at least 3 on the VRS-4 at the time of administration of said antibody.

In some aspects, said patient may exhibit a pain level of at most 2 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

In some aspects, said patient may exhibit a pain level at most 1 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration

5

within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

In some aspects, said patient may not be administered any acute migraine medication within a period of time before and after said administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said administration.

In some aspects, said acute migraine medication may comprise a triptan, an analgesic such as non-opioids or opioids/narcotics, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

In some aspects, said non-opioid analgesic may comprise paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan may comprise use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid may comprise use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication may comprise two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesics may comprise at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/or said combination-analgesic may comprise acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

In some aspects, the patient may be receiving or has received additional migraine medication.

In some aspects, the patient may receive additional migraine medication prior, concurrent or after administration of the anti-CGRP antibody.

In some aspects, the patient may receive additional migraine medication within a period of time before and after said anti-CGRP antibody administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said anti-CGRP antibody administration.

In some aspects, said additional migraine medication may comprise an acute and/or a chronic migraine medication.

In some aspects, said additional migraine medication may comprise a triptan, an analgesic such as non-opioid or opioid/narcotic, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

In some aspects, said non-opioid analgesic may comprise paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan comprises use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid comprises use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication comprises two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesics comprises at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/or said combination-analgesic comprises acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

6

In some aspects, said anti-CGRP antibody may be expressed in or obtained by expression in *Pichia pastoris*.

In some aspects, said anti-CGRP antibody may be expressed in or obtained by expression in CHO cells.

In some aspects, said patient may be administered 100 mg or 300 mg of said anti-CGRP antibody every three months.

In some aspects, said method may result in immediate relief of migraine or headache symptoms.

In some aspects, said method may result in immediate preventative treatment of migraine or headache.

The present disclosure further provides methods of immediate treatment of headache, comprising administering to a patient in need an effective amount of at least one anti-CGRP antibody or antibody fragment or an anti-CGRP-R antibody or antibody fragment or one or more formulations comprising said antibody or antibody fragment as disclosed herein. In some aspects, said antibody may be administered while said patient has a headache. In some aspects, said antibody administration may be initiated within 1-6 hours of the onset of said headache. In some aspects, said headache may comprise migraine, e.g., episodic migraine or chronic migraine. Said headache may comprise medication overuse headache. In some aspects, said anti-CGRP antibody or antibody fragment Ab6 or a Fab fragment thereof, having the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208; or having the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively. In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202. Said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212. Said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567. In some aspects, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells. In some aspects, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions. In some aspects, any of the aforementioned anti-CGRP anti-

bodies or antibody fragments, preferably Ab6, may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. In some aspects, said administered dosage of said antibody may be between about 100 mg and about 300 mg, such as about 100 mg, about 300 mg, 100 mg, or 300 mg. In some aspects, said dosage may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL.

In some aspects, said patient may exhibit less than 25 headache days per month, less than 20 headache days per month, less than 15 headache days per month, or less than 10 headache days per month. For example, said patient may exhibit less than 14 headache days, less than headache 13 days, less than headache 12 days, less than headache 11 days, less than 10 headache days, less than 9 headache days, less than 8 headache days, less than 7 headache days, or less than 6 headache days per month. In some aspects, said patient may exhibit between 2-15 headache days, e.g., 3-14 headache days, 4-13 headache days, 5-12 headache days, 6-11 headache days, or 7-10 headache days/month.

In some aspects, said patient may exhibit less than 10 migraines per month, such as between 1-9 migraines per month, such as between 2-8 migraines per month, between 3-7 migraine per month, between 4-6 migraine per month, or about 5 migraines per month. In some aspects, said patient may exhibit fewer than 1 migraine per month on average, e.g., on average one migraine every 2 months, one every 3 months, one every 4 or 6 months, or intermediate values such as 2 every 3 months, etc. In some aspects, said migraine may be diagnosed in accord with the ICHD-3 guidelines.

In exemplary embodiments, said headache may comprise medication overuse headache. Said medication overuse headache may be determined based on meeting the following criteria: (a) headache occurring on 15 or more days/month in a patient with a pre-existing headache disorder; and (b) overuse for more than 3 months of one or more drugs that can be taken for acute and/or symptomatic treatment of headache.

In some embodiments, said overuse may comprise use of an ergot alkaloid (e.g., ergotamine) on 10 or more days/month, use of a triptan on 10 or more days/month, use of one or more non-opioid analgesics (such as paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic) on 15 or more days/month, use of one or more combination-analgesics (as further described below) on 10 or more days/month, use of one or more opioids on 10 or more days/month, or use of a combination of two or more drug classes (as further described below) on 10 or more days/month.

In the methods herein, said triptan may include, without limitation thereto, any one of or any combination of triptans such as sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, and frovatriptan, among others.

In some aspects, said medication overuse headache may comprise ergotamine-overuse headache, triptan-overuse headache, non-opioid analgesic-overuse headache, opioid-overuse headache, combination-analgesic-overuse headache, medication-overuse headache attributed to multiple drug classes not individually overused, medication-overuse headache attributed to unspecified or unverified overuse of multiple drug classes, or medication-overuse headache attributed to other medication.

In some aspects, said non-opioid analgesic-overuse headache may comprise paracetamol (acetaminophen)-overuse headache, non-steroidal anti-inflammatory drug (NSAID)-overuse headache such as acetylsalicylic acid (aspirin)-overuse headache or ibuprofen-overuse headache, or another non-opioid analgesic-overuse headache.

In some aspects, said ergotamine-overuse headache may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a consequence of regular use of an ergot alkaloid such as ergotamine on 10 or more days/month for more than 3 months.

In the methods herein, said ergot alkaloid may comprise ergotamine, nicergoline, methysergide, or dihydroergotamine, or may comprise an ergot derivative.

In some aspects, said triptan-overuse headache may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a consequence of regular use of one or more triptans on 10 or more days/month for more than 3 months.

In some aspects, said non-opioid analgesic-overuse headache may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a consequence of regular use of one or more non-opioid analgesics (such as paracetamol (acetaminophen), acetylsalicylic acid (aspirin), ibuprofen, another NSAID, or another non-opioid analgesic) on 15 or more days/month for more than 3 months.

In the methods herein, said NSAID may comprise any NSAID or combination thereof, including without limitation thereto, ibuprofen, naproxen, or indomethacin.

In some aspects, said combination-analgesic-overuse headache may comprise headache occurring on 15 or more days/month developing as a consequence of regular use of one or more combination-analgesics on 10 or more days/month for more than 3 months. In the context of medication overuse headache, the term combination-analgesic refers to formulations combining drugs of two or more classes, each with analgesic effects (for example, paracetamol and codeine) or analgesics in combination with agents acting as adjuvants (for example, caffeine). Commonly overused combination-analgesics combine non-opioid analgesics with at least one opioid, barbiturate such as butalbital and/or caffeine. In exemplary embodiments, the combination-analgesic overuse-headache is due to the combination of acetaminophen, aspirin, and caffeine, e.g., EXCEDRIN® or EXCEDRIN MIGRAINE®. Other known combination analgesics comprise an analgesic in combination with at least one non-analgesic, e.g., with a vasoconstrictor drug such as pseudoephedrine for sinus-related preparations, antihistamine drug used to treat allergy sufferers, etc.

In some aspects, said opioid-overuse headache may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a consequence of regular use of one or more opioids 10 or more days/month for more than 3 months.

In some aspects, said medication-overuse headache attributed to multiple drug classes not individually overused may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a result of regular intake of any combination of ergotamine, triptans, non-opioid analgesics and/or opioids on a total of at least 10 days/month for more than 3 months without overuse of any single drug or drug class alone.

In the methods herein, said opioid may be any one or any combination of opioid drugs, including without limitation thereto, oxycodone, tramadol, butorphanol, morphine,

codeine, hydrocodone, thebaine, oripavine, mixed opium alkaloids such as papaveretum, diacetylmorphine, nicomorphine, dipropanoylmorphine, diacetyldihydromorphine, acetylpropionylmorphine, desomorphine, methyl-desomorphine, dibenzoylmorphine, ethylmorphine, heterocodine, buprenorphine, etorphine, hydromorphone, oxymorphone, fentanyl, alphamethylfentanyl, alfentanil, sufentanil, remifentanyl, carfentanyl, ohmefentanyl, pethidine (meperidine), ketobemidone, MPPP, allylprodine, prodine, PEPAP, promedol, diphenylpropylamine, propoxyphene, dextro-propoxyphene, dextromoramide, bezitramide, piritramide, among others.

In some aspects, said medication-overuse headache attributed to unspecified or unverified overuse of multiple drug classes may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a result of regular intake of any combination of ergotamine, triptans, non-opioid analgesics and/or opioids on at least 10 days/month for more than 3 months, wherein the identity, quantity and/or pattern of use or overuse of these classes of drug is not reliably established.

In some aspects, said medication-overuse headache attributed to other medication may comprise headache occurring on 15 or more days/month in a patient with a pre-existing primary headache and developing as a result of regular intake of one or more medications other than those described above, taken for acute or symptomatic treatment of headache, on at least 10 days/month for more than 3 months.

The amount and duration of medication use may be determined utilizing known methods, such as the usage reported by the patient or a relative, a diary, medical records, drug purchase history, prescription fulfilment, biomarkers of medication use, incidence of medication toxicity, incidence of medication overdose, and/or other indicators of a patient's medication use.

The present disclosure provides methods of treating or preventing probable medication overuse headache, comprising administering to a patient in need an effective amount of an anti-CGRP antibody or anti-CGRP antibody fragment or one or more formulations comprising said anti-CGRP antibody or anti-CGRP antibody fragment as disclosed herein. In some aspects, said anti-CGRP antibody Ab6, having the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208; or having the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively. In some aspects, said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202. Said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212. Said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. In some aspects, said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567. In some aspects, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain

polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells. Said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions. Any of the aforementioned anti-CGRP antibodies or antibody fragments, preferably Ab6, may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The administered dosage of said antibody may be between about 100 mg and about 300 mg, such as about 100 mg, about 300 mg, 100 mg, or 300 mg. The dosage may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL. Probable medication overuse headache refers to criteria (a) and (b) not being entirely fulfilled, e.g., having at least 80% or at least 90% of the specified number of headache days and/or medication use days per month, and/or over a shorter time period such as at least 2 months, optionally in the absence of another ICHD-3 diagnosis.

In some aspects, said medication-overuse headache (such as ergotamine-overuse headache, triptan-overuse headache, non-opioid analgesic-overuse headache, opioid-overuse headache, combination-analgesic-overuse headache, medication-overuse headache attributed to multiple drug classes not individually overused, medication-overuse headache attributed to unspecified or unverified overuse of multiple drug classes, or medication-overuse headache attributed to other medication) may be diagnosed according to the third edition of the International Classification of Headache Disorders (ICHD-3). See Headache Classification Committee of the International Headache Society (IHS), The International Classification of Headache Disorders, 3rd edition, Cephalalgia. 2018 January; 38(1):1-211, which is hereby incorporated by reference in its entirety.

Herein, the criterion that a headache occurs "as a consequence of" over use of a medication or medications refers to the apparent association between the medication(s) overuse and the headache, e.g., that the medication(s) overuse and headache are present at the above-specified frequency such that causation may be presumed.

The present disclosure also provides methods of treating chronic migraine, comprising intravenously administering to a patient in need thereof a first dosage comprising between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566, wherein in the first 24 hours after administration of said first dosage the patient exhibits at least a 50% reduction in migraine prevalence.

11

In another aspect, the disclosure provides methods of treating chronic migraine, comprising intravenously administering to a patient in need thereof a first dosage comprising between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566, wherein on the first day following the day of administration the patient exhibits at least a 50% reduction in migraine prevalence.

In some exemplary embodiments, the dosage, e.g., the first dosage, of said anti-CGRP antibody may be 100 mg.

In other exemplary embodiments, the dosage, e.g., the first dosage, of said anti-CGRP antibody may be 300 mg.

In other exemplary embodiments, the method may further comprise intravenously administering 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

In other exemplary embodiments, the method may further comprise intravenously administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

The antibody may be provided or administered in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

Prior to said first dosage, the patient may exhibit between about 10 and about 22 migraine days per month, such as between about 13 and about 19 migraine days per month, such as about 16 migraine days per month.

Prior to said first dosage, the patient may exhibit between about 14 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about 21 headache days per month.

In some embodiments, the patient may have been diagnosed with migraine at least 10 years prior to said first dosage, such as at least 15 years prior to said first dosage, such as at least 18 or at least 19 years prior to said first dosage.

In some embodiments, the patient may have been diagnosed with chronic migraine at least 5 years prior to said first dosage, such as at least 8 years prior to said first dosage, such as at least 11 or at least 12 years prior to said first dosage.

In some embodiments, the patient may have a headache when administered said first dosage.

In some embodiments, the patient may have a migraine, such as a migraine with aura, when administered said first dosage.

In some embodiments, the patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the patient may have a reduction in the number of migraine days by at least 50% in the 12

12

week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

In some embodiments, the method may further comprise administering, e.g., intravenously, a second dose of said anti-CGRP antibody to said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said first dose.

In some embodiments, said first dose may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

In some embodiments, said patient may be a chronic migraine patient or episodic migraine patient at risk of developing medication overuse headache. Said patient may use acute headache medication on at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 day(s) per month. Said patient may use acute headache medication on at least 10 days per month. Optionally said acute medication use is determined over a baseline period of at least 28 days. Said acute medication use may be reported by the patient, a caregiver, or based on records. Said acute medication may comprise use of ergot alkaloids, triptans, non-opioid analgesics, acetaminophen, aspirin, NSAIDs, non-opioid analgesics, combination-analgesics, or opioids.

In some embodiments, prior to said administration, the patient may exhibit between about 15 and about 30 migraine days per month, such as between about 16 and about 28 migraine days per month, such as between about 17 and about 26 migraine days per month, such as about 16 migraine days per month.

In some embodiments, prior to said administration, the patient may exhibit between about 15 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about 21 headache days per month.

In some embodiments, said patient may have been diagnosed with migraine at least 10 years prior to said administration, such as at least 15 years prior to said administration, such as at least 18 or at least 19 years prior to said administration.

In some embodiments, said patient may have been diagnosed with chronic migraine at least 5 years prior to said administration, such as at least 8 years prior to said administration, such as at least 11 or at least 12 years prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

In some embodiments, said method may further comprise administering, e.g., intravenously, a second dose of said anti-CGRP antibody to said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said administration.

In some embodiments, said administration may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

In some embodiments, said anti-CGRP antibody may be aglycosylated or if glycosylated only may contain only mannose residues.

In some embodiments, said anti-CGRP antibody may consist of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. Said anti-CGRP antibody may consist of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain of SEQ ID NO: 222 and/or the variable heavy chain of SEQ ID NO: 202. In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain encoded by SEQ ID NO: 232 and/or the variable heavy chain encoded by SEQ ID NO: 212.

In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain of SEQ ID NO: 221 and/or the heavy chain of SEQ ID NO: 201 or SEQ ID NO: 566. In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain encoded by SEQ ID NO: 231 and/or the heavy chain encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the VL polypeptide of SEQ ID NO: 222 and the VH polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences

encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

In some embodiments, any of the aforementioned anti-CGRP antibodies or antibody fragments may be comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The antibody or fragment may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL.

In some embodiments, about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

In other embodiments, about 100 mg of said anti-CGRP antibody or antibody fragment is administered.

In other embodiments, about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

In exemplary embodiments, the anti-human CGRP antibody or antibody fragment is administered, e.g., intravenously at a frequency which is at most every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks, wherein the antibody dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks. The phrase "the antibody dosage is administered in a single formulation or divided into different formulations" refers to the administration of the recited amount of antibody within a relatively short period of time, e.g., within a period of several hours, e.g., 1 to 8 hours, about one day, within about two days, or within about one week, which may be by the same or different routes (e.g., i.v., i.m., and/or s.c.), sites of administration. The term "different formulations" in this context refers to antibody dosages that are administered at different times and/or at different sites and/or different routes, irrespective of whether the dosages are the same or different with respect to the chemical composition of the pharmaceutical formulation in with each dosage is administered; for example, the concentration, excipients, carriers, pH, and the like may be the same or different between the different administered dosages.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 8 weeks or every 2 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks or every 3 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations

which are administered at a frequency of approximately every 16 weeks or every 4 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 20 weeks or every 5 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 24 weeks or every 6 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 28 weeks or every 7 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 32 weeks or every 8 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 36 weeks or every 9 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 40 weeks or every 8 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 44 weeks or every 9 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 48 weeks or every 10 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 52 weeks or every 11 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 56 weeks or every 12 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 15-18 months.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 18-21 months.

In other exemplary embodiments, the anti-human CGRP antibody dosage or antibody fragment used in the aforementioned methods is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 2 years.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods is administered systemically.

In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment used in the afore-mentioned methods is administered by a mode of administration is selected from intravenous, intramuscular, intravenous, intrathecal, intracranial, topical, intranasal, and oral. In a preferred embodiment, the anti-human CGRP antibody or antibody fragment used in the afore-mentioned methods is administered intravenously.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 10 days.

In other exemplary embodiments, the anti-human CGRP antibody has an in vivo half-life of at least 15 days.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20 days.

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20-30 days.

In other exemplary embodiments, the anti-human CGRP antibody is administered at a dosage of between about 100 mg and about 300 mg has an in vivo half-life of $\pm 20\%$ of at least about (284 $\pm$ 44 hours).

In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods binds to human α - and β -CGRP.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 30 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 60 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in inhibition of vasodilation induced by topically applied capsaicin at least 90 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 120 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 150 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 180 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin more than 180 days after antibody administration.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity, within 5% of the maximal response (I_{max}) (as compared to lower antibody doses).

In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity which is maintained for at least 2-3 months after antibody administration, wherein PK analysis of the anti-human CGRP antibody is derived from plasma concentrations.

In other exemplary embodiments, the administered anti-human CGRP antibody dosage is between about 100 mg and about 300 mg or more which is administered no more frequently than every 2 months.

The present invention is additionally directed to the use of specific antibodies and fragments thereof having binding specificity for CGRP, in particular antibodies having desired epitopic specificity, high affinity or avidity and/or functional properties. A preferred embodiment of the invention is directed to usage of chimeric or humanized antibodies and fragments thereof (including Fab fragments) capable of binding to CGRP and/or inhibiting the biological activities mediated by the binding of CGRP to the CGRP receptor ("CGRP-R") e.g., wherein such antibodies optionally are derived from recombinant cells engineered to express same, optionally yeast or mammalian cells, further optionally *Pichia pastoris* and CHO cells.

In another preferred embodiment of the invention, full length antibodies and Fab fragments thereof are contemplated that inhibit the CGRP- α -, CGRP- β -, and rat CGRP-driven production of cAMP. In a further preferred embodiment of the invention, full length and Fab fragments thereof are contemplated that reduce vasodilation in a recipient following administration.

The invention also contemplates usage of conjugates of anti-CGRP antibodies and binding fragments thereof conjugated to one or more functional or detectable moieties. The invention also contemplates usage of chimeric or humanized anti-CGRP or anti-CGRP/CGRP-R complex antibodies and binding fragments thereof. In one embodiment, binding fragments include, but are not limited to, Fab, Fab', F(ab')₂, Fv, scFv fragments, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIGS. 1A-1F provide the polypeptide sequences of the full-length heavy chain for antibodies Ab1-Ab14 with their framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

FIGS. 2A-2D provide the polypeptide sequences of the full-length light chain for antibodies Ab1-Ab14 with their framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

FIGS. 3A-3P provide exemplary polynucleotide sequences encoding the full-length heavy chain for antibodies Ab1-Ab14 with their framework regions (FR), complementarity determining regions (CDRs), and constant region coding sequences delimited.

FIGS. 4A-4I provide exemplary polynucleotide sequences encoding the full-length light chain for antibodies Ab1-Ab14 with their framework regions (FR), complementarity determining regions (CDRs), and constant region coding sequences delimited.

FIG. 5 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 6 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 7 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of

antibodies Ab1-Ab14 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 8 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 9 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 10 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 11 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length light chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

FIG. 12 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length light chain polypeptide sequences of antibodies Ab1-Ab14 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

FIG. 13 shows the number of subjects in a human clinical trial described in Example 2 who were either treated with Ab6 (treatment group) or placebo groups who showed a 50, 75 or 100% reduction in migraines at each monitoring point throughout the period. The right bar in each group corresponds to patients receiving 1000 mg Ab6 and the left bar in each group corresponds to matched placebo controls. In each response rate group the patients receiving Ab6 had a significantly greater response rate than placebo-treated controls, with p values of 0.0155, 0.0034, and 0.0006 in each respective group as indicated. The administered antibody was produced in *P. pastoris* and consisted of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201.

FIG. 14 shows the median ($\pm$ QR) % change from baseline in the number of migraine days per month in the placebo and Ab6-treated group over the 12 weeks post-treatment. (p=0.0078). The upper (red) line and lower (blue) line show results for placebo-treated controls and patients administered 1000 mg Ab6, respectively.

FIG. 15 shows the median ($\pm$ QR) % change from baseline in the number of migraine episodes per month in the placebo and Ab6-treated group over the 12 weeks post-treatment. The upper (red) line and lower (blue) line show results for placebo-treated controls and patients administered 1000 mg Ab6, respectively.

FIG. 16 shows the median ($\pm$ QR) % change from baseline in the number of migraine hours per month in the placebo and Ab6-treated group over the 12 weeks post-treatment. The upper (red) line and lower (blue) line show results for placebo-treated controls and patients administered 1000 mg Ab6, respectively.

FIG. 17 summarizes the screening of patients, allocation into the treatment and control groups, and loss of patients through follow-up.

FIG. 18 compares the HIT-6 responder analysis for the Ab6-treated and placebo groups at baseline, week 4 after treatment, week 8 after treatment and week 12 after treatment.

FIG. 19 shows the percentage of patients for whom the HIT-6 analysis indicated that the effect of headaches was only “some” or “little/none” at baseline and after Ab6 administration. At baseline most patients had either “substantial” or “severe” impact from migraines. At each subsequent time point, a significantly greater percentage of patients administered 1000 mg Ab6 had only “some” or “little/none” HIT-6 impact (left bar in each group, colored blue) as compared to placebo controls (right bar in each group, colored red).

FIG. 20 contains the pharmacokinetic (PK) profile for Ab6 administered intravenously at a single dosage of 1000 mg.

FIG. 21 contains plasma-free pharmacokinetic (PK) parameters N (number of patients), mean, and standard deviation (SD) for a single 1000 mg intravenous dosage of Ab6. The parameters shown in the table and the units are C_{max} (μg/mL), $AUC_{0-\infty}$ (mg*hr/mL), half-life (days), V_z (L) and C_L (mL/hr).

FIG. 22 shows the change (mean+/-SEM) change from baseline in migraine days per month for Ab6 (1000 mg i.v.) versus placebo as a single dose for the study described in Example 2.

FIG. 23 shows the average migraine days (+/-SD) over time for the full analysis population for the study described in Example 2. Normalization was applied to visit intervals where eDiaries were completed for 21-27 days by multiplying the observed frequency by the inverse of the completion rate.

FIG. 24 shows the distribution of migraine days actual and change for the Ab6 treatment group during weeks 1-4 for the study described in Example 2.

FIG. 25 shows the distribution of migraine days actual and change for the placebo group during weeks 1-4 for the study described in Example 2.

FIG. 26 shows the distribution of migraine days actual and change for the Ab6 treatment group during weeks 5-8 for the study described in Example 2.

FIG. 27 shows the distribution of migraine days actual and change for the placebo group during weeks 5-8 for the study described in Example 2.

FIG. 28 shows the distribution of migraine days actual and change for the Ab6 treatment group during weeks 9-12 for the study described in Example 2.

FIG. 29 shows the distribution of migraine days actual and change for the placebo group during weeks 9-12 for the study described in Example 2.

FIG. 30 shows the 50% responder rate for the Ab6 and placebo treatment groups for the study described in Example 2. Subjects with $\geq 50\%$ reduction in migraine frequency were considered to be a 50% responder. Normalization was applied to visit intervals where eDiary was completed for 21-27 days by multiplying the observed frequency by the inverse of the completion rate.

FIG. 31 shows the 75% responder rate for the Ab6 and placebo treatment groups for the study described in Example 2. Subjects with $\geq 75\%$ reduction in migraine frequency were considered to be a 75% responder. Normalization was applied as described with FIG. 30.

FIG. 32 shows the 100% responder rate for the Ab6 and placebo treatment group for the study described in Example 2. Subjects with 100% reduction in migraine frequency were

considered to be a 100% responder. Normalization was applied as described with FIG. 30.

FIG. 33 shows the mean migraine severity over time for the full analysis population for the study described in Example 2. On the scale used, a mean migraine score of 3 represents “moderate pain.”

FIG. 34 summarizes the change from baseline in measured attributes for the placebo and treatment groups in the study described in Example 2.

FIG. 35 shows the percentages of patients with migraine in the 300 mg, 100 mg, and placebo treatment groups at days 1, 7, 14, 21, and 28 in the clinical trial described in Example 3. The uppermost line shows results for placebo, the lowest line shows results for the 300 mg dosage, and the middle line shows results for the 100 mg dosage.

FIG. 36 show the percentage of patients in the 300 mg and 100 mg treatment groups achieving a 50% reduction in migraine days in month 1, over months 1-3 (after the 1st infusion), and over months 4-5 (after the 2nd infusion) in the clinical trial described in Example 3. In each graph, the data bars, from left to right, show results for the 100 mg, 300 mg, and placebo groups. Statistical significance is as shown. ++ indicates a statistically significant difference from placebo; + indicates a statistically significant difference from placebo (unadjusted); and § indicates a statistically significant difference from placebo (post hoc).

FIG. 37 show the percentage of patients in the 300 mg and 100 mg treatment groups achieving a 75% reduction in migraine days in month 1, over months 1-3 (after the 1st infusion), and over months 4-5 (after the 2nd infusion) in the clinical trial described in Example 3. Data order and statistical significance labels are as indicated with FIG. 36.

FIG. 38 show the percentage of patients in the 300 mg and 100 mg treatment groups achieving a 100% reduction in migraine days in month 1, over months 1-3 (after the 1st infusion), and over months 4-5 (after the 2nd infusion) in the clinical trial described in Example 3. Data order and statistical significance labels are as indicated with FIG. 36.

FIG. 39 summarizes the characteristics of patients in each treatment group in the clinical trial described in Example 3. * According to the American Academy of Neurology/American Headache Society guidelines for migraine preventative treatment (medications identified by clinical review of coded medical data); SD, standard deviation; BMI, body mass index.

FIG. 40. Difference from placebo in change from baseline in mean migraine days (MMD) over months 1-3 by baseline subgroup for a human clinical trial of chronic migraine patients. In the graph, the data point refers to the mean value and the line shows the 95% confidence interval (CI) of the change from placebo for the 100 mg (upper line) or 300 mg (lower line) treatment group, for each subgroup as labeled at the far left.

FIG. 41. Difference from placebo in change from baseline in mean migraine days (MMD) over months 1-3 by baseline subgroup for a human clinical trial of episodic migraine patients. The graph is labeled as in FIG. 40.

FIG. 42. Change from baseline in mean migraine days (MMDs) across 2 dose intervals in chronic migraine patients with at least 1 day of acute medication use per month at baseline. Triangle: placebo (n=366). Circle: 100 mg Ab6 per dose (n=356). Square: 300 mg Ab6 per dose (n=350).

FIG. 43. Mean days with acute medication use in chronic migraine patients with at least one day per month of acute medication use at baseline. Triangle: placebo (n=366). Circle: 100 mg Ab6 per dose (n=356). Square: 300 mg Ab6 per dose (n=350).

FIG. 44. Change from baseline in acute medication use by subgroups of chronic migraine patients with differing baseline days of acute medication use. Solid lines: patients with 10 or more days of acute medication use per month at baseline. Dashed lines: patients with at least 1 and less than 10 days of acute medication use per month at baseline. Triangle: placebo. Circle: 100 mg Ab6 per dose. Square: 300 mg Ab6 per dose.

FIG. 45. Summary of Acute Medication Days by Subgroups of Chronic Migraine Patients with Baseline Acute Medication Use.

FIG. 46. Change from baseline in mean migraine days (MMDs) across 2 dose intervals in episodic migraine patients with at least 1 day of acute medication use per month at baseline. Triangle: placebo (n=222). Circle: 100 mg Ab6 per dose (n=221). Square: 300 mg Ab6 per dose (n=222).

FIG. 47. Mean days with acute medication use in episodic migraine patients with at least one day per month of acute medication use at baseline. Triangle: placebo (n=222). Circle: 100 mg Ab6 per dose (n=221). Square: 300 mg Ab6 per dose (n=222).

FIG. 48. Change from baseline in acute medication use by subgroups of episodic migraine patients with differing baseline days of acute medication use. Solid lines: patients with 10 or more days of acute medication use per month at baseline. Dashed lines: patients with at least 1 and less than 10 days of acute medication use per month at baseline. Triangle: placebo. Circle: 100 mg Ab6 per dose. Square: 300 mg Ab6 per dose.

FIG. 49. Summary of Acute Medication Days by Subgroups of Episodic Migraine Patients with Baseline Acute Medication Use.

FIG. 50. Inclusion of Day -1 in the Migraine Data. Day 0 is defined as the day of the infusion. Thus, the data on Day 0 are indicative of the treatment effect post-infusion

DETAILED DESCRIPTION

Use of anti-CGRP antibodies for treatment of headache is described herein. Additionally, anti-CGRP antibodies are demonstrated herein to be effective for treatment of chronic migraine. The treatment was shown to have a very rapid onset of efficacy, with relief from migraine observed on the first day following administration.

Definitions

It is to be understood that this invention is not limited to the particular methodology, protocols, cell lines, animal species or genera, and reagents described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments, only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims. As used herein the singular forms “a”, “and”, and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a cell” includes a plurality of such cells and reference to “the protein” includes reference to one or more proteins and equivalents thereof known to those skilled in the art, and so forth. All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this invention belongs unless clearly indicated otherwise.

As used herein, the term “medication overuse headache” refers to a headache that meets the criteria for that condition

specified in ICHD-3 (Headache Classification Committee of the International Headache Society (IHS), The International Classification of Headache Disorders, 3rd edition, Cephalalgia. 2018 January; 38(1):1-211). The term includes subtypes of medication overuse headache, as defined in the ICHD-3, such as triptan-overuse headache, non-opioid analgesic overuse headache, opioid overuse headache, etc.

As used herein, the term “reduction in migraine prevalence” refers to a reduction (e.g., a stated percentage reduction, such as 50%) in the likelihood of a patient having a migraine in the stated period, such as the 18 hour, 20 hour, 24 hour, 28 hour, or 30 hour period, preferably the 24 hour period, after a first dosage of an antibody, or on the first day following the day of antibody administration (i.e., on the first full day following the day on which the antibody administration is completed). It is to be understood that a given patient may or may not have a migraine during that period, as the reduction in likelihood may be observable over a large number of patients irrespective of the outcome for an individual patient.

As used herein, the term “chronic migraine” refers to a condition wherein a patient exhibits, on average, at least 15 migraine days and/or headache per month. The term “episodic migraine” refers to a condition wherein a patient exhibits, on average, less than 15 headache and/or migraine days per month.

As used herein, the term “diagnosed with chronic migraine” refers to a patient meeting the clinical criteria for chronic migraine, whether or not a formal diagnosis of that patient was performed.

As used herein, the term “intravenously administering” refers to a mode of administration wherein a substance, e.g., an antibody, is introduced directly into the circulation of that patient, most typically into the venous circulation. The substance may be introduced in a carrier fluid, such as an aqueous solution, e.g., normal saline. The substance may be administered in a single formulation or in multiple formulations, as long as the administration is completed over a short period of time (e.g., within 1 day, preferably within 12 hours, more preferably within 6 hours, and most preferably within 1-2 hours).

As used herein, the term “the baseline number of migraine days” refers to the number of migraine days exhibited by a patient in a specified time period, e.g., prior to treatment. For example, the baseline number of migraine days may be determined over a period of one month, or longer, e.g., by recording each day whether or not a migraine occurred.

As used herein, the term “immediate relief” is intended to mean a relief in headache or migraine symptoms in a patient, e.g., headache or migraine symptoms associated with an acute migraine or chronic/episodic migraine or another headache or migraine condition associated with frequent headache or migraine episodes, wherein said relief of symptoms is experienced rapidly or immediately after anti-CGRP antibody treatment, e.g., relief of one or more symptoms is experienced by the patient within a short time period post-infusion with Ab6, such as within minutes or a few hours, such as within 10 minutes, 20 minutes, 30 minutes, 60 minutes, 1 hour, 2 hours or 6 hours, up to e.g. a day.

As used herein, the term “immediate preventative treatment” is intended to mean prevention of headache or migraine symptoms in a patient, e.g., prevention of headache or migraine symptoms associated with an acute migraine or chronic/episodic migraine or another headache or migraine condition. In this context, “immediate preventative treatment” refers to the prophylactic treatment of a subject who is at risk of developing migraine or headache, resulting in a

decrease in the probability that the subject will develop headache or migraine. Typically, due to a patient history of headache or migraine episodes, there is a high risk of a new headache or migraine episode in the patient. Typically the prevention of symptoms is experienced rapidly or immediately after anti-CGRP antibody treatment, e.g., prevention of one or more symptoms is experienced by the patient within a short time period post-infusion with Ab6, such as within minutes or a few hours, such as within 10 minutes, 20 minutes, 30 minutes, 60 minutes, 1 hour, 2 hours or 6 hours, up to e.g. a day.

As used herein, the terms “4-point scale” or “4 point pain scale” or “VRS” or “VRS-4” refer to the 4-point verbal rating scale (VRS) used to measure pain (VRS-4) (see “The International Classification of Headache Disorders, 3rd edition”, Cephalalgia, 2018, Vol. 38(1) 1-211, at pg. 210 (“intensity of pain”). In the VRS the patient is asked to rate the pain verbally on a 4 point scale (between 0 and 3), with 3 being severe, 2 being moderate, 1 being mild, and 0 being no pain. It may also be scored on a verbal rating scale expressed in terms of its functional consequence: 0, no pain; 1, mild pain, does not interfere with usual activities; 2, moderate pain, inhibits but does not wholly prevent usual activities; 3, severe pain, prevents all activities.

As used herein, the term “migraine days per month” refers to the number of days per month on which a patient has a migraine, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of migraine. The number of migraine days per month may be determined by recording each day whether or not a migraine occurred.

As used herein, the term “headache days per month” refers to the number of days per month on which a patient has a headache, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of a headache. The number of headache days per month may be determined by recording each day whether or not a headache occurred.

Calcitonin Gene Related Peptide (CGRP): As used herein, CGRP encompasses not only the following *Homo sapiens* CGRP-alpha and *Homo sapiens* CGRP-beta amino acid sequences available from American Peptides (Sunnyvale CA) and Bachem (Torrance, CA):

CGRP-alpha: ACDTATCVTHR-
LAGLLSRSGGVVKNFVPTNVGSKAF-NH₂ (SEQ ID
NO: 561), wherein the terminal phenylalanine is amidated;

CGRP-beta: ACNTATCVTHR-
LAGLLSRSGGMVKSNFVPTNVGSKAF-NH₂ (SEQ ID
NO: 562), wherein the terminal phenylalanine is amidated; but also any membrane-bound forms of these CGRP amino acid sequences, as well as mutants (mutiens), splice variants, isoforms, orthologs, homologues and variants of this sequence.

Expression Vector: These DNA vectors contain elements that facilitate manipulation for the expression of a foreign protein within the target host cell, e.g., a yeast or mammalian cell such as *Pichia pastoris* or CHO cells. Conveniently, manipulation of sequences and production of DNA for transformation is first performed in a bacterial host, e.g. *E. coli*, and usually vectors will include sequences to facilitate such manipulations, including a bacterial origin of replication and appropriate bacterial selection marker. Selection markers encode proteins necessary for the survival or growth of transformed host cells grown in a selective culture medium. Host cells not transformed with the vector containing the selection gene will not survive in the culture medium. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, (b) complement auxotrophic deficiencies, or (c) supply critical nutri-

ents not available from complex media. Exemplary vectors and methods for transformation of yeast are described, for example, in Burke, D., Dawson, D., & Stearns, T. (2000). Methods in yeast genetics: a Cold Spring Harbor Laboratory course manual. Plainview, N.Y.: Cold Spring Harbor Laboratory Press.

Expression vectors for use in yeast or mammalian cells will generally further include yeast or mammalian specific sequences, including a selectable auxotrophic or drug marker for identifying transformed yeast strains or transformed mammalian cells. A drug marker may further be used to amplify copy number of the vector in the host cell.

The polypeptide coding sequence of interest is operably linked to transcriptional and translational regulatory sequences that provide for expression of the polypeptide in host cells, e.g., *Pichia pastoris* or CHO cells. These vector components may include, but are not limited to, one or more of the following: an enhancer element, a promoter, and a transcription termination sequence. Sequences for the secretion of the polypeptide may also be included, e.g. a signal sequence, and the like. A yeast or mammalian origin of replication is optional, as expression vectors are often integrated into the host cell genome. In one embodiment of the invention, the polypeptide of interest is operably linked, or fused, to sequences providing for optimized secretion of the polypeptide from yeast diploid cells.

Nucleic acids are “operably linked” when placed into a functional relationship with another nucleic acid sequence. For example, DNA for a signal sequence is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence. Generally, “operably linked” means that the DNA sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading frame. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites or alternatively via a PCR/recombination method familiar to those skilled in the art (Gateway® Technology; Invitrogen, Carlsbad California). If such sites do not exist, the synthetic oligonucleotide adapters or linkers are used in accordance with conventional practice.

Promoters are untranslated sequences located upstream (5') to the start codon of a structural gene (generally within about 100 to 1000 bp) that control the transcription and translation of particular nucleic acid sequences to which they are operably linked. Such promoters fall into several classes: inducible, constitutive, and repressible promoters (that increase levels of transcription in response to absence of a repressor). Inducible promoters may initiate increased levels of transcription from DNA under their control in response to some change in culture conditions, e.g., the presence or absence of a nutrient or a change in temperature.

The promoter fragment may also serve as the site for homologous recombination and integration of the expression vector into the same site in the host genome; alternatively a selectable marker is used as the site for homologous recombination. Examples of suitable promoters from *Pichia* include the AOX1 and promoter (Cregg et al. (1989) *Mol. Cell. Biol.* 9:1316-1323); ICL1 promoter (Menendez et al. (2003) *Yeast* 20(13):1097-108); glyceraldehyde-3-phosphate dehydrogenase promoter (GAP) (Waterham et al. (1997) *Gene* 186(1):37-44); and FLD1 promoter (Shen et al. (1998) *Gene* 216(1):93-102). The GAP promoter is a strong constitutive promoter and the AOX and FLD1 promoters are inducible.

Other yeast promoters include ADH1, alcohol dehydrogenase II, GAL4, PHO3, PHO5, Pyk, and chimeric promoters derived therefrom. Additionally, non-yeast promoters may be used in the invention such as mammalian, insect, plant, reptile, amphibian, viral, and avian promoters. Most typically the promoter will comprise a mammalian promoter (potentially endogenous to the expressed genes) or will comprise a yeast or viral promoter that provides for efficient transcription in yeast systems.

Examples of mammalian promoters include cytomegalovirus (CMV) derived promoters, chicken 3-actin (CBM) derived promoters, adenomatous polyposis coli (APC) derived promoters, leucine-rich repeat containing G protein-coupled receptor 5 (LGR5) promoters, CAG promoter, Beta actin promoter, elongation factor-1 (EF1) promoter, early growth response 1 (EGR-1) promoter, eukaryotic initiation factor 4A (EIF4A1) promoter, simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter, among others. Combinations of two or more of the foregoing promoters may also be used. Further, inducible promoters may be used. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionein promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

The polypeptides of interest may be produced recombinantly not only directly, but also as a fusion polypeptide with a heterologous polypeptide, e.g. a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. In general, the signal sequence may be a component of the vector, or it may be a part of the polypeptide coding sequence that is inserted into the vector. The heterologous signal sequence selected preferably is one that is recognized and processed through one of the standard pathways available within the host cell. The *S. cerevisiae* alpha factor pre-pro signal has proven effective in the secretion of a variety of recombinant proteins from *P. pastoris*. Other yeast signal sequences include the alpha mating factor signal sequence, the invertase signal sequence, and signal sequences derived from other secreted yeast polypeptides. Additionally, these signal peptide sequences may be engineered to provide for enhanced secretion in diploid yeast expression systems. Secretion signals for use in mammalian as well as yeast cells include mammalian signal sequences, which may be heterologous to the protein being secreted, or may be a native sequence for the protein being secreted. Signal sequences include pre-peptide sequences, and in some instances may include propeptide sequences. Many such signal sequences are known in the art, including the signal sequences found on immunoglobulin chains, e.g., K28 preprotoxin sequence, PHA-E, FACE, human MCP-1, human serum albumin signal sequences, human Ig heavy chain, human Ig light chain, and the like. For example, see Hashimoto et. al. Protein Eng 11(2) 75 (1998); and Kobayashi et. al. Therapeutic Apheresis 2(4) 257 (1998).

Transcription may be increased by inserting a transcriptional activator sequence into the vector. These activators are

cis-acting elements of DNA, usually about from 10 to 300 bp, which act on a promoter to increase its transcription. Transcriptional enhancers are relatively orientation and position independent, having been found 5' and 3' to the transcription unit, within an intron, as well as within the coding sequence itself. The enhancer may be spliced into the expression vector at a position 5' or 3' to the coding sequence, but is preferably located at a site 5' from the promoter.

Expression vectors used in eukaryotic host cells may also contain sequences necessary for the termination of transcription and for stabilizing the mRNA. Such sequences are commonly available from 3' to the translation termination codon, in untranslated regions of eukaryotic or viral DNAs or cDNAs. These regions contain nucleotide segments transcribed as polyadenylated fragments in the untranslated portion of the mRNA.

Construction of suitable vectors containing one or more of the above-listed components employs standard ligation techniques or PCR/recombination methods. Isolated plasmids or DNA fragments are cleaved, tailored, and re-ligated in the form desired to generate the plasmids required or via recombination methods. For analysis to confirm correct sequences in plasmids constructed, the ligation mixtures are used to transform host cells, and successful transformants selected by antibiotic resistance (e.g. ampicillin or Zeocin) where appropriate. Plasmids from the transformants are prepared, analyzed by restriction endonuclease digestion and/or sequenced.

As an alternative to restriction and ligation of fragments, recombination methods based on att sites and recombination enzymes may be used to insert DNA sequences into a vector. Such methods are described, for example, by Landy (1989) *Ann. Rev. Biochem.* 58:913-949; and are known to those of skill in the art. Such methods utilize intermolecular DNA recombination that is mediated by a mixture of lambda and *E. coli*-encoded recombination proteins. Recombination occurs between specific attachment (att) sites on the interacting DNA molecules. For a description of att sites see Weisberg and Landy (1983) *Site-Specific Recombination in Phage Lambda*, in *Lambda II*, Weisberg, ed. (Cold Spring Harbor, NY: Cold Spring Harbor Press), pp. 211-250. The DNA segments flanking the recombination sites are switched, such that after recombination, the att sites are hybrid sequences comprised of sequences donated by each parental vector. The recombination can occur between DNAs of any topology.

Att sites may be introduced into a sequence of interest by ligating the sequence of interest into an appropriate vector; generating a PCR product containing att B sites through the use of specific primers; generating a cDNA library cloned into an appropriate vector containing att sites; and the like.

Folding, as used herein, refers to the three-dimensional structure of polypeptides and proteins, where interactions between amino acid residues act to stabilize the structure. Proper folding is typically the arrangement of a polypeptide that results in optimal biological activity, and in the case of antibodies can conveniently be monitored by assays for activity, e.g. antigen binding.

The expression host may be further modified by the introduction of sequences encoding one or more enzymes that enhance folding and disulfide bond formation, i.e. foldases, chaperonins, etc. Such sequences may be constitutively or inducibly expressed in the yeast host cell, using vectors, markers, etc. as known in the art. Preferably the sequences, including transcriptional regulatory elements

sufficient for the desired pattern of expression, are stably integrated in the yeast genome through a targeted methodology.

For example, the eukaryotic PDI is not only an efficient catalyst of protein cysteine oxidation and disulfide bond isomerization, but also exhibits chaperone activity. Co-expression of PDI can facilitate the production of active proteins having multiple disulfide bonds. Also of interest is the expression of BIP (immunoglobulin heavy chain binding protein); cyclophilin; and the like. In one embodiment of the invention, each of the haploid parental strains expresses a distinct folding enzyme, e.g. one strain may express BIP, and the other strain may express PDI or combinations thereof.

The terms “desired protein” or “desired antibody” are used interchangeably and refer generally to a parent antibody specific to a target, i.e., CGRP or a chimeric or humanized antibody or a binding portion thereof derived therefrom as described herein. The term “antibody” is intended to include any polypeptide chain-containing molecular structure with a specific shape that fits to and recognizes an epitope, where one or more non-covalent binding interactions stabilize the complex between the molecular structure and the epitope. The archetypal antibody molecule is the immunoglobulin, and all types of immunoglobulins, IgG, IgM, IgA, IgE, IgD, etc., from all sources, e.g. human, rodent, rabbit, cow, sheep, pig, dog, other mammals, chicken, other avians, etc., are considered to be “antibodies.” A preferred source for producing antibodies useful as starting material according to the invention is rabbits. Numerous antibody coding sequences have been described; and others may be raised by methods well-known in the art. Examples thereof include chimeric antibodies, human antibodies and other non-human mammalian antibodies, humanized antibodies, single chain antibodies (such as scFvs), camelbodies, nobodies, IgNAR (single-chain antibodies derived from sharks), small-modular immunopharmaceuticals (SMIPs), and antibody fragments such as Fabs, Fab', F(ab')₂ and the like. See Streltsov V A, et al., Structure of a shark IgNAR antibody variable domain and modeling of an early-developmental isotype, *Protein Sci.* 2005 November; 14(11):2901-9. Epub 2005 Sep. 30; Greenberg A S, et al., A new antigen receptor gene family that undergoes rearrangement and extensive somatic diversification in sharks, *Nature*. 1995 Mar. 9; 374(6518):168-73; Nuttall S D, et al., Isolation of the new antigen receptor from wobbegong sharks, and use as a scaffold for the display of protein loop libraries, *Mol Immunol.* 2001 August; 38(4):313-26; Hamers-Casterman C, et al., Naturally occurring antibodies devoid of light chains, *Nature*. 1993 Jun. 3; 363(6428):446-8; Gill D S, et al., Biopharmaceutical drug discovery using novel protein scaffolds, *Curr Opin Biotechnol.* 2006 December; 17(6):653-8. Epub 2006 Oct. 19.

For example, antibodies or antigen binding fragments may be produced by genetic engineering. In this technique, as with other methods, antibody-producing cells are sensitized to the desired antigen or immunogen. The messenger RNA isolated from antibody producing cells is used as a template to make cDNA using PCR amplification. A library of vectors, each containing one heavy chain gene and one light chain gene retaining the initial antigen specificity, is produced by insertion of appropriate sections of the amplified immunoglobulin cDNA into the expression vectors. A combinatorial library is constructed by combining the heavy chain gene library with the light chain gene library. This results in a library of clones which co-express a heavy and light chain (resembling the Fab fragment or antigen binding fragment of an antibody molecule). The vectors that carry

these genes are co-transfected into a host cell. When antibody gene synthesis is induced in the transfected host, the heavy and light chain proteins self-assemble to produce active antibodies that can be detected by screening with the antigen or immunogen.

Antibody coding sequences of interest include those encoded by native sequences, as well as nucleic acids that, by virtue of the degeneracy of the genetic code, are not identical in sequence to the disclosed nucleic acids, and variants thereof. Variant polypeptides can include amino acid (aa) substitutions, additions or deletions. The amino acid substitutions can be conservative amino acid substitutions or substitutions to eliminate non-essential amino acids, such as to alter a glycosylation site, or to minimize misfolding by substitution or deletion of one or more cysteine residues that are not necessary for function. Variants can be designed so as to retain or have enhanced biological activity of a particular region of the protein (e.g., a functional domain, catalytic amino acid residues, etc). Variants also include fragments of the polypeptides disclosed herein, particularly biologically active fragments and/or fragments corresponding to functional domains. Techniques for in vitro mutagenesis of cloned genes are known. Also included in the subject invention are polypeptides that have been modified using ordinary molecular biological techniques so as to improve their resistance to proteolytic degradation or to optimize solubility properties or to render them more suitable as a therapeutic agent.

Chimeric antibodies may be made by recombinant means by combining the variable light and heavy chain regions (V_L and V_H), obtained from antibody producing cells of one species with the constant light and heavy chain regions from another. Typically chimeric antibodies utilize rodent or rabbit variable regions and human constant regions, in order to produce an antibody with predominantly human domains. The production of such chimeric antibodies is well known in the art, and may be achieved by standard means (as described, e.g., in U.S. Pat. No. 5,624,659, incorporated herein by reference in its entirety). It is further contemplated that the human constant regions of chimeric antibodies of the invention may be selected from IgG1, IgG2, IgG3, and IgG4 constant regions.

Humanized antibodies are engineered to contain even more human-like immunoglobulin domains, and incorporate only the complementarity-determining regions of the animal-derived antibody. This is accomplished by carefully examining the sequence of the hyper-variable loops of the variable regions of the monoclonal antibody, and fitting them to the structure of the human antibody chains. Although facially complex, the process is straightforward in practice. See, e.g., U.S. Pat. No. 6,187,287, incorporated fully herein by reference.

In addition to entire immunoglobulins (or their recombinant counterparts), immunoglobulin fragments comprising the epitope binding site (e.g., Fab', F(ab')₂, or other fragments) may be synthesized. “Fragment,” or minimal immunoglobulins may be designed utilizing recombinant immunoglobulin techniques. For instance “Fv” immunoglobulins for use in the present invention may be produced by synthesizing a fused variable light chain region and a variable heavy chain region. Combinations of antibodies are also of interest, e.g. diabodies, which comprise two distinct Fv specificities. In another embodiment of the invention, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR are encompassed by immunoglobulin fragments.

Immunoglobulins and fragments thereof may be modified post-translationally, e.g. to add effector moieties such as chemical linkers, detectable moieties, such as fluorescent dyes, enzymes, toxins, substrates, bioluminescent materials, radioactive materials, chemiluminescent moieties and the like, or specific binding moieties, such as streptavidin, avidin, or biotin, and the like may be utilized in the methods and compositions of the present invention. Examples of additional effector molecules are provided infra.

A polynucleotide sequence "corresponds" to a polypeptide sequence if translation of the polynucleotide sequence in accordance with the genetic code yields the polypeptide sequence (i.e., the polynucleotide sequence "encodes" the polypeptide sequence), one polynucleotide sequence "corresponds" to another polynucleotide sequence if the two sequences encode the same polypeptide sequence.

A "heterologous" region or domain of a DNA construct is an identifiable segment of DNA within a larger DNA molecule that is not found in association with the larger molecule in nature. Thus, when the heterologous region encodes a mammalian gene, the gene will usually be flanked by DNA that does not flank the mammalian genomic DNA in the genome of the source organism. Another example of a heterologous region is a construct where the coding sequence itself is not found in nature (e.g., a cDNA where the genomic coding sequence contains introns, or synthetic sequences having codons different than the native gene). Allelic variations or naturally-occurring mutational events do not give rise to a heterologous region of DNA as defined herein.

A "coding sequence" is an in-frame sequence of codons that (in view of the genetic code) correspond to or encode a protein or peptide sequence. Two coding sequences correspond to each other if the sequences or their complementary sequences encode the same amino acid sequences. A coding sequence in association with appropriate regulatory sequences may be transcribed and translated into a polypeptide. A polyadenylation signal and transcription termination sequence will usually be located 3' to the coding sequence. A "promoter sequence" is a DNA regulatory region capable of binding RNA polymerase in a cell and initiating transcription of a downstream (3' direction) coding sequence. Promoter sequences typically contain additional sites for binding of regulatory molecules (e.g., transcription factors) which affect the transcription of the coding sequence. A coding sequence is "under the control" of the promoter sequence or "operatively linked" to the promoter when RNA polymerase binds the promoter sequence in a cell and transcribes the coding sequence into mRNA, which is then in turn translated into the protein encoded by the coding sequence.

Vectors are used to introduce a foreign substance, such as DNA, RNA or protein, into an organism or host cell. Typical vectors include recombinant viruses (for polynucleotides) and liposomes (for polypeptides). A "DNA vector" is a replicon, such as plasmid, phage or cosmid, to which another polynucleotide segment may be attached so as to bring about the replication of the attached segment. An "expression vector" is a DNA vector which contains regulatory sequences which will direct polypeptide synthesis by an appropriate host cell. This usually means a promoter to bind RNA polymerase and initiate transcription of mRNA, as well as ribosome binding sites and initiation signals to direct translation of the mRNA into a polypeptide(s). Incorporation of a polynucleotide sequence into an expression vector at the proper site and in correct reading frame, followed by trans-

formation of an appropriate host cell by the vector, enables the production of a polypeptide encoded by said polynucleotide sequence.

"Amplification" of polynucleotide sequences is the in vitro production of multiple copies of a particular nucleic acid sequence. The amplified sequence is usually in the form of DNA. A variety of techniques for carrying out such amplification are described in a review article by Van Brunt (1990, *Bio/Technol.*, 8(4):291-294). Polymerase chain reaction or PCR is a prototype of nucleic acid amplification, and use of PCR herein should be considered exemplary of other suitable amplification techniques.

The general structure of antibodies in vertebrates now is well understood (Edelman, G. M., *Ann. N.Y. Acad. Sci.*, 190: 5 (1971)). Antibodies consist of two identical light polypeptide chains of molecular weight approximately 23,000 daltons (the "light chain"), and two identical heavy chains of molecular weight 53,000-70,000 (the "heavy chain"). The four chains are joined by disulfide bonds in a "Y" configuration wherein the light chains bracket the heavy chains starting at the mouth of the "Y" configuration. The "branch" portion of the "Y" configuration is designated the F_{ab} region; the stem portion of the "Y" configuration is designated the F_c region. The amino acid sequence orientation runs from the N-terminal end at the top of the "Y" configuration to the C-terminal end at the bottom of each chain. The N-terminal end possesses the variable region having specificity for the antigen that elicited it, and is approximately 100 amino acids in length, there being slight variations between light and heavy chain and from antibody to antibody.

The variable region is linked in each chain to a constant region that extends the remaining length of the chain and that within a particular class of antibody does not vary with the specificity of the antibody (i.e., the antigen eliciting it). There are five known major classes of constant regions that determine the class of the immunoglobulin molecule (IgG, IgM, IgA, IgD, and IgE corresponding to γ , μ , α , δ , and ϵ (gamma, mu, alpha, delta, or epsilon) heavy chain constant regions). The constant region or class determines subsequent effector function of the antibody, including activation of complement (Kabat, E. A., *Structural Concepts in Immunology and Immunochemistry*, 2nd Ed., p. 413-436, Holt, Rinehart, Winston (1976)), and other cellular responses (Andrews, D. W., et al., *Clinical Immunobiology*, pp 1-18, W. B. Sanders (1980); Kohl, S., et al., *Immunology*, 48: 187 (1983)); while the variable region determines the antigen with which it will react. Light chains are classified as either κ (kappa) or λ (lambda). Each heavy chain class can be prepared with either kappa or lambda light chain. The light and heavy chains are covalently bonded to each other, and the "tail" portions of the two heavy chains are bonded to each other by covalent disulfide linkages when the immunoglobulins are generated either by hybridomas or by B cells.

The expression "variable region" or "VR" refers to the domains within each pair of light and heavy chains in an antibody that are involved directly in binding the antibody to the antigen. Each heavy chain has at one end a variable domain (V_H) followed by a number of constant domains. Each light chain has a variable domain (V_L) at one end and a constant domain at its other end; the constant domain of the light chain is aligned with the first constant domain of the heavy chain, and the light chain variable domain is aligned with the variable domain of the heavy chain.

The expressions "complementarity determining region," "hypervariable region," or "CDR" refer to one or more of

31

the hyper-variable or complementarity determining regions (CDRs) found in the variable regions of light or heavy chains of an antibody (See Kabat, E. A. et al., Sequences of Proteins of Immunological Interest, National Institutes of Health, Bethesda, Md., (1987)). These expressions include the hypervariable regions as defined by Kabat et al. ("Sequences of Proteins of Immunological Interest," Kabat E., et al., US Dept. of Health and Human Services, 1983) or the hypervariable loops in 3-dimensional structures of antibodies (Chothia and Lesk, *J Mol. Biol.* 196 901-917 (1987)). The CDRs in each chain are held in close proximity by framework regions and, with the CDRs from the other chain, contribute to the formation of the antigen binding site. Within the CDRs there are select amino acids that have been described as the selectivity determining regions (SDRs) which represent the critical contact residues used by the CDR in the antibody-antigen interaction (Kashmiri, S., *Methods*, 36:25-34 (2005)). In the present invention when specific antibody amino acid or nucleic acid residues are referenced by number this generally refers to its position within a specified amino acid or nucleic acid sequence (i.e., particular sequence identifier) and/or in accordance with Kabat et al numbering.

The expressions "framework region" or "FR" refer to one or more of the framework regions within the variable regions of the light and heavy chains of an antibody (See Kabat, E. A. et al., Sequences of Proteins of Immunological Interest, National Institutes of Health, Bethesda, Md., (1987)). These expressions include those amino acid sequence regions interposed between the CDRs within the variable regions of the light and heavy chains of an antibody.

"Cmax" refers to the maximum (or peak) concentration that an antibody or other compound achieves in tested area (e.g., in the serum or another compartment such as cerebrospinal fluid) after the drug has been administered. For example, serum Cmax may be measured from serum, e.g., prepared by collecting a blood sample, allowing it to clot and separating solid components by centrifugation or other means to yield serum (blood containing neither blood cells nor clotting factors), and then detecting the concentration of the analyte in the serum by ELISA or other means known in the art.

"AUC" refers to the area under the concentration-time curve which is expressed in units of mg/mL*hr (or equivalently mg*hr/ml) unless otherwise specified. "AUC_{0-t}" refers to the area under the concentration-time curve from time=0 to last quantifiable concentration. "AUC_{0-inf}" refers to the area under the concentration-time curve from time=0 extrapolated to infinity.

"I_{max}" refers to the maximal pharmacodynamic response elicited by an anti-CGRP antibody dosage, preferably a dosage of 350 mg or more, more typically at least 750 or 1000 mg, as compared to the response elicited by a lower anti-CGRP antibody doses, e.g., wherein such response may be detected by the inhibition of vasodilation after topical application of capsaicin.

Anti-CGRP Antibodies and Binding Fragments Thereof Having Binding Specificity for CGRP

The invention specifically includes the use of specific anti-CGRP antibodies and antibody fragments referred to herein as Ab1-Ab14 which comprise or consist of the CDR, VL, VH, CL, CH polypeptides sequences identified in FIGS. 1A-12. The polypeptides comprised in an especially preferred anti-CGRP antibody, Ab6 is further described below.

Antibody Ab6

In a preferred exemplary embodiment, the invention includes humanized antibodies having binding specificity to

32

CGRP and possessing a variable light chain sequence comprising the sequence set forth below:

(SEQ ID NO: 222)
 QVLTSPPSSLSASVGDRVTINCQASQSVYHNTYLAWYQQKPGKVPKQLIY
 DASTLASGVPSRFSGSGSGTDFTLTITSSLPEDVATYYCLGSYDCTNGDC
 FVFGGGTKVEIKR.

The invention also includes humanized antibodies having binding specificity to CGRP and possessing a light chain sequence comprising the sequence set forth below:

(SEQ ID NO: 221)
 QVLTSPPSSLSASVGDRVTINCQASQSVYHNTYLAWYQQKPGKVPKQLIY
 DASTLASGVPSRFSGSGSGTDFTLTITSSLPEDVATYYCLGSYDCTNGDC
 FVFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAK
 VQWKVDNALQSGNSQESVTEQDSKDSYSLSSLTLSKADYEKHKVYACE
 VTHQGLSSPVTKSFNRGEC.

The invention further includes humanized antibodies having binding specificity to CGRP and possessing a variable heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 202)
 EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
 IGINATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARGDI
 WGQGTLLVTVSS.

The invention also includes humanized antibodies having binding specificity to CGRP and possessing a heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 201)
 EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
 IGINATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARGDI
 WGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTV
 SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKP
 SNTKVDARVEPKSCDKHTCTCPPELGGPSVFLFPPKPKDTLMISRT
 PEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVL
 TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPVYTLPPSRE
 EMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFL
 YSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK.

Alternatively, the heavy chain of Ab6 may lack the C-terminal lysine of SEQ ID NO: 201, i.e., a heavy chain sequence comprising the sequence set forth below:

(SEQ ID NO: 566)
 EVQLVESGGGLVQPGGSLRLSCAIVSGIDLSGYIMNWVRQAPGKGLEWVG
 IGINATYYASWAKGRFTISRDNKSTTVYLMNSLRAEDTAVYFCARGDI
 WGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTV
 SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKP

-continued

SNTKVDARVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRT
 PEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVSVL
 TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRE
 EMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFL
 YSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPG.

The invention further contemplates antibodies comprising one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221, and/or one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566, or combinations of these polypeptide sequences. In another embodiment of the invention, the antibodies of the invention or fragments thereof comprise, or alternatively consist of, combinations of one or more of the CDRs, the variable heavy and variable light chain sequences, and the heavy and light chain sequences set forth above, including all of them.

The invention also contemplates fragments of the antibody having binding specificity to CGRP. In one embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 222 or SEQ ID NO: 221. In another embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 202 or SEQ ID NO: 201 or SEQ ID NO: 566.

In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

The invention also contemplates antibody fragments which include one or more of the antibody fragments described herein. In one embodiment of the invention, fragments of the antibodies having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following antibody fragments: the variable light chain region of SEQ ID NO: 222; the variable heavy chain region of SEQ ID NO: 202; the complementarity-determining regions (SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228) of the variable light chain region of SEQ ID NO: 222; and the complementarity-

determining regions (SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208) of the variable heavy chain region of SEQ ID NO: 202.

In a particularly preferred embodiment of the invention, the humanized anti-CGRP antibody is Ab6, comprising, or alternatively consisting of, SEQ ID NO: 221 and SEQ ID NO: 201 or SEQ ID NO: 566, and having at least one of the biological activities set forth herein.

In a further particularly preferred embodiment of the invention, antibody fragments comprise, or alternatively consist of, Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the Fab fragment includes the variable light chain sequence of SEQ ID NO: 222 and the variable heavy chain sequence of SEQ ID NO: 202. This embodiment of the invention further contemplates additions, deletions, and variants of SEQ ID NO: 222 and/or SEQ ID NO: 202 in said Fab while retaining binding specificity for CGRP.

In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

In another particularly preferred embodiment of the invention, any of the aforementioned anti-CGRP antibodies or antibody fragments may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

In another embodiment, antibody fragments may be present in one or more of the following non-limiting forms: Fab, Fab', F(ab')₂, Fv and single chain Fv antibody forms. In a preferred embodiment, the anti-CGRP antibodies described herein further comprises the kappa constant light chain sequence comprising the sequence set forth below:

35

(SEQ ID NO: 563)
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGN
SQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKS
FNRGEC.

In another preferred embodiment, the anti-CGRP antibodies described herein further comprises the gamma-1 constant heavy chain polypeptide sequence comprising the sequence set forth below or the same sequence lacking the carboxy terminal lysine residue (SEQ ID NO: 564 and SEQ ID NO: 565, respectively):

(SEQ ID NO: 564)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
KSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
HEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTVLHQDWLNGK
EYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLT
LVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW
QQGNVFCSSVMHEALHNHYTQKSLSLSPGK.

(SEQ ID NO: 565)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
KSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
HEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTVLHQDWLNGK
EYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLT
LVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW
QQGNVFCSSVMHEALHNHYTQKSLSLSPGK.

For clarity, any antibody disclosed herein is intended to include any variant of the disclosed constant region variant sequences, e.g., Ab6 may comprise the constant region of SEQ ID NO: 564 containing the C-terminal lysine or may comprise the constant region of SEQ ID NO: 565 lacking the C-terminal lysine. Thus, every disclosure herein of the heavy chain of SEQ ID NO: 201 also includes a variant lacking the C-terminal lysine residue thereof, i.e., having the heavy chain variable region sequence of Ab6 (SEQ ID NO: 202) and the constant region sequence of SEQ ID NO: 565. For example, the sequence encoding an antibody comprising a C-terminal lysine in the heavy chain may, when expressed in cell lines such as CHO cells, produce an antibody lacking said C-terminal lysine due to proteolysis, or a mixture of heavy chains containing or lacking said C-terminal lysine.

In another embodiment, the invention contemplates use of an isolated anti-CGRP antibody comprising a V_H polypeptide sequence selected from: SEQ ID NO: 2, SEQ ID NO: 42, SEQ ID NO: 82, SEQ ID NO: 122, SEQ ID NO: 162, SEQ ID NO: 202, SEQ ID NO: 242, SEQ ID NO: 282, SEQ ID NO: 322, SEQ ID NO: 362, SEQ ID NO: 402, SEQ ID NO: 442, SEQ ID NO: 482, or SEQ ID NO: 522, or a variant thereof; and further comprising a V_L polypeptide sequence selected from: SEQ ID NO: 22, SEQ ID NO: 62, SEQ ID NO: 102, SEQ ID NO: 142, SEQ ID NO: 182, SEQ ID NO: 222, SEQ ID NO: 262, SEQ ID NO: 302, SEQ ID NO: 342, SEQ ID NO: 382, SEQ ID NO: 422, SEQ ID NO: 462, SEQ ID NO: 502, or SEQ ID NO: 542, or a variant thereof,

36

wherein one or more of the framework residues (FR residues) in said V_H or V_L polypeptide has been substituted with another amino acid residue resulting in an anti-CGRP antibody that specifically binds CGRP. The invention contemplates humanized and chimeric forms of these antibodies. The chimeric antibodies may include an Fc derived from IgG1, IgG2, IgG3, or IgG4 constant regions.

In one embodiment of the invention, the antibodies or V_H or V_L polypeptides originate or are selected from one or more rabbit B cell populations prior to initiation of the humanization process referenced herein.

In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof do not have binding specificity for CGRP-R. In a further embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R. In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R and/or additional proteins and/or multimers thereof, and/or antagonizes the biological effects thereof.

As stated herein, antibodies and fragments thereof may be modified post-translationally to add effector moieties such as chemical linkers, detectable moieties such as for example fluorescent dyes, enzymes, substrates, bioluminescent materials, radioactive materials, and chemiluminescent moieties, or functional moieties such as for example streptavidin, avidin, biotin, a cytotoxin, a cytotoxic agent, and radioactive materials.

Antibodies or fragments thereof may also be chemically modified to provide additional advantages such as increased solubility, stability and circulating time (in vivo half-life) of the polypeptide, or decreased immunogenicity (See U.S. Pat. No. 4,179,337). The chemical moieties for derivatization may be selected from water soluble polymers such as polyethylene glycol, ethylene glycol/propylene glycol copolymers, carboxymethylcellulose, dextran, polyvinyl alcohol and the like. The antibodies and fragments thereof may be modified at random positions within the molecule, or at predetermined positions within the molecule and may include one, two, three or more attached chemical moieties.

The polymer may be of any molecular weight, and may be branched or unbranched. For polyethylene glycol, the preferred molecular weight is between about 1 kDa and about 100 kDa (the term "about" indicating that in preparations of polyethylene glycol, some molecules will weigh more, some less, than the stated molecular weight) for ease in handling and manufacturing. Other sizes may be used, depending on the desired therapeutic profile (e.g., the duration of sustained release desired, the effects, if any on biological activity, the ease in handling, the degree or lack of antigenicity and other known effects of the polyethylene glycol to a therapeutic protein or analog). For example, the polyethylene glycol may have an average molecular weight of about 200, 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500, 10,000, 10,500, 11,000, 11,500, 12,000, 12,500, 13,000, 13,500, 14,000, 14,500, 15,000, 15,500, 16,000, 16,500, 17,000, 17,500, 18,000, 18,500, 19,000, 19,500, 20,000, 25,000, 30,000, 35,000, 40,000, 50,000, 55,000, 60,000, 65,000, 70,000, 75,000, 80,000, 85,000, 90,000, 95,000, or 100,000 kDa. Branched polyethylene glycols are described, for example, in U.S. Pat. No. 5,643,575; Morpurgo et al., *Appl. Biochem. Biotechnol.* 56:59-72 (1996); Vorobjev et al., *Nucleosides Nucleotides* 18:2745-2750 (1999); and Caliceti et al., *Bioconj. Chem.* 10:638-646 (1999), the disclosures of each of which are incorporated herein by reference.

There are a number of attachment methods available to those skilled in the art. See e.g., EP 0 401 384, herein incorporated by reference (coupling PEG to G-CSF). See also Malik et al., *Exp. Hematol.* 20:1028-1035 (1992) (reporting pegylation of GM-CSF using tresyl chloride). For example, polyethylene glycol may be covalently bound through amino acid residues via a reactive group, such as, a free amino or carboxyl group. Reactive groups are those to which an activated polyethylene glycol molecule may be bound. The amino acid residues having a free amino group may include lysine residues and the N-terminal amino acid residues; those having a free carboxyl group may include aspartic acid residues, glutamic acid residues and the C-terminal amino acid residue. Sulfhydryl groups may also be used as a reactive group for attaching the polyethylene glycol molecules. Preferred for therapeutic purposes is attachment at an amino group, such as attachment at the N-terminus or lysine group.

As suggested above, polyethylene glycol may be attached to proteins via linkage to any of a number of amino acid residues. For example, polyethylene glycol can be linked to polypeptides via covalent bonds to lysine, histidine, aspartic acid, glutamic acid, or cysteine residues. One or more reaction chemistries may be employed to attach polyethylene glycol to specific amino acid residues (e.g., lysine, histidine, aspartic acid, glutamic acid, or cysteine) or to more than one type of amino acid residue (e.g., lysine, histidine, aspartic acid, glutamic acid, cysteine and combinations thereof).

Alternatively, antibodies or fragments thereof may have increased in vivo half-lives via fusion with albumin (including but not limited to recombinant human serum albumin or fragments or variants thereof (See, e.g., U.S. Pat. No. 5,876,969, issued Mar. 2, 1999, EP Patent 0 413 622, and U.S. Pat. No. 5,766,883, issued Jun. 16, 1998, herein incorporated by reference in their entirety)) or other circulating blood proteins such as transferrin or ferritin. In a preferred embodiment, polypeptides and/or antibodies of the present invention (including fragments or variants thereof) are fused with the mature form of human serum albumin (i.e., amino acids 1-585 of human serum albumin as shown in FIGS. 1 and 2 of EP Patent 0 322 094) which is herein incorporated by reference in its entirety. Polynucleotides encoding fusion proteins of the invention are also encompassed by the invention.

Regarding detectable moieties, further exemplary enzymes include, but are not limited to, horseradish peroxidase, acetylcholinesterase, alkaline phosphatase, beta-galactosidase and luciferase. Further exemplary fluorescent materials include, but are not limited to, rhodamine, fluorescein, fluorescein isothiocyanate, umbelliferone, dichlorotriazinylamine, phycoerythrin and dansyl chloride. Further exemplary chemiluminescent moieties include, but are not limited to, luminol. Further exemplary bioluminescent materials include, but are not limited to, luciferin and aequorin. Further exemplary radioactive materials include, but are not limited to, Iodine 125 (^{125}I), Carbon 14 (^{14}C), Sulfur 35 (^{35}S), Tritium (^3H) and Phosphorus 32 (^{32}P).

Regarding functional moieties, exemplary cytotoxic agents include, but are not limited to, methotrexate, aminopterin, 6-mercaptopurine, 6-thioguanine, cytarabine, 5-fluorouracil decarbazine; alkylating agents such as mechlorethamine, thiopa chlorambucil, melphalan, carmustine (BSNU), mitomycin C, lomustine (CCNU), 1-methylnitrosourea, cyclophosphamide, mechlorethamine, busulfan, dibromomannitol, streptozotocin, mitomycin C, cis-dichlorodiamine platinum (II) (DDP) cisplatin and

carboplatin (paraplatin); anthracyclines include daunorubicin (formerly daunomycin), doxorubicin (adriamycin), detorubicin, carminomycin, idarubicin, epirubicin, mitoxantrone and bisantrene; antibiotics include dactinomycin (actinomycin D), bleomycin, calicheamicin, mithramycin, and anthramycin (AMC); and antimetabolic agents such as the vinca alkaloids, vincristine and vinblastine. Other cytotoxic agents include paclitaxel (taxol), ricin, pseudomonas exotoxin, gemcitabine, cytochalasin B, gramicidin D, ethidium bromide, emetine, etoposide, tenoposide, colchicin, dihydroxy anthracin dione, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, procarbazine, hydroxyurea, asparaginase, corticosteroids, mytotane (O,P'-(DDD)), interferons, and mixtures of these cytotoxic agents.

Further cytotoxic agents include, but are not limited to, chemotherapeutic agents such as carboplatin, cisplatin, paclitaxel, gemcitabine, calicheamicin, doxorubicin, 5-fluorouracil, mitomycin C, actinomycin D, cyclophosphamide, vincristine and bleomycin. Toxic enzymes from plants and bacteria such as ricin, diphtheria toxin and Pseudomonas toxin may be conjugated to the humanized or chimeric antibodies, or binding fragments thereof, to generate cell-type-specific-killing reagents (Youle, et al., *Proc. Nat'l Acad. Sci. USA* 77:5483 (1980); Gilliland, et al., *Proc. Nat'l Acad. Sci. USA* 77:4539 (1980); Krolick, et al., *Proc. Nat'l Acad. Sci. USA* 77:5419 (1980)).

Other cytotoxic agents include cytotoxic ribonucleases as described by Goldenberg in U.S. Pat. No. 6,653,104. Embodiments of the invention also relate to radioimmunoconjugates where a radionuclide that emits alpha or beta particles is stably coupled to the antibody, or binding fragments thereof, with or without the use of a complex-forming agent. Such radionuclides include beta-emitters such as Phosphorus-32 (^{32}P), Scandium-47 (^{47}Sc), Copper-67 (^{67}Cu), Gallium-67 (^{67}Ga), Yttrium-88 (^{88}Y), Yttrium-90 (^{90}Y), Iodine-125 (^{125}I), Iodine-131 (^{131}I), Samarium-153 (^{153}Sm), Lutetium-177 (^{177}Lu), Rhenium-186 (^{186}Re) or Rhenium-188 (^{188}Re), and alpha-emitters such as Astatine-211 (^{211}At), Lead-212 (^{212}Pb), Bismuth-212 (^{212}Bi) or -213 (^{213}Bi) or Actinium-225 (^{225}Ac).

Methods are known in the art for conjugating an antibody or binding fragment thereof to a detectable moiety and the like, such as for example those methods described by Hunter et al, *Nature* 144:945 (1962); David et al, *Biochemistry* 13:1014 (1974); Pain et al, *J. Immunol. Meth.* 40:219 (1981); and Nygren, J., *Histochem. and Cytochem.* 30:407 (1982).

Embodiments described herein further include variants and equivalents that are substantially homologous to the antibodies, antibody fragments, diabodies, SMIPs, camelbodies, nobodies, IgNAR, polypeptides, variable regions and CDRs set forth herein. These may contain, e.g., conservative substitution mutations, (i.e., the substitution of one or more amino acids by similar amino acids). For example, conservative substitution refers to the substitution of an amino acid with another within the same general class, e.g., one acidic amino acid with another acidic amino acid, one basic amino acid with another basic amino acid, or one neutral amino acid by another neutral amino acid. What is intended by a conservative amino acid substitution is well known in the art.

In another embodiment, the invention contemplates polypeptide sequences having at least 90% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. More preferably, the invention contemplates poly-

peptide sequences having at least 95% or greater sequence homology, even more preferably at least 98% or greater sequence homology, and still more preferably at least 99% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. Methods for determining homology between nucleic acid and amino acid sequences are well known to those of ordinary skill in the art.

In another embodiment, the invention further contemplates the above-recited polypeptide homologs of the antibody fragments, variable regions and CDRs set forth herein further having anti-CGRP activity. Non-limiting examples of anti-CGRP activity are set forth herein.

The present invention also contemplates anti-CGRP antibodies comprising any of the polypeptide or polynucleotide sequences described herein substituted for any of the other polynucleotide sequences described herein. For example, without limitation thereto, the present invention contemplates antibodies comprising the combination of any of the variable light chain and variable heavy chain sequences described herein, and further contemplates antibodies resulting from substitution of any of the CDR sequences described herein for any of the other CDR sequences described herein.

Additional Exemplary Embodiments of the Invention

In another embodiment, the invention contemplates treatment methods using one or more anti-human CGRP antibodies or antibody fragments thereof which specifically bind to the same overlapping linear or conformational epitope(s) and/or competes for binding to the same overlapping linear or conformational epitope(s) on an intact human CGRP polypeptide or fragment thereof as an anti-human CGRP antibody selected from Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab10, Ab11, Ab12, Ab13, or Ab14. In a preferred embodiment, the anti-human CGRP antibody or fragment thereof specifically binds to the same overlapping linear or conformational epitope(s) and/or competes for binding to the same overlapping linear or conformational epitope(s) on an intact human CGRP polypeptide or a fragment thereof as Ab3, Ab6, Ab13, or Ab14.

A preferred embodiment of the invention is directed to treatment methods using chimeric or humanized antibodies and fragments thereof (including Fab fragments) having binding specificity for CGRP and inhibiting biological activities mediated by the binding of CGRP to the CGRP receptor. In a particularly preferred embodiment of the invention, the chimeric or humanized anti-CGRP antibodies are selected from Ab3, Ab6, Ab13, or Ab14.

In another embodiment of the invention, the anti-human CGRP antibody used in the described treatment methods is an antibody which specifically binds to the same overlapping linear or conformational epitopes on an intact CGRP polypeptide or fragment thereof that is (are) specifically bound by Ab3, Ab6, Ab13, or Ab 14 as ascertained by epitopic mapping using overlapping linear peptide fragments which span the full length of the native human CGRP polypeptide.

In another embodiment, the invention is also directed to treatment methods using an isolated anti-CGRP antibody or antibody fragment comprising one or more of the CDRs contained in the V_H polypeptide sequences selected from: 3, 13, 23, 33, 43, 53, 63, 73, 83, 93, 103, 113, 123, or 133, or a variant thereof, and/or one or more of the CDRs contained

in the V_L polypeptide sequences selected from: 1, 11, 21, 31, 41, 51, 61, 71, 81, 91, 101, 111, 121, or 131, or a variant thereof.

In one embodiment of the invention, the anti-human CGRP antibody discussed in the two prior paragraphs comprises at least 2 complementarity determining regions (CDRs) in each the variable light and the variable heavy regions which are identical to those contained in an anti-human CGRP antibody selected from Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab10, Ab 11, Ab12, Ab13, or Ab14.

In a preferred embodiment, the anti-human CGRP antibody used in the described treatment methods comprises at least 2 complementarity determining regions (CDRs) in each the variable light and the variable heavy regions which are identical to those contained in Ab3 or Ab6. In another embodiment, all of the CDRs of the anti-human CGRP antibody discussed above are identical to the CDRs contained in an anti-human CGRP antibody selected from Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab10, Ab11, Ab12, Ab13, or Ab14. In a preferred embodiment of the invention, all of the CDRs of the anti-human CGRP antibody discussed above are identical to the CDRs contained in an anti-human CGRP antibody selected from Ab3 or Ab6.

The invention further contemplates treatment methods wherein the one or more anti-human CGRP antibodies discussed above are aglycosylated or if glycosylated are only mannosylated; that contain an Fc region that has been modified to alter effector function, half-life, proteolysis, and/or glycosylation; are human, humanized, single chain or chimeric; and are a humanized antibody derived from a rabbit (parent) anti-human CGRP antibody. An exemplary mutation which impairs glycosylation comprises the mutation of the Asn residue at position 297 of an IgG heavy chain constant region such as IgG1 to another amino acid, such as Ala as described in U.S. Pat. No. 5,624,821, which is incorporated by reference in its entirety.

The invention further contemplates one or more anti-human CGRP antibodies wherein the framework regions (FRs) in the variable light region and the variable heavy regions of said antibody respectively are human FRs which are unmodified or which have been modified by the substitution of one or more human FR residues in the variable light or heavy chain region with the corresponding FR residues of the parent rabbit antibody, and wherein said human FRs have been derived from human variable heavy and light chain antibody sequences which have been selected from a library of human germline antibody sequences based on their high level of homology to the corresponding rabbit variable heavy or light chain regions relative to other human germline antibody sequences contained in the library.

The invention also contemplates a method of treating or preventing medication overuse headache, e.g., associated with the overuse of anti-migraine drugs and/or associated with triptan and/or ergot and/or analgesic overuse, comprising administering to a patient exhibiting medication overuse headache or at risk of developing medication overuse headache a therapeutically effective amount of at least one anti-human CGRP antibody or fragment described herein. The invention also contemplates that the treatment method may involve the administration of two or more anti-CGRP antibodies or fragments thereof and disclosed herein. If more than one antibody is administered to the patient, the multiple antibodies may be administered simultaneously or concurrently, or may be staggered in their administration. The anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding

41

specificity to CGRP, may also be described by their strength of binding or their affinity for CGRP. In one embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with a dissociation constant (KD) of less than or equal to 5×10^{-7} M, 10^{-7} M, 5×10^{-8} M, 10^{-8} M, 5×10^{-9} M, 10^{-9} M, 5×10^{-10} M, 10^{-10} M, 5×10^{-11} M, 10^{-11} M, 5×10^{-12} M, 10^{-12} M, 5×10^{-13} M, or 10^{-13} M. Preferably, the anti-CGRP antibodies and fragments thereof bind CGRP with a dissociation constant of less than or equal to 10^{-11} M, 5×10^{-12} M, or 10^{-12} M. In another embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to a linear or conformational CGRP epitope.

In another embodiment of the invention, the anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with an off-rate of less than or equal to 10^{-4} S⁻¹, 5×10^{-5} S⁻¹, 10^{-5} S⁻¹, 5×10^{-6} S⁻¹, 10^{-6} S⁻¹, 5×10^{-7} S⁻¹, or 10^{-7} S⁻¹.

In a further embodiment of the invention, the anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, exhibit anti-CGRP activity by preventing, ameliorating or reducing the symptoms of, or alternatively treating, diseases and disorders associated with CGRP. Non-limiting examples of diseases and disorders associated with CGRP are set forth herein and include headache and migraine disorders.

Polynucleotides Encoding Anti-CGRP Antibody Polypeptides

As aforementioned the invention specifically includes the use of specific anti-CGRP antibodies and antibody fragments referred to herein as Ab1-Ab14 which comprise or consist of the CDR, VL, VH, CL, and CH polypeptides having the sequences identified in FIGS. 1A-12. The nucleic acid sequences encoding the foregoing VL, VH, CL, and CH polypeptides comprised in Ab1-Ab14 are also comprised in FIGS. 1A-12. The nucleic acid sequences which encode the CDR, VL, VH, CL, and CH polypeptides of an especially preferred anti-CGRP antibody, Ab6, are further described below.

Antibody Ab6

The invention is further directed to polynucleotides encoding antibody polypeptides having binding specificity to CGRP. In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable light chain polypeptide sequence of SEQ ID NO: 222:

(SEQ ID NO: 232)

CAAGTGCTGaccagctctccatcctccctgtctgcatctgtaggagacag
agtcaccatcAATtgcCAGGCCAGTCAGAGTGTATCATACACCTACC
TGGCCTggtatcagcagaaccagggaagtctcctaagCAActgatctat
GATGCATCCACTCTGGCATCTgggggtcccatctcggttcagtggcagtgg
atctgggacagatttcactctcaccatcagcagcctgcagcctgaagatg
ttgcaacttattactgtCTGGGCAGTTATGATTGTACTAATGGTGATTGT
TTTGTtttcggcggagggaaccaaggtggaatcaaacgt.

42

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the light chain polypeptide sequence of SEQ ID NO: 221:

(SEQ ID NO: 231)

CAAGTGCTGaccagctctccatcctccctgtctgcatctgtaggagacag
agtcaccatcAATtgcCAGGCCAGTCAGAGTGTATCATACACCTACC
TGGCCTggtatcagcagaaccagggaagtctcctaagCAActgatctat
GATGCATCCACTCTGGCATCTgggggtcccatctcggttcagtggcagtgg
atctgggacagatttcactctcaccatcagcagcctgcagcctgaagatg
ttgcaacttattactgtCTGGGCAGTTATGATTGTACTAATGGTGATTGT
TTTGTtttcggcggagggaaccaaggtggaatcaaacgtACGGTGGCTGC
ACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTTGAAATCTGGAA
CTGCCCTCTGTGTGTGCTGCTGAATAACTTCTATCCAGAGAGGCCAAA
GTACAGTGAAGGTGGATAACGCCCTCCAATCGGGTAACCTCCAGGAGAG
TGTCACAGAGCAGGACAGCAAGGACAGCACCTACAGCCTCAGCAGCACCC
TGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGCTACGCCTGCGAA
GTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGAGCTTCAACAGGGG
AGAGTGTTAG.

In another embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable heavy chain polypeptide sequence of SEQ ID NO: 202:

(SEQ ID NO: 212)

gagggtgcagctTgtggagtctggggagggttggtccagcctgggggggtc
cctgagactctcctgtgcaGTCTctggaATCGACCTCagtGGCTACTACA
TGAACTgggtccgtcaggctccagggaaggggctggagtgggtcGGAGTC
ATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCcgatt
caccatctccagagacaattccaagACCACGGTGTatcttcaaatgaaca
gcttgagagctgaggacactgctgtgtatTTCTgtGCTAGAGGGGACATC
tggggccaagggaacctcgctaccgtcTCGAGC.

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 201:

(SEQ ID NO: 211)

gagggtgcagctTgtggagtctggggagggttggtccagcctgggggggtc
cctgagactctcctgtgcaGTCTctggaATCGACCTCagtGGCTACTACA
TGAACTgggtccgtcaggctccagggaaggggctggagtgggtcGGAGTC
ATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCcgatt
caccatctccagagacaattccaagACCACGGTGTatcttcaaatgaaca
gcttgagagctgaggacactgctgtgtatTTCTgtGCTAGAGGGGACATC
tggggccaagggaacctcgctaccgtcTCGAGCGCCTCCACCAAGGGCCC
ATCGGTCTTCCCCCTGGCACCTCCTCCaAGAGCACCTCTGGGGCACAG

43

-continued

CGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTG
TCGTGGAAGTCAAGCGCCCTGACCAGCGCGTGCACACCTTCCCGGTGT
CCTACAGTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAATCTTGTGACAAAAC
TCACACATGCCACCGTGCCAGCACCTGAACTCTGGGGGACCGTCAG
TCTTCTCTTCCCCCAAAACCAAGGACACCTCATGaTCTCCCGGACC
CCTGAGGTACATGCGTGGTGGTGGACGTGAGCCACGAAGACCTGAGGT
CAAGTTCAACTGGTACGTGGACGCGGTGGAGGTGCATAATGCCAAGACAA
AGCCGCGGAGGAGCAGTACGCCAGCACGTACCGTGTGGTCAGCGTCCTC
ACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGAAGGT
CTCCAACAAAGCCCTCCCGACCCCCATCGAGAAAACCATCTCCAAGCCA
AAGGGCAGCCCCGAGAACCACAGGTGTACACCTGCCCCATCCCGGGAG
GAGATGACCAAGAACCAGGTGACCTGACCTGGTCAAAGGCTTCTA
TCCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGCAGCCGAGAACAA
ACTACAAGACCACGCTCCCGTGTGGACTCCGACGGCTCTTCTTCCTC
TACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTT
CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGAGAAGA
GCCTCTCCCTGTCTCCGGTAAATGA.

In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 566:

(SEQ ID NO: 567)
gaggtgcagctTgtggagctctgggggaggttggtccagcctggggggtc
cctgagactctcctgtgcaGTctctggaATCGACCTCagtGGCTACTACA
TGAATcgggtccgtcaggtccaggggaaggggtggagtggtcGGAGTC
ATTGTTATTAATGGTGCCACATACTACGCGAGCTGGCGAAAGGCgatt
caccatctccagagacaattccaagACCAGGTGtatcttcaatgaaca
gcctgagagctgaggacactgctgtgatTTctgtGCTAGAGGGGACATC
tggggccaagggaccctcgtaaccgtcTCGAGCGCTCCACCAAGGGCCC
ATCGTCTTCCCCCTGGCAcCTCTCCaAGAGCACCTCTGGGGGCACAG
CGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTG
TCGTGGAAGTCAAGCGCCCTGACCAGCGCGTGCACACCTTCCCGGTGT
CCTACAGTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAATCTTGTGACAAAAC
TCACACATGCCACCGTGCCAGCACCTGAACTCTGGGGGACCGTCAG
TCTTCTCTTCCCCCAAAACCAAGGACACCTCATGaTCTCCCGGACC
CCTGAGGTACATGCGTGGTGGTGGACGTGAGCCACGAAGACCTGAGGT
CAAGTTCAACTGGTACGTGGACGCGGTGGAGGTGCATAATGCCAAGACAA

44

-continued

AGCCGCGGGAGGAGCAGTACGCCAGCAGTACCGTGTGGTCAGCGTCCTC
ACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGAAGGT
5 CTCCAACAAAGCCCTCCCGACCCCCATCGAGAAAACCATCTCCAAGCCA
AAGGGCAGCCCCGAGAACCACAGGTGTACACCTGCCCCATCCCGGGAG
GAGATGACCAAGAACCAGGTGACCTGACCTGGTCAAAGGCTTCTA
10 TCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGAGAACAA
ACTACAAGACCACGCTCCCGTGTGGACTCCGACGGCTCCTTCTTCCTC
TACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTT
15 CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGAGAAGA
GCCTCTCCCTGTCTCCGGTTGA.

In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the heavy chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

The invention also contemplates polynucleotide sequences including one or more of the polynucleotide sequences encoding antibody fragments described herein. In one embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following polynucleotides encoding antibody fragments: the polynucleotide SEQ ID NO: 232 encoding the light chain variable sequence of SEQ ID NO: 222; the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221; the polynucleotide SEQ ID NO: 212 encoding the heavy chain variable sequence of SEQ ID NO: 202; the polynucleotide SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201; the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566; polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221; and polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218) of the heavy chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

In a preferred embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, polynucleotides encoding Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the polynucleotides encoding the full length Ab6 antibody comprise, or alternatively consist of, the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221 and the polynucleotide

SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201 or the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566.

Another embodiment of the invention contemplates these polynucleotides incorporated into an expression vector for expression in mammalian cells such as CHO, NSO, HEK-293, or in fungal, insect, or microbial systems such as yeast cells such as the yeast *Pichia*. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*. In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6 following expression of the full-length polynucleotides in a suitable host. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression of Ab6 polynucleotides in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

In one embodiment, the invention is directed to an isolated polynucleotide comprising a polynucleotide encoding an anti-CGRP V_H antibody amino acid sequence selected from SEQ ID NO: 2, SEQ ID NO: 42, SEQ ID NO: 82, SEQ ID NO: 122, SEQ ID NO: 162, SEQ ID NO: 202, SEQ ID NO: 242, SEQ ID NO: 282, SEQ ID NO: 322, SEQ ID NO: 362, SEQ ID NO: 402, SEQ ID NO: 442, SEQ ID NO: 482, or SEQ ID NO: 522 or encoding a variant thereof wherein at least one framework residue (FR residue) has been substituted with an amino acid present at the corresponding position in a rabbit anti-CGRP antibody V_H polypeptide or a conservative amino acid substitution.

In another embodiment, the invention is directed to an isolated polynucleotide comprising the polynucleotide sequence encoding an anti-CGRP V_L antibody amino acid sequence of SEQ ID NO: 22, SEQ ID NO: 62, SEQ ID NO: 102, SEQ ID NO: 142, SEQ ID NO: 182, SEQ ID NO: 222, SEQ ID NO: 262, SEQ ID NO: 302, SEQ ID NO: 342, SEQ ID NO: 382, SEQ ID NO: 422, SEQ ID NO: 462, SEQ ID NO: 502, or SEQ ID NO: 542, or encoding a variant thereof wherein at least one framework residue (FR residue) has been substituted with an amino acid present at the corresponding position in a rabbit anti-CGRP antibody V_L polypeptide or a conservative amino acid substitution.

In yet another embodiment, the invention is directed to one or more heterologous polynucleotides comprising a sequence encoding the polypeptides contained in SEQ ID NO: 22 and SEQ ID NO: 2; SEQ ID NO: 62 and SEQ ID NO: 42; SEQ ID NO: 102 and SEQ ID NO: 82; SEQ ID NO: 142 and SEQ ID NO: 122; SEQ ID NO: 182 and SEQ ID NO: 162; SEQ ID NO: 222 and SEQ ID NO: 202; SEQ ID NO: 262 and SEQ ID NO: 242; SEQ ID NO: 302 and SEQ ID NO: 282; SEQ ID NO: 342 and SEQ ID NO: 322; SEQ ID NO: 382 and SEQ ID NO: 362; SEQ ID NO: 422 and SEQ ID NO: 402; SEQ ID NO: 462 and SEQ ID NO: 442; SEQ ID NO: 502 and SEQ ID NO: 482; or SEQ ID NO: 542 and SEQ ID NO: 522.

In another embodiment, the invention is directed to an isolated polynucleotide that expresses a polypeptide containing at least one CDR polypeptide derived from an anti-CGRP antibody wherein said expressed polypeptide alone specifically binds CGRP or specifically binds CGRP when expressed in association with another polynucleotide sequence that expresses a polypeptide containing at least one CDR polypeptide derived from an anti-CGRP antibody wherein said at least one CDR is selected from those contained in the V_L or V_H polypeptides of SEQ ID NO: 22,

SEQ ID NO: 2, SEQ ID NO: 62, SEQ ID NO: 42, SEQ ID NO: 102, SEQ ID NO: 82, SEQ ID NO: 142, SEQ ID NO: 122, SEQ ID NO: 182, SEQ ID NO: 162, SEQ ID NO: 222, SEQ ID NO: 202, SEQ ID NO: 262, SEQ ID NO: 242, SEQ ID NO: 302, SEQ ID NO: 282, SEQ ID NO: 342, SEQ ID NO: 322, SEQ ID NO: 382, SEQ ID NO: 362, SEQ ID NO: 422, SEQ ID NO: 402, SEQ ID NO: 462, SEQ ID NO: 442, SEQ ID NO: 502, SEQ ID NO: 482, SEQ ID NO: 542, or SEQ ID NO: 522.

Host cells and vectors comprising said polynucleotides are also contemplated.

The invention further contemplates vectors comprising the polynucleotide sequences encoding the variable heavy and light chain polypeptide sequences, as well as the individual complementarity-determining regions (CDRs, or hypervariable regions), as set forth herein, as well as host cells comprising said vector sequences. In one embodiment of the invention, the host cell is a yeast cell. In another embodiment of the invention, the yeast host cell belongs to the genus *Pichia*.

Methods of Producing Antibodies and Fragments Thereof

In another embodiment, the present invention contemplates methods for producing anti-CGRP antibodies and fragments thereof. Methods for producing antibodies and fragments thereof secreted from polyploid, preferably diploid or tetraploid strains of mating competent yeast are taught, for example, in U.S. patent application publication no. US 2009/0022659 to Olson et al., and in U.S. Pat. No. 7,935,340 to Garcia-Martinez et al., the disclosures of each of which are herein incorporated by reference in their entireties. Methods for producing antibodies and fragments thereof in mammalian cells, e.g., CHO cells are further well known in the art.

Other methods of producing antibodies are also well known to those of ordinary skill in the art. For example, methods of producing chimeric antibodies are now well known in the art (See, for example, U.S. Pat. No. 4,816,567 to Cabilly et al.; Morrison et al., *P.N.A.S. USA*, 81:8651-55 (1984); Neuberger, M. S. et al., *Nature*, 314:268-270 (1985); Boulianne, G. L. et al., *Nature*, 312:643-46 (1984), the disclosures of each of which are herein incorporated by reference in their entireties).

Likewise, other methods of producing humanized antibodies are now well known in the art (See, for example, U.S. Pat. Nos. 5,530,101, 5,585,089, 5,693,762, and 6,180,370 to Queen et al; U.S. Pat. Nos. 5,225,539 and 6,548,640 to Winter; U.S. Pat. Nos. 6,054,297, 6,407,213 and 6,639,055 to Carter et al; U.S. Pat. No. 6,632,927 to Adair; Jones, P. T. et al, *Nature*, 321:522-525 (1986); Reichmann, L., et al, *Nature*, 332:323-327 (1988); Verhoeven, M., et al, *Science*, 239:1534-36 (1988), the disclosures of each of which are herein incorporated by reference in their entireties).

The term "opioid analgesic" herein refers to all drugs, natural or synthetic, with morphine-like actions. The synthetic and semi-synthetic opioid analgesics are derivatives of five chemical classes of compound: phenanthrenes; phenylheptylamines; phenylpiperidines; morphinans; and benzomorphans, all of which are within the scope of the term. Exemplary opioid analgesics include codeine, dihydrocodeine, diacetylmorphine, hydrocodone, hydromorphone, levorphanol, oxycodone, alfentanil, buprenorphine, butorphanol, fentanyl, sufentanil, meperidine, methadone, nalbuphine, propoxyphene and pentazocine or pharmaceutically acceptable salts thereof.

The term "NSAID" refers to a non-steroidal anti-inflammatory compound. NSAIDs are categorized by virtue of their ability to inhibit cyclooxygenase. Cyclooxygenase 1 and cyclooxygenase 2 are two major isoforms of cyclooxygenase and most standard NSAIDs are mixed inhibitors of the two isoforms. Most standard NSAIDs fall within one of the following five structural categories: (1) propionic acid derivatives, such as ibuprofen, naproxen, naprosyn, diclofenac, and ketoprofen; (2) acetic acid derivatives, such as tolmetin and slindac; (3) fenamic acid derivatives, such as mefenamic acid and meclofenamic acid; (4) biphenylcarboxylic acid derivatives, such as diflunisal and flufenisal; and (5) oxicams, such as piroxim, sudoxicam, and isoxicam. Another class of NSAID has been described which selectively inhibit cyclooxygenase 2. Cox-2 inhibitors have been described, e.g., in U.S. Pat. Nos. 5,616,601; 5,604,260; 5,593,994; 5,550,142; 5,536,752; 5,521,213; 5,475,995; 5,639,780; 5,604,253; 5,552,422; 5,510,368; 5,436,265; 5,409,944; and 5,130,311, all of which are hereby incorporated by reference. Certain exemplary COX-2 inhibitors include celecoxib (SC-58635), DUP-697, flosulide (CGP-28238), meloxicam, 6-methoxy-2 naphthylacetic acid (6-MNA), rofecoxib, MK-966, nabumetone (prodrug for 6-MNA), nimesulide, NS-398, SC-5766, SC-58215, T-614; or combinations thereof.

In some embodiments, aspirin and/or acetaminophen may be taken in conjunction with the subject CGRP antibody or fragment. Aspirin is another type of non-steroidal anti-inflammatory compound.

The subject to which the pharmaceutical formulation is administered can be, e.g., any human or non-human animal that is in need of such treatment, prevention and/or amelioration, or who would otherwise benefit from the inhibition or attenuation of medication overuse headache. For example, the subject can be an individual that is diagnosed with, or who is deemed to be at risk of being afflicted by medication overuse headache. The present invention further includes the use of any of the pharmaceutical formulations disclosed herein in the manufacture of a medicament for the treatment, prevention and/or amelioration of medication overuse headache.

Administration

In one embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of between about 0.1 and 100.0 mg/kg of body weight of recipient subject. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of about 0.4 mg/kg of body weight of recipient subject and/or at a dosage of 100 or 300 mg. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a recipient subject with a frequency of once every twenty-six weeks or six months or less, such as once every sixteen weeks or four months or less, once every eight weeks or two months or less, once every four weeks or monthly or less, once every two weeks or bimonthly or less, once every week or less, or once daily or less. In general the administration of sequential doses may vary by plus or minus a few days from the aforementioned schedule, e.g., administration every 3

months or every 12 weeks includes administration of a dose varying from the schedule day by plus or minus 1, 2, 3, 4, 5, 5, or 7 days.

Fab fragments may be administered every two weeks or less, every week or less, once daily or less, multiple times per day, and/or every few hours. In one embodiment of the invention, a patient receives Fab fragments of 0.1 mg/kg to 40 mg/kg per day given in divided doses of 1 to 6 times a day, or in a sustained release form, effective to obtain desired results.

It is to be understood that the concentration of the antibody or Fab administered to a given patient may be greater or lower than the exemplary administration concentrations set forth above.

A person of skill in the art would be able to determine an effective dosage and frequency of administration through routine experimentation, for example guided by the disclosure herein and the teachings in Goodman, L. S., Gilman, A., Brunton, L. L., Lazo, J. S., & Parker, K. L. (2006). Goodman & Gilman's the pharmacological basis of therapeutics. New York: McGraw-Hill; Howland, R. D., Mycek, M. J., Harvey, R. A., Champe, P. C., & Mycek, M. J. (2006). Pharmacology. Lippincott's illustrated reviews. Philadelphia: Lippincott Williams & Wilkins; and Golan, D. E. (2008). Principles of pharmacology: the pathophysiologic basis of drug therapy. Philadelphia, Pa., [etc.]: Lippincott Williams & Wilkins.

In another embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject in a pharmaceutical formulation.

A "pharmaceutical composition" refers to a chemical or biological composition suitable for administration to a mammal. Such compositions may be specifically formulated for administration via one or more of a number of routes, including but not limited to buccal, epicutaneous, epidural, inhalation, intraarterial, intracardial, intracerebroventricular, intradermal, intramuscular, intranasal, intraocular, intraperitoneal, intraspinal, intrathecal, intravenous, oral, parenteral, rectally via an enema or suppository, subcutaneous, subdermal, sublingual, transdermal, and transmucosal, preferably intravenous. In addition, administration can occur by means of injection, powder, liquid, gel, drops, or other means of administration.

A "pharmaceutically acceptable excipient" or a "pharmaceutically acceptable excipient" is a carrier, usually a liquid, in which an active therapeutic agent is formulated. In one embodiment of the invention, the active therapeutic agent is a humanized antibody described herein, or one or more fragments thereof. The excipient generally does not provide any pharmacological activity to the formulation, though it may provide chemical and/or biological stability, and release characteristics. Exemplary formulations can be found, for example, in Remington's Pharmaceutical Sciences, 19th Ed., Grennaro, A., Ed., 1995 which is incorporated by reference.

As used herein "pharmaceutically acceptable carrier" or "excipient" includes any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents that are physiologically compatible. In one embodiment, the carrier is suitable for parenteral administration. Alternatively, the carrier can be suitable for intravenous, intraperitoneal, intramuscular, or sublingual administration. Pharmaceutically acceptable carriers include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The use of such media and agents for pharmaceutically active substances is well known in the

art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the pharmaceutical compositions of the invention is contemplated. Supplementary active compounds can also be incorporated into the compositions.

Pharmaceutical compositions typically must be sterile and stable under the conditions of manufacture and storage. The invention contemplates that the pharmaceutical composition is present in lyophilized form. The composition can be formulated as a solution, microemulsion, liposome, or other ordered structure suitable to high drug concentration. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof. The invention further contemplates the inclusion of a stabilizer in the pharmaceutical composition. The proper fluidity can be maintained, for example, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants.

In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, or sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent which delays absorption, for example, monostearate salts and gelatin. Moreover, the alkaline polypeptide can be formulated in a time release formulation, for example in a composition which includes a slow release polymer. The active compounds can be prepared with carriers that will protect the compound against rapid release, such as a controlled release formulation, including implants and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, polylactic acid and polylactic, polyglycolic copolymers (PLG). Many methods for the preparation of such formulations are known to those skilled in the art.

An exemplary composition comprises, consists essentially of, or consists of an anti-CGRP antibody or fragment thereof (e.g., Ab6), an excipient such as histidine, an isotonic agent such as sorbitol, and a surfactant such as polysorbate 80 in an aqueous solution. For example, the composition may comprise, consist essentially of, or consist of histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody (e.g., Ab6), about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8, or approximately that constitution, e.g., within 10% of those values, within 5% of those values, within 1% of those values, within 0.5% of those values, or within 0.1% of those values, and water. For example, the pH value may be within 10% of 5.8, i.e., between 5.22 and 6.38. The Ab6 antibody may comprise or consist of the variable light and heavy chain polypeptides of SEQ ID NO: 222 and SEQ ID NO: 202 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively. The composition may be in the form of an aqueous solution, or a concentrate (e.g., lyophilized) which when reconstituted, e.g., by addition of water, yields the aforementioned constitution. An exemplary composition consists of, per mL, 100 mg of the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q.S, or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of

those quantities, within 0.5% of those quantities, or within 0.1% of those quantities. Another exemplary composition consists of, per mL, 100 mg of the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q. S, or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of those quantities, within 0.5% of those quantities, or within 0.1% of those quantities. The composition may be suitable for intravenous or subcutaneous administration, preferably intravenous administration. For example, the composition may be suitable for mixing with an intravenous solution (such as 0.9% sodium chloride) at an amount of between about 100 mg and about 300 mg antibody added to 100 mL of intravenous solution. Preferably the composition may be shelf-stable for at least 1, 3, 6, 12, 18, or 24 months, e.g., showing formation of aggregates of no more than 5% or no more than 10% of the antibody or fragment after storage at room temperature or when refrigerated at 4° C. for the specified duration, or in an accelerated aging test that simulates storage for that duration.

For each of the recited embodiments, the compounds can be administered by a variety of dosage forms. Any biologically-acceptable dosage form known to persons of ordinary skill in the art, and combinations thereof, are contemplated. Examples of such dosage forms include, without limitation, reconstitutable powders, elixirs, liquids, solutions, suspensions, emulsions, powders, granules, particles, microparticles, dispersible granules, cachets, inhalants, aerosol inhalants, patches, particle inhalants, implants, depot implants, injectables (including subcutaneous, intramuscular, intravenous, and intradermal, preferably intravenous), infusions, and combinations thereof.

The above description of various illustrated embodiments, of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed. While specific embodiments, of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. The teachings provided herein of the invention can be applied to other purposes, other than the examples described above.

These and other changes can be made to the invention in light of the above detailed description. In general, in the following claims, the terms used should not be construed to limit the invention to the specific embodiments, disclosed in the specification and the claims. Accordingly, the invention is not limited by the disclosure, but instead the scope of the invention is to be determined entirely by the following claims.

The invention may be practiced in ways other than those particularly described in the foregoing description and examples. Numerous modifications and variations of the invention are possible in light of the above teachings and, therefore, are within the scope of the appended claims.

Certain CGRP antibody polynucleotides and polypeptides are disclosed in the sequence listing accompanying this patent application filing, and the disclosure of said sequence listing is herein incorporated by reference in its entirety.

The entire disclosure of each document cited (including patents, patent applications, journal articles, abstracts, manuals, books, or other disclosures) in the Background of the Invention, Detailed Description, and Examples is herein incorporated by reference in their entireties.

The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure

51

and description of how to make and use the subject invention, and are not intended to limit the scope of what is regarded as the invention. Efforts have been made to ensure accuracy with respect to the numbers used (e.g. amounts, temperature, concentrations, etc.) but some experimental errors and deviations should be allowed for. Unless otherwise indicated, parts are parts by weight, molecular weight is average molecular weight, temperature is in degrees centigrade; and pressure is at or near atmospheric.

ADDITIONAL EXEMPLARY EMBODIMENTS

S1. Use of an anti-CGRP antibody for the manufacture of a medicament for treating migraine or headache in a patient in the need of immediate relief of migraine or headache symptoms or for prevention of migraine or headache in a patient in need of immediate preventative treatment of migraine or headache, wherein said medicament is for intravenous infusion in a dosage of 100 or 300 mg of said anti-CGRP antibody, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

S2. Use of the anti-CGRP antibody of embodiment S1, wherein said medicament is for use in a patient that patient exhibits at least one headache and/or migraine symptom at the time of administration.

S3. Use of the anti-CGRP antibody of embodiment S2, wherein said at least one headache and/or migraine symptom comprises one or more of pain, nausea, photophobia, or phonophobia.

S4. Use of the anti-CGRP antibody of embodiment S3, wherein said pain is head pain.

S5. Use of the anti-CGRP antibody of any one of embodiments S2-S4, wherein the most bothersome symptom is alleviated after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

S6. Use of the anti-CGRP antibody of any one of embodiments S2-S5, wherein said patient no longer has a migraine after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

S7. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

S8. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

52

S9. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

S10. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

S11. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

S12. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

S13. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

S14. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

S15. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

S16. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

S17. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

S18. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

S19. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

S20. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said intravenous administration is infused over a period of approximately 30 min to 60 minutes.

S21. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the headache or migraine symptoms decline or are abolished immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

S22. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of infusion.

S23. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous administration in a dosage of 100 mg of said

anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

S24. Use of the anti-CGRP antibody of any one of the foregoing embodiments S1-S22, wherein said medicament is for intravenous administration in a dosage of 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

S25. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

S26. Use of the anti-CGRP antibody of embodiment S25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

S27. Use of the anti-CGRP antibody of embodiment S25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

S28. Use of the anti-CGRP antibody of embodiment S25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

S29. Use of the anti-CGRP antibody of embodiment S25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

S30. Use of the anti-CGRP antibody of embodiment S25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

S31. Use of the anti-CGRP antibody of any one of the foregoing embodiments S25-S30, wherein said L-Histidine in said formulation comprises a mixture of L-Histidine and L-Histidine monohydrate.

S32. Use of the anti-CGRP antibody of any one of the foregoing embodiments S25-S30, wherein said 3.1 mg of histidine in said formulation comprises a mixture of L-Histidine (1 mg) and L-Histidine monohydrate (2.8 mg), which in the final formulation sums up to 3.1 mg L-histidine free base.

S33. Use of the anti-CGRP antibody of any one of the foregoing embodiments S26-S32, wherein said formulation is comprised in a 100 mg/mL single-dose vial wherein each mL contains 100 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

S34. Use of the anti-CGRP antibody of any one of the foregoing embodiments S26-S32, wherein said formulation is comprised in a 300 mg/mL single-dose vial wherein each mL contains 300 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

S35. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said migraine or headache is selected from the group comprising acute migraine or

headache, migraines with or without aura, chronic migraine, episodic migraine, chronic/episodic migraine, hemiplegic migraines, cluster headaches, migrainous neuralgia, chronic headaches, tension headaches, general headaches, headaches due to an underlying structural problem in the head or neck, sinus headaches (such as for example associated with sinusitis), and allergy-induced headaches or migraines.

S36. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for administration to a patient that exhibits a pain level of at least 2 on the VRS-4 at the time of administration of said antibody.

S37. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for administration to a patient that exhibits a pain level of at least 3 on the VRS-4 at the time of administration of said antibody.

S38. Use of the anti-CGRP antibody of any one of the foregoing embodiments S1-S37, wherein said medicament is for administration to a patient that exhibits a pain level of at most 2 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

S38. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for administration to a patient that exhibits a pain level at most 1 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

S40. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for administration to a patient that is not administered any acute migraine medication within a period of time before and after said administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said administration.

S41. Use of the anti-CGRP antibody of embodiment S40, wherein said acute migraine medication comprises a triptan, an analgesic such as non-opioids or opioids/narcotics, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

S42. Use of the anti-CGRP antibody of embodiment S41, wherein said non-opioid analgesic comprises paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan comprises use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid comprises use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication comprises two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesics comprises at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/

55

or said combination-analgesic comprises acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

S43. Use of the anti-CGRP antibody of any one of embodiments S1-S39, wherein the patient is receiving or has received additional migraine medication.

S44. Use of the anti-CGRP antibody of any one of embodiments S1-S39 or S43, wherein the patient receives additional migraine medication prior, concurrent or after administration of the anti-CGRP antibody.

S45. Use of the anti-CGRP antibody of any one of embodiments S1-S39 or S43-S44, wherein the patient receives additional migraine medication within a period of time before and after said anti-CGRP antibody administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said anti-CGRP antibody administration.

S46. Use of the anti-CGRP antibody of any one of embodiments S44 or S45, wherein said additional migraine medication comprises an acute and/or a chronic migraine medication.

S47. Use of the anti-CGRP antibody of any one of embodiments S44-S46, wherein said additional migraine medication comprises a triptan, an analgesic such as non-opioid or opioid/narcotic, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

S48. Use of the anti-CGRP antibody of embodiment S47, wherein said non-opioid analgesic comprises paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan comprises use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid comprises use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication comprises two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesic comprises at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/or said combination-analgesic comprises acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

S49. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

S50. Use of the anti-CGRP antibody of any of any one of embodiments S1-S48, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

S51. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said patient is administered 100 mg or 300 mg of said anti-CGRP antibody every three months.

S52. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said method results in immediate relief of migraine or headache symptoms.

S53. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said method results in immediate preventative treatment of migraine or headache.

FURTHER EXEMPLARY EMBODIMENTS

E1. An anti-CGRP antibody for use in treating migraine or headache in a patient in the need of immediate relief of

56

migraine or headache symptoms or for use in preventing migraine or headache in a patient in need of immediate preventative treatment of migraine or headache, wherein said anti-CGRP antibody is for intravenous infusion in a dosage of 100 or 300 mg of said anti-CGRP antibody, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

E2. The anti-CGRP antibody for use according to embodiment E1, wherein said anti-CGRP antibody is for use in a patient that patient exhibits at least one headache and/or migraine symptom at the time of administration.

E3. The anti-CGRP antibody for use according to embodiment E2, wherein said at least one headache and/or migraine symptom comprises one or more of pain, nausea, photophobia, or phonophobia.

E4. The anti-CGRP antibody for use according to embodiment E3, wherein said pain is head pain.

E5. The anti-CGRP antibody for use according to any one of embodiments, E2-E4, wherein the most bothersome symptom is alleviated after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

E6. The anti-CGRP antibody for use according to any one of embodiments E2-E5, wherein said patient no longer has a migraine after said administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

E7. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

E8. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

E9. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

E10. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

E11. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

E12. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP

57

antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

E13. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

E14. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

E15. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

E16. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

E17. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

E18. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

E19. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

E20. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said intravenous administration is infused over a period of approximately 30 min to 60 minutes.

E21. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein the headache or migraine symptoms decline or are abolished immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

E22. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of infusion.

E23. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous administration in a dosage of 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

E24. The anti-CGRP antibody for use according to any one of embodiments E1-E22, wherein said anti-CGRP antibody is for intravenous administration in a dosage of 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

E25. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

58

E26. The anti-CGRP antibody for use according to embodiment E25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

E27. The anti-CGRP antibody for use according to embodiment E25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

E28. The anti-CGRP antibody for use according to embodiment E25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

E29. The anti-CGRP antibody for use according to embodiment E25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

E30. The anti-CGRP antibody for use according to embodiment E25, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

E31. The anti-CGRP antibody for use according to any one of embodiments E25-E30, wherein said L-Histidine in said formulation comprises a mixture of L-Histidine and L-Histidine monohydrate.

E32. The anti-CGRP antibody for use according to any one of embodiments E25-E30, wherein said 3.1 mg of histidine in said formulation comprises a mixture of L-Histidine (1 mg) and L-Histidine monohydrate (2.8 mg), which in the final formulation sums up to 3.1 mg L-histidine free base.

E33. The anti-CGRP antibody for use according to any one of embodiments E26-E32, wherein said formulation is comprised in a 100 mg/mL single-dose vial wherein each mL contains 100 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

E34. The anti-CGRP antibody for use according to any one of embodiments E26-E32, wherein said formulation is comprised in a 300 mg/mL single-dose vial wherein each mL contains 300 mg anti-CGRP antibody, L-histidine (1 mg), L-histidine hydrochloride monohydrate (2.8 mg), polysorbate 80 (0.15 mg), sorbitol (40.5 mg), and Water for Injection, USP, at a pH of 5.8.

E35. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said migraine or headache is selected from the group comprising acute migraine or headache, migraines with or without aura, chronic migraine, episodic migraine, chronic/episodic migraine, hemiplegic migraines, cluster headaches, migrainous neuralgia, chronic headaches, tension headaches, general headaches, headaches due to an underlying structural

problem in the head or neck, sinus headaches (such as for example associated with sinusitis), and allergy-induced headaches or migraines.

E36. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for administration to a patient that exhibits a pain level of at least 2 on the VRS-4 at the time of administration of said antibody.

E37. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for administration to a patient that exhibits a pain level of at least 3 on the VRS-4 at the time of administration of said antibody.

E38. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for administration to a patient that exhibits a pain level of at most 2 or 3 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

E39. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for administration to a patient that exhibits a pain level at most 1 on the VRS-4 immediately after administration, such as within the first day after administration, within 12 hours after administration, within 6 hours after administration within 5 hours after administration, within 4 hours after administration, within 3 hours after administration, within 2 hours after administration, or within 1 hour of after administration, within 30 minutes after administration, or such as between 1-6 hours after administration.

E40. The anti-CGRP antibody for use according to any one of the foregoing embodiments, wherein said anti-CGRP antibody is for administration to a patient that is not administered any acute migraine medication within a period of time before and after said administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said administration.

E41. The anti-CGRP antibody for use according to embodiment E40, wherein said acute migraine medication comprises a triptan, an analgesic such as non-opioids or opioids/narcotics, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

E42. The anti-CGRP antibody for use according to embodiment E41, wherein said non-opioid analgesic comprises paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan comprises use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid comprises use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication comprises two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesics comprises at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/or said combination-analgesic comprises acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

E43. Use of the anti-CGRP antibody of any one of embodiments E1-E39, wherein the patient is receiving or has received additional migraine medication.

E44. Use of the anti-CGRP antibody of any one of embodiments E1-E39 or E43, wherein the patient receives additional migraine medication prior, concurrent or after administration of the anti-CGRP antibody.

E45. Use of the anti-CGRP antibody of any one of embodiments E1-S39 or E43-E44, wherein the patient receives additional migraine medication within a period of time before and after said anti-CGRP antibody administration, such as within 15 minutes, within 30 minutes, within 1 hour, within 2 hours, within 3 hours, within 4 hours, within 5 hours, or within 6 hours before and after said anti-CGRP antibody administration.

E46. Use of the anti-CGRP antibody of any one of embodiments E44 or E45, wherein said additional migraine medication comprises an acute and/or a chronic migraine medication.

E47. Use of the anti-CGRP antibody of any one of embodiments E44-E46, wherein said additional migraine medication comprises a triptan, an analgesic such as non-opioid or opioid/narcotic, acetaminophen, an NSAID, a combination medication, an ergotamine, or an ergot derivative.

E48. Use of the anti-CGRP antibody of embodiment E47, wherein said non-opioid analgesic comprises paracetamol (acetaminophen), acetylsalicylic acid (aspirin), another NSAID, or another non-opioid analgesic; said triptan comprises use of one or more of sumatriptan, zolmitriptan, naratriptan, rizatriptan, eletriptan, almotriptan, or frovatriptan; said opioid comprises use of one or more of oxycodone, tramadol, butorphanol, morphine, codeine, and hydrocodone; said combination medication comprises two drugs with analgesic effects (for example, paracetamol and codeine), an analgesic and an adjuvant (for example, paracetamol and caffeine) and/or said combination-analgesics comprises at least one opioid (such as tramadol, butorphanol, morphine, codeine, hydrocodone, or any combination thereof), barbiturate such as butalbital, and/or caffeine, and/or said combination-analgesic comprises acetylsalicylic acid (aspirin), paracetamol and caffeine (EXCEDRIN®, EXCEDRIN MIGRAINE®).

E49. The anti-CGRP antibody for use according to any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

E50. The anti-CGRP antibody for use according to any of any one of embodiments E1-E39, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

E51. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said patient is administered 100 mg or 300 mg of said anti-CGRP antibody every three months.

E52. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said method results in immediate relief of migraine or headache symptoms.

E53. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said method results in immediate preventative treatment of migraine or headache.

EXAMPLES

The following examples are provided in order to illustrate the invention, but are not to be construed as limiting the scope of the claims in any way.

61

Example 1

Preparation of Antibodies that Bind CGRP

The preparation of exemplary anti-CGRP antibodies Ab1-Ab14 having the sequences in FIGS. 1A-12 is disclosed in commonly owned PCT Application WO/2012/162243, published on Nov. 29, 2012, the contents of which are incorporated by reference herein. This application exemplifies synthesis of these antibodies in *Pichia pastoris* cells. The present Applicant further contemplates synthesis of anti-CGRP antibodies Ab1-Ab14, and Ab6 in particular in CHO cells.

Example 2

Human Clinical Study Evaluating the Safety and Efficacy of an Anti-CGRP Antibody According to the Invention Clinical Treatment Protocol

The humanized anti-CGRP IgG1 antibody identified herein as Ab6 was assessed in human subjects for its ability to inhibit, alleviate or prevent the number of, duration, and/or the intensity of migraine episodes. The Ab6 antibody contains the V_L and light chain polypeptides respectively in SEQ ID NO: 222 and SEQ ID NO: 221, and contains the V_H and heavy chain polypeptides respectively in SEQ ID NO: 202 and SEQ ID NO: 201. This antibody comprises an IgG1 constant region that contains a mutation in the heavy chain constant region (replacement of asparagine residue at position 297 with an alanine residue which substantially eliminates glycosylation and lytic activity (see U.S. Pat. No. 5,624,821).

Specifically, the clinical efficacy of the Ab6 antibody was tested in a placebo controlled double-blind, randomized study. The individuals in the study were all selected based on specific criteria. Particularly all were diagnosed as migraine sufferers at ≤50 years of age (ICHD-II, 2004 Section 1), and further had a history of migraine ≥12 months with ≥5 and ≤14 migraine days in each 28 day period in the 3 months prior to screening.

Further, all of the individuals in the study used acute migraine medications ≤14 days per 28 day period and, within those days, ≤10 days of triptan use per 28 day period in the 3 months prior to screening and the 28 day period of completion of eDiary prior to randomization.

Table 1 summarizes the demographic characteristics of the study population.

TABLE 1

Baseline Demographics and Clinical Characteristics		
Characteristic	Placebo iv (n = 82)	Ab6 1000 mg iv (n = 81)
Mean ± SD Age (years)	39.0 (9.6)	38.6 (10.8)
Mean ± SD Weight (kg)	75.4 (14.4)	75.0 (16.5)
Female Gender	66 (80%)	67 (83%)
Race:		
Caucasian	66 (80.5%)	66 (81.5%)
African American	9 (11.0%)	10 (12.4%)
Asian	3 (3.7%)	4 (5.0%)
Other	4 (4.8%)	1 (1.1%)
Baseline (per 28 days):		
Mean ± SD Migraine Days	8.8 (2.7)	8.4 (2.1)
Mean ± SD Migraine Episodes	6.7 (2.4)	6.0 (2.2)
Mean ± SD Headache Frequency	9.6 (2.8)	9.2 (2.6)
Mean ± SD Migraine Hours	72.2 (51.0)	80.1 (49.1)
Mean ± SD HIT-6 Score	64.5 (4.44)	63.8 (5.21)

62

TABLE 1-continued

Baseline Demographics and Clinical Characteristics		
Characteristic	Placebo iv (n = 82)	Ab6 1000 mg iv (n = 81)
Mean ± SD MSQ RFP Score	49.0 (17.9)	49.5 (21.2)
Mean ± SD MSQ RFR Score	61.9 (22.7)	63.9 (24.0)
Mean ± SD MSQ EF Score	59.5 (22.9)	59.8 (27.0)

Throughout the study all of the individuals were required to record their migraine status daily using an e-diary. In the e-diary the subjects in the study were required to record the number of migraine days/month, migraine episodes/month, migraine hours/month, migraine severity, and the use of any abortive medicine such as triptans.

In addition, the study participants were required to use the e-diary to record their migraine status in the 28 day period prior to treatment with antibody or placebo in order to establish a migraine day/hour/episode baseline per month. Also, this allowed the subjects in the study to become familiar with the use of the e-diary.

After the 28-day run-in the subjects in the study were broken into two groups, each including 80 subjects (FIG. 17). In the first group, i.e., the antibody treatment group, (n=80) each subject in the group was administered intravenously a single 1000 mg dose of Ab6. In the second group (n=80), i.e., the placebo group, each of the subjects was given an intravenous injection containing only the aqueous antibody carrier solution.

The individuals in the treated and placebo groups were assessed in the 24 weeks post-dose administration. Initially, a 12 week interim analysis was conducted. Subsequent to the 12 week interim analysis, a refined analysis was conducted. This refined analysis potentially included, for example, addition or removal of patient data in accord with the study protocol, e.g., updating data that had not been fully loaded from the e-diaries. This refinement resulted in slight changes but did not alter the overall conclusions.

The efficacy of the antibody versus the placebo was assessed in part based on the recorded data in the e-diary entries. For example, this analysis included a comparison of the number of recorded migraine days/month, migraine episodes/month, migraine hours/month in the subjects in the treated versus the placebo group. The percentage of responders in each group (i.e., the subjects with 50%, 75%, and 100% reduction in migraine days) in both groups was also compared.

In addition, the responses of the Ab6- and placebo-treated subjects in both groups to MSQ and HIT-6 questionnaires are to be evaluated and compared. MSQ is a frequently utilized disease-specific tool to assess the impact of migraine on health-related quality of life (HRQL). MSQ comprises a 16-item Migraine-Specific Quality-of-Life Questionnaire (Version 1.0), which was developed by Glaxo Wellcome Inc. MSQ is hypothesized to measure 3 parameters: (i) Role Function-Restrictive; (ii) Role Function-Preventive; and (iii) Emotional Function.

The HIT-6 or functional impact (also called the Headache Impact Test or HIT-6) similarly is a well known tool for assessing migraine intensity. This test uses six questions to capture the impact of headache and its treatment on an individual's functional health and well-being.

Also, the pharmacokinetic (PK) properties of the CGRP antibody and immunogenicity are to be assessed in the Ab6 antibody treated subjects.

Clinical Results and Analysis

The results of this human clinical trial and analysis through week 12 in the treated subjects are summarized in the Table 2 below.

TABLE 2

Responder analysis for migraine days				
Time period	% reduction migraine days	Placebo iv	Ab6 1000 mg iv	P value
Week 1-4		n = 80	n = 75	
	50	40 (50.0)	58 (77.3)	p = 0.0005
	75	19 (23.8)	39 (52.0)	p = 0.0005
Week 5-8	100	4 (5.0)	21 (28.0)	p = 0.0001
		n = 80	n = 78	
	50	43 (53.8)	59 (75.6)	p = 0.0048
Week 9-12	75	28 (35.0)	35 (44.9)	p = 0.2555
	100	12 (15.0)	21 (26.9)	p = 0.0791
		n = 77	n = 72	
	50	51 (66.2)	54 (75.0)	p = 0.2827
	75	24 (31.2)	38 (52.8)	p = 0.0083
	100	13 (16.9)	29 (40.3)	p = 0.0019

In addition, the results of the clinical study were compared based on the number of responders in the treatment and placebo groups. As shown in FIG. 13 the number of subjects who showed a 50, 75 or 100% reduction in migraine days for each month of the interim period were compared in the treatment and placebo groups. As shown in the figure, 60% of the Ab6-treated group had at least 50% reduction in headache days, 31% of the Ab6-treated group had at least 75% reduction in headache days and 15% of the Ab6 treated group had 100% reduction in headache days.

By contrast, 33% of the placebo-treated group had at least 50% reduction in headache days, 9% of the placebo-treated group had at least 75% reduction in headache days, and 0% (none) of the placebo-treated group had 100% reduction in headache days.

These results clearly show that the reduction in the number of migraine days was much greater in the Ab6-treated group. But for the significant placebo effect, the difference in these numbers would have been more pronounced. (Elevated placebo effect is not surprising as the phenomenon is often very high for migraine and other neurological drugs).

In addition, the % change from baseline in the number of migraine days per month in the placebo and Ab6-treated group was compared. As shown in FIG. 14, the median ($\pm$ QR) % change from baseline in the number of migraine days per month in the placebo and Ab6-treated group was compared for the 2 groups during the 12 weeks post-treatment. These results which are statistically significant ($p=0.0078$) clearly show the Ab6-treated group had a much greater reduction in the number of headache days per month compared to baseline than the placebo-treated group.

Also, the % change from baseline in the number of migraine episodes per month in the placebo and Ab6-treated group was compared. As shown in FIG. 15 the median ($\pm$ QR) % change from baseline in the number of migraine episodes per month in the placebo and Ab6-treated group was compared during the 12 weeks post-treatment. These results indicate that the Ab6-treated group had a significantly greater reduction in the number of migraine episodes per month compared to baseline than the placebo-treated group.

Further, the % change from baseline in the number of migraine hours per month in the placebo and Ab6-treated group was compared. As shown in FIG. 16, the median ($\pm$ QR) % change from baseline in the number of migraine

hours per month in the placebo and Ab6-treated group was compared for the 2 groups during the 12 weeks post-treatment. These results clearly show the Ab6-treated group had a greater reduction in the number of migraine hours per month compared to baseline than the placebo-treated group.

In addition, the HIT-6 results were compared for both groups. As noted, this questionnaire finds well accepted usage in assessing the migraine status of individuals with frequent/chronic migraine. FIG. 18 compares the HIT-6 responder analysis for the Ab6-treated and placebo groups at baseline, week 4 after treatment, week 8 after treatment and week 12 after treatment. The results at each time point reveal that the Ab6-treated group had a statistically significant improvement in the HIT-6 scores relative to the placebo group, i.e., 54.4% for the Ab6-treated compared to 30% for the placebo at week 4 ($p=0.0023$), 51.3% for the Ab6-treated compared to 38.0% for the placebo at week 8 ($p=0.1094$) and 61.1% for the Ab6-treated compared to 33.3% for the placebo at week 12 ($p=0.0007$). FIG. 19 shows the percentage of patients having a HIT-6 score of some or little/none over time in the placebo and Ab6 treatment groups (statistical significance shown).

In addition, FIG. 20 contains the pharmacokinetic (PK) profile for Ab6 administered intravenously at a single dosage of 1000 mg in mg/mL over the 24 week period following Ab6 administration.

FIG. 21 contains plasma-free pharmacokinetic (PK) parameters N (number of patients), mean, and standard deviation (SD) for a single 1000 mg intravenous dosage of Ab6. The parameters shown in the table and the units are C_{max} (μ g/mL), AUC_{0-28} ($mg \cdot hr/mL$), half-life (days), V_z (L) and C_L (mL/hr).

Further analysis was conducted for patient data between 12-weeks and 24-weeks. The treatment group continued to exhibit decreased migraine days relative to the control group, however, the magnitude of the difference decreased over time. Additionally, the control group exhibited fewer migraine days per month than at baseline. This was thought to result at least in part from "diary fatigue" wherein patients potentially report no migraine on a day in which a migraine actually occurred, in order to avoid the time and effort of answering further queries about the migraine that would result from giving an affirmative answer to the question of whether they had a migraine on a given day.

Further analysis of the study results are shown in FIGS. 22-33. These result include analysis of the change (mean $\pm$ SEM) from baseline in migraine days per month for Ab6 (1000 mg i.v.) versus placebo (FIG. 22), change in average migraine days ($\pm$ SD) over time for the full analysis population (FIG. 23). Additionally, shown are the distribution of migraine days actual and change for the Ab6 treatment group during weeks 1-4 (FIG. 24), distribution of migraine days actual and change for the placebo group during weeks 1-4 (FIG. 25), distribution of migraine days actual and change for the Ab6 treatment group during weeks 5-8 (FIG. 26), distribution of migraine days actual and change for the placebo group during weeks 5-8 (FIG. 27), distribution of migraine days actual and change for the Ab6 treatment group during weeks 9-12 (FIG. 28), and distribution of migraine days actual and change for the placebo group during weeks 9-12 (FIG. 29).

Responder rate analysis was also performed (FIGS. 30-32). These figures respectively show the 50%, 75%, and 100% responder rate for the Ab6 and placebo treatment groups. Subjects with $\geq 50\%$ reduction in migraine frequency were considered to be a 50% responder. Subjects with $\geq 75\%$ reduction in migraine frequency were considered

to be a 75% responder. Likewise, subjects with 100% reduction in migraine frequency were considered to be a 100% responder.

In FIGS. 22 and 30-32, normalization was applied to visit intervals where eDiaries were completed for 21-27 days by multiplying the observed frequency by the inverse of the completion rate.

Migraine severity was also analyzed. FIG. 33 shows the mean migraine severity over time for the full analysis population. On the scale used, a mean migraine score of 3 represents "moderate pain."

FIG. 34 summarizes the change from baseline in migraine days, migraine episodes, migraine hours, average migraine severity, headache frequency, and outcome measures including the HIT-6 score, MSQ (Migraine Specific Quality of Life Questionnaire) RFP (Role Function-Preventative), MSQ RFR (Role Function-Restrictive), and MSQ EF (Emotional Function).

Example 3

Human Clinical Study Evaluating the Safety and Efficacy of an Anti-CGRP Antibody in Chronic Migraine Patients

This example describes a randomized, double-blind, placebo-controlled clinical trial evaluating the safety and efficacy of Ab6 for chronic migraine prevention. In the study, 1,072 patients were randomized to receive Ab6 (300 mg or 100 mg), or placebo administered by infusion once every 12 weeks. To be eligible for the trial, patients must have experienced at least 15 headache days per month, of which at least eight met criteria for migraine. Patients that participated in the trial had an average of 16.1 migraine days per month at baseline. Study endpoints included the mean change from baseline in monthly migraine days, reduction in migraine prevalence at day 1 and over days 1-28, and reduction of at least 50%, 75%, and 100% from baseline in mean monthly migraine days, change from baseline in mean monthly acute migraine-specific medication days, and reductions from baseline in patient-reported impact scores on the Headache Impact Test (HIT-6). The administered antibody, Ab6, is an anti-CGRP antibody consisting of the light chain polypeptide of SEQ ID NO: 221 and heavy chain polypeptide of SEQ ID NO: 201.

Patient characteristics are summarized in FIG. 39, with separate columns for patients receiving placebo, 100 mg of the antibody, or 300 mg of the antibody. Patients had a mean number of years from migraine diagnosis of between 17.0 and 19.0 years, a mean duration of suffering from chronic migraine of between 11.5 and 12.4 years, and between 44.3% and 45.2% of patients utilized at least one prophylactic medication. At baseline, in both antibody treatment groups the mean number of migraine days per month was 16.1, while for the placebo group, the mean number of migraine days per month was 16.2.

The reduction in a specified percentage (50%, 75%, or 100%) from baseline in mean monthly migraine days refers to the number or percentage of patients in a treatment group that exhibited the given percentage reduction in the number of migraine days per month. For example, a patient exhibiting 16 migraine days per month at baseline would be a 75% responder if the number of migraine days per month was decreased by at least 12 days per month over specified period.

The results are shown in FIGS. 35-39. FIG. 35 shows the percentages of patients with migraine in the 300 mg, 100 mg, and placebo treatment groups at days 1, 7, 14, 21, and 28. The uppermost line shows results for placebo, the lowest

line shows results for the 300 mg dosage, and the middle line shows results for the 100 mg dosage.

As shown in FIG. 35, at day 1 the percentage reduction in migraine prevalence was 52% for the 300 mg dosage, 50% at the 100 mg dosage, and 27% for placebo. The decrease was statistically significant compared to the placebo group for both the 100 mg and 300 mg treatment groups.

FIGS. 36-38 show the percentage of patients in the 300 mg and 100 mg treatment groups achieving, respectively, 50%, 75%, and 100% reduction in migraine days in month 1, over months 1-3 (after the 1st infusion), and over months 4-5 (after the 2nd infusion). In each graph, the data bars, from left to right, show results for the 100 mg, 300 mg, and placebo groups. Statistical significance is as shown. ++ indicates a statistically significant difference from placebo; + indicates a statistically significant difference from placebo (unadjusted); and § indicates a statistically significant difference from placebo (post hoc).

Example 4

Baseline Subgroup Analysis for Human Clinical Studies Evaluating the Safety and Efficacy of an Anti-CGRP Antibody in Chronic or Episodic Migraine Patients

In the study of Chronic Migraine described in Example 3, at intake, each patient was assessed for potential medication overuse headache (MOH). MOH was present in 39.9% (139 patients) in the 100 mg treatment group, 42.0% (147 patients) in the 300 mg treatment group, and 39.6% (145 patients) in the placebo group. Assessment of the treatment outcomes in this patient subset indicated that treatment with the anti-CGRP antibody was efficacious for MOH (FIG. 41). Specifically, in the 100 mg treatment group, mean migraine days per month changed by -3.0 days (95% CI, -4.56 to -1.52 days) in the patients having MOH at baseline, compared to MOH patients receiving placebo. Similarly, in the 300 mg treatment group, mean migraine days per month changed by -3.2 days (95% CI, -4.66 to -1.78 days) in the patients having MOH at baseline, compared to MOH patients receiving placebo. By contrast, for patients without MOH at baseline, in the 100 mg treatment group, mean migraine days per month changed by -1.3 days (95% CI, -2.43 to -0.16 days), compared to patients without MOH at baseline receiving placebo. Likewise, for patients without MOH at baseline in the 300 mg treatment group, mean migraine days per month changed by -2.1 days (95% CI, -3.24 to -0.88 days), compared to patients without MOH at baseline receiving placebo. Efficacy for other subgroups was shown as well, including efficacy for patients with mean migraine day (MMD) frequency less than 17 days or greater than or equal to 17 days, patients with an age at diagnosis of less than or equal to 21 years or greater than 21 years, patients having a duration of migraine of less than or equal to 15 years or greater than 15 years, patients suffering from migraine with aura or migraine with no aura, patients with prior prophylactic medication use or no prior prophylactic medication use, patients with concomitant prophylactic medication use or no concomitant prophylactic medication use, ant patients with triptan use on greater than or equal to 33% of days, or less than 33% of days. In each case, efficacy for each subgroup was shown (FIG. 41).

In another human clinical trial of patients with episodic migraine, patients were randomized to receive Ab6 100 mg (n=221), 300 mg (n=222), or placebo (n=222) in a double blind, parallel study. After a 28 day screening period, patients were administered the drug or placebo intravenously every 3 months for 4 total infusions (FIG. 40).

Efficacy was shown over months 1-3 for both the 100 mg and 300 mg treatment groups, with a mean change in migraine days of -3.9 for the 100 mg treatment group and -4.3 days for the 300 mg treatment group, compared to -3.2 days for the placebo group. Efficacy for subgroups of patients was also shown, including efficacy for patients with mean migraine day (MMD) frequency less than or equal to 9 days or greater than 9 days, patients with an age at diagnosis of less than or equal to 21 years or greater than 21 years, patients having a duration of migraine of less than or equal to 15 years or greater than 15 years, and patients suffering from migraine with aura or migraine with no aura.

Example 5

Effects of Ab6 Treatment on Medication Use in Chronic and Episodic Migraine Patients

During the studies of chronic migraine patients described in Example 3 and episodic migraine patients described in Example 4, patients also recorded use of acute medication in a daily eDiary and were allowed to use acute medication at their own discretion. Acute medications for migraine included ergots, triptans, and analgesics (e.g., NSAIDs, opioids, and caffeine-containing combination analgesics).

For further analysis, patients were stratified by the number of days with acute medication use during the 28-day screening period (1-9 or >10 days; "baseline"). Acute medication days were calculated for individual types of acute medications and combined, meaning that if 2 or more types of medications were used on the same calendar days, they were counted as separate medication use days. For example, if a patient took an opioid and a triptan on the same day, it counted as 2 days of acute medication use. These analyses included patients with at least 1 acute medication use day during the 28-day baseline screening period.

In both chronic migraine and episodic migraine patients who used acute medication during the 28-day baseline period, Ab6 treatment resulted in greater average reductions in monthly migraine days and acute medication days than placebo as early as Month 1 after dosing, with similar results across 2 dose intervals over 6 months.

Ab6 consistently demonstrated greater reductions in mean monthly migraine days over 6 months of treatment than placebo in chronic migraine patients taking ≥ 1 day of acute medication use during baseline (FIG. 42). Chronic migraine patients who had at least one day of acute medication use per month during baseline demonstrated greater decreases in acute medication use than placebo as early as month 1 after treatment and across the entire 6 month treatment period (FIG. 43). In the subgroup of chronic migraine patients who were taking 1-9 days of acute medication during baseline, the change from baseline in days of acute medication use was greater in the 300 mg Ab6 group than placebo across 6 months of treatment (FIG. 44). A clear decrease in medication days per month was observed for patients with at least 10 days of medication use per month at baseline for both Ab6 treatment group compared to placebo over the entire 6 month period. FIG. 45 shows the changes in medication use days at Month 1 and Month 6 in the subgroups of chronic migraine patients with ≥ 1 , 1-9, and ≥ 10 days of acute medication use at baseline. With the exception of Ab6 100 mg at month 6 in patients with 1-9 days/month of use at baseline, Ab6 demonstrated a greater treatment effect in reducing acute medication use than placebo.

Similarly, across 2 dose intervals over 6 months, episodic

monthly migraine days with Ab6 than Placebo (FIG. 46). Episodic migraine patients who had at least one day of acute medication use per month during baseline demonstrated greater decreases in acute medication use than placebo as early as month 1 after treatment and across the entire 6 month treatment period (FIG. 47). In the subgroup of episodic migraine patients who were taking 1-9 days of acute medication during baseline, the change from baseline in days of acute medication use was greater with Ab6 than placebo across 6 months of treatment (FIG. 48). A similar pattern was observed in the subgroup of patients who were taking ≥ 10 days of acute medication during baseline, though smaller sample sizes may have contributed to the less consistent pattern over time. FIG. 49 shows the changes in medication use days at Month 1 and Month 6 in the subgroups of episodic migraine patients with ≥ 1 , 1-9, and ≥ 10 days of acute medication use at baseline. With the exception of Ab6 100 mg at Month 6 in patients with ≥ 10 days/month of use at baseline, the reduction in acute medication use was greater in the Ab6 treatment groups than placebo.

The results show that both episodic migraine and chronic migraine patients who were at risk for medication-overuse headache (≥ 10 days/month of acute medication use) demonstrated the greatest reductions in acute medication use, with Ab6 treatment generally resulting in larger decreases in medication use days than placebo.

The most frequently reported acute headache medications in >10% of subjects included Thomapyrin N (44.5%) (a combination of paracetamol, aspirin, and caffeine), ibuprofen (40.6%), sumatriptan (33.6%), paracetamol (acetaminophen) (20.3%), and naproxen sodium (10.2%). The most frequently reported preventive headache medication in >10% of subjects was topiramate (12.5%).

Example 6

Efficacy of Anti-CGRP Antibodies in Subjects Experiencing an Acute Attack of Migraine

This example describes a randomized, double-blind, placebo-controlled clinical trial evaluating the safety and efficacy of Ab6 for the acute treatment of migraine. In the study, approximately 450 patients are randomized 1:1 to receive either 100 mg Ab6 or placebo. During a screening period (approx. 1-8 weeks) patients are assessed for migraine frequency and medication use frequency. Eligible patients have a migraine attack frequency of about 4-15 migraine days per month in the 3 months prior to screening. By history, the subject's typical migraine attack, if untreated, would be associated with headache pain of moderate to severe intensity and a most bothersome symptom of nausea, photophobia, or phonophobia. Subjects must be headache free for at least 24 hours prior to onset of a qualifying migraine in order to participate in the trial. On the day of treatment, the patient will travel to the study site and intravenous infusion of 100 mg Ab6 or placebo will commence between about 1-6 hours from the start of the attack. Patients will not have received any other monoclonal antibody (e.g., any CGRP antagonist antibody) within the 6 month period prior to screening.

Co-Primary Endpoints are time to headache pain freedom and time to absence of most bothersome symptom. Co-Key secondary are headache pain freedom at 2 hours and absence of most bothersome symptom at 2 hours. Secondary endpoints are time to headache pain relief, headache pain freedom at 2 hours with sustained headache pain freedom for 24 and 48 hours, use of rescue medication by 24 hours and by 48 hours, absence of photophobia at 2 hours, absence

of phonophobia at 2 hours, absence of nausea at 2 hours, change from Baseline in Headache Impact Test (HIT 6) at Week 4, and change from Baseline in Migraine Treatment Optimization Questionnaire-6 (mTOQ-6) at Week 4. Exploratory Endpoints are absence of headache pain at all timepoints other than 2 hours, absence of photophobia at all timepoints other than 2 hours, absence of phonophobia at all timepoints other than 2 hours, absence of nausea at all timepoints other than 2 hours, pain relapse when the subject was headache pain-free at 2 hours, patient Global Impression of Change (PGIC) at Week 4, and time to next migraine. Headache pain is collected on a 4-point scale with 3 being severe, 2 being moderate, 1 being mild, and 0 being no pain. Pain freedom is no pain (0) with the absence of rescue medication (note that in the trial rescue medication is not to be used for 2 hours post completion of infusion in order to separate the effects of the antibody from the rescue medication, however, in the course of normal use, rescue medication optionally may be used; any use of rescue medication is collected as data).

Statistical analysis is performed to determine significance of the difference in endpoints between patients receiving Ab6 or placebo, including the time to pain freedom and time to absence of most bothersome symptom, and each of the other aforementioned endpoints.

Use of rescue medication refers to any intervention (medical or device) provided to the subject to provide relief of migraine. In the study this should not be provided sooner than 2 hours following completion of the study drug administration in order to separate the effects of the antibody from the effects of said rescue medication, however, rescue medication is not contraindicated. The proportion of subjects requiring rescue medication use is summarized in the study. Acute rescue medication includes any medication to treat migraine or migraine associated symptoms, e.g., triptans, analgesics such as non-opioids or opioids/narcotics, acetaminophen, NSAIDS, combination medications such as EXCEDRIN® or EXCEDRIN MIGRAINE®, antiemetic medications, ergotamines, ergot derivatives, etc.

Absence of Migraine-Associated Symptoms (Photophobia, Phonophobia and Nausea) refers to the absence or presence of each of the aforementioned migraine-associated symptoms, as reported by the subject. The proportion of subjects absent the symptoms, with no administration of rescue medication, is summarized in the study.

Headache Impact Test (HIT-6) is assessed as the change from baseline of the total score, and is summarized and compared between treatment groups in the study.

Migraine Treatment Optimization Questionnaire-6 (mTOQ-6) is assessed as the change from baseline of the total score and is summarized and compared between the treatment groups in the study.

Time to Headache Pain Relief is assessed as the first time point post completion of infusion at which the subject reports relief of pain meaning their headache pain has gone from moderate or severe (2 or 3) to mild or no pain (1 or 0) with no administration of rescue medication.

Pain Relapse is assessed as the occurrence of headache of any severity within 48 hours of drug administration for a patient who has no headache pain (0) at 2 hours. The proportion of subjects with recurrence of headache pain of any severity is summarized in the study.

The study shows that Ab6 is effective and safe for acute migraine treatment.

Example 7

In the pivotal clinical studies the patients received Ab6 as 100 mg or 300 mg dosages, as described in Example 3. Including day -1 (post infusion of Ab6) in the statistical analysis shows that an apparent treatment effect is present immediately after infusion when the treatment effect is assessed (FIG. 50). In the Figure Day 0 is defined as the day of the infusion and Day -1 data represent the pre-infusion condition. A substantial decrease in the percentage of migraines from Day -1 (baseline, the day prior to infusion) to Day 0 is apparent. Moreover, the magnitude of the effect is greater with the 300 mg dosage than the 100 mg dosage, and both show a greater effect than the placebo group.

SEQUENCE LISTING

```

Sequence total quantity: 567
SEQ ID NO: 1          moltype = AA  length = 439
FEATURE              Location/Qualifiers
REGION              1..439
                    note = Engineered antibody sequence
source              1..439
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1
QSLLESGGRL VTPGTPLTLT CTVSGLDLSS YYMQWVRQAP GKGLEWIGVI GINDNTYYAS 60
WAKGRFTISR ASSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLLTVSSA STKGPSVPPL 120
APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVQLQSSG LYSLSVVTV 180
PSSSLGTQTY ICNVNHPKPN TKVDRKRVPEK SCDKTHTCPP CPAPELLGGP SVFLFPPKPK 240
DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
VKGFPYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM 420
HEALHNHYTQ KSLSLSPGK 439

SEQ ID NO: 2          moltype = AA  length = 109
FEATURE              Location/Qualifiers
REGION              1..109
                    note = Engineered antibody sequence
source              1..109
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 2
QSLLESGGRL VTPGTPLTLT CTVSGLDLSS YYMQWVRQAP GKGLEWIGVI GINDNTYYAS 60
WAKGRFTISR ASSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLLTVSS 109

```

-continued

SEQ ID NO: 3	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = Engineered antibody sequence	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 3		
QSLEESGGRL VTPGTPLTLT CTVSGLDLS		29
SEQ ID NO: 4	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 4		
SYMQ		5
SEQ ID NO: 5	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 5		
WVRQAPGKGL EWIG		14
SEQ ID NO: 6	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 6		
VIGINDNTYY ASWAKG		16
SEQ ID NO: 7	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
REGION	1..31	
	note = Engineered antibody sequence	
source	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
RFTISRASST TVDLKMTSLT TEDTATYFCA R		31
SEQ ID NO: 8	moltype = length =	
SEQUENCE: 8		
000		
SEQ ID NO: 9	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 9		
WGPGLTVTVS S		11
SEQ ID NO: 10	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS		60
GLYLSVVVTP VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDWLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGSEFFLY SKLTVDKSRW		300
QQGNVFCFSV MHEALHNHYT QKSLSLSPGK		330

-continued

```

SEQ ID NO: 11      moltype = DNA length = 1320
FEATURE           Location/Qualifiers
misc_feature       1..1320
                   note = Engineered antibody sequence
source            1..1320
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 11
cagtcgctgg aggagtccgg gggtcgctg gtcacgcctg ggacacccct gacactcacc 60
tgcacagctct ctggactcga cctcagtagc tactacatgc aatgggtccg ccaggctcca 120
gggaaggggc tgggaatggat cggagtcatt ggtattaatg ataacacata ctacgcgagc 180
tgggcgaaaag gccgattcac catctccaga gcctcgctga ccacgggtgga tctgaaaatg 240
accagtctga caaccgagga cacggccacc tatttctgtg ccagagggga catctggggc 300
ccaggcaccg tcgtcacctg ctcgagcgcc tccaccaagg gcccatcggt cttccccctg 360
gcaccctcct ccaagagcac ctctgggggc acacggggcc tgggctgcct ggtcaaggac 420
tacttccccg aaccgggtgac ggtgtcgtgg aactcaggcg cctcgaccag cggcgtgcac 480
accttccccg ctgtctctaca gtctctcagga ctctactccc tcacgacgct ggtgaccgtg 540
ccctccagca gcttggggcac ccagacctac atctgcaacg tgaatcaca gccagcaac 600
accaaggtgg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgccaccg 660
tgccagcaga ctgaactcct ggggggaccg tcagtcttcc tcttcccccc aaaacccaag 720
gacaccctca tgatctcccc gacccctgag gtcacatgag tgggtggtgga cgtgagccac 780
gaagaccctg aggtcaagtt caactggtac gtggacggcg tggaggtgca taatgccaa 840
acaaagccgc gggaggagca gtacgcccag acgtaccgtg tggtcagcgt cctcaccgtc 900
ctgcaccagg actggctgaa tggcaaggag tacaagtgca aggtctccaa caaagccctc 960
ccagccccca tcgagaaaaa catctccaaa gccaaagggc agccccgaga accacaggtg 1020
tacaccctgc ccccatcccc ggaggagatg accaagaacc aggtcagcct gacctgctg 1080
gtcaaaggct tctatccccg gcacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
aacaactaca agaccacgcc tcccgtgctg gactccgacg gctccttctt cctctacagc 1200
aagctcaccg tggacaagag caggtggcag cagggggaac tcttctcatg ctccgtgatg 1260
catgaggctc tgcacaacca ctacacgcag aagagcctct ccctgtctcc gggtaaatga 1320

SEQ ID NO: 12      moltype = DNA length = 327
FEATURE           Location/Qualifiers
misc_feature       1..327
                   note = Engineered antibody sequence
source            1..327
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 12
cagtcgctgg aggagtccgg gggtcgctg gtcacgcctg ggacacccct gacactcacc 60
tgcacagctct ctggactcga cctcagtagc tactacatgc aatgggtccg ccaggctcca 120
gggaaggggc tgggaatggat cggagtcatt ggtattaatg ataacacata ctacgcgagc 180
tgggcgaaaag gccgattcac catctccaga gcctcgctga ccacgggtgga tctgaaaatg 240
accagtctga caaccgagga cacggccacc tatttctgtg ccagagggga catctggggc 300
ccaggcaccg tcgtcacctg ctcgagcgc tccaccaagg gcccatcggt cttccccctg 327

SEQ ID NO: 13      moltype = DNA length = 87
FEATURE           Location/Qualifiers
misc_feature       1..87
                   note = Engineered antibody sequence
source            1..87
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 13
cagtcgctgg aggagtccgg gggtcgctg gtcacgcctg ggacacccct gacactcacc 60
tgcacagctct ctggactcga cctcagtagc tactacatgc aatgggtccg ccaggctcca 87

SEQ ID NO: 14      moltype = DNA length = 15
FEATURE           Location/Qualifiers
misc_feature       1..15
                   note = Engineered antibody sequence
source            1..15
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 14
agctactaca tgcaa 15

SEQ ID NO: 15      moltype = DNA length = 42
FEATURE           Location/Qualifiers
misc_feature       1..42
                   note = Engineered antibody sequence
source            1..42
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 15
tgggtccgcc aggtccagg gaaggggctg gaatggatcg ga 42

SEQ ID NO: 16      moltype = DNA length = 48

```


-continued

FEATURE	Location/Qualifiers	
misc_feature	1..48	
	note = Engineered antibody sequence	
source	1..48	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 16		
gtcatttgta ttaatgataa cacatactac gcgagctggg cgaaaggc		48
SEQ ID NO: 17	moltype = DNA length = 93	
FEATURE	Location/Qualifiers	
misc_feature	1..93	
	note = Engineered antibody sequence	
source	1..93	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 17		
cgattcacca tctccagagc ctgctcgacc acggtggatc tgaaatgac cagtctgaca		60
accgaggaca cgccaccta ttctgtgcc aga		93
SEQ ID NO: 18	moltype = length =	
SEQUENCE: 18		
000		
SEQ ID NO: 19	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Engineered antibody sequence	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 19		
tggggcccag gcaccctcgt caccgtctcg agc		33
SEQ ID NO: 20	moltype = DNA length = 993	
FEATURE	Location/Qualifiers	
misc_feature	1..993	
	note = Engineered antibody sequence	
source	1..993	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 20		
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg		60
ggcacagcgg cctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg		120
tggaactcag gcgcctgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca		180
ggactctact ccctcagcag cgtggtgacc gtgcctccca gcagcttggg caccagacc		240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc		300
aaatcttgtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga		360
ccgtcagtc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct		420
gaggtcacat gcgtggtggt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg		480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgaggagga gcagtacgcc		540
agcacgtacc gtgtggtcag cgtctcacc gtctgcacc aggactggct gaatggcaag		600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccacgcagaa aaccatctcc		660
aaagccaaag ggcagcccgc agaaccacag gtgtacaccc tgccccatc ccgggaggag		720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc		780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg		840
ctggactccg acggtcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg		900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg		960
cagaagagcc tctccctgtc tccgggtaaa tga		993
SEQ ID NO: 21	moltype = AA length = 219	
FEATURE	Location/Qualifiers	
REGION	1..219	
	note = Engineered antibody sequence	
source	1..219	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 21		
QVLTQTASPV SAAVGSTVTI NCQASQSVYD NNYLAWYQQK PGQPPKQLIY STSTLASGVS		60
SRFKGSGSGT QFTLTISDLE CADAATYYCL GSYDCSSGDC FVFGGGTEVV VKRTVAAPSV		120
FIFPPSDEQL KSGTASVVL LNMFYPREK VQWKVDNALQ SGNQSQESVTE QDSKDSTYSL		180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC		219
SEQ ID NO: 22	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
REGION	1..113	
	note = Engineered antibody sequence	
source	1..113	
	mol_type = protein	

organism = synthetic construct	
SEQUENCE: 22	
QVLTQTASPV SAAVGSTVTI	NCQASQSVYD NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDLE	CADAATYYCL GSYDCSSGDC FVFGGGTEVV VKR 113
SEQ ID NO: 23	moltype = AA length = 22
FEATURE	Location/Qualifiers
REGION	1..22
	note = Engineered antibody sequence
source	1..22
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 23	
QVLTQTASPV SAAVGSTVTI	NC 22
SEQ ID NO: 24	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 24	
QASQSVYDNN YLA	13
SEQ ID NO: 25	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
	note = Engineered antibody sequence
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 25	
WYQQKPGQPP KQLIY	15
SEQ ID NO: 26	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = Engineered antibody sequence
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 26	
STSTLAS	7
SEQ ID NO: 27	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 27	
GVSSRFKSGSGTQFTLTIS	DLECAATA YC 32
SEQ ID NO: 28	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 28	
LGSYDCSSGD CFV	13
SEQ ID NO: 29	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 29	
FGGGTEVVVK R	11
SEQ ID NO: 30	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
	note = Engineered antibody sequence

-continued

```

source                1..106
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 30
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREKVQW KVDNALQSGN SQESVTEQDS 60
KDSTYLSST LTLSKADYK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 31         moltype = DNA length = 660
FEATURE               Location/Qualifiers
misc_feature          1..660
                      note = Engineered antibody sequence
source               1..660
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 31
caagtgtga cccagactgc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcgggttca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacctggag 240
tgtgccgatg ctgccactta ctactgtcta ggcagttatg attgtagtag tgggtattgt 300
tttgttttcg gcggaggggac cgaggtggtg gtcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatccagg agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 32         moltype = DNA length = 339
FEATURE               Location/Qualifiers
misc_feature          1..339
                      note = Engineered antibody sequence
source               1..339
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 32
caagtgtga cccagactgc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcgggttca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacctggag 240
tgtgccgatg ctgccactta ctactgtcta ggcagttatg attgtagtag tgggtattgt 300
tttgttttcg gcggaggggac cgaggtggtg gtcaaacgt 339

SEQ ID NO: 33         moltype = DNA length = 66
FEATURE               Location/Qualifiers
misc_feature          1..66
                      note = Engineered antibody sequence
source               1..66
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 33
caagtgtga cccagactgc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

SEQ ID NO: 34         moltype = DNA length = 39
FEATURE               Location/Qualifiers
misc_feature          1..39
                      note = Engineered antibody sequence
source               1..39
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 34
caggccagtc agagtgttta tgataacaac tacctagcc 39

SEQ ID NO: 35         moltype = DNA length = 45
FEATURE               Location/Qualifiers
misc_feature          1..45
                      note = Engineered antibody sequence
source               1..45
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 35
tggatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 36         moltype = DNA length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = Engineered antibody sequence
source               1..21
                      mol_type = other DNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 36
tctacatcca ctctggcatc t                                     21

SEQ ID NO: 37          moltype = DNA length = 96
FEATURE              Location/Qualifiers
misc_feature          1..96
                        note = Engineered antibody sequence
source               1..96
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 37
ggggtctcat cgcggttcaa aggcagtgga tctgggacac agttcactct caccatcagc 60
gacctggagt gtgcgatgc tgccacttac tactgt                               96

SEQ ID NO: 38          moltype = DNA length = 39
FEATURE              Location/Qualifiers
misc_feature          1..39
                        note = Engineered antibody sequence
source               1..39
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 38
ctaggcagtt atgattgtag tagtggtgat tgttttgtt                 39

SEQ ID NO: 39          moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                        note = Engineered antibody sequence
source               1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 39
ttcggcggag ggaccgaggt ggtggtcaaa cgt                       33

SEQ ID NO: 40          moltype = DNA length = 321
FEATURE              Location/Qualifiers
misc_feature          1..321
                        note = Engineered antibody sequence
source               1..321
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 40
acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgcct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtggata acgcctccca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g                                           321

SEQ ID NO: 41          moltype = AA length = 441
FEATURE              Location/Qualifiers
REGION              1..441
                        note = Engineered antibody sequence
source             1..441
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 41
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVGV IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSVV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDKRVK PKSCDKHTHC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAEWESNGQ PENNYKTPP VLDSDGSFFL YSKLTVDKSR WQGNVFSCS 420
VMHEALHNYH TQKSLSLSPG K                                           441

SEQ ID NO: 42          moltype = AA length = 111
FEATURE              Location/Qualifiers
REGION              1..111
                        note = Engineered antibody sequence
source             1..111
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 42
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVGV IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS S          111

SEQ ID NO: 43          moltype = AA length = 30

```

-continued

FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Engineered antibody sequence	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 43		
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS		30
SEQ ID NO: 44	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 44		
SYMQ		5
SEQ ID NO: 45	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 45		
WVRQAPGKGL EWVG		14
SEQ ID NO: 46	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 46		
VIGINDNTYY ASWAKG		16
SEQ ID NO: 47	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 47		
RFTISRDNK TTVYLQMSL RAEDTAVYFC AR		32
SEQ ID NO: 48	moltype = length =	
SEQUENCE: 48		
000		
SEQ ID NO: 49	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 49		
WGQGLTVTVS S		11
SEQ ID NO: 50	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 50		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 51	moltype = DNA length = 1326	

-continued

```

FEATURE                Location/Qualifiers
misc_feature            1..1326
                        note = Engineered antibody sequence
source                  1..1326
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 51
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctctcg caccgtctcg agcgcctcca ccaaggggccc atcggtcttc 360
cccttggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcctgggtc 420
aaggactact tccccgaacc ggtgacgggt tctgtggaact caggcgccct gaccacgggc 480
gtgcacacct tcccgggtgt cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgccct ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctcttt cccccaaaa 720
cccaaggaca cctcctatgat ctcccggacc cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag acctgtaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaaagaaa agccgcggga ggagcagtag gccagcagc accgtgtggt cagcgtcttc 900
accgtcctgc accagagctg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gccctcccag ccccatcgga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca cctggccccc atccggggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttgggtc aaggcttcta tcccagcgac atcgccgtgg agtggggagag caatggggcag 1140
ccggagaaca actacaagac caccgctccc gtgctggact ccgacggctc cttcttcttc 1200
tacagcaagc tcaccgtgga caagacgagg tggcagcagg ggaacgtctt ctcattgtcc 1260
gtgatgcatt aggcctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga                                     1326

SEQ ID NO: 52          moltype = DNA length = 333
FEATURE                Location/Qualifiers
misc_feature            1..333
                        note = Engineered antibody sequence
source                  1..333
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 52
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctctcg caccgtctcg agcgcctcca ccaaggggccc atcggtcttc 333

SEQ ID NO: 53          moltype = DNA length = 90
FEATURE                Location/Qualifiers
misc_feature            1..90
                        note = Engineered antibody sequence
source                  1..90
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 53
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 90

SEQ ID NO: 54          moltype = DNA length = 15
FEATURE                Location/Qualifiers
misc_feature            1..15
                        note = Engineered antibody sequence
source                  1..15
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 54
agctactaca tgcaa                                     15

SEQ ID NO: 55          moltype = DNA length = 42
FEATURE                Location/Qualifiers
misc_feature            1..42
                        note = Engineered antibody sequence
source                  1..42
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 55
tgggtccgtc aggtccagg gaaggggctg gagggggtcg ga                                     42

SEQ ID NO: 56          moltype = DNA length = 48
FEATURE                Location/Qualifiers

```

-continued

```

misc_feature      1..48
                  note = Engineered antibody sequence
source            1..48
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 56
gtcattggta tcaatgataa cacatactac gcgagctggg cgaaaggc      48

SEQ ID NO: 57      moltype = DNA length = 96
FEATURE           Location/Qualifiers
misc_feature      1..96
                  note = Engineered antibody sequence
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 57
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt gctaga                          96

SEQ ID NO: 58      moltype = length =
SEQUENCE: 58
000

SEQ ID NO: 59      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Engineered antibody sequence
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 59
tggggccaag ggaccctcgt caccgtctcg agc                          33

SEQ ID NO: 60      moltype = DNA length = 993
FEATURE           Location/Qualifiers
misc_feature      1..993
                  note = Engineered antibody sequence
source            1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 60
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctggg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc cgaaccggg gacgggtgctg 120
tggaaactcag gcgcctcgac cagcggcgctg cacaccttcc cggtgtcct acagtctctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgccca ccgtgcccag cactgaact cctgggggga 360
ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgagggtcaa gttcaactgg 480
tacgtggacg cgtgggaggt gcataatgcc aagacaaaag cgcggggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctcgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccatcgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctccgtg 840
ctggactccg acggtctctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtcttctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                          993

SEQ ID NO: 61      moltype = AA length = 219
FEATURE           Location/Qualifiers
REGION           1..219
                  note = Engineered antibody sequence
source            1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 61
QVLTQSPSSL SASVGDRVIT NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                      219

SEQ ID NO: 62      moltype = AA length = 113
FEATURE           Location/Qualifiers
REGION           1..113
                  note = Engineered antibody sequence
source            1..113
                  mol_type = protein
                  organism = synthetic construct

```

-continued

SEQUENCE: 62
 QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
 SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGGTKVE IKR 113

SEQ ID NO: 63 moltype = AA length = 22
 FEATURE Location/Qualifiers
 REGION 1..22
 note = Engineered antibody sequence
 source 1..22
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 63
 QVLTQSPSSL SASVGDRVTI NC 22

SEQ ID NO: 64 moltype = AA length = 13
 FEATURE Location/Qualifiers
 REGION 1..13
 note = Engineered antibody sequence
 source 1..13
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 64
 QASQSVYDNN YLA 13

SEQ ID NO: 65 moltype = AA length = 15
 FEATURE Location/Qualifiers
 REGION 1..15
 note = Engineered antibody sequence
 source 1..15
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 65
 WYQQKPGKVP KQLIY 15

SEQ ID NO: 66 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = Engineered antibody sequence
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 66
 STSTLAS 7

SEQ ID NO: 67 moltype = AA length = 32
 FEATURE Location/Qualifiers
 REGION 1..32
 note = Engineered antibody sequence
 source 1..32
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 67
 GVPSRFSGSG SGTDFLTIS SLQPEDVATY YC 32

SEQ ID NO: 68 moltype = AA length = 13
 FEATURE Location/Qualifiers
 REGION 1..13
 note = Engineered antibody sequence
 source 1..13
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 68
 LGSYDCSSGD CFV 13

SEQ ID NO: 69 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Engineered antibody sequence
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 69
 FGGGTKVEIK R 11

SEQ ID NO: 70 moltype = AA length = 106
 FEATURE Location/Qualifiers
 REGION 1..106
 note = Engineered antibody sequence
 source 1..106

-continued

```

mol_type = protein
organism = synthetic construct

SEQUENCE: 70
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
KDSTYLSLST LTLSKADY EK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 71      moltype = DNA length = 660
FEATURE            Location/Qualifiers
misc_feature        1..660
                    note = Engineered antibody sequence
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 71
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggaagag ttctaagca actgatctat tctacatcca ctctggcatc tggggtocca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tgggtgattgt 300
tttgttttcg gcggaggaa caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctgc 420
ctgaataact tctatccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggttaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 72      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 72
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggaagag ttctaagca actgatctat tctacatcca ctctggcatc tggggtocca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tgggtgattgt 300
tttgttttcg gcggaggaa caaggtggaa atcaaacgt 339

SEQ ID NO: 73      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 73
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 74      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 74
caggccagtc agagtgttta tgataacaac tacctagcc 39

SEQ ID NO: 75      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 75
tggtatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 76      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

-continued

SEQUENCE: 76
tctacatcca ctctggcatc t 21

SEQ ID NO: 77 moltype = DNA length = 96
FEATURE Location/Qualifiers
misc_feature 1..96
 note = Engineered antibody sequence
source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 77
gggggcccat ctcgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 78 moltype = DNA length = 39
FEATURE Location/Qualifiers
misc_feature 1..39
 note = Engineered antibody sequence
source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 78
ctaggcagtt atgattgtag tagtggtgat tgttttgtt 39

SEQ ID NO: 79 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = Engineered antibody sequence
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 79
ttcggcggag gaaccaaggt ggaatcaaaa cgt 33

SEQ ID NO: 80 moltype = DNA length = 321
FEATURE Location/Qualifiers
misc_feature 1..321
 note = Engineered antibody sequence
source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 80
acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgcct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtgata acgcctccca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccggt cacaaagagc 300
ttcaacaggg gagagtgtta g 321

SEQ ID NO: 81 moltype = AA length = 441
FEATURE Location/Qualifiers
REGION 1..441
 note = Engineered antibody sequence
source 1..441
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 81
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVG V IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGT LVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHKP SNTKVDARVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQQGNVFS 420
VMHEALHNHY TQKSLSLSPG K 441

SEQ ID NO: 82 moltype = AA length = 111
FEATURE Location/Qualifiers
REGION 1..111
 note = Engineered antibody sequence
source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 82
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS SYMQWVRQA PGKGLEWVG V IGINDNTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGT LVTVS S 111

SEQ ID NO: 83 moltype = AA length = 30
FEATURE Location/Qualifiers

-continued

REGION	1..30	
	note = Engineered antibody sequence	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 83		
EVQLVESGGG LVQPGGSLRL SCAVSGLDLS		30
SEQ ID NO: 84	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 84		
SYMQ		5
SEQ ID NO: 85	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 85		
WVRQAPGKGL EWVG		14
SEQ ID NO: 86	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 86		
VIGINDNTYY ASWAKG		16
SEQ ID NO: 87	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 87		
RFTISRDNK TTVYLQMSL RAEDTAVYFC AR		32
SEQ ID NO: 88	moltype = length =	
SEQUENCE: 88		
000		
SEQ ID NO: 89	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 89		
WGQGLTVTS S		11
SEQ ID NO: 90	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 90		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS		60
GLYSLSVVTP VPSSSLGTQT YICNVNHKPS NTKVDARVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 91	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	

-continued

```

misc_feature      1..1326
                  note = Engineered antibody sequence
source            1..1326
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 91
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgcctcca ccaagggccc atcgggtcttc 360
cccttggcac cctcctccaa gacacactct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tcccgaacc ggtgacggtg tctgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtggtg 540
accgtgcctc ccagcagctt gggcaccagc acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggagcg gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa 720
cccaaggaca cctcatgat cctccggacc cctgaggtea catgcgtggt ggtggacgtg 780
agccacgaag acctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtctc 900
accgtctgcg accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gccctcccag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca ccttgcctcc atcccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aagcttctta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttctctctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcattgctc 1260
gtgatgcatt aggtctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

SEQ ID NO: 92      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature       1..333
                  note = Engineered antibody sequence
source             1..333
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 92
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatca atgataacac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 93      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature       1..90
                  note = Engineered antibody sequence
source             1..90
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 93
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggact cgacctcagt 90

SEQ ID NO: 94      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature       1..15
                  note = Engineered antibody sequence
source             1..15
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 94
agctactaca tgcaa 15

SEQ ID NO: 95      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature       1..42
                  note = Engineered antibody sequence
source             1..42
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 95
tgggtccgtc aggtctcagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 96      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature       1..48

```

-continued

```

source          note = Engineered antibody sequence
                1..48
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 96
gtcattggta tcaatgataa cacatactac gcgagctggg cgaaaggc      48

SEQ ID NO: 97      moltype = DNA length = 96
FEATURE          Location/Qualifiers
misc_feature      1..96
                  note = Engineered antibody sequence
source          1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 97
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt gctaga                          96

SEQ ID NO: 98      moltype = length =
SEQUENCE: 98
000

SEQ ID NO: 99      moltype = DNA length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Engineered antibody sequence
source          1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 99
tggggccaag ggaccctcgt caccgtctcg agc                          33

SEQ ID NO: 100     moltype = DNA length = 993
FEATURE          Location/Qualifiers
misc_feature      1..993
                  note = Engineered antibody sequence
source          1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 100
gcctccacca aggggccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtctctca 180
ggactctact cctccagcag cgtgggtgacc gtgcctctca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacgcgag agttgagccc 300
aaatcttggt acaaaaactca cacatgccca ccgtgccca cactgaaact cctgggggga 360
ccgtcagtc tctcttctcc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag gctctcacc gtcctgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtct caacaaagcc ctcccagccc ccacgcagaa aacctctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggtctctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgct tccgggtaaa tga                          993

SEQ ID NO: 101     moltype = AA length = 219
FEATURE          Location/Qualifiers
REGION          1..219
                  note = Engineered antibody sequence
source          1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 101
QLVTQSPSSL SASVGDRTVI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSL 180
SSTLTLSKAD YEKKHVYACE VTHQGLSSPV TKSPNRGEC                      219

SEQ ID NO: 102     moltype = AA length = 113
FEATURE          Location/Qualifiers
REGION          1..113
                  note = Engineered antibody sequence
source          1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 102

```

-continued

QVLTQSPSSL SASVGDRVTI NCQASQSVYD NNYLAWYQQK PGKVPKQLIY STSTLASGVP	60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSSGDC FVFGGGTKVE IKR	113
SEQ ID NO: 103	moltype = AA length = 22
FEATURE	Location/Qualifiers
REGION	1..22
	note = Engineered antibody sequence
source	1..22
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 103	
QVLTQSPSSL SASVGDRVTI NC	22
SEQ ID NO: 104	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 104	
QASQSVYDNN YLA	13
SEQ ID NO: 105	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
	note = Engineered antibody sequence
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 105	
WYQQKPGKVP KQLIY	15
SEQ ID NO: 106	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = Engineered antibody sequence
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 106	
STSTLAS	7
SEQ ID NO: 107	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 107	
GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC	32
SEQ ID NO: 108	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 108	
LGSYDCSSGD CFV	13
SEQ ID NO: 109	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 109	
FGGGTKVEIK R	11
SEQ ID NO: 110	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
	note = Engineered antibody sequence
source	1..106
	mol_type = protein

-continued

```

        organism = synthetic construct
SEQUENCE: 110
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
KDSTYLSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 111      moltype = DNA length = 660
FEATURE            Location/Qualifiers
misc_feature        1..660
                    note = Engineered antibody sequence
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 111
caagtgtcga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tggtgattgt 300
ttgttttctg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatccagg agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccc tgacgtctg caaagcagac tacgagaaac acaaaagtcta cgctgcgaa 600
gtcaccatcc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 112      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 112
caagtgtcga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatgat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtcta ggcagttatg attgtagtag tggtgattgt 300
ttgttttctg gcggaggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 113      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 113
caagtgtcga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 114      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 114
caggccagtc agagtgttta tgataacaac tacctagcc 39

SEQ ID NO: 115      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 115
tggtatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 116      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 116

```

-continued

tctacatcca ctctggcatc t 21

SEQ ID NO: 117 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Engineered antibody sequence
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 117
 ggggtcccat ctcgtttcag tggcagtga tctgggacag atttcaactct caccatcagc 60
 agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 118 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 118
 ctaggcagtt atgattgtag tagtggtgat tgttttgtt 39

SEQ ID NO: 119 moltype = DNA length = 33
 FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 119
 ttcggcggag gaaccaaggt ggaatcaaa cgt 33

SEQ ID NO: 120 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 120
 acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgect gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aaggtggata acgcccctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
 cacaagctct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
 ttcaacaggg gagagtgtta g 321

SEQ ID NO: 121 moltype = AA length = 439
 FEATURE Location/Qualifiers
 REGION 1..439
 note = Engineered antibody sequence
 source 1..439
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 121
 QSLEESGGRL VTPGTPLTLT CSVSGIDLSG YMMNWVRQAP GKGLEWIGVI GINGATYYAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSSA STKGPSVFPL 120
 APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG LYSLSVVTV 180
 PSSSLGTQTY IICNVNHKPSN TKVDKRVPEK SCDKTHTCP CPAPPELLGGP SVFLFPPKPK 240
 DTLMISRTPE VTCVVVDVSH EDPEVKENWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
 LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
 VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ QGNVFSQCSVM 420
 HEALHNHYTQ KSLSLSPGK 439

SEQ ID NO: 122 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Engineered antibody sequence
 source 1..109
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 122
 QSLEESGGRL VTPGTPLTLT CSVSGIDLSG YMMNWVRQAP GKGLEWIGVI GINGATYYAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGTLVTVSS 109

SEQ ID NO: 123 moltype = AA length = 29
 FEATURE Location/Qualifiers
 REGION 1..29

-continued

source	note = Engineered antibody sequence 1..29 mol_type = protein organism = synthetic construct	
SEQUENCE: 123		
QSLEESGGRL VTPGTPLTLT CSVSGIDLS		29
SEQ ID NO: 124	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
source	note = Engineered antibody sequence 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 124		
GYVMN		5
SEQ ID NO: 125	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
source	note = Engineered antibody sequence 1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 125		
WVRQAPGKGL EWIG		14
SEQ ID NO: 126	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
source	note = Engineered antibody sequence 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 126		
VIGINGATYY ASWAKG		16
SEQ ID NO: 127	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
REGION	1..31	
source	note = Engineered antibody sequence 1..31 mol_type = protein organism = synthetic construct	
SEQUENCE: 127		
RFTISKTSST TVDLKMTSLT TEDTATYFCA R		31
SEQ ID NO: 128	moltype = length =	
SEQUENCE: 128		
000		
SEQ ID NO: 129	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
source	note = Engineered antibody sequence 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 129		
WGPGLVTVS S		11
SEQ ID NO: 130	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
source	note = Engineered antibody sequence 1..330 mol_type = protein organism = synthetic construct	
SEQUENCE: 130		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDRKVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGSEFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 131	moltype = DNA length = 1320	
FEATURE	Location/Qualifiers	
misc_feature	1..1320	

-continued

```

source          note = Engineered antibody sequence
                1..1320
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 131
cagtcgctgg aggagtcggg gggtcgcctg gtcacgcctg ggacaccctt gacactcacc 60
tggtccgtct ctggcatcga cctcagtggc tactacatga actgggtccg ccagggtcca 120
gggaaggggc tgggaatggat cggagtcatt ggtattaatg gtgccacata ctacgcgagc 180
tgggcgaaaag gccgattcac catctccaaa acctcgtcga ccacggtgga tctgaaaatg 240
accagtctga caaccgagga caccggccacc tatttctgtg ccagagggga catctggggc 300
ccgggcaccc tcgtcacctg ctcgagcgcc tccaccaagg gcccatcggt cttcccccgt 360
gcaccctcct ccaagagcac ctctgggggc acagcggccc tgggtcgctt ggtcaaggac 420
tacttccccg aaccggtgac ggtgtcgtgg aactcaggcg ccctgaccag cggcgtgcac 480
accttccccg ctgtcctaca gtctcaggga ctctactccc tcagcagcgt ggtgaccgtg 540
ccctccagca gcttggggcac ccagacctac atctgcaacg tgaatcaca gccacgcaac 600
accaaggtgg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgccaccg 660
tgcccagcac ctgaactcct ggggggaccg tcagtcttcc tcttcccccc aaaacccaag 720
gacaccctca tgatctcccg gacccctgag gtcacatcg tgggtggtgga cgtgagccac 780
gaagaccctg aggtcaagtt caactggtag gtggacggcg tggaggtgca taatgccaa 840
acaaagccgc gggaggagca gtacgcacgc acgtaccgtg tggtcagcgt cctcaccgtc 900
ctgcaccagg actggctgaa tggcaaggag tacaagtga aggtctccaa caaagccctc 960
ccagcccca ctcagaaaac catctccaaa gccaaagggc agccccgaga accacagggt 1020
tacaccctgc ccccatcccg ggaggagatg accaagaacc aggtcagcct gacctgcctg 1080
gtcaaaggct tctatcccg cgacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
aacaactaca agaccacgcc tcccggtgct gactccgacg gctccttctt cctctacagc 1200
aagctcaccg tggacaagag caggtggcag caggggaacg tcttctcatg ctccgtgatg 1260
catgaggctc tgcacaacca ctacacgcag aagagcctct ccctgtctcc gggtaaatga 1320

SEQ ID NO: 132      moltype = DNA length = 327
FEATURE            Location/Qualifiers
misc_feature        1..327
                    note = Engineered antibody sequence
source              1..327
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 132
cagtcgctgg aggagtcggg gggtcgcctg gtcacgcctg ggacaccctt gacactcacc 60
tggtccgtct ctggcatcga cctcagtggc tactacatga actgggtccg ccagggtcca 120
gggaaggggc tgggaatggat cggagtcatt ggtattaatg gtgccacata ctacgcgagc 180
tgggcgaaaag gccgattcac catctccaaa acctcgtcga ccacggtgga tctgaaaatg 240
accagtctga caaccgagga caccggccacc tatttctgtg ccagagggga catctggggc 300
ccgggcaccc tcgtcacctg ctcgagcgc tccaccaagg gcccatcggt cttcccccgt 360

SEQ ID NO: 133      moltype = DNA length = 87
FEATURE            Location/Qualifiers
misc_feature        1..87
                    note = Engineered antibody sequence
source              1..87
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 133
cagtcgctgg aggagtcggg gggtcgcctg gtcacgcctg ggacaccctt gacactcacc 60
tggtccgtct ctggcatcga cctcagtggc tactacatga actgggtccg ccagggtcca 87

SEQ ID NO: 134      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 134
ggctactaca tgaac 15

SEQ ID NO: 135      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 135
tggttcggcc aggtccagg gaaggggctg gaatggatcg ga 42

SEQ ID NO: 136      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48

```

-continued

```

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 136
gtcattggta ttaatgggtgc cacatactac gcgagctggg cgaaaggc 48

SEQ ID NO: 137      moltype = DNA length = 93
FEATURE            Location/Qualifiers
misc_feature        1..93
                    note = Engineered antibody sequence
source              1..93
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 137
cgattcacca tctccaaac ctcgtcgacc acggtggatc tgaaaatgac cagtctgaca 60
accgaggaca cgccacaccta tttctgtgcc aga 93

SEQ ID NO: 138      moltype = length =
SEQUENCE: 138
000

SEQ ID NO: 139      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 139
tggggcccg gcaccctcgt caccgtctcg agc 33

SEQ ID NO: 140      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 140
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg cctgggctg cctgggtcaag gactacttcc ccgaaccggg gacggtgtcg 120
tggaaactcag gcgcctgac cagcggcgtg cacaccttcc cggetgtcct acagtctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgccca ccgtgcccag cacctgaact cctgggggga 360
ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag ccggggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtctcacc gtctcgacc aggactgggt gaatggcaag 600
gagtacaagt gcaaggctct caacaaagcc ctcccagccc ccacgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggctcag ctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtctctt ctctctctac agcaagtca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 141      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION            1..219
                    note = Engineered antibody sequence
source            1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 141
QVLTQTPSPV SAAVGSTVTI NCQASQSVYH NTYLAWYQQK PGQPPKQLIY DASTLASGVP 60
SRFSGSGSGT QFTLTISGVQ CNDAAAYYCL GSYDCTNGDC FVFGGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC 219

SEQ ID NO: 142      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION            1..113
                    note = Engineered antibody sequence
source            1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 142
QVLTQTPSPV SAAVGSTVTI NCQASQSVYH NTYLAWYQQK PGQPPKQLIY DASTLASGVP 60
SRFSGSGSGT QFTLTISGVQ CNDAAAYYCL GSYDCTNGDC FVFGGGTEVV VKR 113

```

-continued

SEQ ID NO: 143	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Engineered antibody sequence	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 143		
QVLTQTPSPV SAAVGSTVTI NC		22
SEQ ID NO: 144	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 144		
QASQSVYHNT YLA		13
SEQ ID NO: 145	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 145		
WYQQKPGQPP KQLIY		15
SEQ ID NO: 146	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 146		
DASTLAS		7
SEQ ID NO: 147	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 147		
GVPSRFGSG SGTQFTLTIS GVQCNDAAAY YC		32
SEQ ID NO: 148	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 148		
LGSYDCTNGD CFV		13
SEQ ID NO: 149	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 149		
FGGGTEVVVK R		11
SEQ ID NO: 150	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 150		

-continued

TVAAPSVFIF	PPSDEQLKSG	TASVVCLLNN	FYPREAKVQW	KVDNALQSGN	SQESVTEQDS	60
KDSTYLSST	LTLSKADYEK	HKVYACEVTH	QGLSSPVTKS	FNRGEC		106

```

SEQ ID NO: 151      moltype = DNA  length = 660
FEATURE             Location/Qualifiers
misc_feature        1..660
                    note = Engineered antibody sequence
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 151
caagtgcgtga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa 120
ccagggcagc ctcccaaaaca actgatctat gatgcatcca ctctggcgctc tgggggtccca 180
tcgcggttca gcggcagtggt atctggggaca cagttcactc tcaccatcag cggcgtgcag 240
tgtaacgatg ctgccgctta ctactgtctg ggcagttatg attgtactaa tgggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga cgggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctgt 420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

```

```

SEQ ID NO: 152      moltype = DNA  length = 339
FEATURE             Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 152
caagtgcgtga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa 120
ccagggcagc ctcccaaaaca actgatctat gatgcatcca ctctggcgctc tgggggtccca 180
tcgcggttca gcggcagtggt atctggggaca cagttcactc tcaccatcag cggcgtgcag 240
tgtaacgatg ctgccgctta ctactgtctg ggcagttatg attgtactaa tgggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga 339

```

```

SEQ ID NO: 153      moltype = DNA  length = 66
FEATURE             Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 153
caagtgcgtga cccagactcc atcccccggtg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

```

```

SEQ ID NO: 154      moltype = DNA  length = 39
FEATURE             Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 154
caggccagtc agagtgttta tcataacacc tacctggcc 39

```

```

SEQ ID NO: 155      moltype = DNA  length = 45
FEATURE             Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 155
tggatcagc agaaaccagg gcagcctccc aaacaactga tctat 45

```

```

SEQ ID NO: 156      moltype = DNA  length = 21
FEATURE             Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 156
gatgcatcca ctctggcgctc t 21

```

-continued

SEQ ID NO: 157 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Engineered antibody sequence
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 157
 ggggtcccat cgcgggttcag cggcagtgga tctgggacac agttcactct caccatcagc 60
 ggcgtgcagt gtaacgatgc tgccgcttac tactgt 96

SEQ ID NO: 158 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 158
 ctgggcagtt atgattgtac taatggtgat tgttttgtt 39

SEQ ID NO: 159 moltype = DNA length = 33
 FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 159
 ttcggcggag ggaccgaggt ggtggtcaaa cgt 33

SEQ ID NO: 160 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 160
 acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aaggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
 cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagagc 300
 tcaacaggag gagagtgtta g 321

SEQ ID NO: 161 moltype = AA length = 441
 FEATURE Location/Qualifiers
 REGION 1..441
 note = Engineered antibody sequence
 source 1..441
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 161
 EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IGINGATYYA 60
 SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS SASTKGPSVF 120
 PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSVV 180
 TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
 PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
 TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
 CLVKGFPYPSD IAVEWESNGQ PENNYKTPPP VLDSDGSFPL YSKLTVDKSR WQQGNVFSKS 420
 VMHEALHNHY TQKSLSLSPG K 441

SEQ ID NO: 162 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = Engineered antibody sequence
 source 1..111
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 162
 EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IGINGATYYA 60
 SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS S 111

SEQ ID NO: 163 moltype = AA length = 30
 FEATURE Location/Qualifiers
 REGION 1..30
 note = Engineered antibody sequence
 source 1..30

-continued

	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 163		
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS		30
SEQ ID NO: 164	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 164		
GYMN		5
SEQ ID NO: 165	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 165		
WVRQAPGKGL EWVG		14
SEQ ID NO: 166	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 166		
VIGINGATYY ASWAKG		16
SEQ ID NO: 167	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 167		
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR		32
SEQ ID NO: 168	moltype = length =	
SEQUENCE: 168		
000		
SEQ ID NO: 169	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 169		
WGQGLVTVS S		11
SEQ ID NO: 170	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 170		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 171	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	
misc_feature	1..1326	
	note = Engineered antibody sequence	
source	1..1326	

-continued

```

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 171
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccttgccac cctcctccaa gacacacctt gggggcacag cggccctggg ctgcttggtc 420
aaggactact tcccgaacc ggtgacgggt tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggtgtg cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgcctt ccagcagctt gggcaccagc acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtgggcaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa 720
cccaaggaca cctcatgat cctccggacc cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac ttggtacgtg acggcgtgga ggtgcataat 840
gccaagacaa agcgcgggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtctctc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcccctccag ccccocatga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca ccttgccccc atcccgagg gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aaggttctta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctctcc gtgctggact ccgacggctc cttctcttc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcattctcc 1260
gtgatgcatt aggtctctga caaccactac acgcagaaga gcctctcctt gtctccgggt 1320
aaatga 1326

SEQ ID NO: 172      moltype = DNA length = 333
FEATURE             Location/Qualifiers
misc_feature         1..333
                     note = Engineered antibody sequence
source               1..333
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 172
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 173      moltype = DNA length = 90
FEATURE             Location/Qualifiers
misc_feature         1..90
                     note = Engineered antibody sequence
source               1..90
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 173
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt 90

SEQ ID NO: 174      moltype = DNA length = 15
FEATURE             Location/Qualifiers
misc_feature         1..15
                     note = Engineered antibody sequence
source               1..15
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 174
ggctactaca tgaac 15

SEQ ID NO: 175      moltype = DNA length = 42
FEATURE             Location/Qualifiers
misc_feature         1..42
                     note = Engineered antibody sequence
source               1..42
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 175
tgggtccgtc aggtccagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 176      moltype = DNA length = 48
FEATURE             Location/Qualifiers
misc_feature         1..48
                     note = Engineered antibody sequence
source               1..48
                     mol_type = other DNA

```


-continued

```

                                organism = synthetic construct
SEQUENCE: 176
gtcatttgta ttaatgggtc cacatactac gcgagctggg cgaaaggc      48

SEQ ID NO: 177      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 177
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg  60
agagctgagg acaactgctg gtatttctgt gctaga                          96

SEQ ID NO: 178      moltype = length =
SEQUENCE: 178
000

SEQ ID NO: 179      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 179
tggggccaag ggaccctcgt caccgtctcg agc                          33

SEQ ID NO: 180      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 180
gcctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg  60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaactcag gcgcctgac cagcggcggtg cacaccttcc cggctgtcct acagtctctca 180
ggactctact ccctcagcag cgtggtgacc gtgcccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttggtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctggggggga 360
ccgtcagtcct tcctcttccc cccaaaaacc aaggacaccc tcatgatctc ccggaccctc 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtaacaagt gcaagggtct caacaaagcc ctcccagccc ccacgagaaa aaccatctcc 660
aaagccaaaag ggcagccccc agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcga tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtcttctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                          993

SEQ ID NO: 181      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 181
QVLTQSPSSL SASVGDRVIT NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP  60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSYL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC                      219

SEQ ID NO: 182      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 182
QVLTQSPSSL SASVGDRVIT NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP  60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGGTKVE IKR      113

```

-continued

SEQ ID NO: 183	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Engineered antibody sequence	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 183		
QVLTQSPSSL SASVGDRVTI NC		22
SEQ ID NO: 184	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 184		
QASQSVYHNT YLA		13
SEQ ID NO: 185	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 185		
WYQQKPGKVP KQLIY		15
SEQ ID NO: 186	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 186		
DASTLAS		7
SEQ ID NO: 187	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 187		
GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 188	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 188		
LGSYDCTNGD CFV		13
SEQ ID NO: 189	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 189		
FGGKVEIK R		11
SEQ ID NO: 190	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 190		
TVAAPSVFIF PPSDEQLKSG TASVVCCLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60

-continued

KDSTYLSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC 106

```

SEQ ID NO: 191      moltype = DNA  length = 660
FEATURE             Location/Qualifiers
misc_feature        1..660
                    note = Engineered antibody sequence
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 191
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat gatgcacca ctctggcatc tgggggccca 180
tctcgtttca gtggcagtg atctgggaca gatttcaact tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tggtgattgt 300
ttgttttcg gcggaggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagtg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagag tgctacagag caggacagca aggacagcac ctacagctc 540
agcagcacc ccagcgtgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcccgtc acaaagagct tcaacagggg agagtgttag 600

```

```

SEQ ID NO: 192      moltype = DNA  length = 339
FEATURE             Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 192
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacctacc tggcctggta tcagcagaaa 120
ccagggaag ttcctaagca actgatctat gatgcacca ctctggcatc tgggggccca 180
tctcgtttca gtggcagtg atctgggaca gatttcaact tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tggtgattgt 300
ttgttttcg gcggaggaac caaggtggaa atcaaacgt 339

```

```

SEQ ID NO: 193      moltype = DNA  length = 66
FEATURE             Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 193
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

```

```

SEQ ID NO: 194      moltype = DNA  length = 39
FEATURE             Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 194
caggccagtc agagtgttta tcataacacc tacctggcc 39

```

```

SEQ ID NO: 195      moltype = DNA  length = 45
FEATURE             Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 195
tggtatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

```

```

SEQ ID NO: 196      moltype = DNA  length = 21
FEATURE             Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 196
gatgcacca ctctggcatc t 21

```

```

SEQ ID NO: 197      moltype = DNA  length = 96

```

-continued

FEATURE Location/Qualifiers
misc_feature 1..96
note = Engineered antibody sequence
source 1..96
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 197
gggggtcccat ctctgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 198 moltype = DNA length = 39
FEATURE Location/Qualifiers
misc_feature 1..39
note = Engineered antibody sequence
source 1..39
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 198
ctgggcagtt atgattgtac taatggtgat tgttttgtt 39

SEQ ID NO: 199 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
note = Engineered antibody sequence
source 1..33
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 199
ttcggcggag gaaccaaggt ggaaatcaaa cgt 33

SEQ ID NO: 200 moltype = DNA length = 321
FEATURE Location/Qualifiers
misc_feature 1..321
note = Engineered antibody sequence
source 1..321
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 200
acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata acgcctctca ctggggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g 321

SEQ ID NO: 201 moltype = AA length = 441
FEATURE Location/Qualifiers
REGION 1..441
note = Engineered antibody sequence
source 1..441
mol_type = protein
organism = synthetic construct

SEQUENCE: 201
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IINGATYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAAALGLV KDYFPEPVTW SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
TVPSSSLGTQ TYICNVNHPK SNTKVDARVE PKSCDKTHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP VLDSDGSFPL YSKLTVDKSR WQGNVFPSCS 420
VMHEALHNNHY TQKSLSLSPG K 441

SEQ ID NO: 202 moltype = AA length = 111
FEATURE Location/Qualifiers
REGION 1..111
note = Engineered antibody sequence
source 1..111
mol_type = protein
organism = synthetic construct

SEQUENCE: 202
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS GYYMNWVRQA PGKGLEWVGV IINGATYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLVTVS S 111

SEQ ID NO: 203 moltype = AA length = 30
FEATURE Location/Qualifiers
REGION 1..30
note = Engineered antibody sequence
source 1..30
mol_type = protein

-continued

organism = synthetic construct	
SEQUENCE: 203	
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS	30
SEQ ID NO: 204	moltype = AA length = 5
FEATURE	Location/Qualifiers
REGION	1..5
	note = Engineered antibody sequence
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 204	
GYMNM	5
SEQ ID NO: 205	moltype = AA length = 14
FEATURE	Location/Qualifiers
REGION	1..14
	note = Engineered antibody sequence
source	1..14
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 205	
WVRQAPGKGL EWVG	14
SEQ ID NO: 206	moltype = AA length = 16
FEATURE	Location/Qualifiers
REGION	1..16
	note = Engineered antibody sequence
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 206	
VIGINGATYY ASWAKG	16
SEQ ID NO: 207	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 207	
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR	32
SEQ ID NO: 208	moltype = length =
SEQUENCE: 208	
000	
SEQ ID NO: 209	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 209	
WGQGLTVTS S	11
SEQ ID NO: 210	moltype = AA length = 330
FEATURE	Location/Qualifiers
REGION	1..330
	note = Engineered antibody sequence
source	1..330
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 210	
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDARVEP KSCDKHTTCP PCPAPELLGG	120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA	180
STYRVVSVLT VHQDWLNGK EYCKKVSNA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE	240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW	300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK	330
SEQ ID NO: 211	moltype = DNA length = 1326
FEATURE	Location/Qualifiers
misc_feature	1..1326
	note = Engineered antibody sequence
source	1..1326
	mol_type = other DNA

-continued

```

organism = synthetic construct

SEQUENCE: 211
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
ccctctggc cctctccaa gagcacctct gggggcacag cggccctggg ctgcctggtc 420
aaggactact tccccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgccct ccagcagctt gggcaccacg acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggagcg gagagttag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa 720
ccaagagaca cctcatgat ctcccgacc cctgaggtca catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agcgcgggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcctcccgag ccccatcga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
cagggtgata cctgcccc cccccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aagcttcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cacgcctccc gtgctggact cgcacggctc cttctcttc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcatg aggcctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aaatga 1326

SEQ ID NO: 212      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature       1..333
                   note = Engineered antibody sequence
source            1..333
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 212
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt ggctactaca tgaactgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtatta atggtgccac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctcgt caccgtctcg agc 333

SEQ ID NO: 213      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature       1..90
                   note = Engineered antibody sequence
source            1..90
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 213
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt 90

SEQ ID NO: 214      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature       1..15
                   note = Engineered antibody sequence
source            1..15
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 214
ggctactaca tgaac 15

SEQ ID NO: 215      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature       1..42
                   note = Engineered antibody sequence
source            1..42
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 215
tgggtccgctc aggcctccagg gaaggggctg gagtgggtcg ga 42

SEQ ID NO: 216      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature       1..48
                   note = Engineered antibody sequence
source            1..48
                   mol_type = other DNA
                   organism = synthetic construct

```

-continued

SEQUENCE: 216
gtcatttgta ttaatggtgc cacatactac gcgagctggg cgaaaggc 48

SEQ ID NO: 217 moltype = DNA length = 96
FEATURE Location/Qualifiers
misc_feature 1..96
 note = Engineered antibody sequence
source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 217
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acactgctgt gtatttctgt gctaga 96

SEQ ID NO: 218 moltype = length =
SEQUENCE: 218
000

SEQ ID NO: 219 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = Engineered antibody sequence
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 219
tggggccaag ggaccctcgt caccgtctcg agc 33

SEQ ID NO: 220 moltype = DNA length = 993
FEATURE Location/Qualifiers
misc_feature 1..993
 note = Engineered antibody sequence
source 1..993
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 220
gcctccacca agggcccatc ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg cctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaaactcag gcgccctgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacgcgag agttgagccc 300
aaatcttgtg acaaaaactca cacatgcccc cctgccccag cacctgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtggtggt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag cgcgggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtcctcacc gtccctgacc aggactggct gaatggcaag 600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccacgcagaa aaccatctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctggcctcag gcttctatcc cagcgacatc 780
gccgtggagt ggggagacaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggtctcct ctctctctac agcaagctca ccgtggacaa gacgaggtgg 900
cagcagggga acgtctctc atgtccctg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 221 moltype = AA length = 219
FEATURE Location/Qualifiers
REGION 1..219
 note = Engineered antibody sequence
source 1..219
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 221
QVLTQSPSSL SASVGDRVTI NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 222 moltype = AA length = 113
FEATURE Location/Qualifiers
REGION 1..113
 note = Engineered antibody sequence
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 222
QVLTQSPSSL SASVGDRVTI NCQASQSVYH NTYLAWYQQK PGKVPKQLIY DASTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCTNGDC FVFGGKVE IKR 113

SEQ ID NO: 223 moltype = AA length = 22

-continued

FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Engineered antibody sequence	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
QVLTQSPSSL SASVGDRVTI NC		22
SEQ ID NO: 224	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 224		
QASQSVYHNT YLA		13
SEQ ID NO: 225	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
WYQQKPGKVP KQLIY		15
SEQ ID NO: 226	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
DASTLAS		7
SEQ ID NO: 227	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
GVPSRFSGSG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 228	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
LGSYDCTNGD CFV		13
SEQ ID NO: 229	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
FGGGTKVEIK R		11
SEQ ID NO: 230	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 230		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYSLSST LTLKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC		106

-continued

```

SEQ ID NO: 231      moltype = DNA length = 660
FEATURE            Location/Qualifiers
misc_feature        1..660
                    note = Engineered antibody sequence
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 231
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacotacc tggcctggta tcagcagaaa 120
ccagggaagag ttcttaagca actgatctat gatgcatcca ctctggcatc tggggtecca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tggtgattgt 300
tttgttttcg gcggagggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatcccg agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag ctaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatcc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 232      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 232
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagag tgtttatcat aacacotacc tggcctggta tcagcagaaa 120
ccagggaagag ttcttaagca actgatctat gatgcatcca ctctggcatc tggggtecca 180
tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtactaa tggtgattgt 300
tttgttttcg gcggagggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 233      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 233
caagtgtga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 234      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 234
caggccagtc agagtgttta tcataacacc tacctggcc 39

SEQ ID NO: 235      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 235
tggtatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 236      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 236
gatgcatcca ctctggcatc t 21

SEQ ID NO: 237      moltype = DNA length = 96
FEATURE            Location/Qualifiers

```

-continued

```

misc_feature      1..96
                  note = Engineered antibody sequence
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 237
gggggtcccat ctcgttttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 238      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                  note = Engineered antibody sequence
source              1..39
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 238
ctgggcagtt atgattgtac taatggtgat tgttttggt 39

SEQ ID NO: 239      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Engineered antibody sequence
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 239
ttcggcggag gaaccaaggt ggaaatcaaa cgt 33

SEQ ID NO: 240      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                  note = Engineered antibody sequence
source              1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 240
acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata acgcctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g 321

SEQ ID NO: 241      moltype = AA length = 440
FEATURE            Location/Qualifiers
REGION              1..440
                  note = Engineered antibody sequence
source              1..440
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 241
QEQLKESGGR LVTPGTS LTL TCTVSGIDLS NHYMQWVRQA PGKGLEWIGV VGINGRTYYA 60
SWAKGRFTIS RTSSTTVDLK MTRLTTEDTA TYFCARGDIW GPGTLVTVSS ASTKGPSVFP 120
LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS GLYSLSSVVT 180
VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG PSVFLFPPKP 240
KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA STYRVVSVLT 300
VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE MTKNQVSLTC 360
LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW QQGNVFSCSV 420
MHEALHNHYT QKSLSLSPGK 440

SEQ ID NO: 242      moltype = AA length = 110
FEATURE            Location/Qualifiers
REGION              1..110
                  note = Engineered antibody sequence
source              1..110
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 242
QEQLKESGGR LVTPGTS LTL TCTVSGIDLS NHYMQWVRQA PGKGLEWIGV VGINGRTYYA 60
SWAKGRFTIS RTSSTTVDLK MTRLTTEDTA TYFCARGDIW GPGTLVTVSS 110

SEQ ID NO: 243      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION              1..30
                  note = Engineered antibody sequence
source              1..30
                  mol_type = protein
                  organism = synthetic construct

```

-continued

SEQUENCE: 243
 QEQLKESGGR LVTPGTSITL TCTVSGIDLS 30

SEQ ID NO: 244 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Engineered antibody sequence
 source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 244
 NHYMQ 5

SEQ ID NO: 245 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Engineered antibody sequence
 source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 245
 WVRQAPGKGL EWIG 14

SEQ ID NO: 246 moltype = AA length = 16
 FEATURE Location/Qualifiers
 REGION 1..16
 note = Engineered antibody sequence
 source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 246
 VVGINGRTYY ASWAKG 16

SEQ ID NO: 247 moltype = AA length = 31
 FEATURE Location/Qualifiers
 REGION 1..31
 note = Engineered antibody sequence
 source 1..31
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 247
 RFTISRTSST TVDLKMTSLT TEDTATYFCA R 31

SEQ ID NO: 248 moltype = length =
 SEQUENCE: 248
 000

SEQ ID NO: 249 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Engineered antibody sequence
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 249
 WGPGLTVTVS S 11

SEQ ID NO: 250 moltype = AA length = 330
 FEATURE Location/Qualifiers
 REGION 1..330
 note = Engineered antibody sequence
 source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 250
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNKKPS NTKVDRKVEP KSCDKHTTCP PCPAPELLGG 120
 PSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYA 180
 STYRVVSVLT VLNQDWSLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
 MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPPV LDSDGSFPLY SKLTVDKSRW 300
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 251 moltype = DNA length = 1323
 FEATURE Location/Qualifiers
 misc_feature 1..1323
 note = Engineered antibody sequence
 source 1..1323
 mol_type = other DNA
 organism = synthetic construct

-continued

```

SEQUENCE: 251
caggagcagc tgaaggagtc cgggggtcgc ctgggtcacgc ctgggacatc cctgacactc 60
acctgcaccg tctctggaat cgacctcagt aaccactaca tgcaatgggt ccgccagggt 120
ccagggaagg ggctggagtg gatcggagtc gttggtatta atggtcgcac atactacgcg 180
agctggggcg aaggccgatt caccatctcc agaacctcgt cgaccacggt ggatctgaaa 240
atgaccaggc tgacaaccga ggacacggcc acctatttct gtgccagagg ggacatctgg 300
ggcccaggca ccttggtcac cgtctcgagc gcctccacca agggcccatc ggtcttcccc 360
ctggcaccct cctccaagag cacctctggg ggcacagcgg ccctgggctg cctggtaag 420
gactacttcc ccgaaccggt gacgggtgctg tggaaactcag gcgccctgac cagcggcgtg 480
cacaccttcc cggtctgctc acagtctctc ggactctact ccctcagcag cgtgggtgac 540
gtgccctcca gcagcttggg caccagacc tacatctgca acgtgaatca caagccagc 600
aacaccaagg tggacaagag agttgagccc aaatcttgtg acaaaactca cacatgccca 660
ccgtgccag cactgaact cctgggggga ccgtcagctc tcctcttccc cccaaaaccc 720
aaggacacc tcactgatct ccggaccctc gaggtcacat gcgtgggtgg ggacgtgagc 780
cacgaagacc ctgaggtcaa gttcaactgg tacgtggagc gcgtggaggt gcataatgcc 840
aagacaaagc cgcgggagga gcagtacgcc agcacgtacc gtgtgggtcag cgtctctacc 900
gtcctgcacc aggactggct gaatggcaag gactacaagt gcaaggtctc caacaaagcc 960
ctcccagccc ccctcgagaa aacctctccc aaagccaaag ggcagccccc agaaccacag 1020
gtgtacaccc tgcctccatc ccgggaggag atgaccaaga accaggtcag cctgacctgc 1080
ctgggtcaaa ctctctatcc cagcgacatc gccgtggagt gggagagcaa tgggcagccg 1140
gagaaacact acaagaccac gccctccgtg ctggactccg acggctcctt ctctctctac 1200
agcaagctca ccgtggacaa gagcaggtgg cagcagggga acgtcttctc atgtccgtg 1260
atgcatgagg ctctgcacaa ccactacacg cagaagagcc tctccctgtc tccgggtaaa 1320
tga 1323

```

```

SEQ ID NO: 252      moltype = DNA length = 330
FEATURE            Location/Qualifiers
misc_feature        1..330
                    note = Engineered antibody sequence
source              1..330
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 252
caggagcagc tgaaggagtc cgggggtcgc ctgggtcacgc ctgggacatc cctgacactc 60
acctgcaccg tctctggaat cgacctcagt aaccactaca tgcaatgggt ccgccagggt 120
ccagggaagg ggctggagtg gatcggagtc gttggtatta atggtcgcac atactacgcg 180
agctggggcg aaggccgatt caccatctcc agaacctcgt cgaccacggt ggatctgaaa 240
atgaccaggc tgacaaccga ggacacggcc acctatttct gtgccagagg ggacatctgg 300
ggcccaggca ccttggtcac cgtctcgagc 330

```

```

SEQ ID NO: 253      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 253
caggagcagc tgaaggagtc cgggggtcgc ctgggtcacgc ctgggacatc cctgacactc 60
acctgcaccg tctctggaat cgacctcagt 90

```

```

SEQ ID NO: 254      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 254
aaccactaca tgcaa 15

```

```

SEQ ID NO: 255      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 255
tgggtccgcc aggtccagg gaaggggctg gagtggatcg ga 42

```

```

SEQ ID NO: 256      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 256

```

-continued

```

gtcgttggtgta ttaatgggtcg cacatactac gcgagctggg cgaaaggc          48

SEQ ID NO: 257          moltype = DNA length = 93
FEATURE
misc_feature           Location/Qualifiers
                        1..93
                        note = Engineered antibody sequence
source                1..93
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 257
cgattcacca tctccagaac ctcgtcgacc acggtggatc tgaaaatgac caggctgaca 60
accgaggaca cggccaccta tttctgtgcc aga                               93

SEQ ID NO: 258          moltype = length =
SEQUENCE: 258
000

SEQ ID NO: 259          moltype = DNA length = 33
FEATURE
misc_feature           Location/Qualifiers
                        1..33
                        note = Engineered antibody sequence
source                1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 259
tggggcccag gcaccctggt caccgtctcg agc                               33

SEQ ID NO: 260          moltype = DNA length = 993
FEATURE
misc_feature           Location/Qualifiers
                        1..993
                        note = Engineered antibody sequence
source                1..993
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 260
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctggg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc cgaaccggg gacggtgtcg 120
tggaaactcag gcgcctcgac cagcggcgctg cacaccttcc cggtgtcct acagtcctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgccca ccgtgcccag cactgaact cctgggggga 360
ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtccctgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccatcgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag ctgcacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gcgctggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 261          moltype = AA length = 219
FEATURE
REGION                Location/Qualifiers
                        1..219
                        note = Engineered antibody sequence
source                1..219
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 261
QVLTQTASPV SAAVGSTVTI NCQASQSVYN YNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSTGDC FVFGGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSYL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSPNRGEC                               219

SEQ ID NO: 262          moltype = AA length = 113
FEATURE
REGION                Location/Qualifiers
                        1..113
                        note = Engineered antibody sequence
source                1..113
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 262
QVLTQTASPV SAAVGSTVTI NCQASQSVYN YNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSTGDC FVFGGGTEVV VKR                               113

SEQ ID NO: 263          moltype = AA length = 22
FEATURE
                        Location/Qualifiers

```

-continued

REGION	1..22	
	note = Engineered antibody sequence	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 263		
QVLQTASPV SAAVGSTVTI NC		22
SEQ ID NO: 264	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 264		
QASQSVYNYN YLA		13
SEQ ID NO: 265	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 265		
WYQQKPGQPP KQLIY		15
SEQ ID NO: 266	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 266		
STSTLAS		7
SEQ ID NO: 267	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 267		
GVSSRPFKGS SGTQFTLTIS DVQCDDAATY YC		32
SEQ ID NO: 268	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 268		
LGSYDCSTGD CFV		13
SEQ ID NO: 269	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 269		
FGGGTEVVVK R		11
SEQ ID NO: 270	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 270		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLSKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC		106

-continued

```

SEQ ID NO: 271      moltype = DNA length = 660
FEATURE            Location/Qualifiers
misc_feature       1..660
                   note = Engineered antibody sequence
source            1..660
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 271
caagtgtctga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttataat tacaactacc ttgcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcgattca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg actgtagtac tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga cgggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctgt 420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtgggataa cgccctccaa 480
tcgggtaact ccagggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctcgcccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 272      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature       1..339
                   note = Engineered antibody sequence
source            1..339
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 272
caagtgtctga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagag tgtttataat tacaactacc ttgcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcgattca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg actgtagtac tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga 339

SEQ ID NO: 273      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature       1..66
                   note = Engineered antibody sequence
source            1..66
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 273
caagtgtctga cccagactgc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

SEQ ID NO: 274      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature       1..39
                   note = Engineered antibody sequence
source            1..39
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 274
caggccagtc agagtgttta taattacaac taccttgcc 39

SEQ ID NO: 275      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature       1..45
                   note = Engineered antibody sequence
source            1..45
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 275
tggatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 276      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature       1..21
                   note = Engineered antibody sequence
source            1..21
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 276
tctacatcca ctctggcatc t 21

SEQ ID NO: 277      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature       1..96

```

-continued

```

source          note = Engineered antibody sequence
                1..96
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 277
ggggtctcat cgcgattcaa aggcagtgga tctgggacac agttcactct caccatcagc 60
gacgtgcagt gtgacgatgc tgccacttac tactgt                               96

SEQ ID NO: 278      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 278
ctaggcagtt atgactgtag tactggtgat tgttttgtt                               39

SEQ ID NO: 279      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 279
ttcgccggag ggaccgaggt ggtggtcaaa cgt                               33

SEQ ID NO: 280      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 280
acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgcct gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata acgcctctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccggt cacaaagagc 300
ttcaacaggg gagagtgtta g                               321

SEQ ID NO: 281      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION              1..441
                    note = Engineered antibody sequence
source              1..441
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 281
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS NHYMQWVRQA PGKGLEWVGV VGINGRTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS SASTKGPSVF 120
PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSV VHTFPAVLQS SGLYSLSSVV 180
TVPSSSLGTQ TYICNVNHKP SNTKVDKRVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQQGNVFS 420
VMHEALHNHY TQKSLSLSPG K                               441

SEQ ID NO: 282      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION              1..111
                    note = Engineered antibody sequence
source              1..111
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 282
EVQLVESGGG LVQPGGSLRL SCAVSGIDLS NHYMQWVRQA PGKGLEWVGV VGINGRTYYA 60
SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WQGTGLTVTS S                               111

SEQ ID NO: 283      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION              1..30
                    note = Engineered antibody sequence
source              1..30
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 283

```


-continued

EVQLVESGGG LVQPGGSLRL SCAVSGIDLS	30
SEQ ID NO: 284	moltype = AA length = 5
FEATURE	Location/Qualifiers
REGION	1..5
	note = Engineered antibody sequence
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 284	
NHYMQ	5
SEQ ID NO: 285	moltype = AA length = 14
FEATURE	Location/Qualifiers
REGION	1..14
	note = Engineered antibody sequence
source	1..14
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 285	
WVRQAPGKGL EWVG	14
SEQ ID NO: 286	moltype = AA length = 16
FEATURE	Location/Qualifiers
REGION	1..16
	note = Engineered antibody sequence
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 286	
VVGINGRTYY ASWAKG	16
SEQ ID NO: 287	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 287	
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR	32
SEQ ID NO: 288	moltype = length =
SEQUENCE: 288	
000	
SEQ ID NO: 289	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 289	
WGQGLVTVS S	11
SEQ ID NO: 290	moltype = AA length = 330
FEATURE	Location/Qualifiers
REGION	1..330
	note = Engineered antibody sequence
source	1..330
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 290	
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG	120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA	180
STYRVVSVLT VLNQDNLNGK EYCKKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE	240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW	300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK	330
SEQ ID NO: 291	moltype = DNA length = 1326
FEATURE	Location/Qualifiers
misc_feature	1..1326
	note = Engineered antibody sequence
source	1..1326
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 291	

-continued

```

gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt aacctactaca tgcaatgggt ccgtcaggct 120
ccagggaagc ggctggagtg ggtcggagtc gttggtatca atggtcgcac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctctgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccttgccac cctcctccaa gagcacctct gggggcacag cggccctggg ctgectggtc 420
aaggactact tcccgaacc ggtgacggtg tctggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggtgt cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgcctt ccagcagctt gggcaccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tottctctt cccccaaaa 720
cccaaggaca cctcatgat cctccggacc cctgaggtea catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtctcgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gccctcccag ccccactga gaaaaccatc tccaaagcca aaggcgagcc ccgagaacca 1020
caggtgtaca ccttgccccc atcccgagg gagatgacca agaaccaggt cagcctgacc 1080
tgctgtgta aaggttcta tcccagcgac atcgccgtg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttctctctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcagtctcc 1260
gtgatgcatg aggtcttgca caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

```

```

SEQ ID NO: 292      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature       1..333
                   note = Engineered antibody sequence
source            1..333
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 292
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt aacctactaca tgcaatgggt ccgtcaggct 120
ccagggaagc ggctggagtg ggtcggagtc gttggtatca atggtcgcac atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgctag aggggacatc 300
tggggccaag ggacctctgt caccgtctcg agc 333

```

```

SEQ ID NO: 293      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature       1..90
                   note = Engineered antibody sequence
source            1..90
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 293
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggc cctgagactc 60
tcctgtgcag tctctggaat cgacctcagt 90

```

```

SEQ ID NO: 294      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature       1..15
                   note = Engineered antibody sequence
source            1..15
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 294
aacctactaca tgcaa 15

```

```

SEQ ID NO: 295      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature       1..42
                   note = Engineered antibody sequence
source            1..42
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 295
tgggtccgtc aggtccagg gaaggggctg gaggggctg ga 42

```

```

SEQ ID NO: 296      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature       1..48
                   note = Engineered antibody sequence
source            1..48
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 296
tgcgttggtg tcaatggctg cacatactac gcgagctggg cgaaaggc 48

```

-continued

```

SEQ ID NO: 297      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 297
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt gctaga                               96

SEQ ID NO: 298      moltype = length =
SEQUENCE: 298
000

SEQ ID NO: 299      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 299
tggggccaag ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 300      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 300
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctctgg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaactcag gcgcctctg cagcggcgctg cacaccttcc cggctgtcct acagtctctca 180
ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgccca ccgtgccag cactgaact cctgggggga 360
ccgtcagtc tctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gtccaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtcctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaaggcttc caacaaagcc ctcccagccc ccacgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 301      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 301
QVLTSQSPSSL SASVGDRVTI NCQASQSVYN YNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSTGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSITYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC                               219

SEQ ID NO: 302      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 302
QVLTSQSPSSL SASVGDRVTI NCQASQSVYN YNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSTGDC FVFGGGTKVE IKR                               113

SEQ ID NO: 303      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION              1..22

```

-continued

source	note = Engineered antibody sequence 1..22 mol_type = protein organism = synthetic construct	
SEQUENCE: 303 QVLTQSPSSL SASVGDRVTI NC		22
SEQ ID NO: 304 FEATURE REGION	moltype = AA length = 13 Location/Qualifiers 1..13 note = Engineered antibody sequence	
source	1..13 mol_type = protein organism = synthetic construct	
SEQUENCE: 304 QASQSVYNYN YLA		13
SEQ ID NO: 305 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Engineered antibody sequence	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 305 WYQQKPGKVP KQLIY		15
SEQ ID NO: 306 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = Engineered antibody sequence	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 306 STSTLAS		7
SEQ ID NO: 307 FEATURE REGION	moltype = AA length = 32 Location/Qualifiers 1..32 note = Engineered antibody sequence	
source	1..32 mol_type = protein organism = synthetic construct	
SEQUENCE: 307 GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 308 FEATURE REGION	moltype = AA length = 13 Location/Qualifiers 1..13 note = Engineered antibody sequence	
source	1..13 mol_type = protein organism = synthetic construct	
SEQUENCE: 308 LGSYDCSTGD CFV		13
SEQ ID NO: 309 FEATURE REGION	moltype = AA length = 11 Location/Qualifiers 1..11 note = Engineered antibody sequence	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 309 FGGGTKVEIK R		11
SEQ ID NO: 310 FEATURE REGION	moltype = AA length = 106 Location/Qualifiers 1..106 note = Engineered antibody sequence	
source	1..106 mol_type = protein organism = synthetic construct	
SEQUENCE: 310 TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS KDSTYLSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC		60 106
SEQ ID NO: 311	moltype = DNA length = 660	

FEATURE	Location/Qualifiers		
misc_feature	1..660	note = Engineered antibody sequence	
source	1..660	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 311			
caagtgtctga	cccagttctcc	atctctccctg	tctgcatctg taggagacag agtcaccatc 60
aattggccagg	ccagtcagag	tgttttacaat	tacaactacc ttgccttgga tcacgagaaa 120
ccagggaaag	ttcctaagca	actgatctat	tctacatcca ctctggcatc tgggggtccca 180
tctcgtttca	gtggcagtg	atctggggaca	gatttcactc tcaccatcag cagcctgcag 240
cttgaagatg	ttgcaactta	ttactgtctg	ggcagttatg attgtagtac tgggtgattgt 300
ttgtttttcg	gcggaggaac	caaggttgga	atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc	cgccacttga	tgagcagttg	aaatctggaa ctgcctctgt tgtgtgctctg 420
ctgaataact	tctatcccg	agaggccaaa	gtacagtgga aggtggataa cgcctccaa 480
tccggtaact	cccaggagag	tgtcacagag	caggacagca aggtggataa ctacagcctc 540
agcagcacc	tgacgtctg	caaagcagac	tacgagaaac acaaatgtcta cgctcgcaa 600
gtcaccatc	agggcctgag	ctgcgccgtc	acaagagct tcaacagggg agagtgttag 660
SEQ ID NO: 312	moltype = DNA length = 339		
FEATURE	Location/Qualifiers		
misc_feature	1..339	note = Engineered antibody sequence	
source	1..339	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 312			
caagtgtctga	cccagttctcc	atctctccctg	tctgcatctg taggagacag agtcaccatc 60
aattggccagg	ccagtcagag	tgttttacaat	tacaactacc ttgccttgga tcacgagaaa 120
ccagggaaag	ttcctaagca	actgatctat	tctacatcca ctctggcatc tgggggtccca 180
tctcgtttca	gtggcagtg	atctggggaca	gatttcactc tcaccatcag cagcctgcag 240
cctgaagatg	ttgcaactta	ttactgtctg	ggcagttatg attgtagtac tgggtgattgt 300
ttgtttttcg	gcggaggaac	caaggttgga	atcaaacgt 339
SEQ ID NO: 313	moltype = DNA length = 66		
FEATURE	Location/Qualifiers		
misc_feature	1..66	note = Engineered antibody sequence	
source	1..66	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 313			
caagtgtctga	cccagttctcc	atctctccctg	tctgcatctg taggagacag agtcaccatc 60
aattgc			66
SEQ ID NO: 314	moltype = DNA length = 39		
FEATURE	Location/Qualifiers		
misc_feature	1..39	note = Engineered antibody sequence	
source	1..39	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 314			
caggccagtc	agagtgttta	caattacaac	taccttgcc 39
SEQ ID NO: 315	moltype = DNA length = 45		
FEATURE	Location/Qualifiers		
misc_feature	1..45	note = Engineered antibody sequence	
source	1..45	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 315			
tggtatcagc	agaaaccagg	gaaagttcct	aagcaactga tctat 45
SEQ ID NO: 316	moltype = DNA length = 21		
FEATURE	Location/Qualifiers		
misc_feature	1..21	note = Engineered antibody sequence	
source	1..21	mol_type = other DNA	
	organism = synthetic construct		
SEQUENCE: 316			
tctacatcca	ctctggcatc	t	21
SEQ ID NO: 317	moltype = DNA length = 96		
FEATURE	Location/Qualifiers		
misc_feature	1..96	note = Engineered antibody sequence	

-continued

```

source                1..96
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 317
gggggtcccat ctcgtttcag tggcagtgga tctgggacag atttcaactct caccatcagc 60
agcctgcagc ctgaagatgt tgcaacttat tactgt                               96

SEQ ID NO: 318        moltype = DNA length = 39
FEATURE              Location/Qualifiers
misc_feature          1..39
                      note = Engineered antibody sequence
source                1..39
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 318
ctgggcagtt atgattgtag tactggtgat tgttttgtt                               39

SEQ ID NO: 319        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = Engineered antibody sequence
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 319
tctggcggag gaaccaaggt ggaatcaaa cgt                                     33

SEQ ID NO: 320        moltype = DNA length = 321
FEATURE              Location/Qualifiers
misc_feature          1..321
                      note = Engineered antibody sequence
source                1..321
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 320
acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtggata acgcccctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacagagaaa 240
cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g                                              321

SEQ ID NO: 321        moltype = AA length = 439
FEATURE              Location/Qualifiers
REGION                1..439
                      note = Engineered antibody sequence
source                1..439
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 321
QSLEESGGRL VTPGTPLTLT CTVSGIGLSS YYMQWVRQSP GRGLEWIGVI GSDGKYYAT 60
WAKGRFTISK TSSTTVDLRM ASLTTEDTAT YFCTRGDIWG PGLTVTVSSA STKGPSVFPL 120
APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPAVLQSSG LYSLSVVTV 180
PSSSLGTQTY ICNVNHKPSN TKVDKRVPEK SCDKTHTCP CPAPPELLGGP SVFLFPPKPK 240
DTLMISRTP EVCVVDVSH EDPEVKENWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM 420
HEALHNHYTQ KSLSLSPGK                                              439

SEQ ID NO: 322        moltype = AA length = 109
FEATURE              Location/Qualifiers
REGION                1..109
                      note = Engineered antibody sequence
source                1..109
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 322
QSLEESGGRL VTPGTPLTLT CTVSGIGLSS YYMQWVRQSP GRGLEWIGVI GSDGKYYAT 60
WAKGRFTISK TSSTTVDLRM ASLTTEDTAT YFCTRGDIWG PGLTVTVSSA 109

SEQ ID NO: 323        moltype = AA length = 29
FEATURE              Location/Qualifiers
REGION                1..29
                      note = Engineered antibody sequence
source                1..29
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 323
QSLEESGGRL VTPGTPLTLT CTVSGIGLS                                         29

```

SEQ ID NO: 324	moltype = AA	length = 5	
FEATURE	Location/Qualifiers		
REGION	1..5		
source	note = Engineered antibody sequence		
	1..5		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 324			
SYMQ			5
SEQ ID NO: 325	moltype = AA	length = 14	
FEATURE	Location/Qualifiers		
REGION	1..14		
source	note = Engineered antibody sequence		
	1..14		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 325			
WVRQSPGRGL EWIG			14
SEQ ID NO: 326	moltype = AA	length = 16	
FEATURE	Location/Qualifiers		
REGION	1..16		
source	note = Engineered antibody sequence		
	1..16		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 326			
VIGSDGKTTY ATWAKG			16
SEQ ID NO: 327	moltype = AA	length = 31	
FEATURE	Location/Qualifiers		
REGION	1..31		
source	note = Engineered antibody sequence		
	1..31		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 327			
RFTISKTSST TVDLRMSLT T	EDTATYFCT	R	31
SEQ ID NO: 328	moltype =	length =	
SEQUENCE: 328			
000			
SEQ ID NO: 329	moltype = AA	length = 11	
FEATURE	Location/Qualifiers		
REGION	1..11		
source	note = Engineered antibody sequence		
	1..11		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 329			
WGPGLTVTS S			11
SEQ ID NO: 330	moltype = AA	length = 330	
FEATURE	Location/Qualifiers		
REGION	1..330		
source	note = Engineered antibody sequence		
	1..330		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 330			
ASTKGPSVFP LAPSSKSTSG	GTAALGLCLKV	DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT	YICNVNHNKS	NTKVDKRVEP	KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRT	EVTCVVVDVS	HEDPEVKFNW	YVDGVEVHNA KTKPREEQYA 180
STYRVVSVLT VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFPYPSDI	AVEWESNGQP	ENNYKTPPV	LDSGDSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT	QKLSLSLSPGK		330
SEQ ID NO: 331	moltype = DNA	length = 1320	
FEATURE	Location/Qualifiers		
misc_feature	1..1320		
source	note = Engineered antibody sequence		
	1..1320		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 331			
caqtcqctqg aqaqgtccqg	qqaqtcqctqg	qtcacqccqg	qqaacacccct qacactcacc 60

-continued

```

tgcacagtct ctggaatcgg cctcagtagc tactacatgc agtgggtccg ccagtctcca 120
gggagggggc tgggaatggat cggagtcatt ggtagtgatg gtaagacata ctacgcgacc 180
tgggcgaaaag gccgattcac catctccaag acctcgtcga ccacggtgga tctgagaatg 240
gccagtctga caaccgagga cacggccacc tatttctgta ccagagggga catctggggc 300
ccgggggacc tcgtcacctg ctcgagcgcc tccaccaagg gcccatcggt cttccccctg 360
gcacctctct ccaagagcac ctctgggggc acagcgcccc tgggctgcct ggtcaaggac 420
tacttccccg aaccggtgac ggtgtcgtgg aactcaggcg ccctgaccag cggcgtgcac 480
accttccccg ctgtctctaca gtctcagga ctctactccc tcagcagcgt ggtgaccgtg 540
ccctccagca gcttggggcac ccagacctac atctgcaacg tgaatcaca gccccagcaac 600
accaaggtgg acaagagagt tgagcccaaa tcttgtgaca aaactcacac atgccaccg 660
tgcccagcac ctgaactcct ggggggaccg tcagtcttcc tcttcccccc aaaacccaag 720
gacacctcta tgatctcccg gacccctgag gtcacatcgc tgggtggtgga cgtgagccac 780
gaagaccctg aggtcaagtt caactggtag gtggacggcg tggaggtgca taatgccaa 840
acaaagccgc tggaggagca gtacgccagc acgtaccgtg tggtcagcgt cctcaccgtc 900
ctgcaccagg actggctgaa tggcaaggag tacaagtgca aggtctccaa caaagccctc 960
ccagccccc cagagaaaaa catctccaaa gccaaagggc agccccgaga accacagggtg 1020
tacacctcgc ccccatcccg ggaggagatg accaagaacc aggtcagcct gacctgctg 1080
gtcaaaggct tctatcccag cgacatcgcc gtggagtggg agagcaatgg gcagccggag 1140
aacaactaca agaccacgcc tcccgctgct gactccgacg gctccttctt cctctacagc 1200
aagctcaccg tggacaagag cagggtggcag tcttctcatg ctccgtgatg 1260
catgaggctc tgcacaacca ctacacgcag aagagcctct ccctgtctcc gggtaaatga 1320

```

```

SEQ ID NO: 332      moltype = DNA length = 327
FEATURE             Location/Qualifiers
misc_feature         1..327
                     note = Engineered antibody sequence
source               1..327
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 332
cagtcgctgg aggagtcggg gggtcgcctg gtcacgcctg ggacaccctt gacactcacc 60
tgcacagtct ctggaatcgg cctcagtagc tactacatgc agtgggtccg ccagtctcca 120
gggagggggc tgggaatggat cggagtcatt ggtagtgatg gtaagacata ctacgcgacc 180
tgggcgaaaag gccgattcac catctccaag acctcgtcga ccacggtgga tctgagaatg 240
gccagtctga caaccgagga cacggccacc tatttctgta ccagagggga catctggggc 300
ccgggggacc tcgtcacctg ctcgagc 327

```

```

SEQ ID NO: 333      moltype = DNA length = 87
FEATURE             Location/Qualifiers
misc_feature         1..87
                     note = Engineered antibody sequence
source               1..87
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 333
cagtcgctgg aggagtcggg gggtcgcctg gtcacgcctg ggacaccctt gacactcacc 60
tgcacagtct ctggaatcgg cctcagtagc 87

```

```

SEQ ID NO: 334      moltype = DNA length = 15
FEATURE             Location/Qualifiers
misc_feature         1..15
                     note = Engineered antibody sequence
source               1..15
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 334
agctactaca tgcag 15

```

```

SEQ ID NO: 335      moltype = DNA length = 42
FEATURE             Location/Qualifiers
misc_feature         1..42
                     note = Engineered antibody sequence
source               1..42
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 335
tgggtccgcc agtctccagg gagggggctg gaatggatcg ga 42

```

```

SEQ ID NO: 336      moltype = DNA length = 48
FEATURE             Location/Qualifiers
misc_feature         1..48
                     note = Engineered antibody sequence
source               1..48
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 336
gtcattggta gtgatggtaa gacatactac gcgacctggg cgaaaggc 48

```

```

SEQ ID NO: 337      moltype = DNA length = 93

```


-continued

```

FEATURE                Location/Qualifiers
misc_feature            1..93
                        note = Engineered antibody sequence
source                  1..93
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 337
cgattcacca tctccaagac ctcgtcgacc acggtggatc tgagaatggc cagtctgaca 60
accgaggaca cggccaccta tttctgtacc aga                               93

SEQ ID NO: 338          moltype =   length =
SEQUENCE: 338
000

SEQ ID NO: 339          moltype = DNA length = 33
FEATURE                Location/Qualifiers
misc_feature            1..33
                        note = Engineered antibody sequence
source                  1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 339
tggggcccg ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 340          moltype = DNA length = 993
FEATURE                Location/Qualifiers
misc_feature            1..993
                        note = Engineered antibody sequence
source                  1..993
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 340
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctggg 60
ggcacagcgg ccctgggctg cctgggtcaa gactacttcc cgaaccggg gacgggtgctg 120
tggaactcag gcgccttgac cagcggcgtg cacaccttcc cggtgtcct acagtccctca 180
ggactctact ccctcagcag cgtggtgacc gtgccttcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccgtgcccag cactgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggaccccc tcatgatctc ccggacccct 420
gaggtcacat gcgtggtggt ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtctcacc gtctcgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccctcgagaa aacctctcc 660
aaagccaaaag ggcagccccg agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtcctt cttctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 341          moltype = AA length = 219
FEATURE                Location/Qualifiers
REGION                  1..219
                        note = Engineered antibody sequence
source                  1..219
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 341
QVLTQTPSPV SAAVGSTVTI NCQASQNVYN NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFRGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSRGDC FVFGGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSPNRGEC                               219

SEQ ID NO: 342          moltype = AA length = 113
FEATURE                Location/Qualifiers
REGION                  1..113
                        note = Engineered antibody sequence
source                  1..113
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 342
QVLTQTPSPV SAAVGSTVTI NCQASQNVYN NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFRGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSRGDC FVFGGGTEVV VKR                               113

SEQ ID NO: 343          moltype = AA length = 22
FEATURE                Location/Qualifiers
REGION                  1..22
                        note = Engineered antibody sequence
source                  1..22

```

-continued

SEQUENCE: 343	mol_type = protein	
QVLTQTPSPV SAAVGSTVTI NC	organism = synthetic construct	22
SEQ ID NO: 344	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
source	note = Engineered antibody sequence	
	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 344		13
QASQNVYNNN YLA		
SEQ ID NO: 345	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
source	note = Engineered antibody sequence	
	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 345		15
WYQQKPGQPP KQLIY		
SEQ ID NO: 346	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
source	note = Engineered antibody sequence	
	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 346		7
STSTLAS		
SEQ ID NO: 347	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
source	note = Engineered antibody sequence	
	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 347		32
GVSSRFRGSG SGTQFTLTIS DVQCDDAATY YC		
SEQ ID NO: 348	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
source	note = Engineered antibody sequence	
	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 348		13
LGSYDCSRGD CFV		
SEQ ID NO: 349	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
source	note = Engineered antibody sequence	
	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 349		11
FGGGTEVVVK R		
SEQ ID NO: 350	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
source	note = Engineered antibody sequence	
	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 350		60
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		
KDSTYLSST LTLKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 351	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	

-continued

```

note = Engineered antibody sequence
source      1..660
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 351
caagtgtctga cccagactcc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagaa tgtttataat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacgtcca ctctggcatc tggggtctca 180
tcgcgattca gaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtcg tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaacgta cgggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 352      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = Engineered antibody sequence
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 352
caagtgtctga cccagactcc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgccagg ccagtcagaa tgtttataat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacgtcca ctctggcatc tggggtctca 180
tcgcgattca gaggcagtg atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtcg tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaacgt 339

SEQ ID NO: 353      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                    note = Engineered antibody sequence
source              1..66
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 353
caagtgtctga cccagactcc atcccccgctg tctgcagctg tgggaagcac agtcaccatc 60
aattgc 66

SEQ ID NO: 354      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 354
caggccagtc agaattgtta taataacaac tacctagcc 39

SEQ ID NO: 355      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = Engineered antibody sequence
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 355
tggtatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 356      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Engineered antibody sequence
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 356
tctacgtcca ctctggcatc t 21

SEQ ID NO: 357      moltype = DNA length = 96
FEATURE            Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96
                    mol_type = other DNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 357
gggggtctcat cgcgattcag aggcagtgga tctgggacac agttcactct caccatcagc 60
gacgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 358      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 358
ctaggcagtt atgattgtag tctgggtgat tgttttgtt 39

SEQ ID NO: 359      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 359
ttcggcggag ggaccgaggt ggtgggtcaaa cgt 33

SEQ ID NO: 360      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 360
acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata acgcccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct acgctgcgca agtcacccat cagggcctga gctcgccctg cacaaagagc 300
ttcaacaggg gagagtgtta g 321

SEQ ID NO: 361      moltype = AA length = 441
FEATURE            Location/Qualifiers
REGION              1..441
                    note = Engineered antibody sequence
source              1..441
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 361
EVQLVESGGG LVQPGGSLRL SCAVSGIGLS SYMQWVRQA PGKGLEWVGV IGSDGKITYYA 60
TWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCTRGDI WQGTGLTVTS SASTKGPSVF 120
PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSVV 180
TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQQGNVFSKS 420
VMHEALHNHY TQKSLSLSPG K 441

SEQ ID NO: 362      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION              1..111
                    note = Engineered antibody sequence
source              1..111
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 362
EVQLVESGGG LVQPGGSLRL SCAVSGIGLS SYMQWVRQA PGKGLEWVGV IGSDGKITYYA 60
TWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCTRGDI WQGTGLTVTS S 111

SEQ ID NO: 363      moltype = AA length = 30
FEATURE            Location/Qualifiers
REGION              1..30
                    note = Engineered antibody sequence
source              1..30
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 363
EVQLVESGGG LVQPGGSLRL SCAVSGIGLS 30

SEQ ID NO: 364      moltype = AA length = 5

```

-continued

FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 364		
SYMQ		5
SEQ ID NO: 365	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 365		
WVRQAPGKGL EWVG		14
SEQ ID NO: 366	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 366		
VIGSDGKTTY ATWAKG		16
SEQ ID NO: 367	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 367		
RFTISRDNK TTVYLQMNLS RAEDTAVYFC TR		32
SEQ ID NO: 368	moltype = length =	
SEQUENCE: 368		
000		
SEQ ID NO: 369	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 369		
WGQTLVTVS S		11
SEQ ID NO: 370	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 370		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVVSVLT VLNQDNLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGSEFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 371	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	
misc_feature	1..1326	
	note = Engineered antibody sequence	
source	1..1326	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 371		
gaggtgcagc ttgtggagtc tgggggaggc ttgtccagc ctgggggggc cctgagactc		60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt cgcgcaggct		120
ccaggaaggg ggctggagtg ggtcggagtc attggtagtg atggtgaagac atactacgcg		180

-continued

```

acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtaccag aggggacatc 300
tggggccaaag ggaccctcgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
cccttgccac cctcctccaa gageacctct gggggcacag cggccctggg ctgcttggtc 420
aaggactact tcccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtggtg 540
accgtgcctt ccagcagctt gggcaccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa 720
cccaaggaca cctcatgat tccccggacc cctgaggtea catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagtccaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcacgt accgtgtggt cagcgtcttc 900
accgtctcgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gccctcccag ccccatcgga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca cctcgcccc atccccggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aaggtctcta tcccagcgac atcgccgtgg agtgggagag caatggggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttctctctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcatt aggtctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga                                           1326

```

```

SEQ ID NO: 372      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 372
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtaaagc atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtaccag aggggacatc 300
tggggccaaag ggaccctcgt caccgtctcg agc                                           333

```

```

SEQ ID NO: 373      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 373
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt                                           90

```

```

SEQ ID NO: 374      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 374
agctactaca tgcaa                                           15

```

```

SEQ ID NO: 375      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 375
tgggtccgtc aggtccagg gaaggggctg gagtgggtcg ga                                           42

```

```

SEQ ID NO: 376      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 376
gtcatttgta gtgatggtaa gacatactac gcgacctggg cgaaaggc                                           48

```

```

SEQ ID NO: 377      moltype = DNA length = 96
FEATURE            Location/Qualifiers

```

-continued

```

misc_feature      1..96
                  note = Engineered antibody sequence
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 377
cgattcacca tctccagaga caattccaag accacgggtgt atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt accaga                               96

SEQ ID NO: 378      moltype =   length =
SEQUENCE: 378
000

SEQ ID NO: 379      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Engineered antibody sequence
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 379
tggggccaag ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 380      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                  note = Engineered antibody sequence
source              1..993
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 380
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctctgg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgccttgac cagcggcggtg cacaccttcc cggctgtcct acagtctctca 180
ggactctact cctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttggt acaaaaactca cacatgcccc ccggtgccag cactgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtctcacc gtctgcacc aggaactgggt gaatggcagg 600
gagtacaagt gcaagggtct caacaaagcc ctcccagccc ccacgagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt ggggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctccct cttctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 381      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                  note = Engineered antibody sequence
source              1..219
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 381
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGKTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSYL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC                               219

SEQ ID NO: 382      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                  note = Engineered antibody sequence
source              1..113
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 382
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGKTKVE IKR                               113

SEQ ID NO: 383      moltype = AA length = 22
FEATURE            Location/Qualifiers
REGION              1..22
                  note = Engineered antibody sequence
source              1..22
                  mol_type = protein

```

-continued

organism = synthetic construct	
SEQUENCE: 383	
QVLTQSPSSL SASVGDRTI NC	22
SEQ ID NO: 384	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 384	
QASQNVYNNN YLA	13
SEQ ID NO: 385	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
	note = Engineered antibody sequence
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 385	
WYQQKPGKVP KQLIY	15
SEQ ID NO: 386	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = Engineered antibody sequence
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 386	
STSTLAS	7
SEQ ID NO: 387	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 387	
GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC	32
SEQ ID NO: 388	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 388	
LGSYDCSRGD CFV	13
SEQ ID NO: 389	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 389	
FGGGTKVEIK R	11
SEQ ID NO: 390	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
	note = Engineered antibody sequence
source	1..106
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 390	
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS	60
KDSTYLSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC	106
SEQ ID NO: 391	moltype = DNA length = 660
FEATURE	Location/Qualifiers
misc_feature	1..660
	note = Engineered antibody sequence

-continued

```

source                1..660
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 391
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcaactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tggtgattgt 300
tttgttttcg gcgagggaac caaggtggaa atcaaacgta cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctc 420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagag tgctcacagag caggacagca aggacagcac ctacagctc 540
agcagcacc ccagcgtgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcctcgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 392        moltype = DNA length = 339
FEATURE              Location/Qualifiers
misc_feature          1..339
                      note = Engineered antibody sequence
source                1..339
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 392
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
tctcgtttca gtggcagtg atctgggaca gatttcaactc tcaccatcag cagcctgcag 240
cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tggtgattgt 300
tttgttttcg gcgagggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 393        moltype = DNA length = 66
FEATURE              Location/Qualifiers
misc_feature          1..66
                      note = Engineered antibody sequence
source                1..66
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 393
caagtgtctga cccagtctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
aattgc 66

SEQ ID NO: 394        moltype = DNA length = 39
FEATURE              Location/Qualifiers
misc_feature          1..39
                      note = Engineered antibody sequence
source                1..39
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 394
caggccagtc agaatgttta caataacaac tacctagcc 39

SEQ ID NO: 395        moltype = DNA length = 45
FEATURE              Location/Qualifiers
misc_feature          1..45
                      note = Engineered antibody sequence
source                1..45
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 395
tggtatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 396        moltype = DNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Engineered antibody sequence
source                1..21
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 396
tctacatcca ctctggcatc t 21

SEQ ID NO: 397        moltype = DNA length = 96
FEATURE              Location/Qualifiers
misc_feature          1..96
                      note = Engineered antibody sequence
source                1..96
                      mol_type = other DNA
                      organism = synthetic construct

```

-continued

SEQUENCE: 397
 ggggtcccat ctcgtttcag tggcagtga tctgggacag atttcaactct caccatcagc 60
 agcctgcagc ctgaagatgt tgcaacttat tactgt 96

SEQ ID NO: 398 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 398
 ctgggcagtt atgattgtag tcgtggtgat tgttttgtt 39

SEQ ID NO: 399 moltype = DNA length = 33
 FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 399
 ttcggcggag gaaccaaggt ggaatcaaaa cgt 33

SEQ ID NO: 400 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 400
 acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgect gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aaggtggata acgcccctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
 cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgccctg cacaaagagc 300
 ttcaacaggg gagagtgtta g 321

SEQ ID NO: 401 moltype = AA length = 439
 FEATURE Location/Qualifiers
 REGION 1..439
 note = Engineered antibody sequence
 source 1..439
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 401
 QSLEESGGRL VTPGGSLLTL CTVSGIDVTN YYMQWVRQAP GKGLEWIGVI GVNGKRYIAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGLTVTVSSA STKGPSVFPL 120
 APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG LYSLSVVTV 180
 PSSSLGTQTY ICNVNHKPSN TKVDKRVPEK SCDKTHTCPP CPAPELLGGP SVFLFPPKPK 240
 DTLMISRTP E VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYAS TYRVVSVLTV 300
 LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM TKNQVSLTCL 360
 VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFFLYS KLTVDKSRWQ QGNVFSQSV 420
 HEALHNHYTQ KSLSLSPGK 439

SEQ ID NO: 402 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Engineered antibody sequence
 source 1..109
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 402
 QSLEESGGRL VTPGGSLLTL CTVSGIDVTN YYMQWVRQAP GKGLEWIGVI GVNGKRYIAS 60
 WAKGRFTISK TSSTTVDLKM TSLTTEDTAT YFCARGDIWG PGLTVTVSS 109

SEQ ID NO: 403 moltype = AA length = 29
 FEATURE Location/Qualifiers
 REGION 1..29
 note = Engineered antibody sequence
 source 1..29
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 403
 QSLEESGGRL VTPGGSLLTL CTVSGIDVT 29

SEQ ID NO: 404 moltype = AA length = 5
 FEATURE Location/Qualifiers

-continued

REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 404		
NYYMQ		5
SEQ ID NO: 405	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 405		
WVRQAPGKGL EWIG		14
SEQ ID NO: 406	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 406		
VIGVNGKRY ASWAKG		16
SEQ ID NO: 407	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
REGION	1..31	
	note = Engineered antibody sequence	
source	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 407		
RPTISKTSSST TVDLKMTSLT TEDTATYFCA R		31
SEQ ID NO: 408	moltype = length =	
SEQUENCE: 408		
000		
SEQ ID NO: 409	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 409		
WGPGLVTVS S		11
SEQ ID NO: 410	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 410		
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG	120
PSVFLFPPKP KDTLMISRTP	EVTCCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA	180
STYRVVSVLT VLHQDWLNGK	EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE	240
MTKNQVSLTC LVKGFYPSDI	AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW	300
QQGNVFSCSV MHEALHNHYT	QKSLSLSPGK	330
SEQ ID NO: 411	moltype = DNA length = 1320	
FEATURE	Location/Qualifiers	
misc_feature	1..1320	
	note = Engineered antibody sequence	
source	1..1320	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 411		
cagtcgctgg aggagtcagg	gggtcgccctg gtcacgcctg gaggatccct gacactcacc	60
tgcacagtcct ctggaatcga	cgctcactaac tactatatgc aatgggtccg ccagggtcca	120
gggaaggggc tggaatggat	cggagtcatt ggtgtgaatg gtaagagata ctacgcgagc	180
tgggcgaaag gccgattcac	catctccaaa acctcgtcga ccacggtgga tctgaaaatg	240

-continued

```

source                1..93
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 417
cgattcacca tctccaaaac ctcgtcgacc acggtggatc tgaaaatgac cagtctgaca 60
accgaggaca cggccaccta tttctgtgcc aga                               93

SEQ ID NO: 418        moltype =   length =
SEQUENCE: 418
000

SEQ ID NO: 419        moltype = DNA length = 33
FEATURE
misc_feature          1..33
                      note = Engineered antibody sequence
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 419
tggggcccgg ggaccctcgt caccgtctcg agc                               33

SEQ ID NO: 420        moltype = DNA length = 993
FEATURE
misc_feature          1..993
                      note = Engineered antibody sequence
source                1..993
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 420
gcctccacca agggcccac c ggtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacacagg ccctgggctg cctgggtcaag gactacttcc cgaaccggg gacgggtgtcg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggtgtcct acagtctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga 360
ccgtcagctc tcctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag cgcgggagga gcagtacgcc 540
agcacgtacc gtgtgggtcag cgtcctcacc gtccctgcacc aggactggct gaatggcaag 600
gagtacaagt gcaaggctct caacaaagcc ctcccagccc ccatcgagaa aaccatctcc 660
aaagccaaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtctcct ctctctctac agcaagctca ccgtgggaca gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 421        moltype = AA length = 219
FEATURE
REGION               1..219
                      note = Engineered antibody sequence
source               1..219
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 421
QVLTQTASPV SPAVGSTVTI NCRASQSVYY NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSNGDC FVFGGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSPNRGEC                               219

SEQ ID NO: 422        moltype = AA length = 113
FEATURE
REGION               1..113
                      note = Engineered antibody sequence
source               1..113
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 422
QVLTQTASPV SPAVGSTVTI NCRASQSVYY NNYLAWYQQK PGQPPKQLIY STSTLASGVS 60
SRFKGSGSGT QFTLTISDVQ CDDAATYYCL GSYDCSNGDC FVFGGGTEVV VKR                               113

SEQ ID NO: 423        moltype = AA length = 22
FEATURE
REGION               1..22
                      note = Engineered antibody sequence
source               1..22
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 423

```

-continued

QVLTQTASPV SPAVGSTVTI NC	22
SEQ ID NO: 424	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 424	
RASQSVYYNN YLA	13
SEQ ID NO: 425	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
	note = Engineered antibody sequence
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 425	
WYQQKPGQPP KQLIY	15
SEQ ID NO: 426	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = Engineered antibody sequence
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 426	
STSTLAS	7
SEQ ID NO: 427	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 427	
GVSSRPFKSG SGTQFTLTIS DVQCDDAATY YC	32
SEQ ID NO: 428	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 428	
LGSYDCSNGD CFV	13
SEQ ID NO: 429	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 429	
FGGTEVVVK R	11
SEQ ID NO: 430	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
	note = Engineered antibody sequence
source	1..106
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 430	
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS	60
KDSTYLSST LTLKADYK HKVYACEVTH QGLSSPVTKS FNRGEC	106
SEQ ID NO: 431	moltype = DNA length = 660
FEATURE	Location/Qualifiers
misc_feature	1..660
	note = Engineered antibody sequence
source	1..660
	mol_type = other DNA

-continued

```

                                organism = synthetic construct
SEQUENCE: 431
cagggtgctga cccagactgc atcccccgctg tctccagctg tgggaagcac agtcaccatc 60
aattgccggg ccagtcagag tgtttattat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcggttca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtaa tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga cggtggctgc accatctgtc 360
ttcatcttcc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctgt 420
ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
tcgggtaact ccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
agcagcaccg tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 432                moltype = DNA length = 339
FEATURE                        Location/Qualifiers
misc_feature                    1..339
                                note = Engineered antibody sequence
source                          1..339
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 432
cagggtgctga cccagactgc atcccccgctg tctccagctg tgggaagcac agtcaccatc 60
aattgccggg ccagtcagag tgtttattat aacaactacc tagcctggta tcagcagaaa 120
ccagggcagc ctcccaagca actgatctat tctacatcca ctctggcatc tggggtctca 180
tcgcggttca aaggcagtggt atctgggaca cagttcactc tcaccatcag cgacgtgcag 240
tgtgacgatg ctgccactta ctactgtcta ggcagttatg attgtagtaa tggtgattgt 300
tttgttttcg gcggaggggac cgaggtgggtg gtcaaactga 339

SEQ ID NO: 433                moltype = DNA length = 66
FEATURE                        Location/Qualifiers
misc_feature                    1..66
                                note = Engineered antibody sequence
source                          1..66
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 433
cagggtgctga cccagactgc atcccccgctg tctccagctg tgggaagcac agtcaccatc 60
aattgc 66

SEQ ID NO: 434                moltype = DNA length = 39
FEATURE                        Location/Qualifiers
misc_feature                    1..39
                                note = Engineered antibody sequence
source                          1..39
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 434
cgggccagtc agagtgttta ttataacaac tacctagcc 39

SEQ ID NO: 435                moltype = DNA length = 45
FEATURE                        Location/Qualifiers
misc_feature                    1..45
                                note = Engineered antibody sequence
source                          1..45
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 435
tggtatcagc agaaaccagg gcagcctccc aagcaactga tctat 45

SEQ ID NO: 436                moltype = DNA length = 21
FEATURE                        Location/Qualifiers
misc_feature                    1..21
                                note = Engineered antibody sequence
source                          1..21
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 436
tctacatcca ctctggcatc t 21

SEQ ID NO: 437                moltype = DNA length = 96
FEATURE                        Location/Qualifiers
misc_feature                    1..96
                                note = Engineered antibody sequence
source                          1..96
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 437
gggggtctcat cgcggttcaa aggcagtgga tctgggacac agttcactct caccatcagc 60

```

-continued

gacgtgcagt gtgacgatgc tgccacttac tactgt 96

SEQ ID NO: 438 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 438
 ctaggcagtt atgattgtag taatgggtgat tgttttgtt 39

SEQ ID NO: 439 moltype = DNA length = 33
 FEATURE Location/Qualifiers
 misc_feature 1..33
 note = Engineered antibody sequence
 source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 439
 ttcggcggag ggaccgaggt ggtgggtcaaa cgt 33

SEQ ID NO: 440 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Engineered antibody sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 440
 acggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
 actgcctctg ttgtgtgctt gctgaataac ttctatccca gagaggccaa agtacagtgg 120
 aaggtggata acgcccctca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
 aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
 cacaaagtct acgcctgcga agtcacccat cagggcctga gctcgccctg cacaaagagc 300
 ttcaacaggg gagagtgtta g 321

SEQ ID NO: 441 moltype = AA length = 441
 FEATURE Location/Qualifiers
 REGION 1..441
 note = Engineered antibody sequence
 source 1..441
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 441
 EVQLVESGGG LVQPGGSLRL SCAVSGIDVT NYMQWVRQA PGKGLEWVGV IGVNGKRYYA 60
 SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS SASTKGPSVF 120
 PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSV 180
 TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
 PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
 TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
 CLVKGFPYSD IAVEWESNGQ PENNYKTPP VLDSDGSFPL YSKLTVDKSR WQQGNVFS 420
 VMHEALHNHY TQKSLSLSPG K 441

SEQ ID NO: 442 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = Engineered antibody sequence
 source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 442
 EVQLVESGGG LVQPGGSLRL SCAVSGIDVT NYMQWVRQA PGKGLEWVGV IGVNGKRYYA 60
 SWAKGRFTIS RDNSKTTVYL QMNSLRAEDT AVYFCARGDI WGQGTLVTVS S 111

SEQ ID NO: 443 moltype = AA length = 30
 FEATURE Location/Qualifiers
 REGION 1..30
 note = Engineered antibody sequence
 source 1..30
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 443
 EVQLVESGGG LVQPGGSLRL SCAVSGIDVT 30

SEQ ID NO: 444 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Engineered antibody sequence

-continued

```

source          1..5
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 444
NYMQ                                                    5

SEQ ID NO: 445      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Engineered antibody sequence
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 445
WVRQAPGKGL EWVG                                         14

SEQ ID NO: 446      moltype = AA  length = 16
FEATURE            Location/Qualifiers
REGION             1..16
                   note = Engineered antibody sequence
source             1..16
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 446
VIGVNGKRY ASWAKG                                         16

SEQ ID NO: 447      moltype = AA  length = 32
FEATURE            Location/Qualifiers
REGION             1..32
                   note = Engineered antibody sequence
source             1..32
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 447
RFTISRDNK TTVYLQMNSL RAEDTAVYFC AR                      32

SEQ ID NO: 448      moltype =      length =
SEQUENCE: 448
000

SEQ ID NO: 449      moltype = AA  length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                   note = Engineered antibody sequence
source             1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 449
WGQGLVTVS S                                              11

SEQ ID NO: 450      moltype = AA  length = 330
FEATURE            Location/Qualifiers
REGION             1..330
                   note = Engineered antibody sequence
source             1..330
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 450
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS  60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG  120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA  180
STYRVVSVLT VLNQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE  240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW  300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK                      330

SEQ ID NO: 451      moltype = DNA  length = 1326
FEATURE            Location/Qualifiers
misc_feature       1..1326
                   note = Engineered antibody sequence
source             1..1326
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 451
gagggtgcagc ttgtggagtc tgggggaggc ttgggtccagc ctgggggggtc cctgagactc  60
tcctgtgcag tctctggaat cgacgtcact aactactaca tgcaatgggt ccgtcaggct  120
ccagggaagc ggctggagtg ggtcggagtc attggtgtga atggtgaag atactacgcg  180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt  240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgccag aggggacatc  300
tggggccaag ggaccctcgt caccgtctcg agcgcctcca ccaagggcc atcggctcttc  360

```

-continued

```

ccccctggcac cctcctccaa gageacctct gggggcacag cggccctggg ctgectggte 420
aaggactact tcccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgcctt ccagcagctt gggcaccacg acctacatct gcaacgtgaa tcacaagccc 600
agcaaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt ccccccaaaa 720
cccaaggaca cctcatgat ctcccggacc cctgaggtea catgcgtggt ggtggacgtg 780
agccacgaag accctgaggt caagttcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtac gccagcacgt accgtgtggt cagcgtctctc 900
accgtctctg accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcccctcccag ccccocatga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
cagggtgtaca ccttgcctcc atcccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aaggtctcta tcccagcgac atcgccgtgg agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctctcc gtgctggact ccgacggctc cttcttctc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcctg aggtctctga caaccactac acgcagaaga gcctctcctt gtctccgggt 1320
aatga

```

```

SEQ ID NO: 452      moltype = DNA length = 333
FEATURE             Location/Qualifiers
misc_feature        1..333
                    note = Engineered antibody sequence
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 452
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacgtcact aactactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtgtga atggtaagag atactacgcg 180
agctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gcctgagagc tgaggacact gctgtgtatt tctgtgccag aggggacatc 300
tggggccaaag ggacctctgt caccgtctcg agc

```

```

SEQ ID NO: 453      moltype = DNA length = 90
FEATURE             Location/Qualifiers
misc_feature        1..90
                    note = Engineered antibody sequence
source              1..90
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 453
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctggggggtc cctgagactc 60
tcctgtgcag tctctggaat cgacgtcact

```

```

SEQ ID NO: 454      moltype = DNA length = 15
FEATURE             Location/Qualifiers
misc_feature        1..15
                    note = Engineered antibody sequence
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 454
aactactaca tgcaa

```

```

SEQ ID NO: 455      moltype = DNA length = 42
FEATURE             Location/Qualifiers
misc_feature        1..42
                    note = Engineered antibody sequence
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 455
tgggtccgtc aggtctccag gaaggggctg gagtgggtcg ga

```

```

SEQ ID NO: 456      moltype = DNA length = 48
FEATURE             Location/Qualifiers
misc_feature        1..48
                    note = Engineered antibody sequence
source              1..48
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 456
gtcattggtg tgaatggtaa gagatactac gcgagctggg cgaaaggc

```

```

SEQ ID NO: 457      moltype = DNA length = 96
FEATURE             Location/Qualifiers
misc_feature        1..96
                    note = Engineered antibody sequence
source              1..96

```

-continued

```

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 457
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg 60
agagctgagg acaactgctgt gtatttctgt gccaga 96

SEQ ID NO: 458      moltype =      length =
SEQUENCE: 458
000

SEQ ID NO: 459      moltype = DNA length = 33
FEATURE             Location/Qualifiers
misc_feature         1..33
                     note = Engineered antibody sequence
source              1..33
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 459
tggggccaag ggaccctcgt caccgtctcg agc 33

SEQ ID NO: 460      moltype = DNA length = 993
FEATURE             Location/Qualifiers
misc_feature         1..993
                     note = Engineered antibody sequence
source              1..993
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 460
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cactctctgg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaactcag gcgccttgac cagcggcgctg cacaccttcc cggctgtcct acagtctctca 180
ggactctact cctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttggtg acaaaaactca cacatgccca ccgtgccag cactgaaact cctgggggga 360
ccgtcagtc tctcttccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtcctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtaacaagt gcaagggtct caacaaagcc ctcccagccc ccacgcagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcaggtgg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga 993

SEQ ID NO: 461      moltype = AA length = 219
FEATURE             Location/Qualifiers
REGION              1..219
                     note = Engineered antibody sequence
source              1..219
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 461
QVLTQSPSSL SASVGDRVTI NCRASQSVYY NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSNGDC FVFGGKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 462      moltype = AA length = 113
FEATURE             Location/Qualifiers
REGION              1..113
                     note = Engineered antibody sequence
source              1..113
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 462
QVLTQSPSSL SASVGDRVTI NCRASQSVYY NNYLAWYQQK PGKVPKQLIY STSTLASGVP 60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSNGDC FVFGGKVE IKR 113

SEQ ID NO: 463      moltype = AA length = 22
FEATURE             Location/Qualifiers
REGION              1..22
                     note = Engineered antibody sequence
source              1..22
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 463
QVLTQSPSSL SASVGDRVTI NC 22

```

-continued

SEQ ID NO: 464	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 464		
RASQSVYYNN YLA		13
SEQ ID NO: 465	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 465		
WYQQKPGKVP KQLIY		15
SEQ ID NO: 466	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 466		
STSTLAS		7
SEQ ID NO: 467	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 467		
GVPSRFGSGG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 468	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 468		
LGSYDCSNGD CFV		13
SEQ ID NO: 469	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 469		
FGGGTKVEIK R		11
SEQ ID NO: 470	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 470		
TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS		60
KDSTYLSST LTLSKADYK HKVYACEVTH QGLSSPVTKS FNRGEC		106
SEQ ID NO: 471	moltype = DNA length = 660	
FEATURE	Location/Qualifiers	
misc_feature	1..660	
	note = Engineered antibody sequence	
source	1..660	
	mol_type = other DNA	
	organism = synthetic construct	

-continued

SEQUENCE: 471
 caagtgtctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgccggg ccagtcagag tgtttactat aacaactacc tagcctggta tcagcagaaa 120
 ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
 tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
 cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtaa tgggtgattgt 300
 ttgttttctg gcgagggaac caaggtggaa atcaaacgta cgggtggctgc accatctgtc 360
 ttcattcttc cgccatctga tgagcagtg aaatctggaa ctgcctctgt tgtgtgctg 420
 ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
 tccggtaact cccaggagag tgtcacagag caggacagca aggacagcac ctacagcctc 540
 agcagcacc tgacgtctg caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
 gtcaccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 472 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 misc_feature 1..339
 note = Engineered antibody sequence
 source 1..339
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 472
 caagtgtctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgccggg ccagtcagag tgtttactat aacaactacc tagcctggta tcagcagaaa 120
 ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tgggggtcca 180
 tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
 cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtaa tgggtgattgt 300
 ttgttttctg gcgagggaac caaggtggaa atcaaacgt 339

SEQ ID NO: 473 moltype = DNA length = 66
 FEATURE Location/Qualifiers
 misc_feature 1..66
 note = Engineered antibody sequence
 source 1..66
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 473
 caagtgtctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgc 66

SEQ ID NO: 474 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 474
 cgggccagtc agagtgttta ctataacaac tacctagcc 39

SEQ ID NO: 475 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Engineered antibody sequence
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 475
 tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 476 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Engineered antibody sequence
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 476
 tctacatcca ctctggcatc t 21

SEQ ID NO: 477 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Engineered antibody sequence
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 477
 ggggtcccat ctcgtttcag tggcagtggg tctgggacag atttcaactc caccatcagc 60
 agcctgcagc ctgaagatgt tgcaacttat tactgt 96

-continued

```

SEQ ID NO: 478      moltype = DNA   length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = Engineered antibody sequence
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 478
ctgggcagtt atgattgtag taatggtgat tgttttgtt      39

SEQ ID NO: 479      moltype = DNA   length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 479
ttcggcggag gaaccaaggt ggaatcaaa cgt              33

SEQ ID NO: 480      moltype = DNA   length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Engineered antibody sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 480
acgggtggctg caccatctgt cttcatcttc ccgccatctg atgagcagtt gaaatctgga 60
actgcctctg ttgtgtgect gctgaataac ttctatccca gagaggccaa agtacagtgg 120
aagggtggata acgccctcca atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca cctacagcct cagcagcacc ctgacgctga gcaaagcaga ctacagagaaa 240
cacaaagtct acgctgcga agtcacccat cagggcctga gctcgcccg cacaagagc 300
ttcaacaggg gagagtgtta g              321

SEQ ID NO: 481      moltype = AA   length = 441
FEATURE            Location/Qualifiers
REGION             1..441
                    note = Engineered antibody sequence
source              1..441
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 481
QSVESGGGL VQPEGSLTLT CTASGFDFSS NAMWWVRQAP GKGLEWIGCI YNGDGSTYYA 60
SWVNGRFSIS KTSSTTVTLQ LNSLTVADTA TYCARDLDL WPGTLVTVS SASTKGPSVF 120
PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSG VHTPPAVLQS SGLYSLSSVV 180
TVPSSSLGTQ TYICNVNHKP SNTKVDKRVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYPSD IAVEWESNGQ PENNYKTPP VLDSDGSFFL YSKLTVDKSR WQQGNVFS 420
VMHEALHNHY TQKSLSLSPG K              441

SEQ ID NO: 482      moltype = AA   length = 111
FEATURE            Location/Qualifiers
REGION             1..111
                    note = Engineered antibody sequence
source              1..111
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 482
QSVESGGGL VQPEGSLTLT CTASGFDFSS NAMWWVRQAP GKGLEWIGCI YNGDGSTYYA 60
SWVNGRFSIS KTSSTTVTLQ LNSLTVADTA TYCARDLDL WPGTLVTVS S          111

SEQ ID NO: 483      moltype = AA   length = 29
FEATURE            Location/Qualifiers
REGION             1..29
                    note = Engineered antibody sequence
source              1..29
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 483
QSVESGGGL VQPEGSLTLT CTASGFDFS                      29

SEQ ID NO: 484      moltype = AA   length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                    note = Engineered antibody sequence
source              1..5

```

-continued

SEQUENCE: 484	mol_type = protein	
SNAMW	organism = synthetic construct	5
SEQ ID NO: 485	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
source	note = Engineered antibody sequence	
	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 485		
WVRQAPGKGL EWIG		14
SEQ ID NO: 486	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
source	note = Engineered antibody sequence	
	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 486		
CIYNGDGSTY YASWNG		17
SEQ ID NO: 487	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
REGION	1..31	
source	note = Engineered antibody sequence	
	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 487		
RFSISKTSST TVTLQLNSLT VADTATYYCA R		31
SEQ ID NO: 488	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
source	note = Engineered antibody sequence	
	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 488		
DLDL		4
SEQ ID NO: 489	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
source	note = Engineered antibody sequence	
	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 489		
WGPGLVTVS S		11
SEQ ID NO: 490	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
source	note = Engineered antibody sequence	
	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 490		
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG	120
PSVFLFPPKP KDTLMISRT	EVTQCVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA	180
STYRVVSVLT VLNQDWLNGK	EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE	240
MTKNQVSLTC LVKGFYPSDI	AVEWESNGQP ENNYKTTTPV LDSGSPFLY SKLTVDKSRW	300
QQGNVFSCSV MHEALHNHYT	QKSLSLSPGK	330
SEQ ID NO: 491	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	
misc_feature	1..1326	
source	note = Engineered antibody sequence	
	1..1326	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 491		
cagtcggtgg aggagtcggg gggaggcctg gtccagcctg agggatccct gacactcacc		60

-continued

```

tgcacagcct ctggattcga cttcagtagc aatgcaatgt ggtgggtccg ccaggtccca 120
gggaaggggc tggagtggat cggatgcatt tacaatgggt atggcagcac atactacgcg 180
agctgggtga atggccgatt ctccatctcc aaaacctcgt cgaccacggt gactctgcaa 240
ctgaatagtc tgacagtcgc ggacacggcc acgtattatt gtgcgagaga tcttgacttg 300
tggggcccg gcacccctgt caccgtctcg agcgctcca ccaagggccc atcggtcttc 360
ccccggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgctgggtc 420
aaggactact tccccgaacc ggtgacggtg tcgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtcc tcaggactct actccctcag cagcgtgggtg 540
accgtgcccc ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacaa gagagttgag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctcttt cccccaaaa 720
cccaaggaca cctcatgat ctcccggacc cctgaggtca catgctggtt ggtggacgtg 780
agccacgaag accctgaggt caagtcaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtc accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcctcccgag ccccatcgga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca cctcgcccc atcccggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggtca aagcttcta tcccagcgac atcgccgtgt agtgggagag caatgggcag 1140
ccggagaaca actacaagac cagcctccc gtgctggact ccgacggctc cttctcttc 1200
tacagcaagc tcaccgtgga caagagcagg tggcagcagg ggaacgtctt ctcatgctcc 1260
gtgatgcatg aggtctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aatga 1326

```

```

SEQ ID NO: 492      moltype = DNA length = 333
FEATURE
misc_feature       1..333
                    note = Engineered antibody sequence
source             1..333
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 492
cagtcggtgg aggagtcagg gggaggcctg gtccagcctg agggatccct gacactcacc 60
tgcacagcct ctggattcga cttcagtagc aatgcaatgt ggtgggtccg ccaggtccca 120
gggaaggggc tggagtggat cggatgcatt tacaatgggt atggcagcac atactacgcg 180
agctgggtga atggccgatt ctccatctcc aaaacctcgt cgaccacggt gactctgcaa 240
ctgaatagtc tgacagtcgc ggacacggcc acgtattatt gtgcgagaga tcttgacttg 300
tggggcccg gcacccctgt caccgtctcg agc 333

```

```

SEQ ID NO: 493      moltype = DNA length = 87
FEATURE
misc_feature       1..87
                    note = Engineered antibody sequence
source             1..87
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 493
cagtcggtgg aggagtcagg gggaggcctg gtccagcctg agggatccct gacactcacc 60
tgcacagcct ctggattcga cttcagtagc 87

```

```

SEQ ID NO: 494      moltype = DNA length = 15
FEATURE
misc_feature       1..15
                    note = Engineered antibody sequence
source             1..15
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 494
agcaatgcaa tgtgg 15

```

```

SEQ ID NO: 495      moltype = DNA length = 42
FEATURE
misc_feature       1..42
                    note = Engineered antibody sequence
source             1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 495
tgggtccgcc aggtccagg gaaggggctg gagggtgatg ga 42

```

```

SEQ ID NO: 496      moltype = DNA length = 51
FEATURE
misc_feature       1..51
                    note = Engineered antibody sequence
source             1..51
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 496
tgcatttaca atgggtgatg cagcacatac tacgcgagct ggggtgaatgg c 51

```


-continued

```

SEQ ID NO: 497      moltype = DNA length = 93
FEATURE            Location/Qualifiers
misc_feature        1..93
                    note = Engineered antibody sequence
source              1..93
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 497
cgattctcca tctccaaaac ctgctcgacc acggtgactc tgcaactgaa tagtctgaca 60
gtcgcgggaca cgccacgta ttattgtgcg aga                               93

SEQ ID NO: 498      moltype = DNA length = 12
FEATURE            Location/Qualifiers
misc_feature        1..12
                    note = Engineered antibody sequence
source              1..12
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 498
gatcttgact tg                                           12

SEQ ID NO: 499      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Engineered antibody sequence
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 499
tggggcccg gcaccctcgt caccgtctcg agc                               33

SEQ ID NO: 500      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Engineered antibody sequence
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 500
gcctccacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaaactcag gcgcccgtgac cagcggcgctg cacaccttcc cggctgtcct acagtccctca 180
ggactctact ccctcagcag cgtgggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacaagag agttgagccc 300
aaatcttgtg acaaaaactca cacatgcccc ccgtgcccag cacctgaact cctgggggga 360
ccgtcagtc tctcttcccc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgg ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtctcacc gtctgcacc aggactggct gaatggcaag 600
gagtacaagt gcaaggtctc caacaagcc ctcccagccc ccacgagaa aaccatctcc 660
aaagccaaag ggcagccccc agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accagggtcag ctgacctgc ctggcacaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactccg acggctcctt ctctctctac agcaagctca ccgtggacaa gagcagggtg 900
cagcagggga acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                               993

SEQ ID NO: 501      moltype = AA length = 219
FEATURE            Location/Qualifiers
REGION              1..219
                    note = Engineered antibody sequence
source              1..219
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 501
AIVMTQTPSS KSPVPGDVT INCQASESLY NNNALAWFQQ KPGQPPKRLI YDASKLASGV 60
PSRFSGGGSG TQFTLTISGV QCDDAATYYC GGYRSDSVDG VAFAGGTEVV VKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC                               219

SEQ ID NO: 502      moltype = AA length = 113
FEATURE            Location/Qualifiers
REGION              1..113
                    note = Engineered antibody sequence
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 502
AIVMTQTPSS KSPVPGDVT INCQASESLY NNNALAWFQQ KPGQPPKRLI YDASKLASGV 60

```

-continued

PSRFSGGGSG TQFTLTISGV QCDDAATYYC GGYRSDSVDG VAFAGGTEVV VKR	113
SEQ ID NO: 503	moltype = AA length = 23
FEATURE	Location/Qualifiers
REGION	1..23
	note = Engineered antibody sequence
source	1..23
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 503	
AIVMTQTPSS KSVPGDITV INC	23
SEQ ID NO: 504	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Engineered antibody sequence
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 504	
QASESLYNN ALA	13
SEQ ID NO: 505	moltype = AA length = 15
FEATURE	Location/Qualifiers
REGION	1..15
	note = Engineered antibody sequence
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 505	
WFQQKPGQPP KRLIY	15
SEQ ID NO: 506	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = Engineered antibody sequence
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 506	
DASKLAS	7
SEQ ID NO: 507	moltype = AA length = 32
FEATURE	Location/Qualifiers
REGION	1..32
	note = Engineered antibody sequence
source	1..32
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 507	
GVPSRFSGGG SGTQFTLTIS GVQCDDAATY YC	32
SEQ ID NO: 508	moltype = AA length = 12
FEATURE	Location/Qualifiers
REGION	1..12
	note = Engineered antibody sequence
source	1..12
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 508	
GGYRSDSVDG VA	12
SEQ ID NO: 509	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Engineered antibody sequence
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 509	
FAGGTEVVVK R	11
SEQ ID NO: 510	moltype = AA length = 106
FEATURE	Location/Qualifiers
REGION	1..106
	note = Engineered antibody sequence
source	1..106
	mol_type = protein
	organism = synthetic construct

-continued

SEQUENCE: 510
 TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
 KDSTYLSST LTLSKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 511 moltype = DNA length = 660
 FEATURE Location/Qualifiers
 misc_feature 1..660
 note = Engineered antibody sequence
 source 1..660
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 511
 gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc 60
 atcaattgcc aggccagtga gagtctttat aataacaacg ccttggcctg gtttcagcag 120
 aaaccagggc agcctcccaa gcgcctgac tatgatgcat ccaaactggc atctggggtc 180
 ccacgcgggt tcagtggcgg tgggtctggg acacagtcca ctctcaccat cagtggcgtg 240
 cagtgtgacg atgctgccac ttactactgt ggaggctaca gaagtgatag tgttgatggt 300
 gttgctttcg ccggaggggc cgaggtggtg gtcaaacgta cggtggctgc accatctgtc 360
 ttcattcttc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
 ctgaataaact tctatccccc agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
 tcgggtaact ccagggagag tgtcacagag caggacagca aggcacagcac ctacagcctc 540
 acgacgaccc tgacgctgag caaagcagac tacgagaaac acaaagtcta cgctgcgaa 600
 gtcacccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 512 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 misc_feature 1..339
 note = Engineered antibody sequence
 source 1..339
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 512
 gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc 60
 atcaattgcc aggccagtga gagtctttat aataacaacg ccttggcctg gtttcagcag 120
 aaaccagggc agcctcccaa gcgcctgac tatgatgcat ccaaactggc atctggggtc 180
 ccacgcgggt tcagtggcgg tgggtctggg acacagtcca ctctcaccat cagtggcgtg 240
 cagtgtgacg atgctgccac ttactactgt ggaggctaca gaagtgatag tgttgatggt 300
 gttgctttcg ccggaggggc cgaggtggtg gtcaaacgt 339

SEQ ID NO: 513 moltype = DNA length = 69
 FEATURE Location/Qualifiers
 misc_feature 1..69
 note = Engineered antibody sequence
 source 1..69
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 513
 gccatcgtga tgacccagac tccatcttcc aagtctgtcc ctgtgggaga cacagtcacc 60
 atcaattgc 69

SEQ ID NO: 514 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 514
 caggccagtg agagtcttta taataacaac gccttggcc 39

SEQ ID NO: 515 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Engineered antibody sequence
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 515
 tggtttcagc agaaaccagg gcagcctccc aagcgctga tctat 45

SEQ ID NO: 516 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Engineered antibody sequence
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 516
 gatgcatcca aactggcatc t 21

SEQ ID NO: 517	moltype = DNA length = 96	
FEATURE	Location/Qualifiers	
misc_feature	1..96	
	note = Engineered antibody sequence	
source	1..96	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 517		
gggggtcccat	cgcggtttcag	tggcggtggg tctgggacac agttcactct caccatcagt 60
ggcgtgcagt	gtgacgatgc	tgccacttac tactgt 96
SEQ ID NO: 518	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	
misc_feature	1..36	
	note = Engineered antibody sequence	
source	1..36	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 518		
ggaggctaca	gaagtgatag	tgttgatggg gttgct 36
SEQ ID NO: 519	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Engineered antibody sequence	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 519		
ttcgccggag	ggaccgaggt	ggtggtcaaa cgt 33
SEQ ID NO: 520	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321	
	note = Engineered antibody sequence	
source	1..321	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 520		
acggtggctg	caccatctgt	cttcattctc cgcctatctg atgagcagtt gaaatctgga 60
actgcctctg	ttgtgtgcct	gtcgaataac ttctatccca gagaggccaa agtacagtgg 120
aaggtggata	acgcctctca	atcgggtaac tcccaggaga gtgtcacaga gcaggacagc 180
aaggacagca	cctacagcct	cagcagcacc ctgacgctga gcaaagcaga ctacgagaaa 240
cacaaagtct	acgcctgcga	agtcacccat cagggcctga gctcgcccggt cacaaagagc 300
ttcaacaggg	gagagtgtta	g 321
SEQ ID NO: 521	moltype = AA length = 441	
FEATURE	Location/Qualifiers	
REGION	1..441	
	note = Engineered antibody sequence	
source	1..441	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 521		
EVQLVESGGG	LVPQGGSLRL	SCAVSGIGLS SYIYQWVRQA PGKGLEWVGV IGSDGKITYYA 60
TWAKGRFTIS	RDNSKTTVYL	QMNSLRAEDT AVYFCTRSDI WQGGLTVTVS SASTKGPSVF 120
PLAPSSKSTS	GGTAALGCLV	KDYFPEPVTV SWNSGALTSG VHTFPAVLQS SGLYSLSSVV 180
TVPSSSLGTQ	TYICNVNHKP	SNTKVDARVE PKSCDKHTC PPCPAPELLG GPSVFLFPPK 240
PKDTLMISRT	PEVTCVVVDV	SHEDPEVKFN WYVDGVEVHN AKTKPREEQY ASTYRVVSVL 300
TVLHQDWLNG	KEYKCKVSNK	ALPAIEKTI SKAKGQPREP QVYTLPPSRE EMTKNQVSLT 360
CLVKGFPYSD	IAVEWESNGQ	PENNYKTPPP VLDSGGSFFL YSKLTVDKSR WQQGNVPSCS 420
VMHEALHNHY	TQKLSLSLSPG	K 441
SEQ ID NO: 522	moltype = AA length = 111	
FEATURE	Location/Qualifiers	
REGION	1..111	
	note = Engineered antibody sequence	
source	1..111	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 522		
EVQLVESGGG	LVPQGGSLRL	SCAVSGIGLS SYIYQWVRQA PGKGLEWVGV IGSDGKITYYA 60
TWAKGRFTIS	RDNSKTTVYL	QMNSLRAEDT AVYFCTRSDI WQGGLTVTVS S 111
SEQ ID NO: 523	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Engineered antibody sequence	

-continued

source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 523		
EVQLVESGGG LVQPGGSLRL SCAVSGIGLS		30
SEQ ID NO: 524	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Engineered antibody sequence	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 524		
SYMQ		5
SEQ ID NO: 525	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Engineered antibody sequence	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 525		
WVRQAPGKGL EWVG		14
SEQ ID NO: 526	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Engineered antibody sequence	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 526		
VIGSDGKTTY ATWAKG		16
SEQ ID NO: 527	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 527		
RFTISRDNK TTVYLQMSL RAEDTAVYFC TR		32
SEQ ID NO: 528	moltype = length =	
SEQUENCE: 528		
000		
SEQ ID NO: 529	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 529		
WGQGLVTVS S		11
SEQ ID NO: 530	moltype = AA length = 330	
FEATURE	Location/Qualifiers	
REGION	1..330	
	note = Engineered antibody sequence	
source	1..330	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 530		
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS		60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDARVEP KSCDKTHTCP PCPAPELLGG		120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYA		180
STYRVSVLT VLHQDWLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE		240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW		300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK		330
SEQ ID NO: 531	moltype = DNA length = 1326	
FEATURE	Location/Qualifiers	
misc_feature	1..1326	
	note = Engineered antibody sequence	

-continued

```

source          1..1326
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 531
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtaaagc atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gccctgagagc tgaggacact gctgtgtatt tctgtaccag agggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agcgcctcca ccaagggccc atcgggtcttc 360
cccttggcac cctcctccaa gagcacctct gggggcacag cggccctggg ctgcttggtc 420
aaggactact tcccgaacc ggtgacggtg tctgtggaact caggcgccct gaccagcggc 480
gtgcacacct tcccggctgt cctacagtc tccagactct actccctcag cagcgtggtg 540
accgtgacct ccagcagctt gggcacccag acctacatct gcaacgtgaa tcacaagccc 600
agcaacacca aggtggacgc gagagttag cccaaatctt gtgacaaaac tcacacatgc 660
ccaccgtgcc cagcacctga actcctgggg ggaccgtcag tcttctctt cccccaaaa 720
cccaaggaca cctccatgat ctcccggacc cctgaggta catgctgggt ggtggacgtg 780
agccacgaag accttgaggt caagtccaac tggtagctgg acggcgtgga ggtgcataat 840
gccaagacaa agccgcggga ggagcagtag gccagcagc accgtgtggt cagcgtcttc 900
accgtcctgc accaggactg gctgaatggc aaggagtaca agtgcaaggt ctccaacaaa 960
gcccctccag ccccctatga gaaaaccatc tccaaagcca aagggcagcc ccgagaacca 1020
caggtgtaca ccttgccccc atcccgggag gagatgacca agaaccaggt cagcctgacc 1080
tgcttggta aaggcttcta tcccagcgac atcgccgtgg agtgggagag caatggggcag 1140
ccggagaaca actacaagac caccgctccc gtgctggact ccgacggctc cttcttctc 1200
tacagcaagc tcaccgtgga caagacgagg tggcagcagg ggaacgtctt ctcattgctc 1260
gtgatgcatg aggtctctga caaccactac acgcagaaga gcctctccct gtctccgggt 1320
aaatga                                             1326

SEQ ID NO: 532      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature       1..333
                  note = Engineered antibody sequence
source            1..333
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 532
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt agctactaca tgcaatgggt ccgtcaggct 120
ccagggaagg ggctggagtg ggtcggagtc attggtagtg atggtaaagc atactacgcg 180
acctgggcga aaggccgatt caccatctcc agagacaatt ccaagaccac ggtgtatctt 240
caaatgaaca gccctgagagc tgaggacact gctgtgtatt tctgtaccag agggggacatc 300
tggggccaag ggaccctcgt caccgtctcg agc 333

SEQ ID NO: 533      moltype = DNA length = 90
FEATURE            Location/Qualifiers
misc_feature       1..90
                  note = Engineered antibody sequence
source            1..90
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 533
gaggtgcagc ttgtggagtc tgggggaggc ttggtccagc ctgggggggtc cctgagactc 60
tcctgtgcag tctctggaat cggcctcagt 90

SEQ ID NO: 534      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature       1..15
                  note = Engineered antibody sequence
source            1..15
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 534
agctactaca tgcaa 15

SEQ ID NO: 535      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature       1..42
                  note = Engineered antibody sequence
source            1..42
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 535
tgggtccgtc aggtccagg gaaggggctg gagggggtcg ga 42

SEQ ID NO: 536      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature       1..48
                  note = Engineered antibody sequence
source            1..48

```

-continued

```

                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 536
gtcattggta gtgatggtaa gacatactac gcgacctggg cgaaaggc      48

SEQ ID NO: 537                moltype = DNA length = 96
FEATURE                      Location/Qualifiers
misc_feature                  1..96
                              note = Engineered antibody sequence
source                       1..96
                              mol_type = other DNA
                              organism = synthetic construct

SEQUENCE: 537
cgattcacca tctccagaga caattccaag accacgggtg atcttcaaat gaacagcctg  60
agagctgagg acaactgctgt gtatttctgt accaga                          96

SEQ ID NO: 538                moltype = length =
SEQUENCE: 538
000

SEQ ID NO: 539                moltype = DNA length = 33
FEATURE                      Location/Qualifiers
misc_feature                  1..33
                              note = Engineered antibody sequence
source                       1..33
                              mol_type = other DNA
                              organism = synthetic construct

SEQUENCE: 539
tggggccaaag ggaccctcgt caccgtctcg agc                          33

SEQ ID NO: 540                moltype = DNA length = 993
FEATURE                      Location/Qualifiers
misc_feature                  1..993
                              note = Engineered antibody sequence
source                       1..993
                              mol_type = other DNA
                              organism = synthetic construct

SEQUENCE: 540
gcctccacca agggcccacgc ggtcttcccc ctggcaccct cctccaagag cacctctggg  60
ggcacagcgg ccctgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgctg 120
tggaaactcag gcgccttgac cagcggcgctg cacaccttcc cggetgtcct acagtcctca 180
ggactctact ccctcagcag cgtggtgacc gtgccctcca gcagcttggg caccagacc 240
tacatctgca acgtgaatca caagcccagc aacaccaagg tggacgcgag agttgagccc 300
aaatcttgtg acaaaaactca cacatgccca ccgtgcccag cacctgaact cctgggggga 360
ccgtcagtc tctcttctcc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaaag ccgggggagga gcagtacgcc 540
agcacgtacc gtgtggtcag cgtctcacc gtctgcacc aggactgggt gaatggcaag 600
gagtacaagt gcaaggtctc caacaaagcc ctcccagccc ccacgagaa aacctctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaag gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa ttggcagccg gagaacaact acaagaccac gcctcccggt 840
ctggactccg acggtcctt ctctctctac agcaagtcga ccgtggacaa gagcaggtgg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                          993

SEQ ID NO: 541                moltype = AA length = 219
FEATURE                      Location/Qualifiers
REGION                      1..219
                              note = Engineered antibody sequence
source                       1..219
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 541
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP  60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC                          219

SEQ ID NO: 542                moltype = AA length = 113
FEATURE                      Location/Qualifiers
REGION                      1..113
                              note = Engineered antibody sequence
source                       1..113
                              mol_type = protein
                              organism = synthetic construct

SEQUENCE: 542
QVLTQSPSSL SASVGDRVTI NCQASQNVYN NNYLAWYQQK PGKVPKQLIY STSTLASGVP  60
SRFSGSGSGT DFTLTISLQ PEDVATYYCL GSYDCSRGDC FVFGGGTKVE IKR          113

```

-continued

SEQ ID NO: 543	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Engineered antibody sequence	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 543		
QVLTQSPSSL SASVGDRVTI NC		22
SEQ ID NO: 544	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 544		
QASQNVYNNN YLA		13
SEQ ID NO: 545	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Engineered antibody sequence	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 545		
WYQQKPGKVP KQLIY		15
SEQ ID NO: 546	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Engineered antibody sequence	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 546		
STSTLAS		7
SEQ ID NO: 547	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
REGION	1..32	
	note = Engineered antibody sequence	
source	1..32	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 547		
GVPSRFGSG SGTDFTLTIS SLQPEDVATY YC		32
SEQ ID NO: 548	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Engineered antibody sequence	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 548		
LGSYDCSRGD CFV		13
SEQ ID NO: 549	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Engineered antibody sequence	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 549		
FGGGTKVEIK R		11
SEQ ID NO: 550	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
REGION	1..106	
	note = Engineered antibody sequence	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 550		

-continued

TVAAPSVFIF PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
 KDSTYLSST LTLSKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 551 moltype = DNA length = 660
 FEATURE Location/Qualifiers
 misc_feature 1..660
 note = Engineered antibody sequence
 source 1..660
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 551
 caagtgcctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
 ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tggggtocca 180
 tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
 cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tgggtgattgt 300
 tttgttttcg gcggaggaa atcaaactgta cgggtggctgc accatctgtc 360
 ttcattcttc cgccatctga tgagcagttg aaatctggaa ctgcctctgt tgtgtgctg 420
 ctgaataact tctatcccag agaggccaaa gtacagtggg aggtggataa cgccctccaa 480
 tcgggttaact ccaggagag tgtaacagag caggacagca aggcacgac ctacagcctc 540
 agcagcaccg tgacgctgag caaagcagac tacgagaac acaaagtcta cgctgcgaa 600
 gtcacccatc agggcctgag ctgcgccgtc acaaagagct tcaacagggg agagtgttag 660

SEQ ID NO: 552 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 misc_feature 1..339
 note = Engineered antibody sequence
 source 1..339
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 552
 caagtgcctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgccagg ccagtcagaa tgtttacaat aacaactacc tagcctggta tcagcagaaa 120
 ccagggaag ttctaagca actgatctat tctacatcca ctctggcatc tggggtocca 180
 tctcgtttca gtggcagtg atctgggaca gatttcactc tcaccatcag cagcctgcag 240
 cctgaagatg ttgcaactta ttactgtctg ggcagttatg attgtagtcg tgggtgattgt 300
 tttgttttcg gcggaggaa caagtgga atcaaactg 339

SEQ ID NO: 553 moltype = DNA length = 66
 FEATURE Location/Qualifiers
 misc_feature 1..66
 note = Engineered antibody sequence
 source 1..66
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 553
 caagtgcctga cccagctctcc atcctccctg tctgcatctg taggagacag agtcaccatc 60
 aattgc 66

SEQ ID NO: 554 moltype = DNA length = 39
 FEATURE Location/Qualifiers
 misc_feature 1..39
 note = Engineered antibody sequence
 source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 554
 caggccagtc agaattgtta caataacaac tacctagcc 39

SEQ ID NO: 555 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Engineered antibody sequence
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 555
 tggatcagc agaaaccagg gaaagttcct aagcaactga tctat 45

SEQ ID NO: 556 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Engineered antibody sequence
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 556
 tctacatcca ctctggcatc t 21

source	1..330				
	mol_type = protein				
	organism = synthetic construct				
SEQUENCE: 564					
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS 60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVEP	KSCDKTHTCP	PCPAPELLGG 120
PSVFLFPPKP	KDTLMISRT	EVTCTVVVDV	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYA 180
STYRVVSVLT	VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE 240
MTKNQVSLTC	LVKGFYPDSI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW 300
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK			330
SEQ ID NO: 565	moltype = AA	length = 329			
FEATURE	Location/Qualifiers				
REGION	1..329				
	note = Engineered antibody sequence				
source	1..329				
	mol_type = protein				
	organism = synthetic construct				
SEQUENCE: 565					
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS 60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVEP	KSCDKTHTCP	PCPAPELLGG 120
PSVFLFPPKP	KDTLMISRT	EVTCTVVVDV	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYA 180
STYRVVSVLT	VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE 240
MTKNQVSLTC	LVKGFYPDSI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW 300
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPG			329
SEQ ID NO: 566	moltype = AA	length = 440			
FEATURE	Location/Qualifiers				
REGION	1..440				
	note = Engineered antibody sequence				
source	1..440				
	mol_type = protein				
	organism = synthetic construct				
SEQUENCE: 566					
EVQLVESGGG	LVPQGGSLRL	SCAVSGIDL	SVYMNWVRQA	PGKGLEWGV	IGINGATYYA 60
SWAKGRFTIS	RDNSKTTVYL	QMNLSRAEDT	AVYFPCARGDI	WQQTGLTVTS	SASTKGPSVF 120
PLAPSSKSTS	GGTAALGCLV	KDYFPEPVT	SWNSGALTSG	VHTFPFPAVLQ	SGLYSLSSVV 180
TVPSSSLGTQ	TYICNVNHKP	SNTKVDARVE	PKSCDKTHTC	PPCPAPELLG	GPSVFLFPPK 240
PKDTLMISRT	PEVTCVVVDV	SHEDPEVKFN	WYVDGVEVHN	AQYTKPREEQY	ASTYRVVSVL 300
TVLHQDWLNG	KEYKCKVSNK	ALPAPIEKT	SKARGQPREP	KKTLLPPSRE	EMTKNQVSLT 360
CLVKGFYPSD	IAVEWESNGQ	PENNYKTTTP	VLDSDGSFFL	YSKLTVDKSR	WQQGNVFSCS 420
VMHEALHNHY	TQKSLSLSPG				440
SEQ ID NO: 567	moltype = DNA	length = 1323			
FEATURE	Location/Qualifiers				
misc_feature	1..1323				
	note = Engineered antibody sequence				
source	1..1323				
	mol_type = other DNA				
	organism = synthetic construct				
SEQUENCE: 567					
gaggtgcagc	tgtgtggagtc	tggggggaggc	tgtgtccagc	ctgggggggctc	cctgagagctc 60
tctctgtgca	tctcttggaa	cgacctcagtc	ggctactaca	tgaactgggtc	ccgtcagggtc 120
ccagggaagc	ggctggagtg	ggctcgaggtc	attggtatta	atggtgccac	atactacgcg 180
agctggggcg	aaggccgatt	caccattctcc	agagacaaatt	ccaagaccac	gggtgatctt 240
caaatgaaca	gcctgagagc	tgaggacact	gcgtgtgatt	tctgtgctag	agggggacatc 300
tggggccaag	ggcgacctgc	caccgtctcg	agcgctccc	ccaagggccc	atcggtcttc 360
ccctctggca	cctcctccaa	gagcaccctc	ggggggcacag	cgggccctggg	ctgectgggtc 420
aaggactact	tcccgcgaac	gggtgacggtg	tcgttgaaact	caggcgccct	gaccagcggc 480
gtgcacacct	tcccgctgtt	ctcacagctc	tcaggactct	actccctcag	cagcgctggtg 540
accgtgcacct	ccagcagctg	gggcaccagc	acctacatct	gcaacgtgaa	tcacaagccc 600
agcaaacacca	aggtggacgc	gagagattgag	cccaaatctt	gtgcacaaaac	tcacacatgc 660
ccaccgtgcc	cagcacctga	actcctggggc	ggaccgtcag	tcttctctct	ccccccaaaa 720
cccaaggaca	ccctcatgat	ctcccggggc	actgagtgca	catgcgtggt	gggtggacgtg 780
agccacgaag	acctgaggtg	caagttcaac	tggtacgtgg	acggcgctgga	gggtgcataat 840
gccaagaacaa	agccgcggga	ggagcagtac	gccagcagct	accgtgtggt	cagcgtctctc 900
accgtctctg	accaggactg	gctggaatggc	aaggagactca	agtgcaaggt	ctccaacaaa 960
gccctcccag	ccccactcga	gaaaaccatc	tccaaagcca	aaggggcagc	ccgagaaccca 1020
caggtgtaca	ccttgccccc	atcccgggag	gagatgacca	agaaccagggt	cagcctgacc 1080
tgcttggtca	aaggcttcta	tcccagcgac	atcgccgtgg	agtgggagag	caatggggcag 1140
ccggagaacaa	actacaagac	cacgctctcc	gtgctggagt	ccgacggctc	ctctctcttc 1200
tcagcaagc	tcaccgtgga	caagagcagg	tggcagcagg	ggaacgtctt	ctcatgctcc 1260
gtatgatcatg	aggctctgca	caaccactac	acgcagaaga	gcctctctct	gtctccgggt 1320
tga					1323

241

What is claimed is:

1. A method of treating migraine in a patient, comprising intravenously administering to the patient an effective amount of an anti-calcitonin gene related peptide (CGRP) antibody comprising: (a) a variable light chain polypeptide comprising the light chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively; and (b) a variable heavy chain polypeptide comprising the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively, wherein said administration starts after the onset of a migraine attack while the patient is exhibiting at least one symptom of the migraine attack.

2. The method of claim 1, wherein said at least one symptom comprises one or more of headache, nausea, photophobia, and phonophobia.

3. The method of claim 2, wherein said administration begins within 1-6 hours of the onset of at least one symptom of said migraine attack.

4. The method of claim 1, wherein said administering begins between about 1-6 hours from the onset of said migraine attack.

5. The method of claim 1, wherein said administering is by intravenous infusion.

6. The method of claim 1, wherein said administering is by intravenous infusion over a period of approximately 30 minutes to 60 minutes.

7. The method of claim 1, wherein said administering is by intravenous infusion over a period of approximately 30 minutes.

8. The method of claim 1, wherein said effective amount is: between about 100 mg and about 300 mg; or is about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg.

9. The method of claim 1, wherein said anti-CGRP antibody is administered in a 0.9% sodium chloride solution, optionally 100 ml of a 0.9% sodium chloride solution.

242

10. The method of claim 1, wherein said administering is by intravenous infusion of a solution comprising about 100 mg to about 300 mg of said anti-CGRP antibody.

11. The method of claim 1, wherein said administering is by intravenous infusion of a 0.9% sodium chloride solution comprising about 100 mg to about 300 mg of said anti-CGRP antibody.

12. The method of claim 1, wherein said administering is by intravenous infusion of 100 ml of a 0.9% sodium chloride solution comprising about 100 mg to about 300 mg of said anti-CGRP antibody over a period of approximately 30 minutes.

13. The method of claim 1, wherein: (a) the variable light chain polypeptide comprises the amino acid sequence of SEQ ID NO: 222; and (b) the variable heavy chain polypeptide comprises the amino acid sequence of SEQ ID NO: 202.

14. The method of claim 1, wherein said anti-CGRP antibody is an IgG molecule.

15. The method of claim 1, wherein said anti-CGRP antibody comprises: (a) the light chain polypeptide of SEQ ID NO: 221; and (b) the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

16. The method of claim 15, wherein said anti-CGRP antibody consists of said light and heavy chain polypeptides.

17. The method of claim 1, wherein said anti-CGRP antibody is expressed in or obtained by expression in: a yeast cell, optionally a *Pichia pastoris* cell; or a mammalian cell, optionally a CHO cell.

18. The method of claim 1, wherein the patient has episodic migraine.

19. The method of claim 1, wherein the patient has chronic migraine.

20. The method of claim 1, wherein said anti-CGRP antibody is an IgG1 molecule.

21. The method of claim 12, wherein said anti-CGRP antibody is an IgG1 molecule.

* * * * *